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Preface

"Was war also das Leben? Es war Wärme, das Wärmeprodukt formerhaltender Bestandlosigkeit, ein Fieber der Materie, von welchem der Prozeß unaufhörlicher Zersetzung und Wiederherstellung unhaltbar verwickelt, unhaltbar kunstreich aufgebauter Eiweißmolekel begleitet war. Es war das Sein des eigentlich Nicht-sein-Könnenden, des nur in diesem verschränkten und fiebrigen Prozeß von Zerfall und Erneuerung mit süß-schmerzlich-genaue Not auf dem Punkte des Seins Balancierenden. Es war nicht materiell und es war nicht Geist. Es war etwas zwischen beidem, ein Phänomen, getragen von Materie, gleich dem Regenbogen auf dem Wasserfall und gleich der Flamme."

Thomas Mann, *Der Zauberberg*

"What then was life? It was warmth, the warmth generated by a form-preserving instability, a fever of matter, which accompanied the process of ceaseless decay and repair of albumen molecules that were too impossibly complicated, too impossibly ingenious in structure. It was the existence of the actually impossible-to-exist, of a half-sweet, half-painful balancing, or scarcely balancing, in this restricted and feverish process of decay and renewal, upon the point of existence. It was not matter and it was not spirit, but something between the two, a phenomenon conveyed by matter, like the rainbow on the waterfall, and like the flame."*

Thomas Mann, *The Magic Mountain*

When I started to teach biophysics to biology students at the Friedrich-Schiller University in Jena in 1965 the question arose: What actually is biophysics? What should I teach? Only one thing seemed to be clear to me: biophysics is neither "physics for biologists" nor "physical methods applied to biology" but a modern field of science leading to new approaches for our understanding of biological functions.

Rashevsky's book on *Mathematical Biophysics* (1960), the classical approaches of Ludwig von Bertalanffy (1968), as well as the excellent book by Katchalsky and Curran on *Nonequilibrium Thermodynamics in Biophysics* (1965), showed me new ways of looking at biological processes. Thus, I came to the conclusion that it would be worthwhile trying to integrate all these various physical and physicochemical approaches to biological problems into a new discipline called "biophysics". The first German edition of this textbook, published in 1971, was developed from these considerations.

Meanwhile, I had moved from Jena to the Humboldt-University in Berlin where I organized courses for biologists specializing in

* Translated by H. T. Lowe-Porter, Penguin Books, 1985, p. 275-276

biophysics. The idea was, why should only physicists find their way to biophysics? Why not help biologists to overcome the "activation energy" barrier of mathematics and physics to discover this fascinating discipline?

In Berlin, a special group was established (1970) in the Department of Biology with the aim of teaching biophysics. This led to a full university degree course of biophysics which has developed successfully and attracts an increasing number of students today.

Consequently, my co-workers and I had the responsibility of organizing not only introductory courses to biophysics for biology students, but also advanced courses in molecular biophysics, biomechanics, membrane biophysics, bioelectrochemistry, environmental biophysics and various aspects of theoretical biophysics.

The evolution of this textbook in the following years was the result of these courses. Innumerable discussions with students, colleagues and friends led to continuous refinement and modification of the contents of this book, resulting in a second, third, and in 1996, a fourth German edition. New topics were added, others updated or even deleted. The only sentences that remained unchanged are those of Thomas Mann at the beginning of the Preface.

The philosophy of this book is that biophysics is not a simple collection of physical approaches to biology, but a defined discipline with its own network of ideas and approaches, spanning all hierarchical levels of biological organization. The paradigm of a holistic view of biological functions, where the biological system is not simply the sum of its molecular components but is rather their functional integration, seems to be the main concept of biophysics.

While it is easier to realize such an integrated view in a 'one-man book', this has, of course, the disadvantage that the knowledge and experience of many specialists cannot be incorporated. However, to a certain degree this problem has been compensated for by discussions with colleagues and friends and by their continuous support over a period of more than three decades. Further problems are the selection of the topics to be included in the book and the emphasis placed on the different aspects, avoiding underestimation of others. Although the author has tried to balance the selection and emphasis of topics by looking at the development of biophysics over the last three decades, he is not sure that he has succeeded. Even if this is the case, this book will at least help to answer the question: What is biophysics? It provides a solid introduction to biophysics. For further reading, books and reviews are recommended at the end of each chapter. The extensive index at the end of the book ensures an easy orientation and will enable this book to be used as a reference work. As mentioned above, this book is written primarily

for biologists and biophysicists with a background in biology. Therefore, some basic knowledge of biology is required, but less knowledge of physics and mathematics is needed. It should encourage biologists to enter the field of biophysics and stimulate further research. The German editions have shown that physicists also will profit from reading this book.

This first English edition is not just a translation of the fourth German edition, but is rather a fully revised fifth edition. For an author, it is impossible to translate his book without substantial rewriting and refining. All chapters have been more or less revised, and results which have been published since the last edition have been integrated. Many figures have been redrawn, some are new; some totally new chapters have also been included.

Last, but not least, I wish to express again my sincere gratitude to all of my colleagues and friends, throughout the world, who helped me before with all previous editions and especially for helping me with this English edition. Thanks are also extended to the staff of Springer-Verlag for encouraging me to write this English version and for correcting my imperfect English.

Berlin, July 2000

ROLAND GLASER

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List of Fundamental Constants and Symbols

The numbers in parentheses indicate equations in the text, where the symbols are explained or defined.

$\equiv$	identical
$=$	equal
$\equiv$	equal by definition
$\approx$	approximately equal
$\sim$	proportional
A	Arrhenius-coefficient [Eq. (2.3.17)]
A	affinity [Eq. (3.1.75)]
A	area
a	chemical activity [Eq. (3.1.34)]
B	magnetic flux density [Eq. (4.4.1)]
b	electrophoretic mobility [Eq. (2.5.13)]
C	electric capacity [Eq. (3.5.6)]
C^*	complex electric capacitance [Eq. (3.5.11)]
C_l	clearance-constant [Eq. (5.2.28)]
C_V	isochoric heat capacity [Eq. (2.4.25)]
c	molar concentration
c_0	speed of light in vacuum = $2.998 \cdot 10^8 \text{ m s}^{-1}$
D	diffusion coefficient [Eq. (3.3.6)]
E	energy (general expression)
E^0	standard redox potential [Eq. (2.2.1)]
E	electric field strength [Eq. (2.4.4)]
e	basis of natural logarithm = 2.71828
e	absolute amount of charge on electron = $1.60218 \cdot 10^{-19} \text{ C}$
F	mechanical force
F	Faraday = $9.6485 \cdot 10^4 \text{ C val}^{-1}$
F	Helmholtz free energy [Eq. (3.1.22)]
f	symbol for an arbitrary function
f	generalized coefficient of friction [Eq. (3.1.52)]
f	activity coefficient [Eq. (3.1.34)]
G	Gibbs free energy [Eq. (3.1.23)]
G	electrical conductivity [Eq. (3.5.2)]
g	specific conductivity [Eq. (3.5.2)]
g	osmotic coefficient [Eq. (3.2.26)]

<i>H</i>	enthalpy [Eq. (3.1.21)]
H	magnetic field strength [Eq. (4.4.1)]
<i>h</i>	Planck's constant = $6.626 \cdot 10^{-34}$ J s
<i>I</i>	sound intensity [Eq. (4.3.3)]
<i>I</i>	information [Eq. (2.3.7)]
<i>I</i>	ionic strength [Eq. (2.4.16)]
<i>I_A</i>	second moment of area [Eq. (3.6.13)]
<i>I_P</i>	polar second moment of area [Eq. (3.6.16)]
i	unit vector in <i>x</i> -direction
j	imaginary unit = $\sqrt{-1}$
j	unit vector in <i>y</i> -direction
<i>j</i>	electric current density [Eq. (4.4.7)]
J	flux [Eq. (3.3.1)]
<i>J</i>	unidirectional flux in kinetic equations
<i>K_p</i>	equilibrium constant of isobaric chemical reactions [Eq. (3.1.71)]
<i>K</i>	bending ($K = 1/R$), (Sect. 3.6.4)
<i>k</i>	Boltzmann's constant = $1.380658 \cdot 10^{-23}$ J K ⁻¹ = $8.6174 \cdot 10^{-5}$ eV K ⁻¹
<i>k</i>	rate constant
<i>L_i</i>	phenomenological coefficient relating a flow to a force [Eq. (3.1.49)]
<i>L</i>	decibel intensity of sound [Eq. (4.3.3)]
<i>l</i>	second (or: azimuthal) quantum number (Sect. 2.1.1)
<i>l</i>	distance, length
<i>M</i>	moment of force [Eq. (3.6.12)]
<i>M</i>	molar mass
<i>m</i>	mass
<i>m</i>	magnetic quantum number (Sect. 2.1.1)
<i>N</i>	Avogadro's number = $6.0221367 \cdot 10^{23}$ mol ⁻¹
<i>n</i>	primary (or: principal) quantum number (Sect. 2.1.1)
<i>n</i>	number of particles, individuals etc.
<i>P</i>	mathematical probability (Sect. 2.3.1)
<i>P</i>	permeability coefficient [Eq. (3.3.9)]
<i>P</i>	electrical power density [Eq. (4.4.7)]
<i>p</i>	pressure
Pa	Pascal
Q	heat
<i>q</i>	electric charge
<i>R</i>	molar gas constant = 8.314510 J K ⁻¹ mol ⁻¹
<i>R</i>	radius of curvature ($R = 1/K$), [Eq. (3.6.8)]
<i>R</i>	resistance coefficient relating a flow to a force [Eq. (3.1.50)]
<i>R</i>	Ohm's resistance (reactance), (Sect. 3.5.3)
Re	Reynolds number [Eq. (3.7.1)]
<i>r</i>	radius, radial distance

r	Donnan ratio [Eq. (3.2.38)]
S	entropy [Eqs. (2.3.4) and (3.1.10)]
s	spin quantum number (Sect. 2.1.1)
T	temperature
t	time
U	internal energy [Eq. (3.1.9)]
V	volume
$\bar{V}$	partial molar volume [Eq. (3.1.8)]
$\mathbf{v}$	velocity
W	thermodynamic probability [Eq. (2.3.4)]
W	work [Eq. (3.1.9)]
X	generalized force [Eq. (3.1.42)]
x	coordinate in an orthogonal system
x	mole fraction [Eq. (3.1.35)]
Y	Young's modulus [Eq. (3.6.7)]
Y^*	electric admittance [Eq. (3.5.1)]
y	coordinate in an orthogonal system
z	coordinate in an orthogonal system
z_i	number of charges
α	electrical polarizability [Eq. (2.4.8)]
β_T	isothermic compressibility [Eq. (2.4.26)]
γ	velocity gradient or shear rate [Eq. (3.6.1)]
γ	surface tension (Section 2.5.1)
δ	difference of length
Δ	sign, indicating a difference between two values
ε	mechanical strain [Eq. (3.6.6)]
ε	dielectric constant or permeability number [Eq. (2.4.1)]
ε_0	dielectric permittivity of vacuum = $8.854187817 \cdot 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}$
ζ	electrokinetic potential [Eq. (2.5.14)]
η	viscosity [(Eq. (3.6.2)]
κ	Debye-Hückel constant [Eq. (2.4.15)]
λ	thermal conductivity [Eq. (4.1.1)]
λ	wavelength
μ	magnetic permeability [Eq. (4.4.1)]
μ_0	magnetic permeability of vacuum = $1.2566370 \cdot 10^{-6} \text{ V s A}^{-1} \text{ m}^{-1}$ [Eq. (4.4.1)]
μ	electric dipole moment [Eq. (2.4.7)]
μ_i	chemical potential of the component i [Eq. (3.1.33)]
$\bar{\mu}_i$	electrochemical potential of the salt i [Eq. (3.1.41)]
ν	stoichiometric number [Eq. (3.1.65)]
ν	kinematic viscosity ($\nu = \eta/\rho$)
ν	frequency in Hz ($\nu = \omega/2\pi$)
ξ	degree of advancement of a chemical reaction [Eq. (3.1.73)]

π	osmotic pressure [Eq. (3.2.14)]
ϱ	density
ϱ	charge density in space [Eq. (2.4.12)]
σ	Stefan-Boltzmann constant [Eq. (4.1.2)]
σ	mechanical stress [Eq. (3.6.5)]
σ	entropy production [Eq. (3.1.63)]
σ_0	surface charge density [Eq. (2.5.15)]
σ	Staverman's reflection coefficient [Eq. (3.2.28)]
τ	time constant
τ	shear stress [Eq. (3.6.3)]
Φ	Rayleigh's dissipation function [Eq. (3.1.64)]
ϕ	fluidity ($\phi = 1/\eta$) (Chapter 3.6.1)
χ	magnetic susceptibility [Eq. (4.4.2)]
ψ	electrical potential [Eq. (2.4.3)]
ω	angular frequency ($\omega = 2\pi\nu$)
ω	coefficient of mobility [Eq. (3.1.52)]

Nature and Subject of Biophysics

The subjects of biophysics are the physical principles underlying all processes of living systems. This also includes the explanation of interactions of various physical influences on physiological functions, which is a special sub-area, called environmental biophysics.

Biophysics is an interdisciplinary science somewhere between biology and physics, as may be concluded from its name, and is furthermore connected with other disciplines, such as mathematics, physical chemistry, and biochemistry. The term "biophysics" was first used in 1892 by Karl Pearson in his book *The Grammar of Science*.

Does biophysics belong to biology, or is it a part of physics? Biology, by definition, claims to be a comprehensive science relating to all functions of living systems. Hence, biophysics, like genetics, biochemistry, physiology etc., should be considered as a specialized sub-area of biology. This view has not remained undisputed by physicists, since physics is not confined to subjects of inanimate matter. Biophysics can be considered, with equal justification, as a specialized part of physics. It would be futile to try to balance those aspects against each other. Both of them are justified. Biophysics cannot flourish unless cooperation is ensured between professionals from either side.

Delimitation of biophysics from clearly unrelated areas has appeared to be much easier than its definition. Biophysics, for example, is by no means some sort of a melting pot for various physical methods and their applications to biological problems. The use of a magnifying glass, the most primitive optico-physical instrument, for example, has just as little to do with biophysics as the use of most up-to-date optical or electronic measuring instruments. Biophysical research, of course, requires modern methods, just as other fields of science do. The nature of biophysics, however, is actually defined by the scientific problems and approaches rather than by the applied methods.

Biophysical chemistry and bioelectrochemistry can be considered as specialized sub-areas of biophysics. Medical physics, on the other hand, is an interdisciplinary area which has its roots in biophysics but has ramifications of far-reaching dimensions, even with medical engineering.

In terms of science history, biophysical thought, according to the above definition, can be traced back to early phases of philosophical speculations on nature, that is back to antiquity. This applies to the earliest mechanistic theories of processes of life and to insights into their dynamics, for example of Heraclitus in the 5th century B.C. The promotion of scientific research in

the Renaissance also includes biophysical considerations. Leonardo da Vinci (1452–1519), for example, investigated mechanical principles of bird flight in order to use the information for engineering design; research which would be termed bionics today. A remarkably comprehensive biomechanical description of functions, such as mobility of limbs, bird's flight, swimming movement, etc., was given in a book by Alfonso Borelli (1608–1679) *De motu animalium* published in Rome, as early as 1680. The same Borelli founded a school in Pisa of *iatro-mathematics* and *iatro-physics* in which the human body was perceived as a mechanical machine, and where attempts were made to draw medical conclusions from that perception (*Iatric* – Greek term for medical art). *Iatro-physics* has often been considered as a mechanistic forerunner of medical biophysics.

Parallels to processes of life were established not only in the area of tempestuous progress of mechanics but at all levels throughout the development of physics. Reference can be made, in this context, to the frog experiments undertaken by Luigi Galvani (1737–1798). The physics of electricity was thus studied in direct relationship with phenomena of electrophysiology. Worth mentioning is the strong controversy between Luigi Galvani and Alessandro Volta (1745–1827) about the so-called *elettricità animale* (animal electricity), which had serious personal consequences for both.

It is well-known that medical observations played a role in the discovery of the first law of thermodynamics by J. R. Mayer (1814–1878). Calorimetric studies of heat generation of mammals were conducted in Paris by A. L. Lavoisier (1743–1794) and P. S. de Laplace (1749–1827) as early as about 1780. Reference should also be made, in this context, to investigations of Thomas Young (1773–1829), and later Hermann v. Helmholtz (1821–1894) on the optical aspects of the human eye and on the theory of hearing. These activities added momentum to the development of physiology which thus became the first biological platform for biophysics.

The development of physical chemistry around the turn of this century was accompanied by applications of these discoveries and insights in understanding various functions of living cells. There have also been many instances in which biologically induced problems had stimulating effects upon progress in physics and physical chemistry. Brown's motion, discovered in pollen grains and subsequently calculated by A. Einstein, is an example. Research on osmotic processes, as well, were largely stimulated by the botanist W. Pfeffer. The temperature dependence of rate constants of chemical reactions was initially formulated in terms of phenomenology by S. Arrhenius (1859–1927), and has, ever since, been applied to a great number of functions of life, including phenomena as sophisticated as processes of growth. Studies of physiochemical foundations of cellular processes have continued to be important in biophysical research, especially after the introduction of the principles of nonequilibrium thermodynamics. In particular, biological membranes, as highly organized anisotropic structures, are always attractive subjects for biophysical investigations.

A decisive impetus has been given to biophysical research through the discovery of X-rays and their application to medicine. It was attributable to close cooperation between physicists, biologists, and medical scientists which paved the way for the emergence of radiation biophysics which not only opened up possible new approaches to medical diagnosis and therapy but also made substantive contributions to the growth of modern molecular biology.

The year 1948 saw the publication of Norbert Wiener's book *Cybernetics* dealing with control and communications in men and machines. While regulation and control of biological systems had been subjects of research before, biocybernetics has given further important inspiration to biophysics. In the 1970s, biological system theory moved very close to thermodynamics. It should be borne in mind, in this context, that the expansion of classical thermodynamics to cover nonequilibrium systems with non-linear equations of motion was strongly stimulated by biological challenges. Supported by the works of A. Katchalsky, I. Prigogine, H. Haken and many others, elements of thermodynamics are found to be closely interconnected to those of kinetics within the theory of non-linear systems.

The word "bionics" was coined by a synthesis of "biology" and "technics" at a conference in Dayton, USA, in 1960. More specific shape was thus given to the millennial quest of man to look at nature's complete technological design. Biophysics, and especially biomechanics, play a substantial role in these attempts.

This brief view of the history and the development of biophysics allows us now to draw the following conclusions about its nature and relevance: biophysics seems to be quite a new branch of interdisciplinary science, but, in fact, biophysical questions have always been asked in the history of science. Biophysics relates to all levels of biological organization, from molecular processes to ecological phenomena. Hence, all the other biological sub-areas are penetrated by biophysics, including biochemistry, physiology, cytology, morphology, genetics, systematics, and ecology.

Biological processes are among the most intricate phenomena with which scientists find themselves confronted. It is, therefore, not surprising that biologists and other scientists have repeatedly warned against schematism and simplifications. Such warning is justified and is a permanent reminder to the biophysicist of the need for caution. Yet, on the other hand, there is no reason to conclude that biological phenomena are too sophisticated for physical calculation. Despite the fact that at present we are not able to explain all biological reactions, no evidence has ever been produced that physical laws were no longer valid, when it comes to biological systems.

Further reading: Rowbottom and Susskind (1984)

Molecular Structure of Biological Systems

This section starts with quantum-mechanical approaches, which allow us to explain molecular bonds and processes of energy transfer. Later we will explain the physical basis of thermal noise, which, finally, will lead us to the problem of self organization, or self assembly of supramolecular structures. It is the intention of this section to make the reader familiar with some specific physical properties of biological systems at the molecular level. The leading idea of this section is the controversy between thermal fluctuation against the forces of molecular orientation and organization.

Two kinds of physical behavior meet at the molecular level of biological structures: on the one hand, there are the characteristic properties of *microphysical* processes, based on the individual behavior of single small particles like atoms, molecules or supramolecular structures. These processes are mostly stochastic. On the other hand, there are reactions which resemble “*macrophysical*” properties, the kind of behavior of “large” bodies. The “macrophysics” is ruled by the laws of classical physics, for example classical mechanics. Our daily experiences with macrophysical systems teach us that their behavior is generally deterministic.

To explain this difference, let us consider a simple mechanical wheelwork. The knowledge of its design and construction allows a precise prediction of the behavior of the system. This prediction is based on the laws of classical mechanics. In contrast to this, a chemical reaction with a small number of molecules in a homogeneous phase depends on stochastic collisions of the individual molecules with each other. Since this process is stochastic, it is only predictable in a statistical way.

This stochastic behavior of molecular systems can be transformed into a deterministic one, if the number of participating stochastic events is large, or if the degrees of freedom of the single reactions are extremely limited. The increase of stochastic events can be realized either by an increasing number of participating molecules, by enlarging the volume for example, where the reaction takes place, or by an increase of the time interval of observation. This consideration indicates an interesting interplay between volume, time constants, and reliability of a biochemical reaction.

The limitation of the degree of freedom of a biochemical reaction is realized by a property of the system which is called *anisotropy*. In contrast to isotropic systems, like simple solutions, in anisotropic systems the mobility of molecules in various directions is not identical, but is restricted in some directions, and

promoted in others. This, for example, is the case for enzymatic reactions, where the participating enzymes are orientated in membranes, or if the reactions of charged or polar reactants occur in strong electric fields of electrical double layers.

In many fields the biological organism works as an amplifier of the microphysical stochastics. A molecular mutation, for examples, leads to a reaction chain, which finally ends with a phenomenological alteration of the organism. Or, as another example: a few molecular events in the pigments of optical receptors can lead to perception and to reaction in behavior.

During the first step in considering molecular mechanisms of biological systems, a further aspect is taken into consideration. Unfortunately, biologists often ignore the fact that a qualitative jump has to be made in the transition from the “visible” macrophysical structures, to the microphysical systems such as atoms or molecules. This includes not only the above-mentioned transition from the deterministic behavior of macroscopic systems to the stochastic behavior of single molecules, but many more aspects as well. The biologists, for example, must acknowledge that the term “structure” receives a new meaning. The visible “biological structure”, as known in the fields of anatomy, morphology and histology, now appears as concentration profiles or as systems of electric charges or electromagnetic fields. Instead of visible and measurable lengths, diameters or distances, as common in the visible world, in the microphysical world so called *effective parameters* are used. These sorts of parameters are exactly defined and they can be measured with arbitrary exactness, but they do not correspond to some visible boundaries. A single ion, for example, has no diameter in the sense of the diameter of a cell, or a cell nucleus, which can be measured by a microscopic scale. In the following sections we will define effective parameters like crystal radius, hydration radius and Debye-Hückel radius, which really are important parameters for functional explanations.

It is not the intention of this book to describe the topics of molecular biology. However, the theoretical foundations and principles will be explained to make possible a link between structure and function at the molecular level and current biological thinking in these dimensions.

2.1

Intramolecular Bonds

Any representation of the dynamics of molecular and supramolecular structures has to begin with the atom, its organization and energy states and with interactions between atoms in a molecule. The molecule, as described in the next chapter, is initially assumed to be thermally unaffected. The thermal energy of movement will be introduced as an additional parameter in Section 2.7.

2.1.1

Some Properties of Atomic Orbitals

The Schrödinger equation is the theoretical basis for calculation of the wave functions of electrons and the probability of their presence at a particular point in space. This is a link between wave mechanics and the atomic model postulated by Niels Bohr. The latter concept, however, is substantially extended by the incorporation of the wave properties of elementary particles.

The quantification of electric energy by Planck's theory which has already been postulated ad hoc in the atomic model of Niels Bohr, results directly from the solution of Schrödinger's equation in the wave mechanical model. For some considerations, it will be more convenient as a kind of approximation, and quite legitimate to postulate the particular nature of the electron for the purpose of model construction. The limitation of this model is defined by Heisenberg's Uncertainty Principle.

The energy state of electrons and their distribution in space are expressed by so-called *quantum numbers*. Every single electron is characterized by four quantum numbers which are interconnected with each other in a well-defined way.

The *primary* (or *principal*) *quantum number* (n), according to Bohr's model, expresses which electron shell the electron belongs to and can assume the following values: $n = 1, 2, 3, 4, \dots$

The *second* (or *azimuthal*) *quantum number* (l) determines the distribution of charge density in space. It is dependent on n , as it can assume only the following values: $l = 0, 1, 2, 3, \dots (n - 1)$. This implies that the possible values of l are limited by n . Only value 0 and 1, therefore, can be assumed by l , if $n = 2$. Electrons with azimuthal quantum numbers $l = 0, 1, 2$ are defined as: *s-electrons*, *p-electrons*, and *d-electrons* respectively.

The *magnetic quantum number* (m) results from the fact that a moving electron, similar to an electric current in a coil, generates a magnetic field and, consequently, can also be influenced by an external magnetic field. The following values can be assumed by the magnetic quantum number: $m = -l, \dots 0 \dots +l$. With $n = 2$, and $l = 0$ or 1 ; m , therefore, can only be $-2, -1, 0, +1, +2$.

The *spin quantum number* (s) of the electron describes its direction of rotation around its own axis. There are only two conditions possible: clockwise or anti-clockwise rotation. These situations are denoted by $s = +1/2$, and $s = -1/2$. The quantum number has no effect on electron energy, unless magnetic fields are involved.

It can be easily calculated by the combination of quantum numbers, especially those with high n -values, that a large number of electron states can be achieved. In 1926 W. Pauli made a postulation which, so far, has not been invalidated. According to this, the so-called *Pauli exclusion principle*, it is impossible for two electrons with identical quantum numbers to occur in the same atom. This principle limits the number of possible electron orbits and is of great importance for various applications in quantum mechanics.

The points discussed so far can best be demonstrated using the hydrogen atom. Only the values: $n = 1$, $l = 0$, and $m = 0$ are possible. The energy of the electron can be calculated for this condition by the following equation:

$$E = -\frac{e^4 m}{8\epsilon_0^2 n^2 h^2} = -2.18 \cdot 10^{-18} \text{ J} = -13.6 \text{ eV}, \quad (2.1.1)$$

where m is the mass of an electron ($m = 9.109 \cdot 10^{-31} \text{ kg}$), e is the charge of the electron ($e = 1.60218 \cdot 10^{-19} \text{ C}$), h is Planck's constant ($h = 6.626 \cdot 10^{-34} \text{ J s} = 4.136 \cdot 10^{-15} \text{ eV s}$). The electric constant (also referred to as the permittivity constant), $\epsilon_0 = 8.854 \cdot 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}$, is just a factor which converts the charge as an electrical variable into the mechanical variables: length and force. It will be explained in detail in Section 2.4.1. Here, eV, as a unit of energy is introduced. It will be used frequently in the following text. It stands for "electron volt" and is identical with the kinetic energy of an electron, which has been accelerated by a potential difference of 1 V. Its relation to other energy units is as follows:

$$\begin{aligned} 1 \text{ eV} &= 1.602177 \cdot 10^{-19} \text{ J} = 4.45049 \cdot 10^{-23} \text{ Wh} \\ &= 1.634 \cdot 10^{-20} \text{ kpm} = 1.602 \cdot 10^{-12} \text{ erg} \end{aligned} \quad (2.1.2)$$

The energy of an electron E , as calculated for example by Eq. (2.1.1) for the valence electron of hydrogen, is defined as the amount of work which is required to lift it from its orbit, against the attractive force of the positively charged nucleus, and to move it to an infinite distance from the nucleus. It is also possible to speak of an energy potential, called *ionization potential* because the formation of an ion results when an electron is lost from an atom. In Sections 2.2.2 and 4.5 reference to this idea will be made, describing ionization through interaction with electromagnetic radiation.

The wave function describing the stationary state of an electron is called the *orbital*. Sometimes the term *orbital cloud* is used, which gives a better illustration of the statistical nature of the electron distribution. Spherical orbitals, as in the case of the hydrogen atom, occur when $l = 0$; $m = 0$. Such charge clouds are called *s-orbitals*.

For: $n = 2$, not only $l = 0$, but additionally, the azimuthal quantum number $l = 1$ is possible. In this case, the corresponding orbital resembles a double sphere. The position of the common axis of this double sphere will be determined by the magnetic quantum numbers $m = -1$; $m = 0$ and $m = +1$. These shapes are called *p-orbitals*, with the variants p_x , p_y , and p_z (see Fig. 2.1).

When the azimuthal quantum number is $l = 2$ then there are five possibilities for m , namely the numbers -2 , -1 , 0 , $+1$, and $+2$. Accordingly, in this case five different orbitals are possible. These so-called *d-orbitals* play an important role in the ligand field theory of coordinative bonds which will be explained in Section 2.1.4.

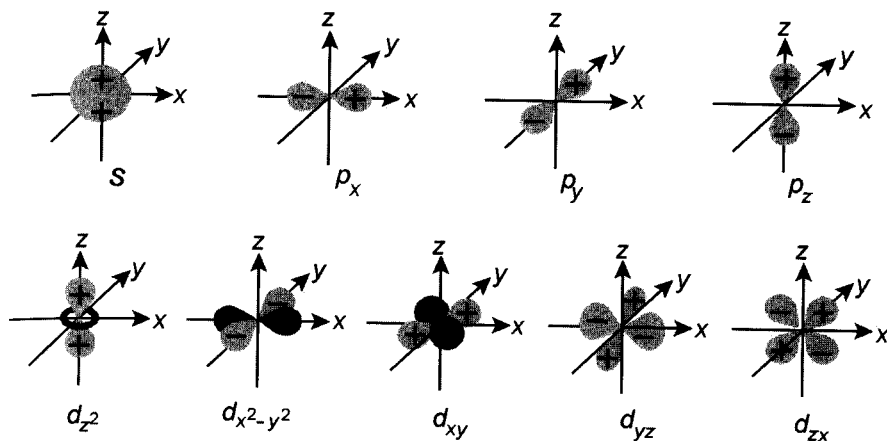


Fig. 2.1. Schematic representation of the s -, p -, and d -orbitals. For clear demonstration of the geometric conditions, the clouds are cut at certain amounts of probability. Therefore, the sharp borders of the orbitals do not reflect their realistic structure

It must be underlined that the spin quantum number does not influence either the shape, or the size of the orbitals. This parameter is important in connection with the pairs of valence electrons when considering the Pauli principle.

2.1.2

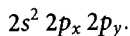
Covalent Bonds, Molecular Orbitals

The calculation of molecular orbitals provides the basis for the theoretical interpretation of atomic interactions. A mutual approach of atoms is accompanied by an overlapping of their electromagnetic fields and, consequently, by a change in the wave function of their electrons. The energy levels of the wave functions, modified in such a way, can sink lower than the sum of the levels of energy of the undisturbed atoms. In this case the connection of the two atoms becomes a stable chemical bond.

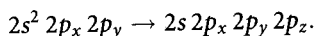
The intramolecular bonding energy of atoms consists of several components. The kinetic energy of the electrons, the electrostatic interaction among the electrons, and the interactions between electrons and nuclei, for example, are to be included in this calculation. These components have different signs, and their interaction forces are determined by various functions of distance. If at a certain distance the sum of all of these energy functions becomes a minimum then this distance will determine the bonding distance between two atoms in the molecule. This function has been accurately calculated just for H_2 , the simplest molecule, which consists of two proton nuclei and one common electron. Yet serious problems already occur when calculating other, more complicated, diatomic molecules.

The most important chemical bond is the *covalent bond*. It can only occur if two mutually approaching atoms have unpaired electrons in their valence shells. This means that both relevant orbits are each occupied by one electron only, thus leaving space for another electron. In this case the Pauli exclusion principle is not violated. Figuratively speaking, an electron pair with anti-parallel spin quantum numbers is formed, being common to both atoms, and forming a type of molecular orbital.

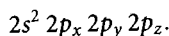
This may be explained using carbon, the atom of greatest importance for biological systems. The distribution of the valence of bonding electrons of the carbon atoms can be characterized by the following formula:



Consequently, there are three occupied orbitals which belong to shell $n = 2$. The p_x - and p_y -orbitals are each occupied by one unpaired electron (no exponent, i.e., exponent = 1), but the s -orbital is occupied by an electron pair (exponent = 2). Hence, only two of the four bond, or valence electrons are able to form bonds. It is, however, a peculiarity of the carbon atom that it jumps to another energy state immediately prior to a reaction by a relatively low energy input (some 250 kJ/mol). For this, one of the two electrons of the s -orbital moves to a completely unoccupied p_z -orbit:



This is a specific property of the carbon atom, and it is just one of the many other physiochemical peculiarities which enabled the emergence of life at all. The nitrogen atom, in contrast to this, is not capable of such modification. Its valency electrons may be formulated as follows:

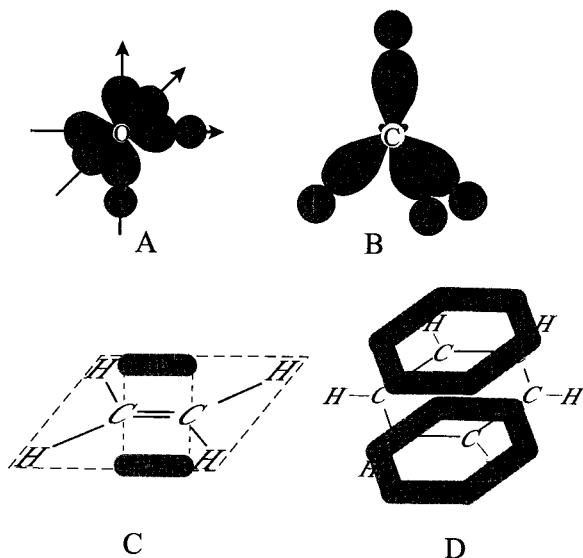


Consequently, the nitrogen atom remains trivalent in its reaction.

As a first approach, molecular geometry can be represented by a simple overlapping of the orbitals of the involved atoms. The configuration of the H_2O molecule, using this principle, is depicted in Fig 2.2A. Oxygen has two unpaired electrons with $2p_x$ - and $2p_y$ -orbitals. Hence, the $1s$ -orbitals of the two hydrogen atoms can only form bonds with the two electrons if they come from two defined directions. This leads to an electron cloud built up by one electron from each atom (with different spin numbers!) which is common to both atoms. Accordingly, this is defined as an *sp-bond*.

The above example illustrates the angular orientation of the covalent bond. Measurements have shown, however, that significant deviations can occur from the models which had been obtained by simple geometrical considerations. For example, in the water molecule, the bonding angle between the two hydrogen atoms is 104.5° (Fig. 2.2A) and not 90° , as might have been expected from the orientation of the $2p_x$ -orbital of oxygen toward its $2p_y$ -orbital (Fig. 2.2A). Such

Fig. 2.2. A H_2O -molecule (simplified construction using the overlapping of the atomic orbitals); B sp^3 -hybrid-orbital of CH_4 ; C π -orbital of a double bond; D π -orbitals of benzene (after Christen 1968)



shifts of bonding angles are the result of mutual electrostatic repulsion of the valence electrons. These divergences can be even stronger in other molecules. The four bonding orbitals of the carbon atom in the promoted state have already been discussed. Correspondingly, in methane (CH_4), the three hydrogen atoms should form sp -bonds at right angles to one another, while the ss -bond should be fully undirected. In reality, however, the four hydrogen atoms are found to be in a precise tetrahedral arrangement (Fig. 2.2B).

These examples show that simple geometrical constructions are not sufficient to indicate a realistic picture of the molecule orbitals. For accurate analysis, the Schrödinger equation must be resolved for the entire molecule, or at least for some parts of it. Approximative calculations of this kind result in so-called *hybrid orbitals*, which reflect all of the interactions within the given molecule. The tetrahedrally arranged bond angles of methane, for example, correspond to so-called sp^3 -orbitals. The number 3 in the exponent, in this context, indicates that a hybridization of one s -orbital occurred with three p -orbitals.

The sp -orbital, and the sp^3 -hybrid orbital indicate rotational symmetry. Bonds of this kind are called σ -bonds. They may be subjected to thermal rotations, as described in detail in Section 2.3.6. The electrons involved in such bonds are called σ -electrons. In case of double bonds, so-called π -orbitals occur with corresponding π -electrons. Such orbitals are not symmetrical with regard to their bonding orientation, as may be seen from Fig. 2.2C.

From Table 2.1 it can be seen that, in a double bond, the interatomic distance is shortened when compared with that of a single bond. The total energy of a double bond is smaller than the sum of energies of two single bonds. In contrast to the classical model of Kekulé, the benzene ring must not be perceived as a

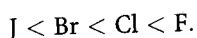
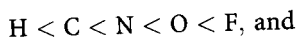
Table 2.1. Properties of biologically important bonds. (After Pullman and Pullman 1963)

Kind of bond	Bond distance (nm)	Bond energy (kJ mol ⁻¹)
C—C	0.154	348
C=C	0.133	615
C≡C	0.121	812
N—N	0.145	161
N=N	0.124	419
N≡N	0.110	946
C—N	0.147	292
C=N	0.126	615
C≡N	0.115	892
C—O	0.143	352
C=O	0.114	716 (for aldehydes) 729 (for ketons)
C—S	0.182	260
C=S		477
C—Cl	0.176	329
O—H	0.096	463
N—H	0.101	391
S—H	0.135	340
C—H	0.109	414

static arrangement of alternate single and double bonds, but rather as a continuous π -orbital in which the π -electrons can move around the ring.

Molecule orbitals may sometimes result in a considerable charge displacement within the molecule. Simply, this can be illustrated by considering that the electron pairs common to two atoms in a covalent bond are distributed asymmetrically. In other words, the molecular orbital of the bond is shifted toward one of the two covalently bonded atoms. The atom with the greater probability of the presence of the electron pair is more strongly “electronegative” than the other one.

In this respect, a series of atoms with an increasing degree of *electronegativity* can be constructed. In the periodic system of elements this series is directed toward increasing atomic numbers within the periods as well as within the groups. Hence, the following relation applies:



In this way, the displacement of bonding angles within a water molecule can be explained. Because of the strong electronegativity of the oxygen atom in relation to the hydrogen atom, a dipole O—H results and, consequently, the two hydrogen atoms will repel each other (for more details on the structure of water and the importance of this polarization effect, see Sect. 2.4.2). This polarization

effect contributes directly to the bonding energy. About 22% of the bonding energy in a C—O bond, and 39% in a H—O bond are attributable to this effect.

2.1.3

Ionic Bonds

If the polarization effect of covalent bonds, as described in the previous section, is pushed to the extreme, it is no longer possible to refer to a molecular orbit, nor to bonding electrons. There occurs a total transmission of an electron from the valency orbital of one atom to that of the other. This causes a full separation of charges. Ions are generated, which attract each other electrostatically. With the loss of the molecular orbitals there also occurs a loss of the molecular identity. Strictly speaking, it makes no sense to use the term NaCl-molecule. In solutions, the molecular character of this salt is just expressed by the stoichiometric relation of the number of anions and cations. A crystal of NaCl, on the other hand, can be considered as a super molecule, because in this case the ions are arranged in an electrostatic lattice.

The ionic bond, in contrast to the covalent bond, can be considered simply from an electrostatic point of view. The basic formula of the electrostatics is *Coulomb's law*. It defines the force (F) with which two points, carrying electric charges (q_1) and (q_2), repel each other at a distance (x) in a vacuum.

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 x^2}. \quad (2.1.3)$$

(The meaning of the electric field constant $\epsilon_0 = 8.854 \cdot 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}$ will be explained in Chap. 2.4.1). This equation also makes it possible to calculate the bonding force, as the negative value of the force ($-F$) which is required to separate two ions. This force, however, is not measurable directly. It is better therefore to transform this equation in such a way that it can provide information on the bonding energy or, which is the same, on the energy of ionization. The *bonding energy* is equivalent to the work which is necessary to move a given ion from its bonding place up to an infinite distance from the counter ion. A work differential (dW), which has a very small, but finite value, can be calculated from the product of force (F) and a corresponding distance differential (dx):

$$dW = -\frac{q_1 q_2}{4\pi\epsilon_0 x^2} dx. \quad (2.1.4)$$

The sign of dW depends on the point of view or consideration, and thus is a matter of definition. Work applied to a system (e.g., by combination of the ions) means a gain of work by the system, and at the same time a loss of work by the environment. As in the majority of textbooks, we will consider here positive dW as a gain of work by the system. Furthermore, positive dx means an enlargement, negative dx , on the other hand, a diminution of the distance

between the charges. In the case under review [Eq. (2.1.4)], dW becomes positive if the signs of q_1 and q_2 are equal, and both charges are moved toward each other (this means: $dx < 0$).

The overall work (W) which would be necessary to move the particles from the distance $x = \infty$ closer to the bond distance $x = x_1$ can be obtained by integration of Eq. (2.1.4) within these limits.

$$W = \int_{\infty}^{x_1} dW = \int_{\infty}^{x_1} -\frac{q_1 q_2}{4\pi\epsilon_0 x^2} dx = \frac{q_1 q_2}{4\pi\epsilon_0 x_1}. \quad (2.1.5)$$

As Eq. (2.1.5) applies to any value of x , no index is required for these variables. Bonding energy (E) we defined above as the energy needed to interrupt the bond, this means, to move the particle from $x = x_1$ to a distance $x = \infty$. Hence, $E = -W$.

Let us illustrate this situation by the following consideration. What is the value of the bonding energy between a Na^+ , and a Cl^- ion? The charge of either ion is based on the excess or the lack of one electron respectively. The electrostatic charge of a single electron is $1.602 \cdot 10^{-19}$ C. This value, furnished with the appropriate sign, must be inserted in Eq. (2.1.5). Now, information on the bonding distance (x) is still required. On account of their electrostatic attractive force, both ions would fully fall into each other, however, at the minimum distance of approximation a strong electrostatic repulsion is generated. This repulsive force is related to the structure of their inner electronic orbits. The minimum distance of approximation of two ions is, in fact, the sum of their Van der Waals radii. The *van der Waals radius* (sometimes also called *Bohr's radius*) can be determined by means of X-ray diffraction diagrams from the position of the atoms in the crystal. It is also referred to as the *crystal radius*. For the sodium ion this value is 0.098 nm (see Table 2.3, page 60), for chloride: 0.181 nm. Consequently, the distance between both ions in the NaCl molecule (i.e., in the crystal) amounts to: $x = 2.79 \cdot 10^{-10}$ m.

Substituting these values in Eq. (2.1.5), we obtain:

$$E = -W = -\frac{(-1.602 \cdot 10^{-19}) \cdot (1.602 \cdot 10^{-19})}{4 \cdot \pi \cdot (8.85 \cdot 10^{-12}) \cdot (2.79 \cdot 10^{-10})} = 8.27 \cdot 10^{-19} \text{ J}, \quad (2.1.6)$$

or, according to the conversion, given in Chap. 2.1.1

$$E = (8.27 \cdot 10^{-19}) \cdot (6.242 \cdot 10^{18}) = 5.31 \text{ eV}.$$

Usually, the unit electron volt (eV) is used to characterize the energy of single bonds in a molecule. For macroscopic considerations the unit Joule (J) is used. For calculations of the molar bonding energy, the *Avogadro number* is required, which gives the number of molecules per mol:

$$E = (8.27 \cdot 10^{-19}) \cdot (6.02 \cdot 10^{23}) = 4.98 \cdot 10^5 \text{ J mol}^{-1},$$

$$E = 498 \text{ kJ mol}^{-1}.$$

The electrostatic nature of ionic bonds as described in this section, automatically indicates that in contrast to the covalent bonds, no valence angle is predicted.

2.1.4

Coordinative Bonds, Metallo-Organic Complexes

The stability of a series of biologically important molecules is mediated by polyvalent metals and transition elements. This stability cannot be explained either by single covalent bonds, or by ionic bonds. For these types of bonds, which occur mainly in inorganic chemistry, the terms *coordinative bonds* or *complex bonds* have been introduced.

These complexes, which may also be charge-carrying ions, consist of a *central atom* which is surrounded by ligands, i.e. molecular complexes, in geometrically defined arrangement. There are also polynuclear complexes with several central atoms. Ligands are directly bound to the central atom but form no bonds among each other. If a ligand has two or more bonds to a central atom it is called *chelate complex*.

Most of the central atoms are elements with incompletely occupied *d*-orbitals, i.e. elements with bonding electrons having principal quantum numbers $n > 2$. Some trace elements of biological systems are of special interest in this context, for example Fe (in the porphyrin complex of hemoglobin), Mg (in chlorophyll), Co (in B₁₂-vitamin). The same phenomenon is of importance for the Ca-Mg-antagonism in cell physiological processes.

An explanation for these bonding mechanisms is given by the *ligand field theory*. The approach of the ligands to the central atom leads to a re-orientation of the electron orbitals. This interaction of the central atom with the ligands is caused by the so-called *ligand field*. As may be seen from Fig. 2.1, according to the differences in the magnetic quantum number (m), five different *d*-orbitals are possible. Considering that each orbital can be occupied by two electrons with different spin quantum numbers, the total number of possible electron states will be higher than the number of electrons actually present.

An isolated Fe³⁺-ion, for example, occupies in free solution each of these orbitals simply with one unpaired electron (Fig. 2.3). Hence, all five electrons are in the same energy state. When a ligand now approaches this atom, some of the electrons will assume a higher energy level, and some of them a lower one, depending on the direction of the approach, relative to the orbital coordinates. For reasons of quantum mechanics, the mean energy of all these levels has to remain constant. This means that the amount by which one level is elevated must be the same as that by which the other level has been lowered.

If the field is still weak, the differences in the energy levels are too low to induce changes of the spin quantum numbers which would be necessary to avoid conflicts with Pauli's exclusion principle (see Sect. 2.1.1). Such a situation is called a *high spin state*, since the number of unpaired electrons remains at a

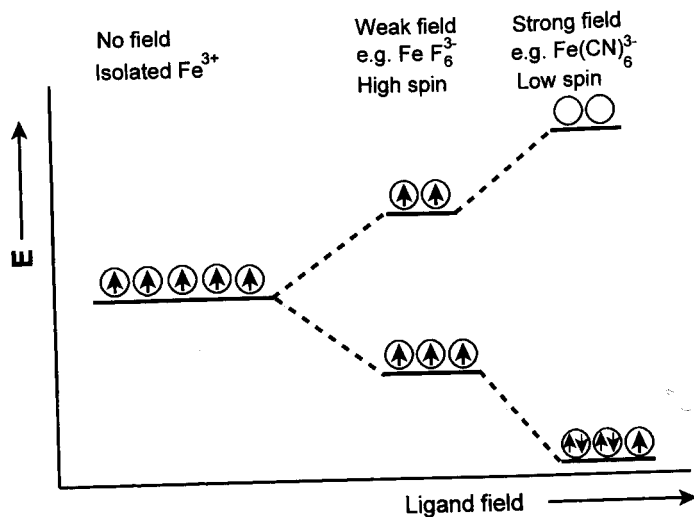


Fig. 2.3. The occupation of the orbitals of Fe^{3+} influenced by weak or strong ligand fields. The arrows in the circles indicate the orientation of the electron spins. Circles without arrows indicate empty orbital positions

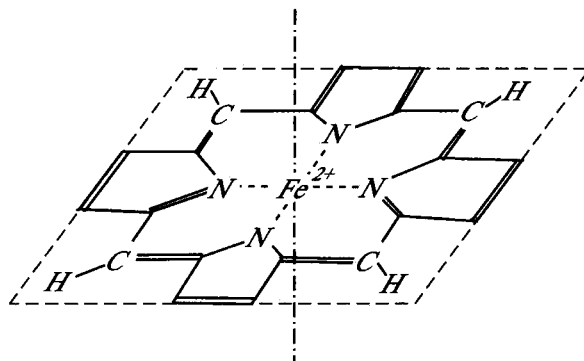
maximum. Strengthening the field will cause all of the electrons to collect together at the lowest level of energy, with some of them undergoing spin flip. This results in a *low spin state*. A favorable energy state has thus been reached for the central atom, as may be seen in Fig. 2.3. The stability of the complex is guaranteed by the existence of the difference between the new and the old energy levels.

The hem-complex, which is a prosthetic group of a number of important proteins (hemoglobin, myoglobin, cytochrome, etc.), as depicted in Fig. 2.4 is an example of a chelate complex. Here, the central iron atom is under the influence of the ligand field of porphyrin. This molecule consists of four pyrrole nuclei, located in one plane and kept together by methine bridges. Four nitrogen atoms are coordinated to the iron atom. The remaining bond orbitals of the iron are perpendicular to this plane and can be occupied by other ligands. Chlorophyll is a porphyrin complex with magnesium as the central atom.

Free bonding sides of the central atom can also be occupied by water molecules. A major role in the ligand binding process is played by the competition between ligands with different affinities. This competition explains many inhibitory effects in metabolism. The action of cyanide, a respiratory poison, may be mentioned as an example. In this case, water is displaced by CN^- from its coordination bonds with the iron atom of hemoglobin.

The highly specific nature of reactions involving complex formation in biological systems cannot be explained, however, simply by differences in the bond affinity. In some cases, steric aspects of macromolecular structures are

Fig. 2.4. The structure of the hem complex. The Fe^{2+} is bound four times with the pyrrole ring (chelate bond). Both free valences of the central atom (Fe^{2+}) are perpendicular in relation to the plane of the pyrrol-ring system



responsible for this. Because of the dynamic properties of biological structures, in other cases the rate constants of the processes of complexation are essential. For example, the equilibrium constant of the complexation of ATP (adenosine triphosphate) with magnesium is nearly the same as with calcium. Nevertheless, the Ca-ATP complex occurs in larger concentration, because the rate constant of its formation is about 100 times faster than for the Mg-ATP complex.

2.1.5

Hydrogen Bond

A brief reference will be made at this point to a special kind of dipole-dipole interaction which is of great importance in molecular biology. The contact of a hydrogen atom with an electro-negative partner leads to the formation of a polar molecule in which the hydrogen atom is the positive pole (see Sect. 2.1.2). This dipole can attract another polar molecule which will then turn its negative pole towards the bonded hydrogen atom. The two dipoles may approach each other very closely. These distances between 0.26 and 0.31 nm are even below Van der Waals radii. This suggests that in the formation of such bonds there is a covalent contribution in addition to the electrostatic interaction. Once the two molecules have moved sufficiently close to each other, the hydrogen atom can no longer be unambiguously assigned to one of them. It belongs, quasi, to both molecules simultaneously and, consequently, constitutes a so-called *hydrogen bridge*.

To calculate the bonding energy of a hydrogen bridge, aspects of wave mechanics have to be taken into consideration. This energy is between 13 and 25 kJ mol^{-1} and is a function of bond distance. The hydrogen bond, therefore, falls into the category of “weak” bonds, which can easily be split by thermic collision in the range of biological temperatures (cf. Sect. 2.3.5). The two molecules connected by a hydrogen bridge face each other with their negative poles and therefore repel each other; thus the hydrogen bond between them becomes stretched. Thus, hydrogen bonds which are formed on large molecules are orientated at defined angles.

Hydrogen bridges may be formed within large molecules as well as between them. More details on the importance of this type of bond will be given in the following sections, in the context of the formation and stabilization of the secondary structure of proteins and nucleic acids (Sect. 2.4.5) and with regard to the structure of water (Sect. 2.4.2).

2.2

Molecular Excitation and Energy Transfer

From the thermodynamic point of view, the living organism can be considered as an open system in nonequilibrium state. This nonequilibrium is maintained permanently by a continuous uptake of free energy. The biosphere assimilates this energy from photons of sunlight through the process of photosynthesis. This assimilated energy is then stored and transferred in molecules with energy rich bonds at first within plants, and then through the food chain into animals.

This long process, starting with energy assimilation in the thylakoid membranes of chloroplasts and ending, for example, in energy transformation and dissipation in humans and other animals, is associated with numerous elementary processes which can be described in terms of quantum mechanics. In the following chapter we will briefly discuss this variety of processes.

2.2.1

Mechanisms of Photon-Induced Molecular Excitation

Whenever an assessment of the effect of any radiation energy is made, it should always be borne in mind that the effective amount of energy is only that which is actually absorbed by the system, and not that part which penetrates it. This is the so-called *Grotthus-Draper principle*. The Grotthus-Draper principle is applicable for all kinds of radiation. It plays a particular role in photobiology because of the specific spectral absorption properties of light.

The wavelength of visible light lies between 400 and 800 nm. This corresponds to quantum energies between 3.1 eV and 1.6 eV respectively. (cf. Sect. 4.4, Fig. 4.13). The quantum energy of the photons of visible light, therefore, is much lower than the ionization energy of water (12.56 eV), a value which is arrived only in short-wave ultraviolet light (UVC). The energy of thermic noise (kT), which we will consider in detail in Section 2.3.4, in relation to this, is much smaller. At temperatures of biological importance it amounts only to about 0.025 eV. For photosynthesis the organism uses exactly this gap between the quantum energy of thermic noise on the one hand, and the quantum energies causing ionization of water and organic molecules on the other hand, using the energy of "visible" light. The photons of light are strong enough to be used for an effective process of energy conversion, but they do not endanger the stability of biomolecules. How narrow this biologically usable part of the electromagnetic spectrum in fact is, can be seen from Fig. 4.13.

Lets consider first the process of molecular excitation in general. If a molecule absorbs a quantum of energy of light, a sequence of processes will be induced. This is illustrated in Fig. 2.5. In a first step, the absorption of a photon leads to the raising of an electron to an orbit of higher quantum number. This can occur in the framework of the following two series of energetic states, which are qualitatively different: $S_0, S_1, S_2, \dots$, and $T_1, T_2, T_3, \dots$. Between these steps with additionally increasing quantum numbers, small energetic steps of thermic excitations are positioned. In the case of *singlet states* (S), the electrons of a pair have anti-parallel orientated spins; the spin quantum numbers, therefore, have different signs. In the case of *triplet states* (T), the spins of the electrons of the pair have antiparallel orientation, their spin quantum numbers are identical. As already stated, the occurrence of electrons in which all quantum numbers are equal is ruled out by Pauli's exclusion principle (cf. Sect. 2.1.1). Thus, if the triplet state represents an electron pair with identical spin quantum numbers, these two electrons must differ with regard to other energy parameters although their orbitals can be energetically very similar. A triplet is a so-called *degenerated state*.

These two states of excitation differ substantially in their life span. Figure 2.5 shows that a setback, $S_1 \rightarrow S_0$ occurs under emission of *fluorescent light* within 10^{-9} to 10^{-5} s, whereas a setback, $T_1 \rightarrow S_0$, recordable as *phosphorescence*, will occur at a much slower rate. Consequently, triplet states are characterized by an increased stability when compared with excited singlet states.

2.2.2

Mechanisms of Molecular Energy Transfer

All processes of life need a driving energy, which originally comes from the quantum energy of visible light, emitted by the sun and absorbed by pigments of photosynthetic units. After this process of molecular excitation, the absorbed

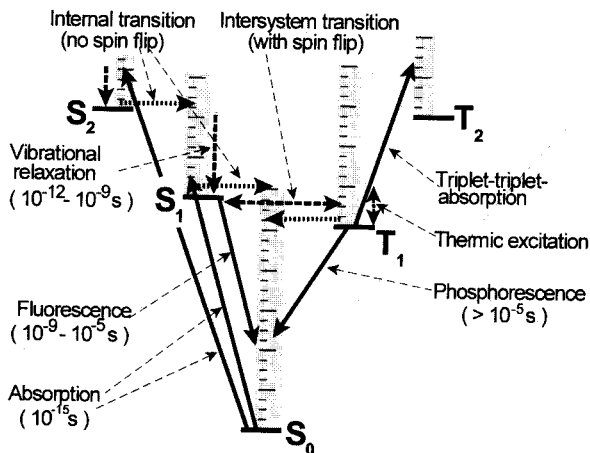


Fig. 2.5. Energy diagram of ground and excited states of organic molecules

energy must be accumulated and transmitted to other parts of the cell, to other parts of the plant, and finally to other organisms which are not able to obtain energy by photosynthesis directly.

For this, the energy of molecular excitation will be transformed to the chemical energy of so-called *high-energy* compounds. The most common accumulator of chemical energy in the cell is adenosine triphosphate (ATP), formed in the process of photosynthesis and used in nearly all processes of energy conversion in other cells. The hydrolysis of ATP, producing adenosine diphosphate (ADP) and catalyzed by special enzymes, the ATPases, allows the use of this stored energy for ionic pumps, for processes of molecular synthesis, for production of mechanical energy and many others. The amount of energy, stored by the $\text{ADP} \rightleftharpoons \text{ATP}$ reaction in the cell, however, is limited just because of osmotic stability. Therefore, other molecules, like sugars and fats, are used for long-term energy storage. The free energy of ATP is used to synthesize these molecules. Subsequently, in the respiratory chain, these molecules will be decomposed, whereas ATP is produced again.

These metabolic steps are the subject of biochemistry. Here, the processes of energy transfer will be discussed in their most general form. The following nomenclature of electrochemistry will be used: the energy-supplying molecule will be referred to as *donor*, the energy receiving molecule as *acceptor*, no matter what the actual mechanism of energy transfer is.

In general, the following mechanisms of intermolecular energy transfer must be considered:

- energy transfer by radiation
- energy transfer by inductive resonance
- energy transfer by charged carriers

Energy transfer by radiation can be envisioned in the following way: the excited molecule emits fluorescent radiation which matches precisely the absorption spectrum of the neighboring molecule and, consequently, excites it. Such mechanisms are capable of transferring energy over distances which are large compared with the other processes described in this context. However, the efficiency of this process is quite low, it declines sharply with increasing distance. In fact, such mechanisms do not play a significant role in biological processes.

Of particular importance, primarily in the process of photosynthesis, is the transfer of energy by inductive processes, namely the so-called *resonance transfer*. This form of molecular energy transfer is a *non-radiant energy transfer*, since fluorescent light does not occur in this process. This mechanism can be envisioned as some sort of coupling between oscillating dipoles. The excited electron of the donor molecule undergoes oscillations and returns to its basic state thus inducing excitation of an electron in the acceptor molecule. This process requires an overlapping of the fluorescent bands of the donor with the absorption band of the acceptor, i.e., on the resonance of both oscillators. The smaller the difference between the characteristic frequencies of donor and

acceptor, the faster the transfer will be. Singlet, as well as triplet states can be involved in such processes. So-called strong dipole-dipole couplings are possible to distances of up to 5 nm. This is a process in which a $S_1 \rightarrow S_0$ transition in the donor molecule induces an $S_0 \rightarrow S_1$ excitation in the acceptor. Inductions through triplet states are only effective over shorter distances.

Energy transfer by charge carriers is the most common reaction in metabolic processes. It is a process which can take very different courses. The redox process is a classical example of this. It consists basically in the transfer of one or two electrons from the donor to the acceptor molecule. In this way, the donor becomes oxidized and the acceptor reduced. This apparently simple scheme however conceals a number of complicated sub-routines which have not yet been completely resolved.

For this process of electron transfer, donor and acceptor molecules must be in exactly defined position to each other and at a minimum distance, so that overlapping of respective electron orbitals can occur. In the first place, donor and acceptor will form a complex of highly specific steric configuration, a so-called *charge transfer complex*. This process of complex formation which occasionally requires steric transformations of both molecules, causes the actual transfer. It takes place at lower rates than energy transfer by induction as discussed earlier. Hence, the charge-transfer complex is an activated transition state which enables redox processes to take place between highly specific reaction partners in the enzyme systems of the cellular metabolism. Because of the oscillating nature of electron transfer, this coupling of two molecules is strengthened by additional electrostatic forces sometimes called *charge-transfer forces*.

In the process of energy transfer, differences between energetic potentials of donor and acceptors play an important role. An uphill transfer of electrons is only possible through an input of external radiation energy. Actually, these differences are slight, when compared with the absolute values of the ionization energy. They are only in the region of about 1.5 eV.

What is the scale of these energy gradients? Szent-Györgyi in his remarkable "Study in Cellular Regulations, Defense, and Cancer" in 1968 proposed to introduce a scale of so-called *biopotentials*. This scale is based on the ionization energy of water (12.56 eV), the basic molecule of life, and has the opposite direction to the scale of ionization energy (see Fig. 2.6). This scale has never really come to the fore, probably because the name is somewhat confusing as the same term had already been used for a completely different parameter, namely for the electrical transmembrane potential. On the other hand, it is quite illustrative.

In electrochemistry, a more common scale is used, namely the scale of redox potentials. This is based on measurements with hydrogen electrodes, i.e. platinum electrodes, surrounded by hydrogen gas. A potential of 0.82 V exists between a hydrogen electrode and an oxygen electrode, generating water. This therefore corresponds to the reference point of Szent-Györgyi's biopotentials.

The energies, depicted in Fig. 2.6, refer to bonds which are involved in the corresponding reactions. If chains of biochemical reactions are studied, the term

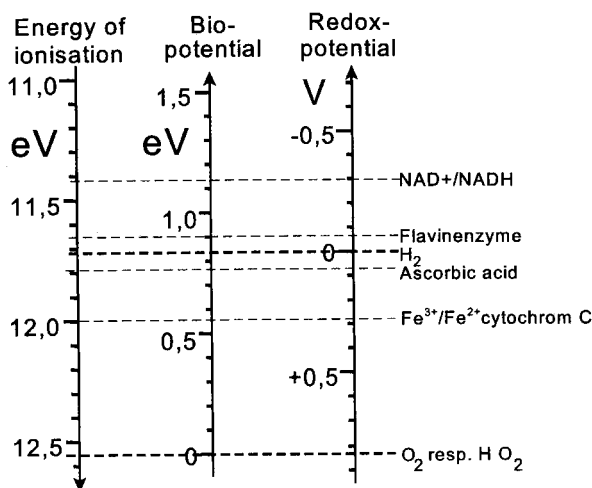


Fig. 2.6. Correlations of the scales of ionization energy, redox-potential and Szent-Györgyi's biopotential

transfer potential is used. The redox potential would thus be an electron transfer potential. Similarly, the transition of ATP → ADP is referred to as a phosphate transfer potential. In general terminology, these are all group transfer potentials. As we will see later, in the terminology of phenomenological thermodynamics it is a standard free Gibbs energy of formation (ΔG^0) with the following relationship to the standard redox potential (E^0):

$$\Delta G^0 = -nFE^0, \quad (2.2.1)$$

where n is the number of transmitted electrons and F is the Faraday constant.

Redox processes are of great importance in biological metabolism, since oxygen has through the ages progressively invaded what was initially an anaerobic world. The electron transfer from O₂ to its end product of reduction – H₂O is a complicated process and is attributed to the formation of free radicals. A radical is an electrically neutral species with an unpaired electron. These radicals can occur either on a photodynamic way, or by catalysis of transition metals or enzymes. A special role in cell metabolism is played by the so-called *reactive oxygen species* (ROS). They trigger chain reactions capable of damaging the different constituents of the cell. The organism uses different anti-oxidant defense systems against this toxicity of oxygen. We will come back to the role of ROS when discussing primary processes of radiation chemistry of water (Sect. 4.5.2).

In connection with the processes of charge transfer, the *semi-conductor effects* shall be mentioned. In solid state physics, a semi-conductor is defined as a material the electric resistance of which is higher than that of a metallic conductor indicating a very specific temperature dependence. Semi-conductor phenomena can be based on mechanisms which differ significantly from each other. This is particularly true for organic semi-conductors.

Evidence for the existence of semi-conductor phenomena in biological structures is very difficult to obtain. Objects which contain water usually exhibit an ion conductivity which masks the semi-conductor effect. Water-free macromolecules, on the other hand, may be considered to be denaturated. In spite of these difficulties, semi-conductor properties have been verified from proteins, nucleic acids and lipids. Its role in biological functions, however, is still unclear. In some cases also the effect of supra-conductivity was also proposed to occur in biological molecules. Supra-conductivity is to be considered as an electron transition with negligible loss of energy.

In relation to the above mentioned mechanisms, reference must also be made to *piezoelectric effects* in biological matter. These comprise translocations of charges in crystals or crystalloid macromolecular structures resulting from mechanical loading, such as compression, elongation or bending. Piezoelectric properties have been found in bones, wood and various tissues (including tendons), as well as in isolated biomacromolecules. It should be noted that the piezoelectric properties of bones are not the result of mechanical tension in the inorganic crystals, but rather of the protein components of the bone. The biological role of this piezoelectricity is unclear. In connection with the streaming potentials occurring in living bones, there may be a connection with processes of bone remodeling (cf. Sects. 3.5.2 and 3.5.6).

Further Reading: For free radicals and biochemical redox-systems: Buettner 1993; Bucala 1996; for supra-conductivity of biological materials: Tributsch and Pohlmann 1993; for piezoelectricity: Fukada 1983; Guzelsu and Walsh 1993

2.2.3

Photosynthesis as Process of Energy Transfer and Energy Transformation

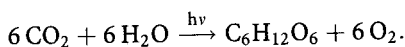
Molecular processes of energy transformation take place in many metabolic systems. Of special interest, however, are the primary reactions in which the quantum energy of sunlight is transformed into the energy of chemical bonds. This is the case in photosynthesis, as well as in photoreception. Photosynthesis is the first step of energy gain in the biosphere. Photoreception on the other hand, is a process of signal transduction.

About 0.05% of the total 10^{22} kJ energy which reaches the earth every year from the sun is assimilated by photosynthesis. This is the general energetic pool for all living processes of the earth. The efficiency of photosynthesis is very high, compared with our recent technical equipment. A large amount of the energy of sunlight, absorbed by the photosynthetic reaction centers of the green plants is transformed by the primary process of photosynthesis. During subsequent metabolic processes of energy transfer, however, an additional loss of energy occurs. The total efficiency of the process of photosynthesis, in fact, is assumed to be about 5%.

In eucaryotic plants the process of photosynthesis occurs in the chloroplasts, especially in the thylakoids located there. *Thylakoids* are flat vesicles with a

diameter of about 500 nm, stacking together to form so-called *grana*. About 10^3 thylakoids are located in one chloroplast. Every thylakoid contains about 10^6 pigment molecules. A precondition for the efficiency of energy transfer in the process of photosynthesis is the high degree of supramolecular organization.

In general, photosynthesis can be considered as a reaction during which water is split, driven by the energy of photons, producing O_2 , and transferring hydrogen to the redox system NADPH/NADP⁺, the nicotinamid-adenin-dinucleotid phosphate. Simultaneously, a proton gradient across the thylakoid membrane is generated. Within a separate process, this leads to the synthesis of ATP. Subsequently, an ATP-consuming synthesis of carbohydrates occurs. This process is usually called the *dark reaction* of photosynthesis occurring in the stroma of chloroplasts. In general, photosynthesis can be considered as the reversal of respiration and can be characterized roughly by the following scheme:



This process is characterized by a standard free Gibbs energy of reaction of $\Delta G^0 = +2868 \text{ kJ mol}^{-1}$ [the basic thermodynamic parameters will be explained in detail in Section 3.1.2; see Eq. (3.1.31)].

As indicated in Fig. 2.7, the absorption of the photons $h\nu_1$ and $h\nu_2$ occur at two points: at photosystem I, and at photosystem II. Each of these photosystems has its own reaction center, the place where the photochemical processes actually occur, and an antenna system, that occupies the largest part of the complex. These antennae contain polypeptides with light absorbing pigments. They are responsible for the absorption of light of various wavelengths and, correspondingly, quantum energies, and for their transport in the form of so-called *excitons* to the reaction center. This energy transmission is realized by processes of non-radiant energy transfer. One reaction center corresponds to nearly 300 antenna molecules. This antenna system, in fact, enlarges the diameter of optical effectiveness of the photoactive pigments nearly by two orders of magnitude. Depending on the species of plant, chlorophyll and various pigments (e.g., carotenoids like xanthophylls) may be part of these antennae. The size, composition and structure of the antennae are quite different for bacteria, algae and various higher plants. This is understandable, considering the large differences of intensities and of spectral characteristics of the light leading to photosynthesis in organisms ranging from submarine algae, to tropical plants. Probably, the antenna molecules of photosystem I and photosystem II are interconnected to each other. If photosystem II, for example, is overloaded, absorbed energy will be transferred to photosystem I by a so-called *spillover process*. There are further mechanisms to protect the photosystem from over exposition.

In Fig. 2.7. the primary process of photosynthesis is depicted schematically. The complexes: photosystem I (PSI), photosystem II (PSII), as well as the cytochrome b/f complex (Cyt b/f) are seen as components of the thylakoid

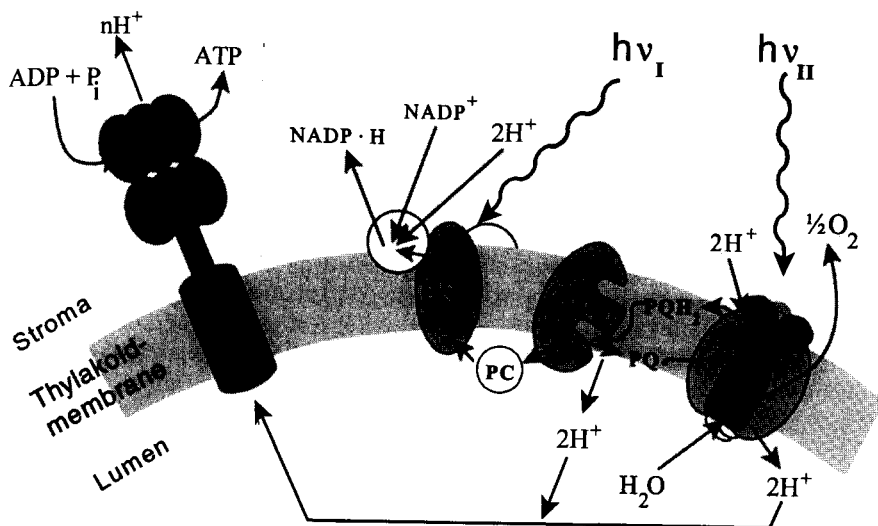


Fig. 2.7. Structural and functional organization of primary processes of photosynthesis in the thylakoid membrane. Explanation in text. (From Renger 1994, redrawn)

membrane. The light-induced splitting of water occurs in the so-called water oxidizing multienzyme complex, structurally connected to photosystem II, where the molecular oxygen becomes free, but the hydrogen, in contrast, is bound by plastoquinone (PQ), forming plastoquinone (PQH₂). The cytochrome b/f complex mediates the electron transport between PSII and PSI, reducing two molecules of plastocyanin (PC), using the hydrogen of PQH₂. This light-independent process at the same time extrudes two protons into the internal volume of the thylakoid. In a second photochemical process, the photosystem I (PSI), using the redox potential of plastocyanin, transfers one proton to NADP⁺, producing NADPH, one of the energy rich products of photosynthesis.

As a general result of this process, starting with molecular excitations by light, including a system of extremely complicated redox reactions, protons are pumped from the stroma into the internal volume of the thylakoids. This electrochemical gradient of protons is the energy source of an ATP-synthetase, which finally releases ATP as a mobile accumulator of energy into the stroma.

These mechanisms of absorption and transport of energy in the system of photosynthesis, were recently investigated using modern methods of photometry with laser flashes, the durations of which were as short as 10⁻¹⁴ s. To study the mechanisms of primary photobiological reactions, many investigations were performed on bacteria rhodopsin, a pigment-protein complex which is also located in a special system of membranes.

In addition to photosynthesis as the primary process of energy conversion, a number of other photobiological processes exist, which are responsible for light

reception and therefore use photons as a source of information. This concerns the following physiological functions:

- *Visual perception* of photons to obtain a more or less comprehensive image of the environment
- *Phototaxis* – movement of an organism by orientation towards light
- *Phototropism* – movement of parts of an organism by orientation towards light (for example: growth direction of a plant)
- *Photonastic movement* – movement of parts of an organism by orientation towards light with the help of performed structures (for example: daily orientation of blossoms)
- *Photodinesis* – light-induced alterations of plasma flow (for example: in algae)
- *Photomorphogenesis* – light controlled differentiation.

Of course, in these functions, as in photosynthesis, the capture of photons by receptor molecules means a gain in energy. This gain, however, is low compared with the energy turnover resulting from subsequent reactions. Just one single photon is sufficient to excite the visual cell of a vertebrate eye. These reactions are possible only by processes of amplification, which need a considerable input of metabolic energy. This high sensitivity of the receptors is associated with a high tolerance of the system to overexposure. The visual cell which is even responsive to the absorption of one single photon tolerates a maximal irradiation of 10^5 quanta. We will mention this problem during the discussion of formal theories of radiation effects in Section 4.5.5.

Further reading: Barber 1992; Deisenhofer and Norris 1991; Govindjee 1975; Hall and Rao 1994

2.3

Thermal Molecular Movement, Order and Probability

In this section, the biophysics of molecular organization of biological systems will be discussed in the context of processes of thermal movements. Having presented atomic processes, which are best described by the equations of quantum and wave mechanics, we now come to topics where a statistical mechanics approach can be applied. Stochastic phenomena are of great importance in molecular biophysics. Here, we are immediately confronted with the dialectics of arbitrary distribution on the one hand, and organized order on the other. This touches on problems of biological self-organization and stability of the resulting structures.

2.3.1

Thermodynamic Probability and Entropy

In 1854, Rudolf J. E. Clausius introduced the *entropy* (S) as a parameter of phenomenological thermodynamics, and defined it as the heat, added to a

system in a reversible way in relation to the temperature [see Eq. (3.1.10)]. Later, in 1894, Ludwig Boltzmann used this parameter in the framework of statistical thermodynamics. In this context, entropy and the second principle of thermodynamics become more imaginable. Entropy appears as a kind of measure of disorder, or as a degree of random distribution, i.e. of missing order. The correlation between order and probability and, as will be explained later – information – is of great importance for the understanding of the principles of biological organization.

Let us start with the assumption that entropy is a measure of randomization of a given distribution. We will consider a system of maximal entropy as a system in maximal disorder. Furthermore, let us demand that the entropy be an extensive parameter. Therefore, like volume, or mass, but in contrast for example to temperature or density, the entropies S_1 and S_2 of two systems can be added, if these systems come together:

$$S_1 + S_2 = S. \quad (2.3.1)$$

How can we now define a parameter which indicates the degree of randomization or, on the contrary, the degree of order? What does order of organization mean? Of course, our daily experience shows that an ordered system spontaneously transforms into a disordered one, but not vice versa. This, actually, is the consequence of the second principle of thermodynamics.

Let us consider a very simple structure, just the distribution of four distinguishable spheres in two compartments of a box (Fig. 2.8). Let each of these spheres, independently of the three others, fall just by chance into one or the other compartment of the box. All of the 11 possibilities of the distribution, as indicated in Fig. 2.8, therefore, have the same degree of probability, because the probability of each sphere individually falling into compartment 1 or into compartment 2 is equal. Summarizing the patterns of distribution, there is only one way to realize the distributions 0:4 and 4:0. In contrast, there are four ways to realize the distribution 3:1 and 1:3, and, finally, six ways for equal distribution: 2:2.

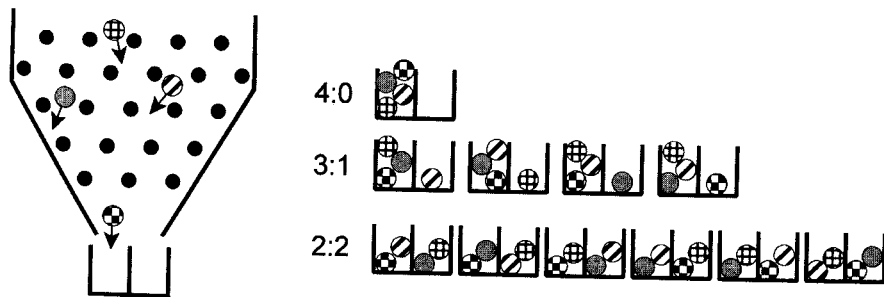


Fig. 2.8. All possibilities of the statistical distribution of four spheres in two compartments of a box

Let us now ignore the fact that the spheres are distinguishable. Let us simply ask: how large is the probability that just by stochastic distributions one of the relations 4:0, 3:1, 2:2, 1:3, or 0:4 occurs? Apparently, the probability of any kind of distribution will be larger if it can be realized in a larger number of ways. The distribution mode 2:2, for example, is six times more likely than the distribution 4:0, or 0:4. The number of ways which lead to the realization of a definite situation, in fact, seems to be a measure of the probability of the occurrence of it. We will designate this number of ways by the parameter W which we will call *thermodynamic probability*. The amount of W can be at least 1 and at maximum ∞ , in contrast to the mathematical probability (P), which we will use in Chapter 2.3.2, and which ranges between 0 and 1.

Now we come to the following sequence of conclusions: if W really is a measure of the probability of getting a definite distribution, and if an increase of the degree of order is the most uncertain result of a stochastic distribution and finally, if the entropy (S) is a parameter, indicating the degree of disorder, then S should be a function of W . If two situations with relative probabilities W_1 and W_2 are connected together, then the probability of this combined situation results from the product ($W_1 \cdot W_2$). Using Eq. (2.3.1), this means:

$$S = f(W) = S_1 + S_2 = f(W_1) + f(W_2) = f(W_1 \cdot W_2). \quad (2.3.2)$$

This demand is met by the logarithmic function:

$$\ln A + \ln B = \ln(A \cdot B). \quad (2.3.3)$$

Hence entropy is proportional to logarithm of W :

$$S = k \ln W. \quad (2.3.4)$$

This is the Boltzmann equation of entropy. Boltzmann's constant k was defined as a universal constant later by Max Planck. It must have the same unit of measurement as entropy, and is as follows:

$$k = 1.380658 \cdot 10^{-23} \text{ J K}^{-1} = 8.6174 \cdot 10^{-5} \text{ eV K}^{-1}.$$

This explanation has been based on the simplest experiment where four spheres are distributed randomly over two compartments. One step towards serious thermodynamics can be taken if, for example, the compartments of this box are thought of as molecules of a system, and the spheres as quanta of energy, distributed among them. This complication, of course, means a transition to the handling of larger numbers. If the number of elements and classes are increased, W cannot be evaluated just by simple trial. It is possible to calculate this value using the following equation:

$$W = \frac{n!}{n_1! \cdot n_2! \cdot n_3! \cdot \dots \cdot n_m!}, \quad (2.3.5)$$

where n is the total number of all elements in the system (in case of Fig. 2.8, the total number of spheres: $n = 4$); n_i (for $i = 1, \dots, m$) is the number of elements in each class of state (this means the number n_1 in compartment 1 and n_2 in compartment 2); and m is the number of classes of state (namely: number of compartments in the box).

2.3.2

Information and Entropy

C. E. Shannon introduced a parameter into information theory which was to some extent, by its very substance, related to entropy; this and was named *information*. The information of a message depends on the effort required to guess it by a highly set system of questions. Hence, information is some sort of degree of the actuality of a message.

It is not difficult to guess the result of the toss of a coin, since there are only two possibilities of equal probability. To guess a certain card in a full deck of playing cards is much more difficult. In this case, a much greater uncertainty factor has to be taken into account. Using a more systematic approach, a large number of yes-no questions have to be answered. Hence, the information content of a playing card is higher than that of a tossed coin. Should a deck consist of cards which are all the same, and this is known by the challenged person, guessing will not make sense at all. The information content of each of these cards is zero. The probability by which possibilities are turned into reality, consequently, seems to become a measure of information.

In contrast to thermodynamics, in information theory the mathematical term probability (P) is used, which is defined as follows:

$$P = \frac{\text{number of favorable cases}}{\text{greatest possible number of cases}}.$$

On average, coins tossed a hundred times will land with heads up in 50 instances. Hence, the probability of heads facing up may be expressed as follows:

$$P = \frac{50}{100} = \frac{1}{2}.$$

On the other hand, the probability of throwing a "six" with some dice is only $P = 1/6$, whereas the probability of throwing one of the three even numbers would be higher: $P = 3/6 = 1/2$.

Whereas the thermodynamic probability (W) is always larger than 1, (cf. Sect. 2.3.1), the value of the mathematical probability lies between 0 and 1 ($0 \leq P \leq 1$). In this context, $P = 0$ means an impossibility, while $P = 1$ expresses absolute certainty.

The logical conclusions which led to the derivation of the Boltzmann equation (cf. Sect. 2.3.1) are the same as those on which the Shannon relation is based. Information (I) is a function of mathematical probability (P):

$$I = f(P). \quad (2.3.6)$$

The conditions for the function f in Eq. (2.3.6) are again satisfied by a logarithmic function, since here too, the multiplication rule for the calculation of probabilities must be valid [cf. Eq. (2.3.3)]. Therefore:

$$I = K \ln P. \quad (2.3.7)$$

This is the *Shannon equation* of information theory. The unit of I is determined by the unit of the factor K . The bit ("binary digit") is most commonly used. It expresses the number of binary yes-no decisions which are needed to determine a given message. For example, the one side of the coin can be guessed by one single decision, its information value, consequently, is 1 bit. Five binary decisions will be sufficient to guess a card from a deck. Hence, the information value of one card is 5 bits. The factor $K = -1/\ln 2 = -1.443$ must be used to calculate I in bits. In information theory the logarithm to the base 2 ($\log_2$) is occasionally used:

$$I = -1.443 \ln P = -\log_2 P \quad (I \text{ in bit}). \quad (2.3.8)$$

If Boltzmann's constant (k) is used ($K = -k$), then information is obtained formally in entropy units:

$$I = -k \ln P \quad (I \text{ in } J K^{-1}) \quad (2.3.9)$$

The similarity of Eqs. (2.3.9) and (2.3.4) is evident. The formal interconnection of Shannon's definition of information with entropy triggered off an intense discussion about the relation between these parameters. The starting point for this discussion was the characterization of entropy as the measure of disorder according to the second principle of thermodynamics. Isolated systems spontaneously try to reach a state of maximum disorder. This can be seen in the process of information transmission where a message loses some of its information value.

The idea of a link between information and entropy was first suggested by Boltzmann and was then developed by other authors. E. Schrödinger (1944) made the frequently quoted statement: "The living system feeds on negative entropy". This is the reason why sometimes the term "negentropy" is used.

What is the real importance of Shannon's information equation in biophysics? In principle, it is possible to calculate the information content of a protein by the above approach. The requirements are, firstly, a statistical record of the frequency of the occurrence of the individual amino acids in proteins. This will provide the probability (P) for the presence of a given amino acid at a certain locus in the protein. Subsequently, using Eqs. (2.3.8) or (2.3.9) the information content of each monomer can be calculated. Subsequently, the information content of a whole protein can be obtained by addition of the values of its monomers.

The information content of a nucleic acid can be obtained in the same way. One mammalian DNA molecule consists on average of 15 000 pairs of nucleotides. Assuming that the four possible types of nucleoside bases have an equal probability of occurrence, then the information content of each single nucleotide will, consequently, have a value of 2 bits. The information capacity of one DNA molecule, in this case, amounts to 30 000 bits.

The problems of this kind of calculation may be illustrated in the following example: the information content of an English text can be calculated from the frequency of the use of individual letters. In this way one can derive the information content of a word, a sentence, and, consequently, even of this textbook. It is, however, obvious that this parameter does not reveal anything about the "information value" of the book as generally understood. The same information (I) would be given by any other book with the same number of meaningless strings of English words.

This situation, however, does not invalidate Shannon's information concept. Everybody knows how important the calculation of information is today in the field of computer science. So, for example, for the author of this book it is quite important to know which sort of discs he needs to store the text and the figures of this book. But, obviously, this is just a question of its volume, of the codes used, but in no case of its "information value" in the common sense.

Does this mean that it is impossible to quantify biologically important information? Does it mean that the information concept is not applicable to biological systems at all? In fact, there is no reason for scepticism. Consequently, a distinction has to be made between a syntactic measure of information, and a semantic measure.

The *syntactic* information content of a DNA molecule, as calculated above, provides some information on the maximum storage capacity of the genome. The amount of information actually stored is even lower, if the redundancy in the storage of genetic information which is required for the protection of the information is taken into account. Estimates of information capacity in the genome vary between $3 \cdot 10^2$ bit and 10^{12} bit.

The *semantic* information, in contrast to the syntactic information, really should contain some kind of validation of the content. Despite many attempts, quantification of semantic information has not yet been achieved. In spite of the problems of giving an exact and meaningful quantification of information, it is doubtless a quality of matter and has a place in information theory and thermodynamics.

The interconnection of information with entropy in terms of thermodynamics may be illustrated best by a thought experiment conceived by James Clerk Maxwell in 1881 and still discussed today (Fig. 2.9). Maxwell proposed a room, which is connected with another by an opening. This opening can be closed by means of a slide. Both rooms are filled with a gas which is, in the beginning, in equilibrium, e.g. of equal pressure and temperature. An intelligent creature, later called "Maxwell's demon", is able to handle the slide between the two rooms with ease. This "demon" can observe accurately the direction and the velocity of the molecules in his room. Velocity and direction of these particles are

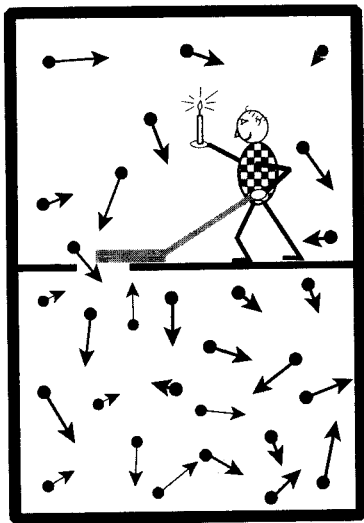


Fig. 2.9. Maxwell's demon

statistically distributed. If a particle flies accidentally toward the opening, the demon opens the slide to let the particle pass. As a result of such sorting, the pressure in the lower room would rise. The demon could also take another approach. For example, he could separate fast from slow particles. In this case, a difference in the temperature between the two rooms would occur. In both cases, the entropy of the whole system would be reduced and energy might be generated by an appropriate turbine. The result would be a "perpetuum mobile", a perpetual motion machine of the second order, as it would contradict the second principle of thermodynamics.

This apparent contradiction was subsequently the subject of a large number of scientific publications. The problem, however, may be resolved by the following consideration: the "demon" requires information to carry out the sorting. He collects this information by "watching" the molecules. In order to "see" he needs light. For this, the demon in Fig. 2.9 symbolically carries a lit candle. Yet, a body will only be able to emit light in a state of nonequilibrium relative to its environment. This, however, contradicts the equilibrium condition at the beginning of the experiment. The same would apply to any other approach to acquisition of information. This resolves the apparent contradiction with the second law of thermodynamics.

Why do we discuss this thought experiment here, if it is clear that it will not work as a perpetual moving machine? What has this to do with biophysics? In fact, independently of this discussion about the virtual violation of the second law of thermodynamics, Maxwell's demon demands particular interest in biophysics because of its analogy to various functions of living systems. The living cell, too, reduces its entropy at the expense of its environment, using information processing. Yet, in this case it is not the energy of statistical

fluctuations which is used. The biological system selects molecules from its environment which are rich in free Gibbs energy of formation, and correspondingly have a low content of entropy. It uses this energy and extrudes molecules with lower free energy and larger entropy. The basic information for this process of selection, in other words, the “software” for this process, is stored in the structure information of the proteins, which are responsible for the recognition of these molecules, and eventually for their metabolism. These proteins get their information during the process of synthesis via the RNA, from the DNA of the genome.

This example raises the question: what is the threshold value of information which is required to control the processes of living systems? Or translated into the language of modern computer science: how large must the simplest software for this sort of a biological Maxwell demon be? What is the threshold information that carries out not only the metabolic function of the primordial organism but additionally its replication? How did the first accumulation of information in a molecule occur? A purely accidental combination of monomers to build their first functional macromolecule must be ruled out. The probability for this occurrence is too low by far. Today, so-called *prebiotic evolution* is assumed, i.e. chemical selection of developing polymers even before the stage of the biopolymer (see also Sect. 5.3.3).

Further reading: Shannon and Weaver 1962; bioinformatics in molecular biology: Kauffman 1993; Strait and Dewey 1996; Volkenstein 1994; Yockey (1992); Maxwell’s demon: Leff and Rex (1990)

2.3.3

Biological Structures: General Aspects

In the previous section we introduced expressions like *order*, *structure* and *organization* and discussed them in relation to entropy and information, as well as with the statements of the second law of thermodynamics. This touches on a set of questions which are of central interest in biophysics and which will be mentioned in many sections of this textbook. Therefore it is necessary at this point to explain some basic definitions and principal ideas.

What, really, is a structure? To the biologist, the term “structure”, is usually related to the macroscopically or microscopically visible organization of an organism. This means, for example, the structure of an animal skeleton, structure of a cell, of a mitochondrion etc. The term “molecular structure” already lies outside the limits of this view. It refers to a certain arrangement of atoms, without defined contours, which can be described just by means of wave mechanics. The same applies for the concentration profile of an electrical double layer (Fig. 2.42), and also for so-called time structures, namely special time courses, like oscillations of a biological system (Figs. 5.4 and 5.19), like the shape of an electrocardiogram (Fig. 3.33), or the sonogram of a bat’s cry (Fig. 4.12). This means that the definition of the term “structure”, which is used

in biophysics has to be broader than that of the morphologists and cytologists. It must include these structures as well as those of metabolic networks, ecosystems, or others.

The best, and most generalized definition of this term is given by the set theory of mathematics. Therefore: *a system is an aggregate of elements with certain interrelations between them.* The totality of these interrelations is called the *structure* of the system. This definition does not prescribe at all what kind of elements, and what kind of interrelations these are. It is applicable to all kinds of systems including biological systems and structures. In biophysics, we are especially interested in *dynamic systems*, i.e. where the interrelations between their elements are *interactions*. In contrast to this, in *static systems* the elements have no interaction at all, but are just interrelated by formal *relations*. Examples for static systems, are the system of natural numbers in mathematics, or the system of animal species or of plants in biology.

The elements of a metabolic network are the metabolites, and the interrelations between them, i.e. their interactions, are the steps of biochemical reactions. Correspondingly, an ecosystem is to be considered as an interaction of individuals and populations depending on abiotic conditions.

Biological macromolecules can be considered as systems too. In this case, the following levels or organization occur, depending on what will be assumed as their elements.

- *Primary structure* - linear sequence of monomers (=elements) in the linear molecular chain (=system). For example, a polypeptide chain: ...-serine-alanine-lysine-arginine-...
- *Secondary structure* - positioning in space of monomers (=elements) in a part of the molecule (=system) relative to each other. For example, the α -helix, or the β -sheet structure of an amino acid sequence in a protein
- *Tertiary structure* - position in space of molecular regions of homogeneous secondary structures (=elements) in a molecule (=system). For example, intramolecular coordination of the position of several helical regions relative to each other or to a β -sheet
- *Quaternary structure* - position in space of macromolecules (=elements) in a supramolecular complex (=system). For example, single proteins in a multi-enzyme complex

When a salt is crystallized, a *periodic structure* occurs, which is characterized by a periodic arrangement of the elements. E. Schrödinger (1944) called the biological structure an *aperiodic crystal*. This means a highly organized structure, the elements of which are not simply repeated periodically. Sometimes one tries to evaluate this structural organization as *structural information*. As we pointed out in the previous section, however, it is hard to quantify this parameter. Probably, the structural information should be measured as the effort which is necessary to describe such a structure perfectly.

Consequently, the processes of structure formation of biological systems, be it the development of life, its reproduction, or simply the biosynthesis of a macromolecule, are all accompanied by reduction of entropy. This appears to be contrary to the second law of thermodynamics and has given rise to heated philosophical discussions in the past. The second law of thermodynamics actually postulates that in spontaneous processes occurring in isolated systems, the entropy strives towards a maximum. Yet, neither an organism nor its environment, i.e. the earth as a whole can be considered as an isolated system. The earth is constantly absorbing energy from the sun and is emitting this energy again. That continuous flow of energy maintains a permanent nonequilibrium state which manifests itself not only in a direct way in photosynthesis with subsequent metabolism of heterotrophic organisms, but also in the environment of life, for example, in flowing water, in alternation of light and darkness, and in changes in temperature, humidity etc.

In fact, structures are also formed in inanimate nature under loss of entropy. Basically, a distinction must be made between two kinds of structure formation which at the same time explain the two ways of self-organization in living systems.

- *Equilibrium structures*, for example a salt crystal, formed spontaneously during evaporation of solvent
- *Nonequilibrium structures* (or: *dissipative structures*), for example an arrangement of cirrus clouds as the result of processes of air convection under special meteorological conditions

The genesis of life which is based on the prebiotic formation of a first bio-macromolecule, and the subsequent evolution of all organisms, can be understood as the result of complicated processes which occur far from equilibrium. In this context, a major role is played by dissipative structures of the inanimate environment, such as periodic temperature variations, tides etc. Substantial importance also has to be attributed to such nonequilibrium structures in the course of life itself. They are frequently represented by time structures, such as for example the heart rate, or other kinds of oscillations, which in some cases are associated with the so-called biological clock. (for further explanations see Sects. 3.1.4, 5.3.2 and 5.3.3).

Equilibrium structures, such as inorganic salt crystals with a very primitive geometry, become very complex and variable in shape, when based on the sophisticated pattern of the primary structure of bio-macromolecules rather than on the relatively simple field of interaction of spherico-symmetric ions. The spontaneous folding of proteins and their arrangement to supramolecular structures, such as complex enzyme systems of even ribosomes, must be viewed in this way. Such processes are also referred to as *self assembly*. More specific aspects relating to the formation of equilibrium and nonequilibrium structures will be given in subsequent chapters of this textbook.

Further reading: Eigen (1971, 1992); Meinhardt (1982); Kauffman (1993); Strait and Dewey (1996)

2.3.4

Distribution of Molecular Energy and Velocity at Equilibrium

The Boltzmann equation of entropy [Eq. (2.3.4)] as derived in Section 2.3.1, helps to illustrate the second law of thermodynamics, according to which isolated systems spontaneously approach a state of maximum entropy. We demonstrated there that one can also express this in the following way: at equilibrium isolated systems reach a state of highest realization probability (maximum of W). Now we will ask the question: what are the mean properties of the molecules at this equilibrium state? This is of fundamental importance for further considerations.

It is possible to answer this question with the help of statistical thermodynamics, which helps to make predictions on the probability of distribution of energy among the elements of a system in thermodynamic equilibrium. It should, however, be emphasized that a deterministic description of the properties of single particles lies outside the limits of this discipline. Nevertheless, it will be seen that even a statistical statement allows the drawing of valuable conclusions regarding reaction rates, stability of molecules and more.

Let us imagine a space which contains gas molecules of uniform mass (m), which at the beginning all have the same velocity (v), and consequently, the kinetic energy (E) which can be calculated by the following equation:

$$E = \frac{m}{2} v^2. \quad (2.3.10)$$

This equality of the kinetic energy of all molecules is, in fact, a highly improbable state. In this case, using Fig. 2.8, all molecules would belong to a single box, or a single class of properties n_i . Thus, according to Eq. (2.3.5): $W = n!/n_i! = 1$. This situation will change instantaneously. The molecules would exchange their energy by elastic collisions with each other, and soon a great number of energy states would be occupied. A simple mathematical example shows that W , and according to Eq. (2.3.4) also the entropy (S), will increase with a growing number of (m) of state classes, provided the following relation applies:

$$\sum_{i=1}^m n_i = n.$$

Because of the law of conservation of energy (first principle of thermodynamics), the following condition must be satisfied at the same time:

$$\sum_{i=1}^m n_i E_i = \text{const.}$$

The energy of this system, therefore, can be distributed randomly among all of its molecules. The total energy of the system, however, must always remain constant.

Now, let us look for a state with maximum probability; this means with maximum entropy, and corresponding to the second principle of thermodynamics this is actually the thermodynamic equilibrium. Translated for our example: how will the energy and, correspondingly, the velocity be distributed between the particles, if the system is isolated, and if it is left to itself for a sufficiently long period of time? Even under equilibrium conditions, of course, the energy of individual molecules will change permanently, but nevertheless the statistical mode of energy distribution, or of the distribution of various degrees of molecular velocity at equilibrium becomes stationary, i.e. time independent (for a detailed description of various kinds of stationary states see Sect. 3.1.4).

Considerations of this type lead to *Maxwell's equation of velocity distribution*:

$$\frac{dn(v)}{n_0 dv} = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2kT} \right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}}. \quad (2.3.11)$$

The left part of this equation contains a relation which expresses the relative number (dn/n_0) of those molecules which are related to a particular velocity interval (dv). Its unit is $s\ m^{-1}$. This is a function expressing the probability of the distribution of the velocity, where m is the mass of a molecule [not to be confused with the numeral m in Eq. (2.3.5)], and k is the Boltzmann constant. Multiplying in Eq. (2.3.11) the denominator and numerator of the expression within the brackets as well as that in the exponent with the Avogadro number ($N = 6.023 \cdot 10^{23}\ \text{mol}^{-1}$) introduces molar, instead of molecular parameters:

$$M = N \cdot m, \quad (2.3.12)$$

and

$$R = N \cdot k, \quad (2.3.13)$$

where M is the molar mass, and $R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$ gas constant.

In Fig. 2.10, as an example, the velocity distribution of molecules of oxygen is depicted ($M = 0.032\ \text{kg mol}^{-1}$). This curve is not symmetrical. The mean velocity (weighed arithmetic mean value) in general will be higher than the maximum value. The following relation applies:

$$v_{\max} = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}}; \quad \bar{v} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}. \quad (2.3.14)$$

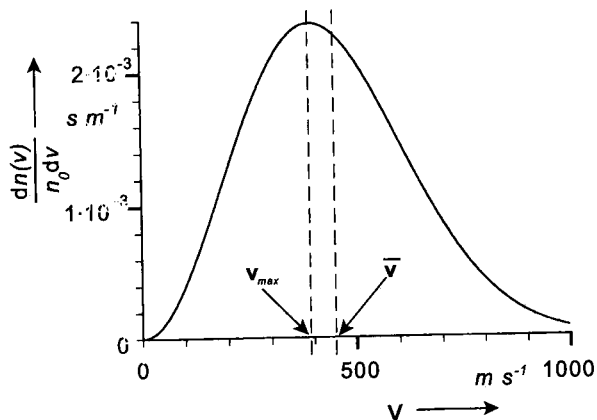


Fig. 2.10. Velocity distribution of O_2 molecules at $37^\circ C$ based on Eq. (2.3.11)

These values increase with increasing temperature (T) and decrease with increasing molar mass (M).

Maxwell's velocity distribution is based on the Boltzmann function of energy distribution. It determines the number (n_i) of those particles which under equilibrium conditions of the system have an energy level of E_i :

$$n_i = C e^{-\frac{E_i}{kT}}, \quad (2.3.15)$$

(C is a factor of standardization). The exponent of this equation, as in Eq. (2.3.11), expresses the relation between energy E_i of the particle and the mean energy of thermal noise (kT). The Boltzmann constant (k) should be replaced by the gas constant (R) in cases where E_i expresses not the energy (in: J) of a single molecule but is a molar parameter (in: $J\ mol^{-1}$), as is customary in chemical thermodynamics.

The Boltzmann equation predicts the distribution of any form of energy. It expresses the situation wherein lower states of energy tend to occur more often than states of higher energetic levels. Related to the total number of particles, it may also be written in the following form:

$$\frac{n_i}{n_{\text{total}}} = \frac{e^{-\frac{E_i}{kT}}}{\sum_{i=0}^{\infty} e^{-\frac{E_i}{kT}}}. \quad (2.3.16)$$

This equation provides the basis for many calculations in statistical thermodynamics. For example, it can be used to determine the percentage of molecules in a given system which are able to overcome a certain energy barrier (E_i). This aspect will be discussed in greater detail in the following chapter.

The use of this equation is allowed only in cases of thermodynamic equilibrium. At first glance it seems to limit its applicability to biological

systems which are often far from equilibrium. The process of the adjustment of the Boltzmann distribution is, however, so fast that an imbalance of this kind has to be considered only in rare cases. On many occasions in this book we will indicate that in fact subsystems could be in equilibrium even in cases where the whole system is far from it (see especially Sect. 3.1.4).

2.3.5

Energy of Activation, Theory of Absolute Reaction Rate

The temperature dependence of the rate of a chemical reaction was described by S. Arrhenius by the following equation:

$$k_R = Ae^{-\frac{E_A}{RT}}. \quad (2.3.17)$$

Here, k_R is the rate constant. The energy E_A was defined as *activation energy*; A is an empirical factor.

This equation describes not only the temperature dependence of chemical reactions. It can also be used to describe reaction rates of other physiochemical processes, such as diffusion, kinetics of phase transitions etc. At certain temperature intervals, complicated biological processes, like growth rate, or heart rate of poikilothermic animals etc., can also be fitted in this way.

To evaluate experiments in which time dependent parameters like rate constants, reaction rates, frequencies of oscillations etc. were measured as functions of temperature, a so-called *Arrhenius plot* may be applied (see. Fig. 2.11). In this case the logarithmic form of Eq. (2.3.17) is used:

$$\ln k_R = \ln A - \frac{E_A}{RT} \quad (2.3.18)$$

Applying this kind of evaluation, $y = -\ln k_R$ is plotted against $x = 1/T$. If in the investigated process the activation energy itself is temperature independent, then the measured points are located on a straight line:

$$y = ax + b \quad (2.3.19)$$

The slope of this straight line is, according to Eq. (2.3.18): $a = -E_A/R$. The relation: $b = \ln A$ and, consequently, the factor of A can be read from the extrapolated value of $x = 1/T = 0$. Direct derivation of the logarithmic function to dT is more complicated. We need it later:

$$\frac{d \ln k_R}{dT} = \frac{E_A}{RT^2}. \quad (2.3.20)$$

Real biological processes usually consist of a large number of single reactions, each with different activation energies. Hence, a temperature dependence must

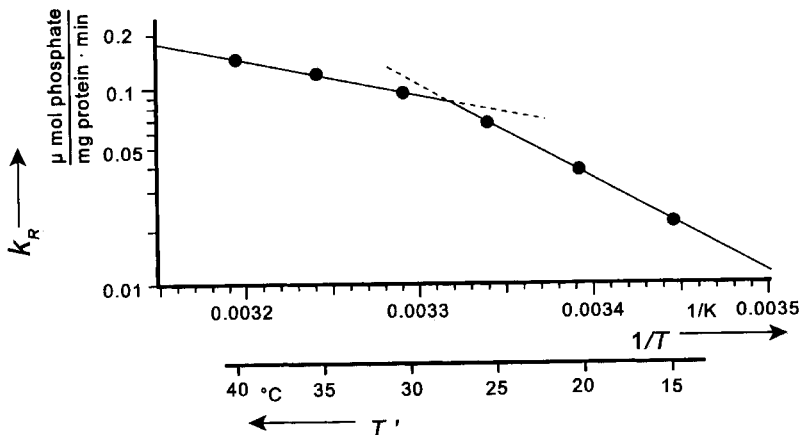


Fig. 2.11. An example for an Arrhenius-plot: the logarithm of the activity of isolated ATPase (A) is plotted versus the reciprocal value of the temperature in a region between 15 and 40 °C. (Parameters from Blank and Soo 1992)

be expected which is much more complicated than that expressed by the Arrhenius equation [Eq. (2.3.17)]. Surprisingly, however, even in these cases functions will still be obtained which can be described by straight lines in the Arrhenius plot. This is caused by the fact that in complicated processes the rate of the overall process is determined by a simple rate limiting reaction. The Arrhenius plot just reflects the activation energy of this rate limiting process.

In some physiological papers, a so called *Van't Hoff rule* of temperature dependence is used. According to this the rate of a reaction increases by two to three times, if the temperature rises by 10 °C. This factor is referred to as Q_{10} value. This rule, however, is not supported by thermodynamic considerations. Even in cases where the energy of activation is constant, according to the Arrhenius equation, the Q_{10} value will increase with increasing temperatures.

In some reactions the energy of activation (E_A) itself is a function of temperature. In these cases the straight line in the Arrhenius plot gets kinks (Fig. 2.11). The reasons for this can vary. In complex processes it is possible that at different temperatures, different reactions with different energies of activation will become rate limiting. In other cases such transitions may occur as the result of conformational changes of one of the components of the reaction, probably a conformational change in the enzyme itself. In reactions of membrane-bound enzymes, sometimes phase transitions of the enzyme near lipids may cause such effects.

The statistical mechanics allow a direct derivation of the Arrhenius equation. This is based on the *theory of absolute reaction rate* (or *theory of transmission states*) as developed by H. Eyring. Here we can explain just some basic ideas of this approach.

Chemical reactions presuppose splitting of chemical bounds. To analyze such processes it is necessary to know the interaction energy of a given atom in the molecule as a function of the distance to each of its neighbors. This is an intricate function which can be expressed mathematically as an n -dimensional space corresponding to n possible ways of geometrical approach. In this space a line may be found which shows the way from state A to state B over the smallest peaks of the activation energies. This line is called the *reaction coordinate*. It is possible to represent the energy level along this line in a two-dimensional graph.

In Fig. 2.12 such a function with two minima is plotted, representing the stationary states A and B. In these positions, the molecule will be in a stable state, which is however continuously deflected in a direction of the reaction coordinate by thermic collisions (the term *stability* will be explained in detail in Sects. 3.1.4 and 3.5). The effectiveness of those collisions depends on their energy i.e. on the relation of their energy to the activation energy E_A . The energy of collisions, in fact, is thermic energy, which can be expressed by the factor kT (or in molar units, correspondingly in RT). The Boltzmann equation [Eq. (2.3.16)] allows the calculation of the relative number of those molecules (n_i/n_{total}) which are able to overcome the energy barrier (E_i), which in our case is identical with the energy of activation (E_A). The relation E_i/kT [Eq. (2.3.16)] of course decreases with increasing temperature (T). This is illustrated in Fig. 2.13. While 36.8% of all particles have reached the energy level $E_i/kT = 1$, a value twice as high is only reached by 13.5%, and for $E_i/kT = 3$ there are only 5% reaching this level. The percentages drop fast with rising values of E_i/kT .

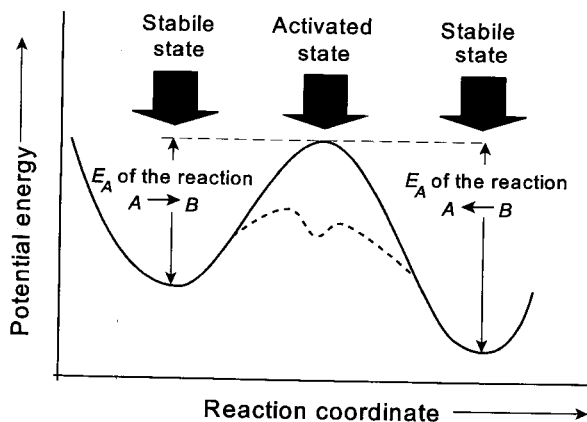


Fig. 2.12. Schematic potential energy diagram of a molecule in the states A and B taken along the reaction coordinate. The *solid curve* represents the reaction without a catalyst; all the explanations correspond to this curve. The *dashed line* indicates how the energy of activation is modified in the presence of a catalyst (enzyme). In this case, the curve with a single maximum, dividing the states A and B is transformed into another with two maxima, between which the minimal energy of an enzyme-substrate complex is positioned

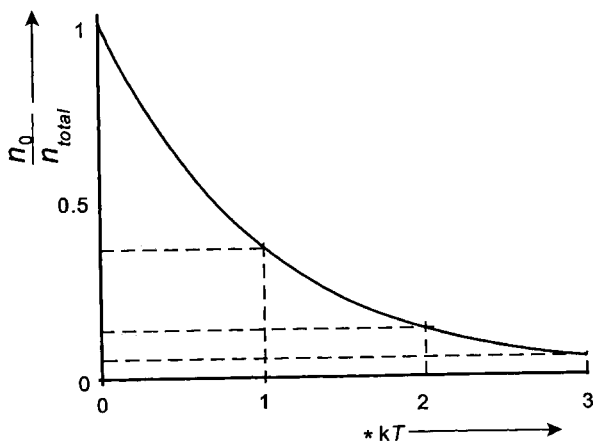


Fig. 2.13. The relative number (n_0/n_{total}) of molecules as a function of the energy in kT -units. This curve corresponds to Eq. (2.3.16)

The theory of absolute reaction rate is based on the assumption that an activated transition state between the two states A and B exists. Its potential energy is identical with the energy maximum in Fig. 2.12. This activated state is, of course, highly unfavorable and unstable from the energetic point of view. At any point in time it will be assumed only by a very small number of molecules which will drop back from the energy maximum into the stable states of an energy minimum to the right and left of this maximum. An equilibrium constant (K^*) can be defined which characterizes the relation between activated and stable states according to the mass action law. The proportionality between the rate constant (k_R) and the number of molecules in the activated state can then be expressed as follows:

$$k_R = q \frac{kT}{h} K^*. \quad (2.3.21)$$

The symbol $*$ indicates the reference to an activated state. The relation between kT and Planck's constant h comes from kinetic quantum theory as derived by Wilson-Sommerfeld. The factor q gives the fraction of activated molecules which leave to the right or left of the energy maximum. For the case of symmetry of the relations, it becomes $q = 1$.

The phenomenological thermodynamics allows calculation of the relation between the molar free Gibbs energy of reaction (ΔG) and the equilibrium constant (K):

$$\Delta G = -RT \ln K. \quad (2.3.22)$$

In the case of a reaction $A \rightarrow B$, i.e. for a reaction in which the total number of molecules remains constant, and under the above-mentioned condition of $q = 1$, it is possible to calculate the conditions for the activated state (using the symbol $*$):

$$k_R = \frac{kT}{h} e^{-\frac{\Delta G^*}{RT}}. \quad (2.3.23)$$

If one further considers the following relation:

$$\Delta G = \Delta H - T\Delta S, \quad (2.3.24)$$

it follows that:

$$k_R = \frac{kT}{h} e^{\frac{\Delta S^*}{R}} e^{-\frac{\Delta H^*}{RT}}. \quad (2.3.25)$$

The relation of this formula with the Arrhenius equation [Eq. (2.3.17)] can be derived by modification of the above equation in the following way:

$$\ln k_R = \ln T + \ln \left(\frac{k}{h} e^{\frac{\Delta S^*}{R}} \right) - \frac{\Delta H^*}{RT}, \quad (2.3.26)$$

and:

$$\frac{d \ln k_R}{dT} = \frac{1}{T} + \frac{\Delta H^*}{RT^2} = \frac{RT + \Delta H^*}{RT^2}. \quad (2.3.27)$$

Comparing Eq. (2.3.27) with Eq. (2.3.20) the following correlation is obvious:

$$E_A = RT + \Delta H^*. \quad (2.3.28)$$

For other values of q , e.g. for reactions of synthesis or lysis, this relation can be modified.

In Eq. (2.3.25) the temperature independent exponential part is of particular interest, which contains the *entropy of activation* (S^*). The entropy of activation indicates conformations of the molecules which occur during the reaction. This is important, especially in biochemical reactions in which macromolecules are included. The parameter ΔS^* can become positive as well as negative. This depends on whether the molecule during the reaction gets a larger ($\Delta S^* < 0$) or a lower ($\Delta S^* > 0$) degree of order. On the other hand, the entropy changes of the water molecules during this reaction must be considered too. We will discuss the entropy modifications of water during reactions of hydration later in detail (Sect. 2.4.2; Fig. 2.26).

The acceleration of biochemical reactions by enzymes is based on a change in the reaction coordinate. The energy levels of the starting and the end product of the reaction will remain unaltered by this. However, with the help of an enzyme as a biological catalyst the transition from one state into another is possible with lower barriers of activation energy. This is indicated by the dashed curve in Fig. 2.12. First, an enzyme-substrate complex is formed, which requires less activation energy than that needed to form the activated complex without the

enzyme. This enzyme-substrate complex is then broken up again by a small input of activation energy, and substance B will be released. It goes without saying that the reaction $A \rightarrow B$ cannot take place spontaneously unless the energy level of B is below that of A even if an enzyme is involved.

The concept of the theory of absolute reaction rate is not only applicable to processes of chemical reactions. Processes of diffusion can be described in a similar manner. Diffusion of a substance can be imagined as a reaction consisting entirely of location jumps from one position in the vicinity of a molecule to another. In this situation the reaction coordinate will largely coincide with the direction of diffusion. Investigations of activation energy of diffusion processes may provide the path of diffusion, for example, the mechanism of permeability of a molecule through a membrane.

Let us now discuss the problem of the life span of a single bound, or a whole molecule. Using the Arrhenius equation, it is also possible to calculate the stability of a given molecule. In this case, however, it is not the reaction rate of the molecular transformation which is of interest, but its reciprocal value, namely the mean time during which a bond with a given bonding energy will resist the attack of the energy of thermic noise (kT , resp. RT). In this case the energy of activation of a decay reaction (E_D) is of interest. This mean time of stability (t) is inversely proportional to the rate of transformation. Thus:

$$t = \tau e^{\frac{E_D}{kT}}. \quad (2.3.29)$$

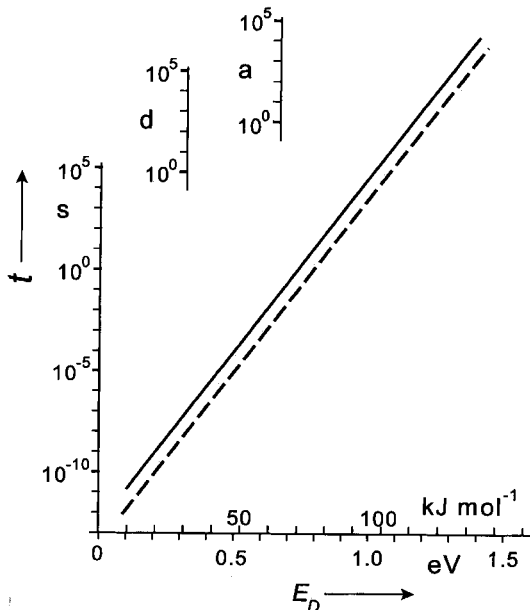
The time constant τ in this equation, as well as the value A in Eq. (2.3.17) depends on a number of structural properties of the molecule. That is why only values relating to several orders of magnitude are available. For smaller molecules, τ will lie between 10^{-14} and 10^{-13} s. The characteristics of this function are illustrated in Fig. 2.14. Semi-logarithmic plotting ($\ln t$ against E_D) will result in a straight line. It can be clearly seen that even small changes of the activation energy of decomposition (E_D) will lead to dramatic changes of the life span (t). The range may stretch from 10^{-12} s to geological periods of time amounting to 10^5 years.

Considering the energies of covalent bonds (Table 2.1, page 12), one understands that they practically cannot be destroyed by thermic impacts. Hydrogen bonds, on the other hand, with bonding energies between 13 and 25 kJ mol^{-1} (see Sect. 2.1.5) have only a short lifetime. When we discuss the structure of water (Sect. 2.4.2), we will show that in these cases oscillations of bonds occur.

In all investigations of the stability of covalent bonds one must of course also consider the possibility of their destruction by energy rich quanta of radiation (ionizing radiation). This we will consider later in connection with the target theory of radiation biophysics (Sect. 4.5.5).

The mean life span of a molecule is a typical variable of statistical thermodynamics. It is only a value of probability which cannot predict the concrete destiny of an individual molecule. The assumption can be made that by biological selection only DNA molecules with monomers of high bonding energy

Fig. 2.14. The mean lifetime (t) of a bond depending on the energy of activation of decay (E_D), according to Eq. (2.3.29) ($T = 293\text{ K}$; solid line: $\tau = 10^{-13}$; dashed line: $\tau = 10^{-14}\text{ s}$)



are left. Nevertheless, spontaneous molecular transformations take place which are known as mutations. A mutation results in a new molecule whose stability has not yet been approved by selection. The life span of this molecule can be substantially changed by a slight displacement of its bond energy. This may explain why remutations or mutations of already mutated molecules occur more often than could be expected statistically.

Further reading: Blumenfeld and Tikhonov 1994; Westerhoff and van Dam 1987. The theory of absolute reaction rate: Eyring and Urry in: Waterman and Morowitz 1965

2.3.6

Thermal Molecular Movement

The disadvantage of the method of illustrating structures of organic molecules by using mechanical models composed of spheres and connecting wire axes, as well as of computer simulations of molecular structures, is their suggestion of a more or less static point of view. In reality, however, in liquids as well as liquid-crystalline phases, the molecules are always in a state of vigorous thermal motion. The following three forms of molecular movements are to be considered: vibrations, rotations, and translations.

Vibrations are oscillations in the binding distances between the atoms in a molecule. The term *rotations* means not only the rotation of the whole molecule,

but additionally, the spin of individual atoms or atomic groups around the axes of their bonds. Of course, as already explained in Section 2.1.2., in this case the intramolecular interactions of the atoms are to be considered. So, for example, considering a simple molecule of ethane, the rotation around the $\text{CH}_3\text{—CH}_3$ bond cannot be homogeneous. Energy minima of rotations occur periodically every 120° . Hence, rotation is “locked” at certain angles of low energy levels.

A covalent bound, rotating around its angle relative to another, consequently describes a cone. The next bond rotates, as well, but it does so quasi on the spot, moving itself in an orbit of that lateral cone surface (Fig. 2.15). In this way chain molecules can assume a stochastic orientation unless other strong attracting or repelling interactions between the monomers are preventing this. The molecular structures resulting in this way can be calculated by the laws of statistical thermodynamics.

The so-called random flight model allows the simplest description of the behavior of a long molecule. In this case, the real molecule is described as a sequence of segments with length l . These segments are connected to each other in such a way that the orientation of each is independent from the orientation of the former segment. This random flight chain is characterized by the length (l) and the number (n) of its segments. The positions of the end points of the segments can be expressed as vectors in a polar coordinate system which is based on the center of gravity of the molecule (Fig. 2.16). The distance between the endpoints of such a molecule (d_{max}) at maximal elongation is:

$$d_{\text{max}} = nl. \quad (2.3.30)$$

The real distance of the endpoints of a random flight chain with large enough amounts of n segments, and after a sufficiently long time of observation, however, is distributed according to a Gauss statistic. As shown in Fig. 2.10, the mean of this distance is not identical with its maximum, i.e. the value of maximal probability. The mean value of this distance between the end points of the chain can be calculated by the following equation:

$$d = \sqrt{\langle r^2 \rangle} = l\sqrt{n}. \quad (2.3.31)$$

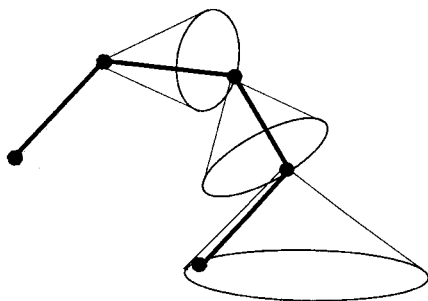
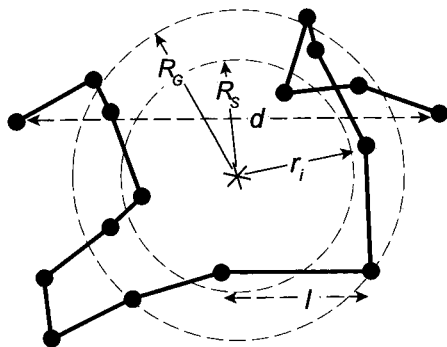


Fig. 2.15. The rotation cones of the atomic groups in a chain molecule

Fig. 2.16. Projection of a random flight chain, composed of 15 segments with a characteristic length l . * Center of gravity of the molecule, d distance between the two ends of the chain, r_i distance of the i -th point from the center of gravity, R_G radius of gyration, R_S Stock's radius. This model corresponds to an unfolded polypeptide, composed of 150 amino acids, or a double strand DNA. (According to data from G. Damaschun)



In this case the expression $\langle r^2 \rangle$ is the mean of the squares of the distances as defined as follows:

$$\langle r^2 \rangle = \frac{\sum r_i^2}{n}. \quad (2.3.32)$$

Comparing Eqs. (2.3.30) and (2.3.31) it can be seen that the mean distance (d) between the endpoints of the chain is smaller by $\sqrt{n}$ in relation to its maximum ($d_{\max}$). Considering a chain consisting of 100 segments, the mean distance, therefore, will be only 10% of the maximal length, for chains of 1000 segments, it will be only 3.2%.

The geometric dimension of such a molecule can be described by the mean of the square distance of all atoms of the molecule from the common center of gravity of the molecule. This gives the so-called *radius of gyration* (R_G). For a random flight chain with $n \rightarrow \infty$ it gives:

$$R_G = \sqrt{\frac{\langle r^2 \rangle}{6}} = l \sqrt{\frac{n}{6}}. \quad (2.3.33)$$

The radius of gyration, therefore, is proportional to the square root of length of the chain ($\sqrt{n}$). For compact molecules, however, in which the monomers are not arranged in this way, one gets $R_G \sim \sqrt[3]{n}$, and for rod-shaped molecules, $R_G \sim n$.

Furthermore, it is possible to define a so-called *Stockes' radius* (R_S) for macromolecules, which is measurable experimentally. This is an example of a typical effective parameter. It predicts that the molecule moves in the water like a macroscopic sphere with a hydrophilic surface. In this case, Stockes' law can be applied. It gives the frictional force (F) of a sphere with radius r , moving in a fluid with viscosity η and with a velocity v :

$$\mathbf{F} = 6\pi\eta r \mathbf{v}, \quad (2.3.34)$$

(the bold letters emphasize the vector natures of force and velocity).

The Stokes' radius, therefore, is a measurable parameter which, in the same way as the radius of gyration, does not correspond to any "visible" radius of the molecule. Both are effective parameters which, however, are well defined (see Fig. 2.16). We will come back to this point when considering the hydration radius of ions in Section 2.4.2. The relationship between the experimentally determined parameters R_G and R_S gives important hints about the configuration of a given macromolecule. If the molecule really resembles a random flight chain, one gets: $R_G/R_S = 1.51$. If the molecule has a spherical but compact structure, this relation becomes 0.8. In the case of root-shaped molecules, this parameter depends on the relation of radius to length of it.

Real molecular chains, composed of covalently bound atoms (Fig. 2.15) are usually not as flexible as predicted by the random flight model. Nevertheless, this model is also applicable in these cases. For this, monomeric constituents were connected together as segments, matching the conditions of the random flight model. The stiffer the molecules are, the larger the characteristic length of the segments which must be chosen. In case of a quite flexible molecule of poly-L-glycine, the characteristic segments are formed by approximately four amino acids with a characteristic length of the segments of $l = 1.38$ nm. For the more stiffer molecule of poly-L-alanine, in contrast, twelve amino acids must be put together in one segment with $l = 4.10$ nm.

A repulsive force between the atoms, e.g. by electrical charges of equal sign, results in an additional expansion of coiled molecules. A similar limitation of the random flight model is given by the exclusion principle, i.e. by the fact that two atoms cannot be at the same location at the same time. Therefore, the characteristic length of the segment of a DNA molecule as a strong polyelectrolyte is $l = 90$ nm.

If such a chain molecule is fixed at two points (Fig. 2.17) then attractive or repulsive forces are developed if the distance between the points is larger or smaller, respectively, than the parameter d [Eq. (2.3.31)]. This is the so-called rubber or entropy elasticity. It is caused by the decrease of the entropy of the molecular chain during stretching. In entropy-elastic materials, the points of fixation are built by interconnections of the molecules. This property is the most common type of elasticity of biological materials. We will come back to this in Section 3.6.3.

The third kind of molecular movement mentioned above, *translation*, means a movement in space. This translational movement, driven by thermic hits, is important not only for single molecules, but also for microscopically visible particles. This was first observed in 1827 by the Scottish botanist Robert Brown, observing the stochastic movement of gymnosperm pollen in suspension. This movement later was named *Brownian motion*. The reason for this stochastic translocation of microscopically visible particles are the hits, resulting from thermic translations of the surrounding molecules. The movement results from the vector sum of these hits (Fig. 2.18). If the particles are sufficiently small, then a real vector occurs as the sum. For larger particles, this mean vector force becomes zero.

Fig. 2.17. The molecular mechanism of rubber elasticity. If the distance between the two points of fixation increases, the entropy content of the molecule decreases (schematically illustrated in the elongation of the chain in the lower part of the figure). A force is generated with the tendency to recover the previous situation (the state of maximal entropy). (According to data from G. Damaschun)

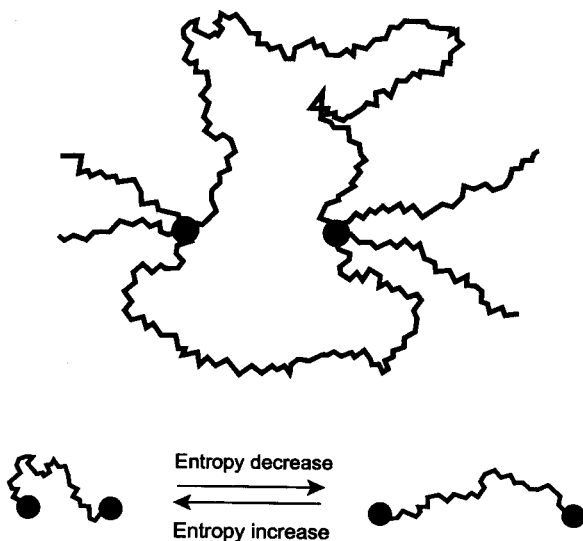
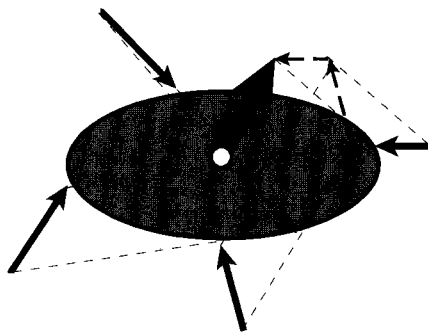


Fig. 2.18. A snapshot indicating the thermic hits by surrounding molecules on a microscopically small particle. The individual hits, drawn as fine vectors, induce by vector addition a resting force (large vector) which effects the Brownian motion of the particle



The spatial translocation of the particle, in fact, is stochastic as well as the molecular translocations themselves. Observing the position of the particle during a sufficiently long time, there will be no translocation at all. As a measure of the Brownian motion, therefore, the mean of the squares of the translocations are used. For this, the position of the particle will be registered at definite time intervals Δt (Fig. 2.19). Subsequently, when the distance x_i is measured, the particle has moved in a projection of an arbitrarily drawn axis. If these values are measured n times, the parameters x_i for $i = 1, \dots, n$ are obtained. The square of displacement $\langle x^2 \rangle$ can be calculated in analogy to Eq. (2.3.32) in the following way:

$$\langle x^2 \rangle = \frac{\sum x_i^2}{n}. \quad (2.3.35)$$

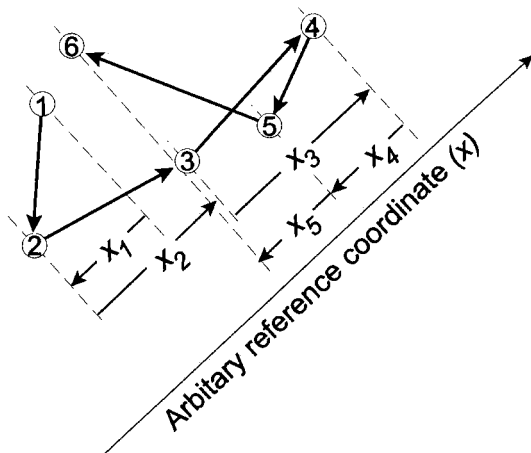


Fig. 2.19. Method to evaluate Brownian motion. At definite intervals of time (Δt) the position of the particle was marked. This two dimensional picture was subsequently projected on an arbitrarily drawn axis, which allows measurement of the distances x_i

This parameter corresponds to the square of the mean path length of the moving particle:

$$\bar{x} = \sqrt{\langle x^2 \rangle}. \quad (2.3.36)$$

The intensity of Brownian movement depends on shape and size of the particle as well as on the temperature and the viscosity of the medium. Einstein and Smoluchowski derived an equation for spherical particles, meeting the conditions of the Stockes' equation [Eq. (2.3.34)]:

$$\langle x^2 \rangle = \frac{RT\Delta t}{3N\pi\eta r} = \frac{kT\Delta t}{3\pi\eta r}. \quad (2.3.37)$$

The Einstein-Smoluchowski Eq. (2.3.37) therefore connected the measurable square of displacement ($\langle x^2 \rangle$) of Eq. (2.3.35) in relation to the radius of the particle (r), the viscosity of the fluid (η) and the temperature (T), this means by the energy of the thermic movement ($RT/N = kT$). Δt is the time interval for which the values x_i were measured.

This equation indicates that the square of the mean translocation length which was achieved by Brownian motion per time interval was inversely proportional to the particle radius (r) and the viscosity (η). A particle with diameter $r = 1 \mu\text{m} = 10^{-6} \text{ m}$ moves in water ($\eta = 0.00089 \text{ N s m}^{-2}$) at $T = 298 \text{ K}$ during $\Delta t = 1 \text{ s}$ at a mean distance of:

$$\bar{x} = \sqrt{\frac{1.38 \cdot 10^{-23} \cdot 298 \cdot 1}{3 \cdot \pi \cdot 0.00089 \cdot 10^{-6}}} = 0.7 \cdot 10^{-6} \text{ m}.$$

The size of the particle (r), as well as the mean translocation ($\bar{x}$) are just in the range of resolution of a normal microscope. Measuring $\bar{x}$, and knowing r , it is

possible to measure the viscosity (η). This was sometimes used to measure the viscosity of cytoplasm. On the other hand, it is possible to measure the diameter of very small particles by dark field microscopy in fluids with known viscosity, measuring the parameter $\bar{x}$.

The Einstein-Smoluchowski equation is not only important for using these methods. With the help of this equation it is also possible to calculate the time a particle needs to cross a certain distance by diffusion. If, for example, at one point in the cell a vesicle was created and is bound at a definite other place by specific receptors, the mean time this particle needs to cross this distance by stochastic movements can be calculated. Phenomenologically such an event appears to be the result of directed transport. (This, by the way, is an example of a kind of biological Maxwell demon, as discussed in Sect. 2.3.2!).

This translational movement is the basis of all processes of diffusion. Einstein found the connection between the square of displacement ($\langle x^2 \rangle$) and the diffusion coefficient (D) of a molecule in solution:

$$D = \frac{\langle x^2 \rangle}{2\Delta t}. \quad (2.3.38)$$

Introducing Eq. (2.3.37), it gives

$$D = \frac{RT}{6N\pi\eta r} = \frac{kT}{6\pi\eta r}. \quad (2.3.39)$$

We will come back to deriving the Fick's law of diffusion in Section 3.3.1 using the approaches of phenomenological thermodynamics.

This equation allows the conclusion that the diffusion constants of two molecules with similar molecular shapes are inversely proportional to the square root of their molar masses (M):

$$\frac{D_1}{D_2} = \frac{\sqrt{M_2}}{\sqrt{M_1}}. \quad (2.3.40)$$

Further reading: Cantor and Schimmel 1980; Pain 1994

2.4

Molecular and Ionic Interactions as the Basis for the Formation of Biological Structures

In Section 2.3 the relationship between order and life was discussed, and in this context, the molecular basis of biological structures was considered from the perspective of statistical thermodynamics. Emphasis has been placed on the energy of thermal movement and the thermal noise (kT). These stochastic forces of movement are opposed by intermolecular forces which tend to create order

and build up molecular and supramolecular structures. Therefore, the expression E_i/kT [see Eqs. (2.3.16), (2.3.17), (2.3.29)] characterizes the relationship between any sort of structure forming energy and the destroying energy of thermic noise.

Now, we will consider in particular those forces which contribute to the formation of molecular and supramolecular equilibrium structures. The forces operating in these so-called self-assembling processes are mostly weak bonds, i.e. bonds with energies of the same order as thermic noise (kT). These bonds, therefore, can be split by thermic hits in a realistic period of time. Examples of such weak bonds, are hydrogen bonds as well as the whole complex of Van der Waals interactions.

2.4.1

Some Foundations of Electrostatics

Because electrical phenomena are of major importance at all levels of biological organization an account will be given of some fundamental parameters, definitions, and implications.

The basis of electrostatic considerations are electric charges. In the CGS-System the electric charges are defined by mechanical parameters. So, for example, the electrostatic unit (eu) is an amount of charge which interacts with an opposite polarized charge of the same size at a distance of 1 cm by an attractive force of 1 dyn. This definition eliminates the need for defining specific electrical units. In the *Système International d'Unités* (SI), however, the application of mechanical parameters has been abandoned, and the charge is defined by the unit for current – the ampere (A), as coulomb (C) = ampere (A) · second (s). In this book, we will use this now generally accepted international system of units.

In electrochemical as well as in biophysical calculations, the smallest electric charge is the charge of a univalent ion or a corresponding charged group. The *Faraday constant* (F) gives the number of charges per mole of singly charged ions, an amount which is usually called *val*:

$$F = 9.6485 \cdot 10^4 \text{ C val}^{-1}.$$

Dividing this molar amount by the Avogadro number (N), gives the absolute value of the charge of a single ion (e), the same value as that of a positron:

$$e = \frac{F}{N} = 1.6021 \cdot 10^{-19} \text{ C}.$$

Using the units of the SI system, a conversion factor must be applied to calculate mechanical parameters resulting from electrical interactions. This conversion factor is the so-called *permittivity of free space* (ϵ_0):

$$\epsilon_0 = 8.854 \cdot 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}.$$

The Coulomb's law makes it possible to calculate the repulsion force (F) of two charges (q_1, q_2) at a distance r from each other:

$$F = \frac{q_1 q_2}{4\pi\epsilon_0\epsilon r^2}. \quad (2.4.1)$$

We already used this equation in Section 2.1.3 to calculate the force of interaction of two ions. In contrast to Eq. (2.1.3) which indicates the force of two charges in vacuum, Eq. (2.4.1) additionally contains the *dielectric constant* (ϵ), which is also called *permittivity*. In contrast to ϵ_0 , which unfortunately is written with the same symbol, the dielectric constant ϵ is just a number without dimension. It is a factor indicating how much the interaction force between two charges will be diminished, if not a vacuum, but a real substance is in between. The dielectric constant for pure water at 18 °C, for example, amounts to 81.1. For various kinds of glass, it is between 5 and 7, and for arranged layers of lipids about 3.5.

Using the Coulomb's law [Eq. (2.4.1)] one can also calculate the energy of electrostatic binding (E) of two charges in a medium with $\epsilon > 1$. As already mentioned (Sect. 2.1.3), with the inverted sign, it equals the work required to move both charges from the distance $r = \infty$ to the distance $r = r_i$:

$$E = -W = \int_{\infty}^{r_i} \frac{q_1 q_2}{4\pi\epsilon_0\epsilon r^2} dr = -\frac{q_1 q_2}{4\pi\epsilon_0\epsilon r}. \quad (2.4.2)$$

Any charge in space generates an electric field which is characterized by the gradient of the electric potential. The *electric potential* (ψ) is a state parameter of a particular point in space. It is defined as the work which is required to move a positive charge from infinite distance to the point r . The unit of the potential is the volt (V). In the case of the electric field around a charge q , the electric potential is a radial function of the distance r from this point. According to this definition and Eq. (2.4.2), the potential ψ_i at the distance r_i from the charge q is given by:

$$\psi_1 = \frac{q_1}{4\pi\epsilon_0\epsilon r_1}. \quad (2.4.3)$$

It is important to stress that the electrical potential is a scalar parameter in space, similar, say, to pressure or temperature. Later we will discuss surface potentials, donnan potentials, diffusion potentials etc. These are all just electrical potentials. Their names only indicate the reason for their occurrence. At any given time, at one and the same point in the space, there can be only one value of an electrical potential, as there can be only one single temperature at one time at one point!

Points of equal potential in space can be interconnected by *equipotential lines*. Perpendicular to these equipotential lines electrostatic forces occur. Correspondingly, an electric field strength (E) as a vector parameter can be defined as follows:

$$\mathbf{E} = -\text{grad } \psi = -\nabla \psi. \quad (2.4.4)$$

The differential operator “grad” as well as the so-called *Nabla operator* ∇ means: derivation of the potential ψ to all three dimensions of space. It transforms the scalar ψ into the vector $\mathbf{E}$.

Figure 2.20 indicates the equipotential lines and the field vectors around a positively charged point. The direction of these vectors corresponds with the vector of the electric field strength ($\mathbf{E}$). According to a convention, these arrows are always directed from the positive, toward the negative pole. Similar illustrations can be seen in Figs. 3.31 and 3.32.

Considering the gradient of the electric field just in one direction (x) of the space, Eq. (2.4.4) becomes:

$$E_x = -\frac{d\psi}{dx} \mathbf{i}. \quad (2.4.5)$$

In this case $\mathbf{i}$ is the unit vector in the direction of the x coordinate.

In Fig. 2.21 the potential and the field strength in the x -direction are depicted in the vicinity of a homogeneous membrane which separates two different electrolyte solutions (phase I and phase II). This demonstrates that points of discontinuity of the field strength can occur, with a reversal of the field direction. In this case, the reason for these discontinuities are surface charges on the phase boundaries. The field vectors and the equipotential lines in this example would indicate a set of parallels, perpendicular to each other. Actually, the biological membrane is not homogeneous, but it is a mosaic of various compounds with different surface charges and various dielectric constants. Even its cross section is not homogeneous. Its electrostatic structure is far more complicated, as we will see in Section 2.5.5, Fig. 2.48.

The unit of field strength, according to its definition [Eqs. (2.4.4) and (2.4.5)] is V m^{-1} . As an example, let us calculate the field strength across a cell membrane with a thickness of $7 \text{ nm} = 7 \cdot 10^{-9} \text{ m}$. If the electrical potential

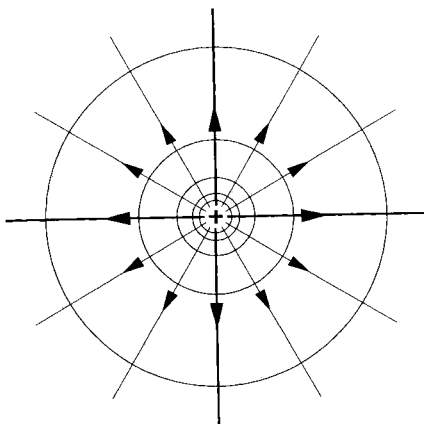
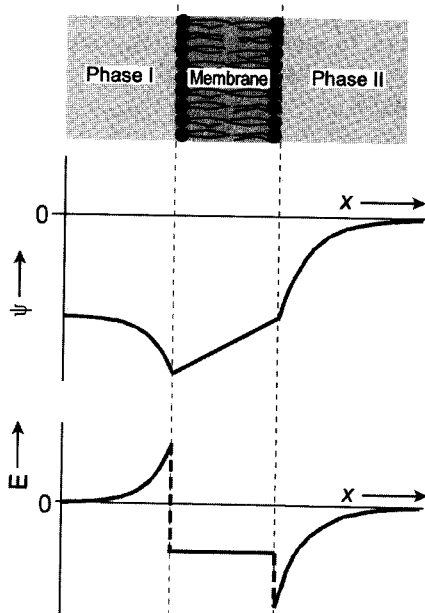


Fig. 2.20. Equipotential lines (—) and field vectors (—→) around a positively charged point

Fig. 2.21. The electrical potential [$\psi(x)$] and the electric field strength [$E(x)$] near a homogeneous membrane which separates two electrolyte solutions with different composition



difference between the two surfaces amounts to $\Delta\psi = 50 \text{ mV} = 5 \cdot 10^{-2} \text{ V}$, the following field strength results:

$$|E_i| = \frac{5 \cdot 10^{-2}}{7 \cdot 10^{-9}} = 7.14 \cdot 10^6 \text{ Vm}^{-1}.$$

The term $|E_i|$ means the magnitude of the field strength in the x -direction. Later we will discuss the biophysical consequence of this large electric field strength (Sects. 2.5.5 and 3.4.2).

Using Eqs. (2.4.1), (2.4.3), and (2.4.4) one can calculate the force (F), acting on a charge (q) in an electric field (E):

$$F = Eq. \quad (2.4.6)$$

In addition to charges, dipoles play an important role in electrostatics. A dipole is a structure which contains two equal charges (q) of opposite polarity, positioned at a certain distance (l) from each other. Consequently, an electric dipole, as a whole, is electrically neutral. A dipole moment (μ) is defined as the product of charge (q) and distance (l) between them:

$$\mu = ql. \quad (2.4.7)$$

Dipoles can be formed by ampholytic molecules, this means in molecules simultaneously carrying positive as well as negative charges. They may also result from polarization phenomena of covalent bonds, as described in Section 2.1.2.

According to Eq. (2.4.7), the unit of a dipole is C m or, correspondingly: A s m. In molecular studies, sometimes the Debye-unit (D) is preferred because its amount is in the order of real molecules. The relation between these two units is:

$$1 \text{ D} = 3.3 \cdot 10^{-30} \text{ C m.}$$

In Table 2.2. for demonstration, dipole moments of several compounds are listed. The dipole moments of chloroform, methanol and water are caused exclusively by polarizations of covalent bonds. In the case of proteins and nucleic acids, the values listed in this Table are to be considered just as orders of magnitude. In these cases charged groups are responsible, whose distance apart, according to the size of these molecules, can be considerable. Single α -helices of proteins have a dipole moment corresponding to a charge $q = e/2$ on both ends of them. Considering a mean length of such an α -helix of $l = 1.3 \text{ nm}$ corresponding to Eq. (2.4.7), a value of $\mu = 1.602 \cdot 10^{-19} \cdot 1.3 \cdot 10^{-9} = 2.08 \cdot 10^{-28} \text{ C m} = 63.1 \text{ D}$ results.

Ampholytic molecules exhibit electrostatic neutrality in the vicinity of the isoelectric point, since just at this pH value equal numbers of negative and positive charges are dissociated (for example: $-\text{COO}^-$ and $-\text{NH}_3^+$ groups, see Sect. 2.4.6). Because of the reversibility of this process, however, a rapid alteration of dissociation and association may take place. A molecule of this type can be considered as a *fluctuating dipole*, the charges of which oscillate in a frequency range of about 10^7 s^{-1} .

If a noncharged molecule is influenced by a strong electric field, shifts of electrons may take place, and an *induced dipole* occurs. The resulting dipole moment depends on the external field strength (E) and on the ability of the molecule to polarize, a property which is described by a factor α :

$$\mu_{\text{ind}} = \alpha E. \quad (2.4.8)$$

Such kinds of induction are measurable as *protein electric response signals*.

At the beginning of this chapter, the dielectric constant (ϵ) had already been introduced. Now we can characterize this parameter in greater detail on the

Table 2.2. Dipole moments of various substances. (After Netter 1959, ¹after Raudino and Mauzerall 1986, ²see text, ³after Pethig 1979, ⁴after Takashima 1963)

	D	$\cdot 10^{-30} \text{ C m}$
Chloroform	1.05	3.47
Methanol	1.68	5.54
Water	1.85	6.11
Urea	8.6	28.4
Glycocoll	15	50
Phosphatidylcholine headgroup ¹	20	66
α -helix ($l = 1.3 \text{ nm}$) ²	63.1	208
Proteins ³	100–1000	330–3300
DNA from calf thymus ⁴	32 000	105 600

basis of dipole properties. The electric field strength (E) of a plate capacitor is proportional to its surface charge density (σ):

$$E = \frac{\sigma}{\epsilon_0 \epsilon} \quad (2.4.9)$$

The surface charge density expresses the charges per area ($\sigma = q/A$). Introducing a dielectric between the two plates of the capacitor means an increase of the dielectric constant (ϵ). As can easily be seen in Eq. (2.4.9), an increase of the dielectric constant (ϵ) has the same effect as a decrease of the surface charge density (σ). This is not only a mathematical relation. In fact, it is caused by the polarization of the dielectrics under the influence of the electric field of the capacitor. Therefore, there is a correlation between the dielectric constant and the dipole moment of a substance.

Further reading: On protein response signal: Keszethelyi and Omos 1989

2.4.2

The Water Structure, Effects of Hydration

The physicochemical properties of water are quite different from those of chemically similar compounds, for example H_2Te , H_2Se or H_2S . Astonishing values could be predicted extrapolating the properties of these substances to H_2O . This would predict a melting point of $-100^\circ C$ and boiling point of $-30^\circ C$. Water, therefore, would be a gas under the normal conditions. The same deviations between predicted and actually existing values would be found in relation to other physicochemical properties, such as molar heat of evaporation, heat capacity, surface tension, dielectric constant etc.

Another speciality is the so-called *temperature anomaly* of water, which is of particular relevance for the generation and preservation of life. As illustrated in Fig. 2.22, water has its highest density at a temperature of $4^\circ C$. This is the

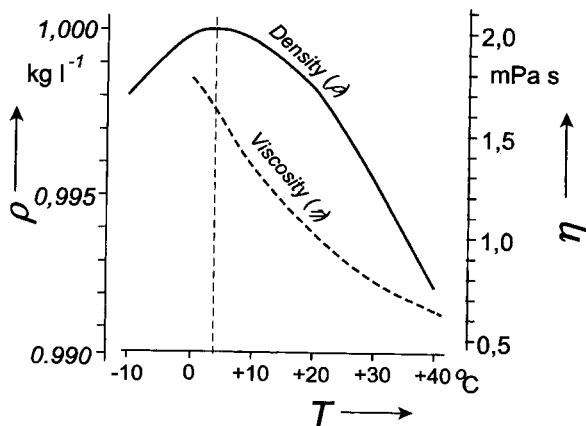


Fig. 2.22. Density (ρ) and viscosity (η) of water as function of temperature (T)

reason why, in contrast to other melting substances, ice floats on the surface, and the temperature of water near the bottom of lakes will not become colder than 4 °C. If water had the same properties as other substances, the cold water would always sink and the lakes would freeze downside up. Life in this lake could not survive the winter.

There are two molecular properties which are responsible for these effects: the dipole moment of water, and the ability of water molecules to build intermolecular hydrogen bonds. In Section 2.1.2 we discussed the electronegativity of elements in covalent bonds and found the following sequence:



Hence, in both H—O-bonds of the water molecule a strong polarization will occur. The common valence electrons in these bonds are strongly attracted by the oxygen atom, which therefore becomes negative in relation to the protons. The positive charges of the two H atoms lead to a repulsion of each other. This increases the angle between them up to a value of 104.5°. The water molecule in all becomes a dipole which is directed towards the bisector of this angle.

This polarization of the H—O-bonds in the water molecule not only has consequences for its electrostatic behavior but is also responsible for its ability to build up hydrogen bonds. They can be built between two water molecules as well as between water and other molecules. This is the reason for the occurrence of the water structure in ice as well as in the fluid water.

The structure of ice has received most attention by investigators. As shown in Fig. 2.23, a tetrahedral structure is built by oxygen atoms which are always

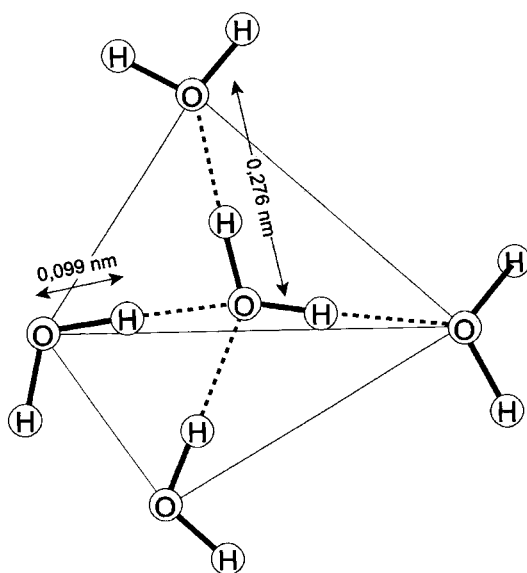


Fig. 2.23. The tetrahedral structure of water in ice

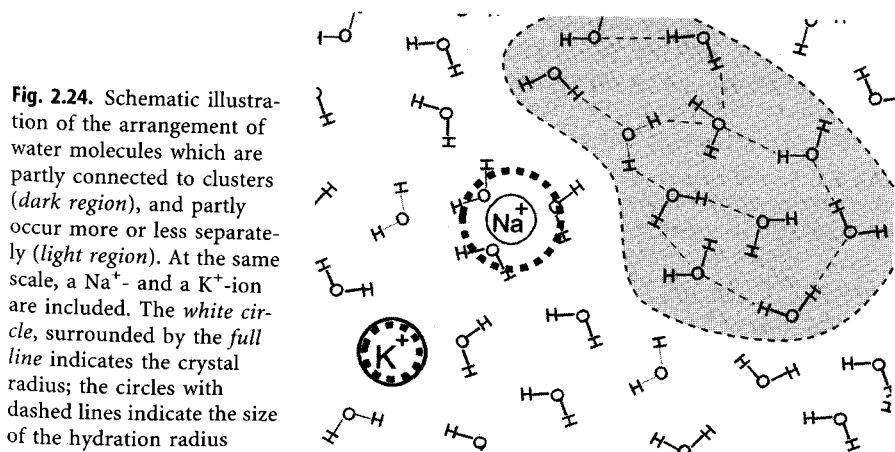
connected by H-bonds with two hydrogen atoms of other molecules. The continuation of this kind of structure builds up a 3-dimensional network which finally leads to the crystalline structure of ice.

Thermochemical measurements as well as investigations with infrared spectroscopy indicate that these elements of the ice structure will not be destroyed fully if the ice melts. Structural water clusters exist even at temperatures above the melting point of ice. These clusters consist of interconnected water molecules with a structure similar to that of ice. The size of these clusters diminishes at increasing temperatures. Near the melting point, about 90–650 water molecules are clustered together, near the boiling temperature, there are only 25–75 molecules connected in such a cluster (the numbers differ according to different authors). This is the reason for the temperature dependence of the viscosity of water as depicted in Fig. 2.22.

This property of water molecules is enclosed in the mixed cluster model of fluid water as proposed by Nemethy and Scheraga (Fig. 2.24). This model considers that the tetrahedral structure of water, as depicted in Fig. 2.23 is not the only possible geometrical arrangement of the cluster. The strongly orientated crystalline structure of ice is disturbed in floating water by thermic noise. Thus, irregularities and structures on non-tetrahedral geometry occur.

These structures are to be considered as being in permanent movement. An H-bond oscillates with a frequency of about $0.5 \cdot 10^{13}$ Hz. The mean lifetime of a single cluster lasts approximately 10^{-10} – 10^{-11} s. Therefore, during the oscillation every H-bond will form 100–1000 times with the same oxygen atom, before it connects to another one. Because of these dynamics, the name *flickering cluster* is used.

In biological systems the water structure is influenced to a great extent by interactions of the water with ions and organic molecules. The interaction of these compounds with the structure of water occurs electrostatically as well as by hydrogen bonds.



The interaction of ions with water molecules is called *hydration of the first order*. Inorganic ions interact with water molecules exclusively electrostatically. In Table 2.2 (page 56) the dipole moment of water is listed as 1.85 D. The electric field around an ion, therefore, will lead to a more or less strong orientation of water molecules. This orientating force from the electric field of the ion is in competition with influences from other water molecules, and of course with the destructive activity of thermic noise. As we discussed in Section 2.4.1 (Fig. 2.20), the electric field strength decreases with increasing distance from the center of charge. Correspondingly, two areas at different distances from the ion must be considered: near the ion is a region of *primary hydration*. This is a small number of water molecules which are strongly orientated in the electric field of the ion. In Fig. 2.24 this is illustrated in case of the three water molecules near the Na^+ -ion. The region of *secondary hydration* follows. At this distance the electric field is always too weak to arrange an orientation of water molecules. On the other hand, it is strong enough to disturb the normal water structure. In Fig. 2.24 this is indicated by the destruction of the darkly marked water cluster near the Na^+ -ion. In this, it is like a fault zone without any defined water structure.

To quantify this situation several parameters can be used. They are illustrated in Table 2.3 NMR measurements, and also investigations of coefficients of self diffusion of water, including its activation energy, makes it possible to determine the relative residence time of water molecules in the vicinity of ions. *Self diffusion* means the diffusion of water in water, or in solutions. If t_i stands for the mean time in which the H_2O molecule resists near an ion, and t stands for the time interval during which such a molecule will permanently stand at any point near another water molecule, the relation t_i/t characterizes the degree of demobilization of these molecules near an ion. If $t_i/t > 1$ then obviously an increase of the structural degree of water near the ion occurs, if $t_i/t < 1$ the ion is a structure breaking agent.

An important parameter to characterize the hydration shell around an ion is the *hydration radius*, or *Stockes' radius*. It corresponds to the effective radius which we introduced in Section 2.3.6 for spherical molecules. Measuring the equivalent conductivity of an ion at infinite dilution, it is possible to calculate its electrophoretic mobility, i.e. its mobility in relation to the applied field strength [see: Eq. (2.5.9)]. Equation (2.4.6) allows us to calculate the force which is

Table 2.3. Several characteristic parameters of alkali ions

Ions	Relat. atomic mass M	Crystall radius r_K (nm)	Hydration radius r_H (nm)	Equivalent conductivity at infinite dilution ($\text{S m}^2 \text{ val}^{-1}$)	Relative residence time of a H_2O molecule t_i/t
Li^+	6.94	0.069	0.240	0.00387	2.60
Na^+	22.99	0.098	0.185	0.00501	1.27
K^+	39.10	0.133	0.126	0.00735	0.54
Rb^+	85.48	0.148	0.119	0.00778	
Cs^+	132.91	0.169	0.120	0.00772	0.59

applied to an ion in a given electric field (E). Under conditions of stationary movement, this driving force is fully compensated by the friction which results from the interaction of the moving ion with the surrounding water. Now let us postulate that this frictional force can be calculated by Stokes' law [Eq. (2.3.34)]. This means the ion is taken as a macroscopic sphere with a hydrophilic surface. "Macroscopic" means large enough that water can be considered as a continuum, and not as a molecular structure. In this case, knowing the viscosity (η), the driving force (F), and the mobility (v) of the ion, the Stokes equation makes it possible to calculate an equivalent radius (r) of it as a sphere. This radius is per definition the hydration radius.

This hydration radius is a typical example of an effective parameter. It is exactly defined by its way of measurement and calculation. On the other hand, it does not exist as a visible or at least imaginable border. As indicated in Fig. 2.24, it is in no case identical for example to the thickness of a single layer of water molecules. In fact, the Stokes equation is not the proper law to use to calculate the real movement of an ion in water. The conditions listed above are not realized in this case. Nevertheless this parameter, because of its exact definition, is an important measure for the hydration properties of ions. When using it, for example, to explain membrane permeability for ions, or to discuss other similar mechanisms, it is always necessary to be aware of the conditions of its definition.

As illustrated in Table 2.3, the hydration radius of the alkali ions decreases with increasing atomic mass. In the case of lithium, the hydration radius is much larger than the crystal radius, whereas for potassium it is already smaller. For this, K^+ and Cs^+ are called *structure breaking* ions. The region of secondary hydration, i.e. the fault zone of broken water structure, is larger than that of the electrically orientated molecules. In contrast to this, in small ions, like Li^+ and Na^+ , the region of primary hydration is far larger. These are *structure building* ions. This category also includes the two valent cations Mg^{2+} and Ca^{2+} which are not mentioned in the Table. Sometimes, instead of the terms "structure breaking" or "structure building", the expressions "chaotropic" or "cosmotropic" are used.

What is the reason for these differences? Both ions, Na^+ as well as K^+ , carry the same single positive charge. The Bohr's radius of these ions, i.e. the effective radius of the outer electron shell which is the same as the crystal radius, is, however, larger for K^+ than for Na^+ , because of the additional electron orbital in the case of potassium. With a larger distance from the charged nucleus of the ion, the electric field strength will become lower.

Hydration effects which are not caused directly by electrostatic interactions are called *hydrations of the second order*. In this case hydrogen bonds between organic molecules and water are the reason for the structure of a water layer in the direct vicinity of the surface. In view of these sorts of interaction, two types of molecular surfaces must be considered: proton donator and proton acceptor surfaces. As depicted in Fig. 2.25, this difference is responsible for the orientation of the attached water molecules. This orientation may be transmitted by several layers of water molecules. This kind of hydration can also lead to an interaction between two surfaces. Surfaces with identical orientation of water

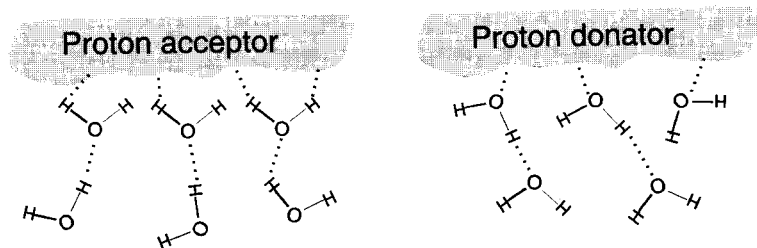


Fig. 2.25. Water orientation near proton-accepting and proton-donating surfaces

molecules repel each other, surfaces with opposite orientation of water show attractions at short distances.

Hydrophobic molecules, in general, behave in relation to the surrounding water like an air bubble. From the phenomenological point of view, water molecules near this surface exhibit a larger amount of energy than molecules in the water bulk. This energy of interfacial molecules in connection with interfacial tension will be described in detail in Section 2.5.1.

There are some peculiarities of the physical properties of water near surfaces in relation to the bulk water. In a layer of 0.1 to 0.2 nm, for example, the effective dielectric constant of water decreases dramatically up to the value of 10 (Takashima et al. 1986). Furthermore, because of the water structure, this region may become anisotropic. This means, for example, that diffusion rates become higher then the average value in the direction parallel to the surface and lower in the direction normal to it (Makarov et al. 1998).

Second order hydration can play a predominant role in conformational changes of molecules and in the formation of supramolecular structures. In these processes not only the entropy of the macromolecules changes, but also the entropy of the surrounding water. Because of the structure-building properties of some molecular surfaces, the entropy of the water near the surface is lower than that in the bulk phase. In the case of conformational changes of macromolecules, or of processes of supramolecular organization, the surface available for interaction with water molecules may decrease. This happens for example if lipid molecules aggregate to form membranes or vesicles. In this case the water between the molecules will be pushed away. Considering these sorts of processes thermodynamically, a higher degree of organization results in the entropy of the macromolecules decreasing, but because of the loss of structure-building surfaces the entropy of water increases. In general, therefore, even if a macromolecular structure of a higher degree of organization results, the entropy of the whole system increases. According to the second principle of thermodynamics, therefore, these kinds of processes may occur spontaneously.

In the same way the spontaneous formation of secondary and tertiary structures of macromolecules can be considered (Fig. 2.26). These kinds of processes are called *entropy driven*.

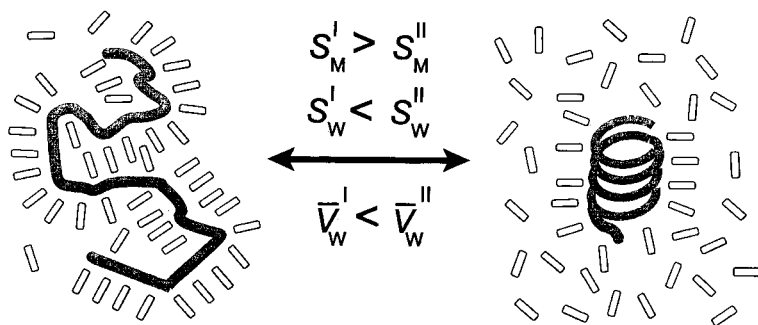


Fig. 2.26. The orientation of water molecules, schematically drawn as *bars*, near the surfaces of macromolecules. In contrast to the decrease of entropy of the macromolecule during formation of the helix ($S_M^I > S_M^{II}$), the entropy of water during this process increases ($S_W^I < S_W^{II}$). This leads to an increase of the entropy in the whole system. At the same time the partial molar volume of water increases

It was found that the desorption of water from surfaces leads to a certain increase in its partial molar volume (Fig. 2.26). Obviously, the more highly organized water molecules at the surfaces are packed closer together. In this respect, we should remain with the so-called Le Châtelier principle. It predicts that if a constraint is applied to a system in equilibrium, the system adjusts to a new equilibrium that tends to counteract the constraint. An increase of the hydrostatic pressure, in this sense, could be counteracted by the decrease of the partial volume of water in the system. In Fig. 2.26 this would mean an increase of pressure would lead to an increase of hydration, and therefore to a decrease in organization of the macromolecular, or the supramolecular structure. Hence, an increase of hydrostatic pressure should destroy structures which are connected by processes of hydration, i.e. by so-called *hydrophobic bonds*. In fact, extremely high pressures in the order of 10 to 100 MPa are necessary for these reactions. In these cases membranes, microtubules and other supramolecular structures will be destroyed. In some cases high hydrostatic pressures were applied to study physiological properties of membranes. In this context it must be remembered that in the deepest parts of the ocean, at depths of 10 km, a hydrostatic pressure of 100 MPa occurs!

Figure 2.27 illustrates how the shift of the partial volume of water can be measured directly during a dissociation-association reaction. For this, in a vessel with exactly controlled temperature, the macromolecular sample enclosed in a dialysis sac hangs on a very sensitive quartz spring. The surrounding solution of this dialysis sac is exchanged continuously. A modification of the density of the macromolecules will be reflected in a change of the length of the spring. As indicated in the figure, the observable changes are minimal. In this case the change during the pH dependent polymerization of proteins of the tobacco mosaic virus is demonstrated. The time which is necessary to arrive at the equilibrium in this process is quite long.

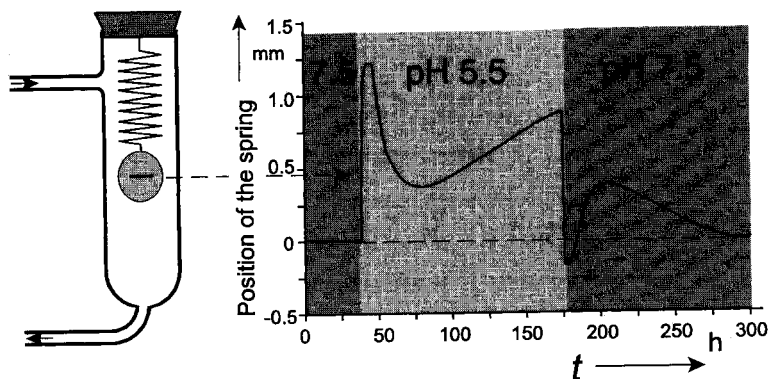


Fig. 2.27. Measurement of the kinetics of a hydration process, using a dialysis sac hanging on a quartz spring. The curve indicates results obtained on the protein of the tobacco mosaic virus. According to data from Lauffer 1975

Hydrophobic bonds, in fact, are not true interatomic connections like covalent bonds. Molecules or parts of them rather were pressed together by a force generated by the entropy increase in the involved water molecules.

Further reading: Benz and Conti 1986; Collins and Washabaugh 1985; Horn 1972; Israelachvili 1994; Lauffer 1975; Makarov et al. 1998; Wiggins 1990

2.4.3

Ions in Aqueous Solutions, the Debye-Hückel Radius

In the previous sections we discussed the electric field, which was induced by a single charged point, and which influences molecules in its vicinity. When considering extremely short distances, calculating for example force and energy of ionic bonds, or the influence on water dipoles nearby, it is possible to consider only this single charged point and to use Coulomb's law [Eq. (2.4.3)] for the calculation. In the case of larger distances from the source of the field, however, the influence of other charges must be considered. In a 100 mM solution of NaCl, for example, the mean distance between the ions only amounts to about 2 nm. This means that under physiological conditions the interactions of ions must be considered as a set of charged points.

These kinds of calculations are possible on the basis of the theory of strong electrolytes, the *Debye-Hückel theory*. This theory considers an ionic cloud around an arbitrarily chosen central ion. This central ion is taken as the center of polar coordinates. It attracts ions with opposite charges (counter-ions) and repels them with identical charges (co-ions). In this way an electric field will build up, generated by the central ion as well as by the ions surrounding it. This field is spherically symmetrical and can be described by the function $\psi(r)$.

At a distance r from the central ion, other ions will be attracted or repelled by the central ion, depending on their charge. We will denote the number of charges of an ion by z_i , for example: $z_{\text{Na}} = +1$, $z_{\text{SO}_4} = -2$ etc. The electrostatic energy of an ion at a particular point in space with an electrical potential ψ amounts to $z_i e \psi$.

Boltzmann's law of energy distribution [Eq. (2.3.16)] makes it possible to calculate the concentration (c_i) of the ions i at a point with the potential ψ . In this case, the energy of an ion in the electric field ($z_i e \psi$) opposes the energy of thermic noise (kT). This means that an orientation in the common electric field opposes the stochastic distribution of the ions by thermic fluctuations.

$$c_i(\psi) = c_{i0} e^{-\frac{z_i e \psi}{kT}}. \quad (2.4.10)$$

In this equation, c_{i0} is the concentration of the ion i , far away from the influence of the central ion, which means in the bulk solution. This equation, however, will not answer our question. We do not know yet the real amount of the potential ψ at this point, because it depends itself on the concentration of the ions in the ionic cloud. Furthermore, we are not interested in the function $c_i(\psi)$, but would rather know the concentration and potential distribution in space, i.e. the functions $c_i(r)$ and $\psi(r)$. This is possible using the Poisson equation. This is a partial differential equation of the second order, which gives the function $\psi(r)$ in space as a result of a charge density $\rho(r)$:

$$\nabla^2 \psi = -\frac{1}{\epsilon_0 \epsilon} \rho. \quad (2.4.11)$$

$\nabla^2 \psi$ means the second derivation of the potential by the coordinates of the space. This Nabla operator has already been described in Section 2.4.1. For ϵ the dielectric constant of water is used.

The charge density (ρ , in C m^{-3}) can be calculated from the ion concentration c_i of the ions in the cloud by:

$$\rho = \sum_{i=1}^n c_i z_i N e = F \sum_{i=1}^n c_i z_i. \quad (2.4.12)$$

Introducing Eq. (2.4.10) in (2.4.12), and both in (2.4.11), one gets the *Poisson-Boltzmann equation*:

$$\nabla^2 \psi = -\frac{F}{\epsilon_0 \epsilon} \sum_{i=1}^n c_{i0} z_i e^{-\frac{z_i e \psi}{kT}}. \quad (2.4.13)$$

The expression (2.4.13) is a partial differential equation, which can be solved analytically only under simplified conditions. Let us consider the following situation: let the solution contain only a single kind of ions i , let the potential be very low ($z_i e \psi \ll kT$) and simply a spherical symmetric function of the central ion. For this simplified case, the solution of Eq. (2.4.13) becomes:

$$\psi_r = \frac{z_i e}{4\pi\epsilon_0\epsilon r} e^{-\kappa r}. \quad (2.4.14)$$

Comparing this equation with Eq. (2.4.3), it shows that the influence of the potential of the central ion will be diminished by the factor $e^{-\kappa r}$ ($e^{-\kappa r} = 1/e^{\kappa r}$). The extent of this decrease depends on the concentration of the ions in the solution (see Fig. 2.28). This becomes understandable when considering the definition of the *Debye-Hückel parameter* (κ) in following equation:

$$\kappa = \sqrt{\frac{e^2 N}{\epsilon_0 \epsilon kT} \sum_{i=1}^n c_{i0} z_i^2} = \sqrt{\frac{F^2}{\epsilon_0 \epsilon RT} \sum_{i=1}^n c_{i0} z_i^2} = \sqrt{\frac{2F^2 I}{\epsilon_0 \epsilon RT}}. \quad (2.4.15)$$

The first expression of this equation can easily be transformed into the second one, using the relation between the Faraday constant (F) and the charge of a single electron (e) as: $F = eN$, as well as the relation: $R = kN$. In the last expression of Eq. (2.4.15), the *ionic strength* (I) is introduced.

$$I = \frac{1}{2} \sum_{i=1}^n c_{i0} z_i^2. \quad (2.4.16)$$

When calculating the value of κ for real conditions it can be seen that this Debye-Hückel parameter has the measure: m^{-1} (For this of course, the concentration measure: mole per m^3 must be used!). The parameter $1/\kappa$ is therefore a measure of a distance. Accordingly, it is called *Debye-Hückel length* or *Debye Hückel radius* of the ionic cloud. Considering Fig. 2.28 it is obvious that this value $1/\kappa$ again is not a “visible” radius, but just an effective parameter.

In the following example we will demonstrate the calculation of the ionic force of a solution: Let us consider a physiological Ringer's solution with the following composition: 105 mM NaCl, 5 mM KCl, 25 mM Na_2HPO_4 , 2 mM $CaCl_2$

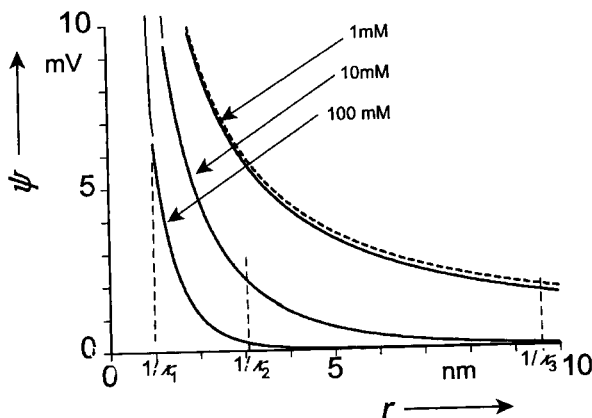


Fig. 2.28. The electrical potential (ψ) as function of the distance (r) from a central ion. The *dashed line* ignores the influence of the ionic cloud according to Eq. (2.4.3); the *solid lines* show the potential $\psi(r)$ which is diminished by the ionic clouds in solutions of various concentrations of a one-one-valent salt ($z = \pm 1$) according to Eq. (2.4.14)

(mM means the usual measure of concentration: mmol l^{-1}). Let us assume a full dissociation of the Na_2HPO_4 . Using Eq. (2.4.16), one gets:

$$\begin{aligned} I &= \frac{1}{2}(c_{\text{Na}}z_{\text{Na}}^2 + c_{\text{K}}z_{\text{K}}^2 + c_{\text{Cl}}z_{\text{Cl}}^2 + c_{\text{HPO}_4}z_{\text{HPO}_4}^2 + c_{\text{Ca}}z_{\text{Ca}}^2) \\ &= \frac{1}{2}(0.155 \cdot 1^2 + 0.005 \cdot 1^2 + 0.114 \cdot 1^2 + 0.025 \cdot 2^2 + 0.002 \cdot 2^2) \\ &= 0.191 \text{ mol l}^{-1}. \end{aligned}$$

According to Eq. (2.4.15), the Debye-Hückel radius (in nm) at temperature $T = 298 \text{ K}$ can easily be calculated by the following equation, using as a concentration measure $\text{mol} \cdot \text{l}^{-1} = M$:

$$\frac{1}{\kappa} = \frac{0.304}{\sqrt{I}} \quad (\text{in nm}). \quad (2.4.17)$$

This means for the case of the above considered solution:

$$\frac{1}{\kappa} = \frac{0.304}{\sqrt{0.191}} = 0.696 \quad (\text{in nm}). \quad (2.4.18)$$

Summarizing, we conclude that with the Poisson-Boltzmann equation [Eq. (2.4.13)] and its integration, the electrical potential can be calculated depending on the distance from a central ion. In this case the screening effect of the ionic cloud is considered. This screening, in fact, increases with increasing ionic strength of the solution. As a measure of the thickness of the ionic cloud, the Debye-Hückel radius ($1/\kappa$) is defined, which, in contrast to the crystal radius or the hydration radius, is not a property of a specific ion but just depends on the ionic strength of the solution. The Debye-Hückel radius is an effective parameter as well as the others (see Sect. 2.4.2).

2.4.4

Intermolecular Interactions

Self-assembly of biological structures is an intricate process consisting of an interplay of a variety of different intermolecular forces. In Section 2.4.2 we explained the role of hydrogen bonds for building up the structure of water, for interactions among ions and between ions and water dipoles, as well as the role of hydrophobic bonds in the process of self organization of macromolecules. In the following, the spectrum of these interactions will be extended to include additional electrostatic and electrodynamic forces. In fact, any kind of interaction between two macromolecules in an aqueous medium is the sum of several forces which differ by their sign (i.e. repulsive or attractive), and by their distance function.

Let us first consider a dipole interacting with an ion. In this case, the energy of electrostatic binding (E_{ID}), can be calculated using Coulomb's law [Eq. (2.4.2)]. It simply results from the sum of all electrostatic components, namely the interaction of the ion (q_1) with one ($-q_2$), as well as with the other ($+q_2$)

charge of the dipole. Considering the simplest case, namely the linear orientation of the dipole in relation to the ion (Fig. 2.29A), it gives:

$$E_{ID} = \left[-\frac{q_1(-q_2)}{4\pi\epsilon_0\epsilon r} \right] + \left[-\frac{q_1q_2}{4\pi\epsilon_0\epsilon(r+l)} \right] \quad (2.4.19)$$

We will consider simply that the dipole $\mu = q_2l$ contains two charges with equal size but opposite signs. A simple algebraic rearrangement of Eq. (2.4.19), and the introduction of the dipole moment (μ) results in:

$$E_{ID} = \frac{q_1q_2l}{4\pi\epsilon_0\epsilon r(r+l)} = \frac{q_1\mu}{4\pi\epsilon_0\epsilon r(r+l)}. \quad (2.4.20)$$

Assuming: $r \gg l$, the Eq. (2.4.20) will be simplified to:

$$E_{ID} \approx \frac{q_1\mu}{4\pi\epsilon_0\epsilon r^2}. \quad (2.4.21)$$

In contrast to the energy of interaction of two ions [Eq. (2.1.5)] as a function of $1/r = r^{-1}$, this equation shows that the interaction energy of an ion with a dipole decreases with r^{-2} .

The mutual interaction between two dipoles (E_{DD}) can be calculated in a similar way. Let us again look at the simplest case of collinear orientation of both dipoles, as depicted in Fig. 2.29 B. Using the same approach, one gets:

$$E_{DD} = -\frac{1}{4\pi\epsilon_0\epsilon} \left[\frac{q_1q_2}{l_1+r} + \frac{q_1(-q_2)}{l_1+r+l_2} + \frac{(-q_1)q_2}{r} + \frac{(-q_1)(-q_2)}{r+l_2} \right]. \quad (2.4.22)$$

Again we will rearrange this equation and introduce the dipole moments μ_1 and μ_2 :

$$\begin{aligned} E_{DD} &= \frac{q_1l_1q_2l_2}{4\pi\epsilon_0\epsilon} \cdot \frac{l_1+l_2+2r}{(l_1+r)(l_2+r)(l_1+l_2+r)r} \\ &= \frac{\mu_1\mu_2}{4\pi\epsilon_0\epsilon} \cdot \frac{l_1+l_2+2r}{(l_1+r)(l_2+r)(l_1+l_2+r)r}. \end{aligned} \quad (2.4.23)$$

Using the same approach: $r \gg l$, it gives:

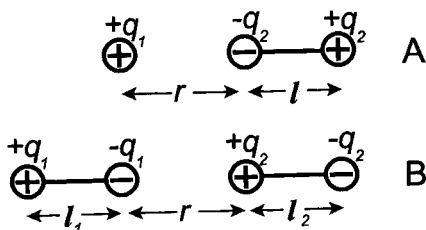


Fig. 2.29. Ion-dipole (A) and dipole-dipole (B) interaction. q - charge, l - length of the dipoles, r - distance

$$E_{DD} \approx \frac{\mu_1 \mu_2}{4\pi\epsilon_0\epsilon} \cdot \frac{2r}{r^4} = \frac{2\mu_1 \mu_2}{4\pi\epsilon_0\epsilon r^3}. \quad (2.4.24)$$

This sequence of possible electrostatic interactions can be extended including interactions with induced dipoles (see Sect. 2.4.1). In this case, a more complicated function of the distance must be expected because the induction of a dipole itself depends on the electric field [Eq. (2.4.8)]. For this case, a function can be calculated, including r^{-4} . If an interaction of two dipoles is considered, inducing each other, a function with r^{-6} will be obtained.

This brings us to the wide field of *van der Waals interactions*. That the expression is in plural indicates that it includes a number of different intermolecular forces. Van der Waals interactions in general are attractive forces between molecular components, whole molecules, or supramolecular particles which are not simply based on electrostatic interactions. Rather they are attributed to electromagnetic interactions, occurring by fluctuations of charges. The reasons for such charge fluctuations in molecules may be different. There can be thermal molecular translations, or oscillations in the thermic level of electrons, as illustrated in Fig. 2.5. Furthermore, fluctuations occur in the electron structure of molecules which can be calculated by quantum mechanics. This is the basis of so-called dispersion forces, or *London-dispersion forces*, sometimes also called *London-van der Waals interactions*. These oscillations occur at frequencies similar to that of visible light.

The understanding of van der Waals forces has been substantially advanced by F. London, H. C. Hamaker and E. M. Lifschitz. Nevertheless, many aspects are unclear even today. The central question concerns the character of this interaction, and its energy as a function of inter-particle distance (r). Furthermore, it still seems unclear whether in these kinds of interactions a molecular specificity can exist which might be related to the frequency of interaction.

The energy of van der Waals interactions is ruled by the *Hamaker constant*, which results from quantum mechanical calculations. The distance relation of this energy depends strongly on particle geometry. This is illustrated in Fig. 2.30. There is a difference as to whether it is an interaction between a spherical and/or a cylindrical particle, and additionally there are different functions $f(r^{-n})$ depending on the total distance between the particles. This means at the same time that simple hyperbolic functions of this kind are only rough approximations.

To understand the relevance of van der Waals forces, it is necessary to consider them in context with electrostatic interactions. Here we touch a central concept of colloid chemistry, especially the concept which is connected with the names of B. V. Derjaguin, L. D. Landau, E. J. W. Verwey, and J. Th. G. Overbeek. It is therefore called the *DLVO-theory* of colloidal interactions. The basic ideas of this approach, considering the Debye-Hückel theory of the ionic clouds, as well as the properties of van der Waals, can be listed as follows:

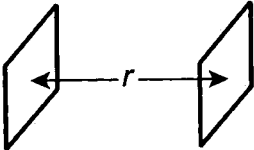
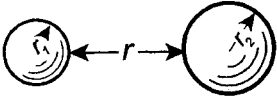
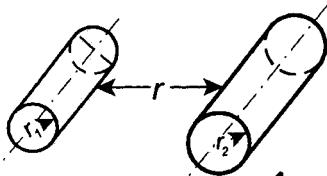
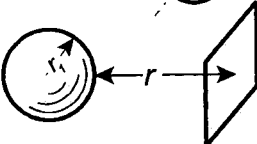
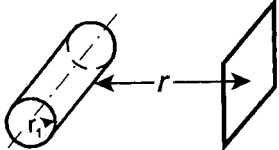
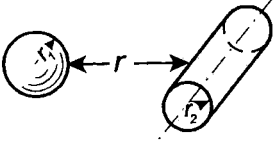
Geometric conditions	Relation of distances	
	$r_1, r_2 \gg r$	$r_1, r_2 \ll r$
	r^{-2}	
	r^{-1}	r^{-6}
	$r^{-3/2}$	r^{-5}
	r^{-1}	r^{-3}
	$r^{-3/2}$	r^{-3}
	r^{-1}	r^{-5}

Fig. 2.30. The distance function $[f(r^{-n})]$ of the van der Waals energy of interaction between particles of various shape. (Parameters after Israelachvili 1974)

- The interaction force between two particles results from the sum of the particular forces, each one decreasing with increasing distances in different ways.
- In contrast to generally attractive van der Waals forces, electrostatic interactions can also become repulsive.
- The intensity of electrostatic interactions, in contrast to van der Waals forces, depends to a high degree on the environmental conditions.

To illustrate the consequences of these postulates, we must imagine that the distance function of the interaction energy $[E(r)]$ results as a sum of various functions, schematically drawn in Fig. 2.31. There are so-called *short distance interactions*, decreasing with r^{-5} , or r^{-6} , and *long distance interactions*, the curves of which do not decrease as fast (r^{-1}). Some of these curves can be looked at upside down, representing repulsive forces, which consequently must be subtracted by the others. Additionally, the dependence of the electrostatic forces on the ionic strength of the solution is to be considered, as illustrated in Fig. 2.28, as well as the various distance functions of the van der Waals forces, as depicted in Fig. 2.30. Hence, the shape of the resulting sum can be quite complicated.

Figure 2.32 illustrates this situation for a more or less simple case of the interaction of two particles in a solution. In this case the interaction is supposed to be the sum of an electrostatic repulsion (r^{-3}) and two kinds of van der Waals attraction forces, one a long- (r^{-1}), another a short-distance interaction (r^{-6}). Considering the sophisticated function of the electrical potential near a charge in electrolyte solutions, as depicted in Fig. 2.28, and furthermore the circumstance that the function of van der Waals interactions itself depends on distance (Fig. 2.30), it is understandable that the real conditions of interactions are much more complicated. Additionally, steric barriers have to be considered when particles are in closer contact as well as changes in the hydration property and many others.

Nevertheless, the curves of Fig. 2.32 demonstrate at least qualitatively some basic properties of macromolecular interactions. It shows that two particles may attract each other at larger distances, even if they carry charges of the same sign, if additionally a long ranging van der Waals attraction occurs. This is demonstrated in Fig. 2.32 at distance $r > r_{\min}$. The point $r_{\min}$ marks the position of minimum interaction energy. The attractive force ($F = dE/dr$) at this point becomes zero. If this distance is decreased a repulsion force

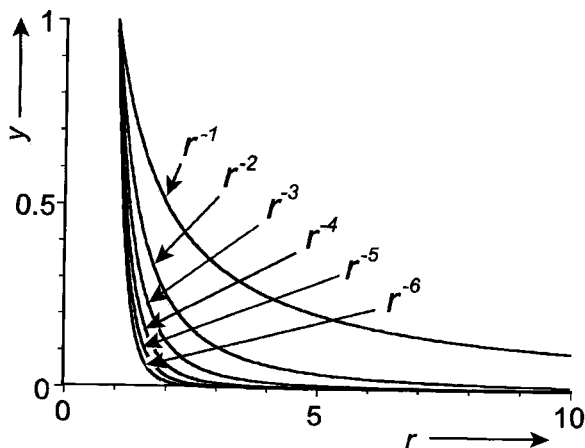


Fig. 2.31. Functions: $y = r^{-n}$ to illustrate the character of short- and long-distance interactions

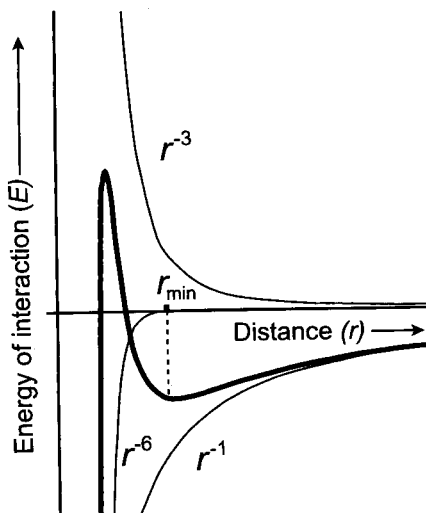


Fig. 2.32. The total energy of interaction (E) (thick line) between two particles as a sum of three functions (fine lines): one sort of electrostatic repulsion (r^{-3}), and two kinds of van der Waals attractions (r^{-1} and r^{-6}). The point $r_{\min}$ indicates the distance of the secondary minimum

appears. The particles, therefore, will rest at this point, which is called the distance of the *second minimum*. If in any way this repulsion force were to be overcome, attractive interactions again would take place and the particles would aggregate at the distance of the *primary minimum*. In Fig. 2.32 this primary minimum is not shown. This would predict an inclusion of further components of interactions at very short distances. So, for example, the influence of Bohr's radii must be considered in this case, since the extremely strong repulsion forces arising from electron shell interactions come into play.

As already mentioned, the electrostatic interaction strongly depends on the ionic conditions of the medium. This is understandable in view of Eq. (2.4.14) (Sect. 2.4.3), as well as Fig. 2.28. By increasing the ionic strength, for example, the energy barrier in the region $r < r_{\min}$ will become diminished. Finally, thermic hits could be sufficient to let the particles cross this barrier and to agglutinate. This effect has been experimentally verified. Increasing the salt concentration of a colloid solution leads to the agglutination of the particles. This is known as the *salt-out effect*, in contrast to the *salt-in* process, leading to a re-stabilization of a colloidal system.

In molecular biology this interplay between the various forces involved in intermolecular interactions is very important. This is one way to control supramolecular processes. The influence is possible either by the control of the dielectric constant, or by changing the ionic concentration. The local dielectric constant depends on the arrangement of molecules, of their polarization and of the water structure in their vicinity. These processes, are mainly controlled by changing local ionic compositions.

In this context the influence of multivalent ions are of particular interest. Let us mention the role of calcium in muscle contraction, nerve excitation, exocytosis, cell adhesion, and many other processes. In colloid chemistry the Schulze-Hardy-

rule is known, an empirically found relation which later was also verified theoretically. This rule predicts that the concentration of a salt, which would be necessary to overcome the energy barrier between the second and the first minimum, is inversely proportional to the 6th degree of its valency. This means that the critical concentration produced by a univalent ion relates to this concentration of a bivalent one, like 1^{-6} to 2^{-6} , which means: 1 to $1/64$. Consequently, 2 mM Ca^{++} would be as effective, as 128 mM K^{+} or Na^{+} . This is a relation which reminds us of the real concentrations of *in vivo* conditions. It must however, be considered that this would not explain the difference in the behavior of Ca^{++} in contrast to Mg^{++} . Other reasons must also be taken into consideration.

Further reading: For van der Waals interactions in molecular biology: Jehle 1969; Israelachvili 1994

2.4.5

Structure Formation of Biomacromolecules

In the preceding sections, problems relating to the information content of biomacromolecules were discussed, the idea of random coil formation by molecular chains as a state of maximum entropy was introduced, as well as the role of water in the entropy balance of macromolecular systems, and the broad spectrum of intermolecular interactions. A knowledge of these aspects is an important pre-requisite to understanding the dynamic structure of proteins and nucleic acids in cells.

Investigations of the 3-dimensional structure of biological macromolecules, their genetic coding, and especially the processes of self assembling of these structures, are the main problems of molecular biophysics. There is no doubt that the total information on the stearic structure of proteins is stored in its primary structure, i.e. in the sequence of the amino acids. The question arises as to how one can decode the 3-dimensional structure of a protein, reading its amino acid sequence and furthermore, how the process of self assembly, which leads spontaneously and reversibly to this genetically predicted protein structure, really works?

The proteins are composed of α -L-amino acids which are connected to each other, forming long chains. In these connections the carboxyl groups of the α -S-Atoms are condensed with the amino group of the following monomer, splitting up a molecule of water. In Fig. 2.33.A the distances and the angles of the bonds of the basic structure of a polypeptide are illustrated. The symbols R mean the side chains characterizing the 20 amino acids appearing in living nature.

These molecular chains have a high degree of flexibility. During Brownian movement, interconnections of various parts of these chains are possible. This occurs mostly by hydrogen bonds connecting atoms of two peptide bonds, or atoms of a peptide bond with those of a side chain of an amino acid, or finally of two side chains together. Covalent bonds between side chains are also possible. This is mostly done by disulfide bridges between the SH-groups of cysteins. Electrostatic and hydrophobic interactions also take place.

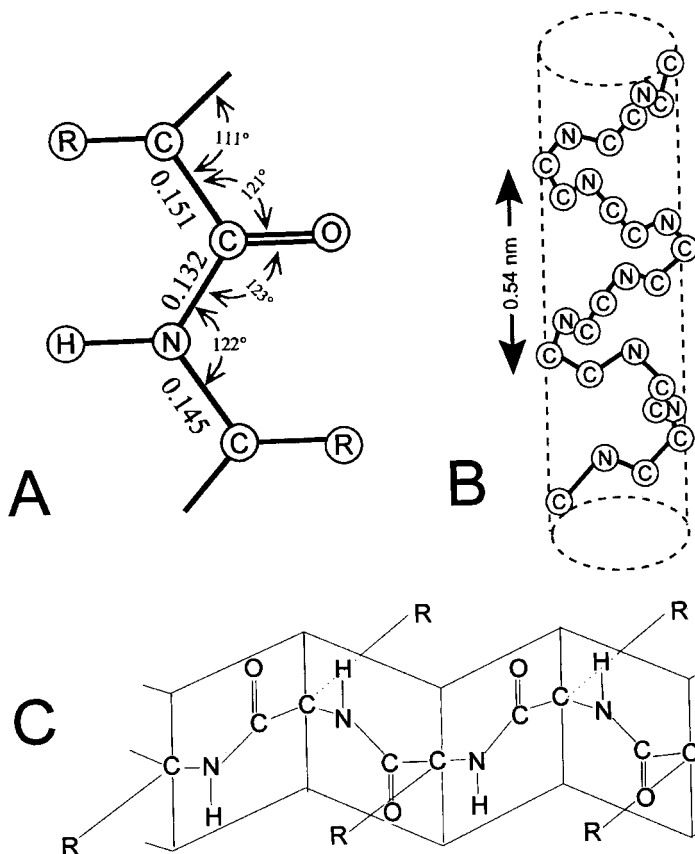


Fig. 2.33. Elements of the protein structure: **A** angles and distances in a peptide chain; **B** parameters of an α -helix (for more clarity, only the basic atoms of the polypeptide chain are depicted); **C** β -sheet

Let us start to consider possible mechanisms of self assembly, hypothesizing that the molecule will move stochastically, reversibly connecting and disconnecting bridges with monomers of its own chain. This proceeds as long as the polypeptide chain does not become a structure of minimal conformational energy, which will be identical with the active structure of the natural protein. This hypothesis can easily be reduced to absurdity, considering the time which would be necessary for this process. Let us take a protein with 300 amino acids, each of which could have 8 rotational positions. In this case $8^{300} = 10^{270}$ positions of the whole molecule are possible. Even in a case where, by reason of steric hindrance, this amount were to be limited, and even if the conformational changes were extremely fast, the time needed to arrive at the correct structure with this procedure would be astronomically long. Finally, it would be

questionable whether the absolute minimum was really found, and whether this really represents the proper structure of the protein.

In fact, however, the number of possible configurations are much more strongly limited. There are always two types of secondary structures which occur, and which are predicted by the interaction of neighboring monomers in the sequence of the chain. These are the α -helix, and the β -sheet. The α -helix is a right-hand spiral, the parameters of which are illustrated in Fig. 2.33B. This structure mostly occurs in proteins which are embedded in the lipid leaflet of membranes (see also Fig. 2.38). In these hydrophobic surroundings they are obviously more stable than in aqueous solutions. In such solutions complexes are sometimes built consisting of several helices, which stabilize each other.

In β -sheets the groups of 6 atoms (C_α -CONH- C_β ; see Fig. 2.33A) are positioned in a plane. Schematically such an arrangement of planar monomers is depicted in Fig. 2.33C. Up to nine monomers are usually arranged in such a β -sheet. Mostly, these planar elements are not simply folded in a plane, but the folded elements themselves form a right-hand spiral. In contrast to the α -helix where the side chains (R) are orientated perpendicular to the helical axis (not depicted in Fig. 2.33B!), in the case of the β -sheet they are orientated perpendicular to the plane of the monomer (Fig. 2.33C). The stability of these sheets increases by parallel or by antiparallel arrangements of several sheets which are connected together by hydrogen bonds.

Beside these two types of arrangements of polypeptides, others are sometimes possible, as for example α -helices with another pitch, a different amount of amino acids per turn (3_{10} -helix, π -helix), or even left-hand turned structures.

Apparently there are specific sequences of amino acids which tend to generate helical structures, and others forming β -sheets or unstructured regions connecting them. There is always a search for correlations between amino acid sequences and their secondary structures, to derive rules predicting the occurrence of them. Collagen, as a typical structure protein for example, forms a triplet-helix as a ternary structure. This is already marked in the primary structure by a repeating sequence $(Gly-X-Y)_n$. In this case Gly stands for glycine, whereas X and Y demand some other amino acids. Mostly these are proline and hydroxyproline. This repeated occurrence of glycine in each third position stabilizes the triplet-helix by hydrogen bonds. In some cases amino acids with the ability to form disulfide bridges build a sort of starting point for continuing the formation of super helices. In this case, as in the formation of α -helices, the process resembles a zip fastener. In a similar way myosin forms a left-hand double super-helix. The turn of this helix is predicted by the arrangement of amino acids in the corresponding α -helices of the secondary structure.

Recently a number of enzymes were found significantly accelerating the process of structure formation of proteins, for example the group of *chaperons*. They mostly help to avoid wrong ways of folding, binding reversibly to specific sequences of the primary structure of the polypeptide. For this reason it is not correct to use the term structure forming enzymes for them. In contrast, protein-disulfide-isomerases and poly-cis/trans-isomerases directly promote the

process of folding. This is because all peptide bonds are primarily synthesized in trans-form, but 7% actually occur in cis-form. This trans-cis transformation in general occurs quite slowly, but will be accelerated significantly by these enzymes.

A correctly folded protein, i.e. a native protein, can become reversibly denaturated by shifts of pH, temperature or by modifications of the ionic conditions in the solution. In this way it becomes a so-called *molten globe*. The way back again toward the native protein is marked by various intermediates, where transient H-bonds or even disulfide bonds are established, which will be released again on the way to the next intermediate state. Therefore, the path of this reaction has apparently been determined, but is not yet fully understood.

The difference in free molar Gibbs energy (ΔG) between the native, versus the denaturated status of the proteins, is sometimes lower than 50 kJ mol^{-1} . Considering that about 100 connections between amino acids of neighboring chains must be loosened during this process, their binding energies must be lower than the energy of thermic motion ($RT = 2.5 \text{ kJ mol}^{-1}$ for $T = 300 \text{ K}$). The stability of the protein structure, therefore, can be explained only as a phenomenon of cooperativity. Additionally, this estimation illustrates the dynamics of the molecule. Obviously, in the native protein small conformational transitions must continuously occur, caused by thermic fluctuations.

These thermic fluctuations of protein structure are, in fact, prerequisites for their function. Such modifications, for example, can lead to activation or inactivation of enzymes. The enzyme activity can therefore be controlled in this way. Similar modifications can be caused by ligands, by changes of the local ionic conditions, or even under the influence of electric fields. These field effects occur for example during the control of Na- and K-channels in nerve cells, causing the action potential (see Sect. 3.4.3).

These molecular fluctuations may modify the shape, the volume, as well as the energetic state of the molecules. Analysis of these fluctuations makes it possible to calculate the average square of displacement of the energy ($\langle \Delta U^2 \rangle$) as well as that of the volume of the molecule ($\langle \Delta V^2 \rangle$):

$$\langle \Delta U^2 \rangle = kT^2 m C_V, \quad (2.4.25)$$

$$\langle \Delta V^2 \rangle = kTV\beta_T. \quad (2.4.26)$$

Hence, these parameters depend on the energy of thermic noise (kT), on the molecular mass (m), on the molecular volume (V), on the isochoric heat capacity (C_V), as well as on the isothermic volume compressibility (β_T). Considering a globular protein with a molecular mass $M = 25\,000$, it results:

$$m = \frac{M}{N} = 4.15 \cdot 10^{-20} \text{ g} = 4.15 \cdot 10^{-23} \text{ kg}.$$

Let us consider the following case: $V = 3.2 \cdot 10^{-26} \text{ m}^3$, $C_V = 1.34 \text{ kJ kg}^{-1}$, $T = 298 \text{ K}$, as well as $\beta_T = 2 \cdot 10^{-11} \text{ Pa}^{-1}$. From Eq. (2.4.25) results

$\Delta U = 2.61 \cdot 10^{-19}$ J per molecule. Multiplying this by the Avogadro number ($N = 6.022 \cdot 10^{23} \text{ mol}^{-1}$), would mean 157 kJ mol^{-1} in a case where all molecules would fluctuate synchronously. Already this is enough energy to denaturate the protein. The deviations in the volume of this molecule result from Eq. (2.4.26) for: $\Delta V = 5.1 \cdot 10^{-29} \text{ m}^3$. This is about 0.16% of the total volume of the molecule. Nevertheless, these deviations would be sufficient to create transient local hollow spaces in the molecule, or to allow the penetration of ions in preformed pores. Hence, this kind of statistical fluctuation should be included in the consideration of protein functions.

It must be added that we considered here the proteins as structures in equilibrium. This corresponds to the situation of isolated molecules. During biological processes however, the proteins are usually considered as far from equilibrium. Therefore, many other structural modifications occur, mostly concerning special functional groups of the molecule. This is a field of further research.

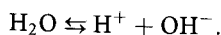
Further reading: Cantor and Schimmel 1980; Creighton 1992; Cooper 1976

2.4.6

Ampholytes in Solution, the Acid-Base Equilibrium

We already learned that charges in biomolecules significantly influence the formation of their molecular and supramolecular structures. This aspect will be underlined in the next section when we discuss properties of interfaces and biological membranes. Furthermore, in Section 3.2.3 we will discuss the role of fixed charges in the cytoplasm to explain the occurrence of Donnan potentials. These aspects demand knowledge about the pH dependence of these sorts of charges.

An acid, per definition, is a proton donor, i.e. a compound which is dissociating hydrogen ions from particular groups. A base, on the other hand, is a proton acceptor and thus is able to associate it. According to this definition, water is a weak acid, dissociating with the following scheme:



This equilibrium leads to the definition of the ionic product of water and, finally, to the pH-value:

$$\text{pH} = -\log c_{\text{H}} \quad (2.4.27)$$

Strictly of course, it is not the concentration (c_{H}), but rather the activity of the protons that must be considered, but in view of the extremely small concentration of these protons, this makes no difference.

In the same way, the dissociation of any other acid can be written:



The equilibrium constant of this reaction may be written as:

$$K = \frac{c_A c_H}{c_{HA}} \quad (2.4.28)$$

In the same way, as in the case of pH, it is possible to define a pK-value:

$$\text{pK} = -\log K = -\left(\log c_H + \log \frac{c_A}{c_{HA}}\right) \quad (2.4.29)$$

or, using Eq. (2.4.27):

$$\text{pK} = \text{pH} - \log \frac{c_A}{c_{HA}} \quad (2.4.30)$$

This is the *Henderson-Hasselbalch equation* which makes it possible to understand the buffer properties of various substances and their titration curves.

Substances which contain acid as well as basic groups are called *ampholytes*. As a first example, there are the amino acids. The buffer capacity, as well as the dissociation property of these substances can be investigated by titration experiments. Furthermore, this allows calculation of the dynamics of electrical charges of these molecules. In Fig. 2.34 the charge of glycine is demonstrated as a function of the pH in the solution. For this, the mean number of elementary charges (z) per molecule is used. In the preceding sections we learned that dissociation and association are stochastic processes. Hence these amounts of z can be considered as an average in time or as a mean value of a large number of molecules.

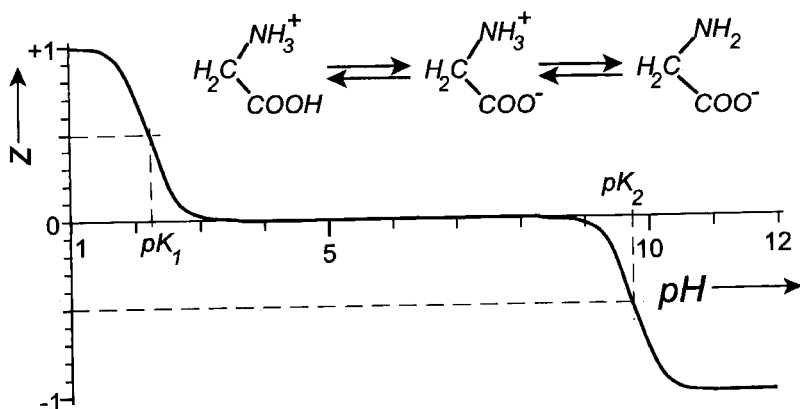


Fig. 2.34. The average number of charges (z) of glycine plotted against the pH in the solution. ($\text{pK}_1 = 2.35$, $\text{pK}_2 = 9.78$)

Glycine is a cation at extremely low pH, and in contrast an anion at extremely high pH. In the pH region in between, both groups of this molecule are dissociated, demonstrating the zwitterionic character of this molecule. In this case it has a considerable dipole moment and a maximal number of charges but on average it is electroneutral. This curve allows us to determine two pK-values which always correspond to the pH of the medium, where 50% of the charges are dissociated (in Fig. 2.34 corresponding to $z = +1/2$, and $z = -1/2$), as well as an isoelectric point which is positioned exactly in the middle between these two pH-values.

The pH dependent charge of ampholytes of course also strongly influences the structure of the surrounding water. This is the reason why the effective volume of these molecules was often modified by the pH of the medium (see Sect. 2.4.2, Fig. 2.27).

Ampholytes are typical constituents of buffer solutions. Buffers are substances which are able to bind protons, or release them, in this way stabilizing the pH of the solution. A buffer capacity can be defined which corresponds to the slope of the titration curve. It is maximal near the pK-values.

Whereas the Henderson-Hasselbalch equation can easily be applied for binary ampholytes with two pK-values at considerable distance from each other, this becomes impossible in the case of polyampholytes, such as proteins with a large number of different amino acids. This is illustrated in Fig. 2.35, showing the titration curve of hemoglobin. This curve can be considered as a sum of many sigmoidal curves with different properties. Near the isoelectric point, this curve can be fitted by the simple function:

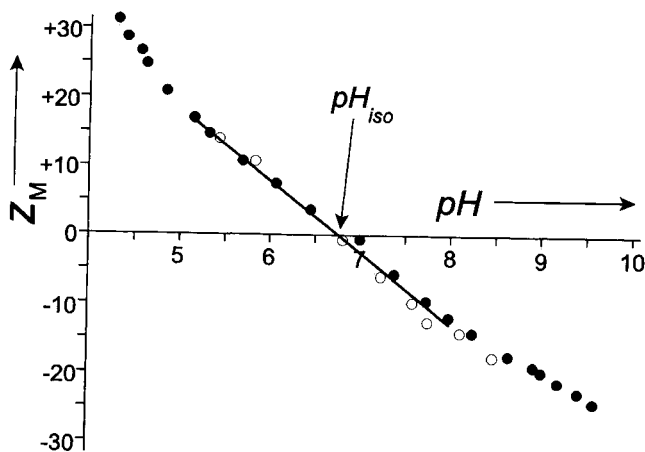


Fig. 2.35. The mean number of charges of horse hemoglobin as a function of the pH. $\circ$ – points for oxidized hemoglobin, deviating from others. The line between pH 5 and pH 8 corresponds to Eq. (2.4.31) with a buffer capacity of $z_{M0} = 10.2 \text{ val mol}^{-1}$, and an isoelectric point $\text{pH}_{\text{iso}} = 6.68$. (According to measurements of German and Wyman 1937)

$$Z_M = -Z_{M0}(\text{pH} - \text{pH}_{\text{iso}}) \quad (2.4.31)$$

(Z_{M0} – buffer capacity, pH_{iso} – isoelectric point).

The titration curve, of course, indicates only the effective charges of molecules as a function of the pH in the solution. In the case of polymer molecules, the interaction of their monomers must be considered, depending on their mutual distance in the molecule. Furthermore, near the place of dissociation, a local pH may exist, differing from that of the bulk solution significantly. This is caused by fixed charges and therefore can be predicted by local electric potentials in the sense of the Boltzmann or Nernst equations [Eq. (3.2.43)]. In this case, the term *intrinsic* pK is used, meaning the true pK of specific dissociable groups in a polymer, considering the real local proton activity. This, in fact, is possible to measure. Using nuclear magnetic spin resonance (NMR) for example, the dissociation of specific carboxyl groups can be studied, traced by the isotope ^{13}C . Investigations of polypeptides indicated that considerable differences may exist between the effective and the intrinsic pK of some groups even if the influences of neighboring amino acids are quite low.

Further reading: On electrostatic properties of proteins: Matthew 1985; Pethig and Kell 1987

2.5

Interfacial Phenomena and Membranes

After explaining various kinds of molecular interactions in the previous section, we will now come to the next level of biological organization, considering self assembly and molecular dynamics of supramolecular structures, especially of membranes.

The cell membrane is a well-organized structure fulfilling a broad spectrum of physiological functions:

- As a surface, it forms a dynamic matrix for enzymatic reactions, receptor processes, and immunological recognition
- As a barrier of diffusion it controls the ionic composition of the cytoplasm by highly specific transporters
- As an electrically isolating leaflet it contains a mosaic of various passive and active electric devices, controlling membrane potential as well as near-membrane electrodynamic conditions
- As a mechanical structure it guarantees the integrity of the cell and influences its shape and movement as well as the displacement of organelles

We will discuss these biophysical mechanisms of membrane functions in many sections of this book in detail. Here, we will consider the molecular dynamics and the processes of self organization. Unfortunately, in some cases it is necessary to deal with some topics which will not be explained in detail until later.

2.5.1

Surface and Interfacial Tensions

Each molecule of a homogeneous isotropic phase interacts energetically with all its surrounding neighbors. If these forces are averaged for a sufficiently long period of time, all the moments of interaction are abolished. Considering, however, a molecule not in the bulk phase, but at the phase boundary, or moreover, at the liquid-air surface, the situation is quite different (Fig. 2.36). In this case the forces of interaction from one side differ from those the other side. A molecule of a liquid for example at the liquid-gas surface feels a stronger attraction to its neighbor molecules in the liquid than to the molecules of the gas phase. The energy of the surface molecules is higher than that of the molecules in the bulk phase. This is the reason why water droplets in air spontaneously form a sphere as the shape of minimal relative surface and therefore minimal surface energy.

To increase the surface of a liquid against air or against a gas which is saturated by the vapor of this liquid, it is necessary to heave more molecules to this higher level of energy. The energy which is required to enlarge the surface of a liquid phase in this way by 1 m^2 is named *specific surface energy*, which from the physical point of view, is the same as the *surface tension* γ ($\text{J} \cdot \text{m}^{-2} = \text{N} \cdot \text{m} \cdot \text{m}^{-2} = \text{N} \cdot \text{m}^{-1}$).

Water as a polar liquid with strong intermolecular interactions (see Sect. 2.4.2) has a high surface tension which decreases with increasing temperature. At 25°C it amounts to 0.0728 N m^{-1} . The surface tension of organic liquids, for example of benzene, is only 0.0282 N m^{-1} , and of ethanol only 0.0223 N m^{-1} at the same temperature.

For biological reactions the surface tension at the liquid-air interface is interesting in two ways: on one hand it is important for organisms living directly at the surface of a lake or the sea. These are microorganisms and small algae, forming the so-called neuston, as well as various higher plants, and finally

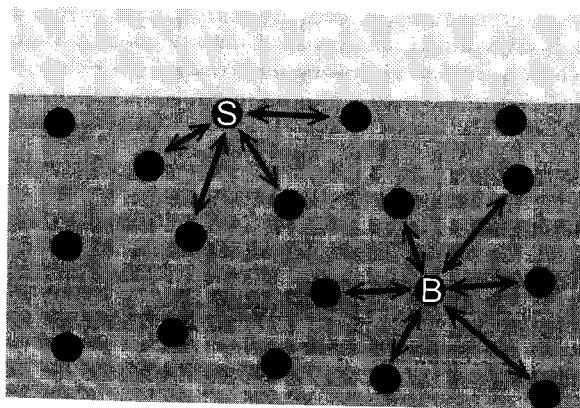


Fig. 2.36. Interaction forces of molecules at surface (S) and in the bulk solution (B)

various insects using hydrophobic properties of their legs to run on the water surface. On the other hand surface tension plays an important role in the biomechanics of the lung, where a highly solvated surface is in contact with the air.

To explain this situation the behavior of bubbles or vesicles formed by a liquid must be considered. In the same way that surface tension tends to decrease the surface of a droplet, bringing it into a spherical shape, the surface tension of a liquid-formed vesicle tends to decrease its inner volume, generating pressure against the included air (Fig. 2.37A). The pressure difference $\Delta p = p_i - p_e$ can be calculated by the Laplace equation:

$$\Delta p = \frac{2\gamma}{r}. \quad (2.5.1)$$

This means that the internal pressure of a vesicle is proportional to the surface tension of the liquid (γ) which forms it, and inversely proportional to its radius (r). The alveoli of the lung are vesicles of various diameters, connected each other and changing their diameter periodically. As indicated in Fig. 2.37B, according to Eq. (2.5.1), the pressure in the smaller vesicles would be higher than in the larger vesicles. This would lead to a collapse of the smaller vesicle in favor of a larger one if they were connected to each other. To avoid this, the surface tension of the water-air interface in the alveoli of the lung is strongly reduced by a surface-active material known as *pulmonary surfactant*. This layer consists primarily of phospholipids. The periodic change of the alveolar volume during inhalation additionally requires the property of this layer to be expanded quickly and reversibly. For this, a reservoir is formed, reversibly folding and unfolding these layers (Fig. 2.37C). This is a complicated process stabilized by special surfactant-associated proteins.

In most cases it is not the *surface tension* of the liquid-air interface, but rather *interfacial tensions* between two liquid phases which are important, or, in other words, the specific energy of interfacial molecules. In the ideal case the

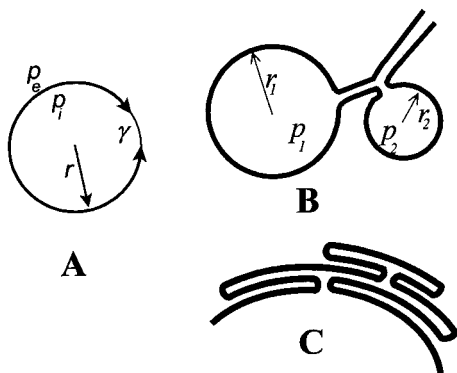


Fig. 2.37. A A vesicle with a surface tension γ tends to decrease the internal volume and generates an internal pressure $\Delta p = p_i - p_e$; B Two vesicles, built by a liquid of the same surface tension generate different internal pressures which are reciprocally proportional to their radius. In case of communication, the smaller vesicle should collapse in favor of the larger one. C The change in the surface of the surfactant layer by folding

interfacial tension of two liquids is equal to the difference between their surface tensions.

The attempt to measure the surface tension of living cells and to use this parameter to explain various physiological properties such as phagocytosis, exocytosis, cell division, amoeboid movement, cell shape etc. is quite old. In 1898 L. Rhumbler demonstrated that a drop of chloroform in water could "phagocytose" a thin, shellac-covered glass fibre in a manner similar to that of an amoeba ingesting a filamentous alga. In the same way cell shapes in a tissue were used to try to explain a result of surface tensions, comparing the tissue to a lather. D'Arcy Thompson, in his classical book *On Growth and Form*, first published in 1917 discussed this problem in detail.

Those comparisons, however, as visually impressive as they may be, are not very helpful in explaining these biological processes. In the case of a liquid droplet, surface molecules can always be drawn back into the bulk, or extruded again in the surface layer very quickly. This process is reversible; the mechanically deformed drop will re-assume its spherical shape as soon as the external mechanical disturbance is discontinued. In contrast, a living cell is surrounded by a membrane. An increase of the membrane area by inclination of additional molecules from the cytoplasm is a process which is much slower.

Further reading: Historical papers: Rhumbler 1898; Thompson 1966; for the surfactant of the lung: Batenburg and Haagsman 1998; Galla et al. 1998

2.5.2

Self Assembly and the Molecular Structure of Membranes

The spontaneous orientations of molecules in phase boundaries are of particular interest for questions of self-organization and stability of biological membranes. Let us first consider the behavior of molecules in the boundary between two phases, one being hydrophilic (water), the other hydrophobic (for example oil). A protein tends to orient its polar groups as much as possible toward the aqueous phase, and its nonpolar, i.e. hydrophobic groups toward oil (Fig. 2.38A). This leads to an orientation of the molecule at the phase boundary. If the polar groups of the molecule are distributed homogeneously over the whole molecule, then it could become unfolded by orientation in such a boundary. The protein, now being a fibril will spread over this interface.

The most important constituents of biological membranes are phospholipids. This is a group of compounds which consists of two fatty acids and a phosphate group, all attached to a polyhydric alcohol such as glycerol. The phosphate group is usually esterified with a nitrogen-containing alcohol. The structure and formula of a simple phospholipid is depicted schematically in Fig. 2.39. Two long hydrophobic fatty acid chains and a single phosphate group esterified with an ethanolamine residue are linked to a central glycerol molecule. The phosphate and the amino-group of the ethanolamine residue represents the hydrophilic part of the molecule.

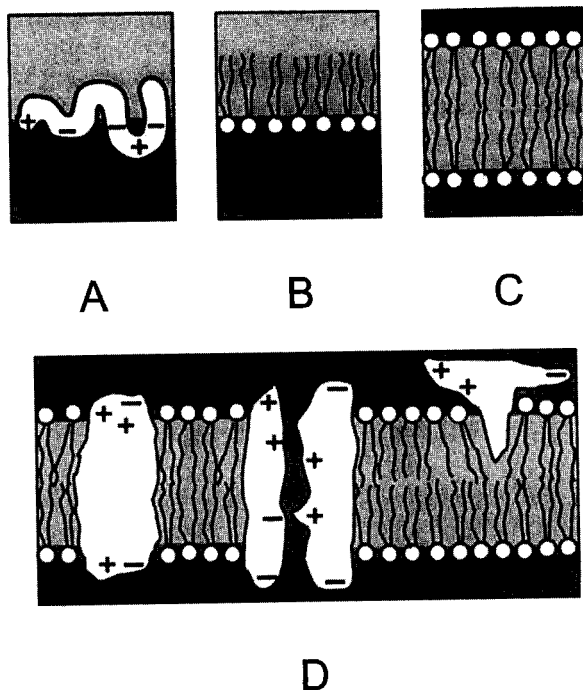


Fig. 2.38. Arrangement of molecules in phase boundaries (*light area* – hydrophobic phase, *dark area* – hydrophilic phase): **A** orientation of a protein with hydrophobic and hydrophilic regions; **B** orientation of a monolayer of phospholipids (this corresponds also to the lipid layer at a water-air surface); **C** a lipid bilayer in a homogeneous aqueous phase. In the region of the fatty-acid chains, orientated against each other, a hydrophobic micro-phase is formed; **D** orientation of proteins in such a phase, where the hydrophilic parts of the molecules are orientated towards water and the polar head groups of the lipids. In this way the formation of hydrophilic membrane pores is also possible

The large variety of biological lipids is realized by the diversity of the fatty acids as well as by their different head groups. In the case of phosphatidylcholine, the head group is polar but does not carry net charges under physiological conditions. The negative charge of the phosphate is compensated by a positive charge of the amino group. Phosphatidyl-serine, in contrast to this, contains an additional negative charge, namely that of a carboxyl group. This is the reason why it significantly influences the surface charges of biological membranes.

Phospholipids strongly orientate themselves at interfaces. In this case the hydrophobic chains of fatty acids are orientated parallel to each other and perpendicular to the interface (2.38B). In this way, monomolecular layers, so-called *monolayers*, are formed.

In case of homogeneous aqueous phases, phospholipids reach a state of minimal surface tension if they orientate their hydrophobic portions toward one

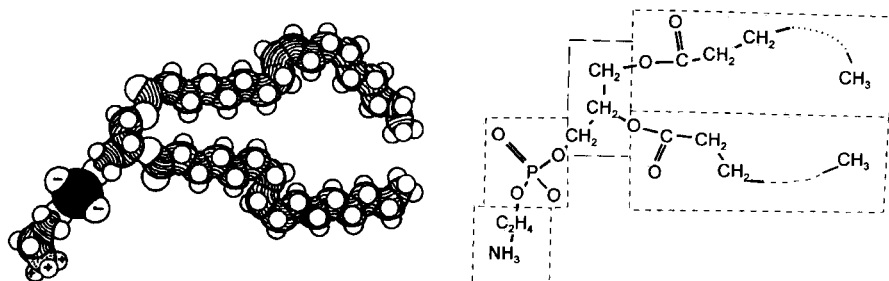
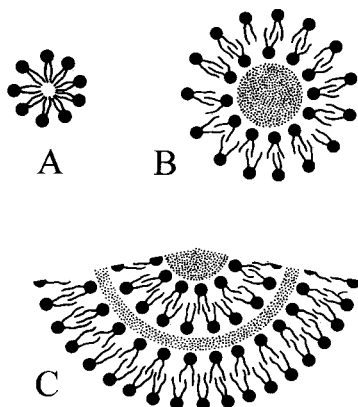


Fig. 2.39. Model and formula of a simple phospholipid (phosphatidylethanolamine with olein- and palmitin acid-rests. (After Haggis 1965 redrawn)

Fig. 2.40. Self-assembly of lipids in aqueous solutions. A Micelle; B Unilamellar liposome; C Sector of a multilamellar liposome. The *dotted areas* always indicate local micro phases of water



another (Fig. 2.40A). There is a large diversity in lipid aggregates in aqueous solutions. Using special methods, for example sonications of lipid suspensions with ultrasound, it is possible to produce structures with lipid double layers (*bilayers*) and vesicles, containing aqueous solutions (*liposomes*). There are two types of liposomes: unilamellar liposomes covered only by a single bilayer (Fig. 2.40B), and multilamellar liposomes (Fig. 2.40C) containing numerous lipid bilayers, separated from each other by a thin water layer.

It is also possible to prepare planar lipid membranes in aqueous phases (Fig. 2.38C). If such a membrane was formed, its fatty acid chains would have built a hydrophobic micro phase. In this microphase other hydrophobic molecules can be included and orientated, as illustrated for the case of intrinsic membrane proteins in Fig. 2.38D.

In this context it must be mentioned that surface tension is, at least partly, the phenomenological expression of processes which on the molecular level are the result of the dynamics of water structure and hydrophobic bonds. All the lipid structures mentioned here can be destroyed by extremely high pressure in the sense of the Le Châtelier principle, as discussed in Section 2.4.2.

During recent decades ingenious methods have been developed to produce bimolecular membranes with high structural and chemical specificity. These are the so-called BLMs which is an abbreviation for black lipid membrane, or bimolecular lipid membrane.

Experiments with such membranes have significantly promoted our knowledge on self-assembly of biological membranes, their molecular dynamics and processes of membrane transport. Standard experiments are the measurement of electrical conductivity of these membranes, using chambers connected by a hole, which is covered by a BLM. With such equipment, transport properties of these membranes can also be investigated. For this, specific transport proteins are inserted into these membranes to study their specific properties *in vitro*.

Measurement of electric conductivity of cell suspensions indicated a long time ago that the cell must be surrounded by an electrically isolating layer with a specific capacity of about 10 mF m^{-2} (in detail, see Sects. 2.5.5 and 3.5.3). This would correspond to a double layer of lipids with a thickness of 8 nm. Singer and Nicolson in 1972 proposed a *fluid-mosaic model* of the cell membrane. Meanwhile, this model has been substantially developed, and today it forms the fundament of our knowledge about the molecular structure and the dynamics of biological membranes. According to this model the membrane consists of a highly organized dynamic mosaic of proteins, embedded in a double layer of phospholipids.

The proteins are organized in the membrane according to their hydrophobic (i.e. apolar), and hydrophilic (polar) regions (see Fig. 3.38D). Many proteins proceed to the external surface of the membrane as *glycoproteins*. They are located with their C-terminal end in the cytoplasm. Their N-terminal end, which is modified by various carbohydrates, extends over a long distance into the outer environment of the cell. These carbohydrates in some cases represent the receptor-sides of the proteins and are the location of immunological reactions. Preferentially, at the end of the protein chains, monomers of the N-acetylneuraminic acid (also called sialic acid) are located, which carry a dissociable carboxyl group. In this way, these are the most important carriers of fixed negative surface charges of the cell (see Fig. 2.48). Depending on the type of cell, there are between 1 and 10 groups of sialic acid per 1 nm^2 membrane area. Hence, the glycoproteins form a loose external layer of the cell, the so-called *glycocalyx* or surface coat. The charges, located at the ends of these filaments modify the thickness of this structure electrostatically (for detail, see Sect. 2.5.5).

An important role for the lateral membrane structure is played by a network of cytoplasmatic proteins as a part of the cytoskeleton with connections to the inner side of the membrane. It is best investigated in human erythrocytes, where it consists mostly of spectrin. This is a heterodimeric protein (molecular masses: 260 and 220 kDa) which forms rods of approximately 100 nm length, aggregating head to head, to form tetramers of the double length. These spectrin rods are connected to each other by the globular actin molecules (molecular mass: 100 kDa). Other proteins, the ankyrins, connect this network

with the intrinsic membrane proteins. The cytoskeleton is a dynamic structure which controls the lateral distribution of the proteins in the membrane.

The bilayer of membrane lipids forms a two-dimensional phase with amphiphilic properties. This can be thought of as a fluid matrix in which the proteins are embedded. The lipid composition of the inner and outer membrane leaflet is highly specific. This concerns the specificity of their fatty acids, as well as their polar head groups (see Fig. 2.39). The fluidity of the membrane, i.e. the lateral movement of its constituents is strongly determined by the length of the aliphatic fatty acids chains, as well as by their degree of saturation (see also Sect. 3.6.3). The property of the polar head groups determines their mutual distance in the membrane surface, i.e. their density of packing. The heterogeneity of the lipid composition in the membrane in some cases may lead to formation of specific lipid domains. These are lipid clusters which can be considered as sort of lateral microphases. Specific lipid domains also surround the proteins. Furthermore, the lipids are distributed asymmetrically between the two leaflets of the membrane. The negatively charged phosphatidylserine, for example, in animal cells is located nearly exclusively at the inner leaflet of the membrane, in contrast to phosphatidylcholine, which is concentrated mostly in its outer leaflet. This asymmetric distribution of lipids is controlled by specific lipid transport systems.

Further reading: Bloom et al. 1991; Hianik and Passechnik 1995

2.5.3

Mechanical Properties of Biological Membranes

The mechanical properties of the biological membrane are very important for understanding a number of cell physiological functions such as cell movement, cell division, vesiculation, and membrane fusion. In the same way, the mechanical properties of tissues, as well as the streaming properties of the blood, which we will discuss in Section 3.7, are controlled by the viscoelastic properties of cell membranes.

It must be stressed that physical parameters, for example viscosity, elasticity, strain etc. are defined for homogeneous phases and materials (see Sect. 3.6). Hence, they are not proper parameters of highly organized supramolecular structures like biological membranes. In some cases it may be convenient to use these parameters, but then they must be considered as *effective parameters*, which we explained in the introduction to Section 2, and which we have already used in many other cases.

The effective membrane viscosity, for example, is a parameter which was measured using specific effects, as for example the rotation or translocation of a spin-marker in the membrane. This is possible using special fluorescence methods or electron spin resonance techniques (ESR). Considering the membrane as a homogeneous phase, and the marker molecule as a macroscopic body with a particular shape, one can calculate the viscosity, applying usual phenomenological equations for movement. Knowing the method of measure-

ment, as well as the applied equations, this measurement can be reproduced any time, and the measured parameter is exactly defined. The problem which arises from using an equation, which is not adequate for the movement of a molecule in a heterogeneous phase, is not relevant because it is included in the definition. On the other hand, however, it means that by measuring these effective viscosities by different methods, one gets different parameters but not a viscosity according to the original physical definition. This is the difference between molecular, and the usual macroscopic kinds of measurement, where viscosity is independent of the measurement method.

Applying different markers, the parameters measured in this way depend on the kind of marker molecule (protein, lipid), and on the regions of the molecule where the marker was bound; for example, different locations in the fatty acid of a lipid. These differences indicate the inhomogeneity in the plane of the membrane, as well as the anisotropy of the mechanical properties of the biological membrane in general.

The lipids in biological membranes show a high lateral mobility because they can easily exchange their positions. The mobility of proteins, in contrast, is strongly limited by their fixation on the cytoskeleton. Local enrichments of proteins to particular locations in the membrane can lead to inhomogeneities of its bending property.

Lipids may also exchange between the two leaflets of the membrane. This is a so-called flip-flop process. The rate of this exchange strongly depends on the type of lipids. Half life times of between minutes up to many hours are known. We already mentioned that there are special mechanisms of lipid transport, catalyzing this process.

Summarizing all these aspects, the following mechanical properties for the biological membrane are to be expected: it is easily deformable by shearing forces (Fig. 2.41C), independently of the position of the intrinsic proteins which occasionally are fixed by the cytoskeleton (Fig. 2.41D). In contrast to this shearing deformation which occurs without an increase area, it is nearly impossible to extend the area of the membrane. The stress (σ') which must be applied to extend a membrane area A by the amount ΔA can be calculated by:

$$\sigma' = Y' \frac{\Delta A}{A}. \quad (2.5.2)$$

Y' is the specific modulus of elasticity, a coefficient which is related to the thickness of the membrane. The symbol Y stands for Youngs modulus, another name for the same parameter (see Sect. 3.6.3). For the membrane of erythrocytes $Y' = 0.45 \text{ N m}^{-1}$, for lymphocytes 0.64 N m^{-1} . To find a relationship between membrane properties and properties of macroscopic materials, like those in Table 3.2 (page 213), Y' must be divided by the thickness of the membrane (d), which is approximately $8 \cdot 10^{-9} \text{ m}$. In this case, for erythrocytes the result is:

$$Y = Y'/d = 5.6 \cdot 10^7 \text{ N m}^{-2} = 56 \text{ Mpa}$$

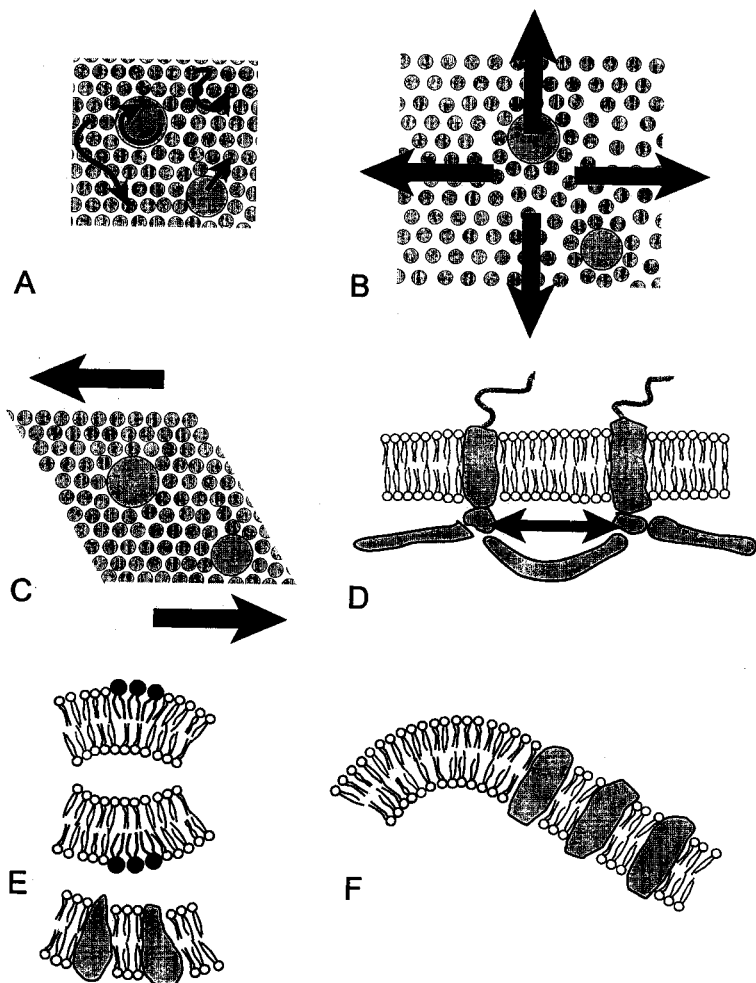


Fig. 2.41. Possible kinds of molecular displacement and mechanical deformations of biological membranes. **A** Translation and rotation of membrane components; **B** planar extension of the membrane area; **C** shear deformation; **D** translocation of membrane proteins by the spectrin network; **E** bending of the membrane, induced by asymmetric lipids or proteins, or by inclination of additional molecules; **F** extrusion of membrane components with lower flexibility from highly bended areas

Relating this value to the parameters of Table 3.2 one can see that it already tends toward the values of steel-elastic materials. The membrane will rupture if it is expanded by more than 1–2% of its area. This is understandable considering that any increase of the membrane area leads to an increase of the distance between the lipid head groups.

The membrane therefore resembles a material with a high degree of flexibility in plan, but with minimal ability for area extension. Hence, it has properties very unlike a rubber sheet, which can easily be expanded on account of its thickness. Living cells, therefore, can swell only by spherulation, by unfolding of the membrane, or by inclusion of new membrane material. Each swelling of the cell, which extends the surface of a sphere by more than 1–2% of its area or an unfolding beyond its capacity, will lead to immediate cytolysis.

As we will see later, using the elastic modulus, it is also possible to calculate forces, which oppose the bending of a material (Sect. 3.6.4). Applying this to a biological membrane, again, the inhomogeneity of the membrane must be considered. According to the heterogeneity of the lipid composition and the differences in the distribution of the proteins, the bending resistance of the membrane differs from place to place. It is possible to calculate forces, acting to extrude small areas of enlarged stiffness, like protein patches, out from strongly bent regions (Fig. 2.41F). In shapes with regions of extremely different membrane bending, for example in the case of crenated erythrocytes, (so-called echinocytes), it could be shown that the number of proteins is lower in strongly bent regions.

In some cases membranes are actively bent by themselves. This is done by a bilayer couple mechanism, if one of the membrane leaflets increases its area by inclusion of additional molecules. Another possibility is the introduction of conically shaped molecules (Fig. 2.41E). A factor (f) characterizes this property. It relates the area of the head group (A_H) to the mean area (A_F) which is required for the region of fatty acids (Fig. 2.42). A_F can be calculated by the mean length of the fatty acids (l) and the effective volume of the region of fatty acids (V_F) by: $A_F = V_F/l$. The shape factor f , therefore, is defined as:

$$f = \frac{V_F}{lA_H}. \quad (2.5.3)$$

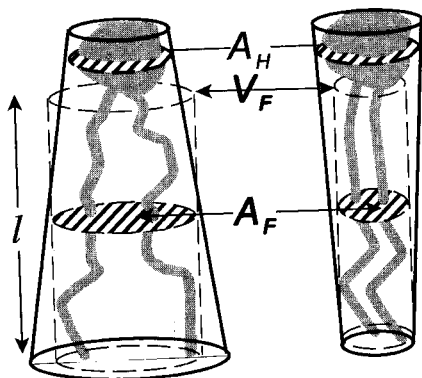


Fig. 2.42. The effective shape of a phospholipid (explanations in text)

In contrast to the area of the head group (A_H), which is more or less constant, the effective volume of the fatty acid chains (V_F) strongly depends on their phase conditions, i.e. on their degree of order. There are dynamic variations possible as well as phase transitions by temperature modifications. Other parameters of the molecule depend directly on their composition. So, for example, lysophospholipids, containing only one single fatty acid chain, are to a large degree conical with a dominating area A_H . In this case, $f < 1/3$. In contrast, phosphatidylethanolamine, usually containing a large amount of unsaturated fatty acids, gives $f > 1$, with dominating A_F . Mostly lipids are more or less cylindrical, with $1/2 < f < 1$.

Further reading: Bereiter-Hahn et al. 1987; Bloom et al. 1991; Hianik and Passechnik 1995; Mohandas and Evans 1994; Schmid-Schönbein et al. 1986; Silver 1985

2.5.4

Electrical Double Layers and Electrokinetic Phenomena

An interface carrying fixed charges induces in its vicinity an electric field, and accordingly, modifies the local ionic conditions, analogous to a point charge, as described in Section 2.4.3. In the simplest case, a double layer occurs with the fixed charges on one side, and the electrostatically attracted mobile counterions on the opposite side. This concept was originally put forward by H. von Helmholtz. It only applies for extremely high concentrations of mobile ions in the solution. In general, thermic movement repels a great part of the ions from their positions near the fixed charges. This leads to the formation of a *diffuse double layer*, the electrical potential of which declines exponentially with the distance from the surface.

Figure 2.43 illustrates the generalized model of Stern, which includes the Helmholtz double layer as well as its diffuse part. To avoid overloading of the picture, only the fixed charges (in this case proposed as negative), and the compensating mobile counterions are depicted. Of course, there are also mobile anions present, and there is a great number of both cations and anions that should be included to give a true picture. This picture indicates an enlarged concentration of cations near the surface for the case of negative surface charges, and the electroneutrality of the total system, including the bulk solution.

Despite the thermic noise, some of the mobile ions directly oppose the fixed charges, forming the so-called Helmholtz layer. Between the surface and the end of the Helmholtz layer, the potential is proposed to decline linearly from ψ_0 to ψ_H . Further, the potential declines according to the model of the diffuse double layer which was calculated by Gouy and Chapman.

The theory of the diffuse double layer again is based on the Poisson-Boltzmann equation, which we explained in relation to the Debye-Hückel theory of ion clouds (Sect. 2.4.3). This equation makes it possible to calculate the distribution of mobile charges in a given electric field:

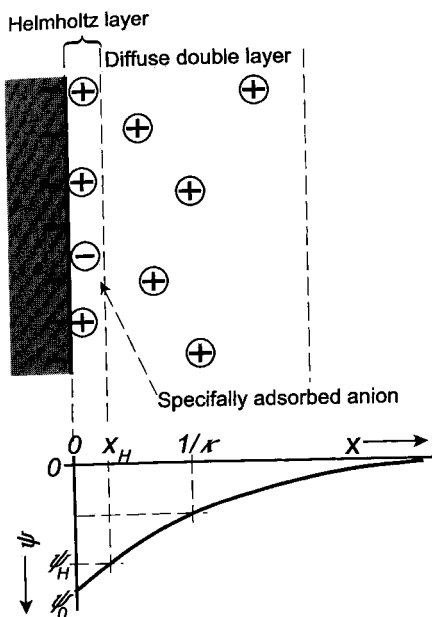


Fig. 2.43. Schematic illustration of an electrical double layer. *Above* fixed charges and mobile counter-ions near a surface; *beneath* the function of the electrical potential, according to the model of Stern. The charge of the surface in this particular case is enforced additionally by a chemically adsorbed negatively charged ion. ψ_0 – surface potential, ψ_H – potential at the end of the Helmholtz layer, x_H – thickness of the Helmholtz layer, $1/\kappa$ – Debye-Hückel length as a measure of the effective thickness of the diffuse double layer

$$\nabla^2 \psi = -\frac{F}{\varepsilon_0 \varepsilon} \sum_{i=1}^n c_{i0} z_i e^{-\frac{z_i e \psi}{kT}}. \quad (2.5.4)$$

In this case the amount of fixed charges is compensated by the excess of charges in the whole double layer.

Applying this equation for one-one-valent electrolytes, like KCl or NaCl, it reduces to:

$$\nabla^2 \psi = -\frac{Fc_0}{\varepsilon_0 \varepsilon} \left(e^{-\frac{e\psi}{kT}} - e^{+\frac{e\psi}{kT}} \right) = \frac{2Fc_0}{\varepsilon_0 \varepsilon} \sinh \frac{e\psi}{kT}. \quad (2.5.5)$$

In this case, a rearrangement was made, using the function

$$\sinh x = (e^x - e^{-x})/2.$$

To solve the Poisson-Boltzmann equation, i.e. to calculate the function $\psi(x)$, one needs to define two initial conditions. For our case this means:

$$\psi(\infty) = 0,$$

$$\psi(0) = \psi_H,$$

whereas the end of the Helmholtz layer (ψ_H) is taken as the starting point of the function $[\psi(0)]$. The algebraic integration of this equation again, is possible only after linearization. For $x > x_H$ one gets the following function $\psi(x)$:

$$\psi(x) = \psi_H e^{-\kappa x}. \quad (2.5.6)$$

In this equation, κ is again the Debye-Hückel parameter, which was introduced in Section 2.4.3 [Eq. (2.4.15)]. The Debye-Hückel length ($1/\kappa$) is the distance, over which the potential ψ_H is dropped by the factor e :

$$\frac{\psi(1/\kappa)}{\psi_H} = \frac{1}{e} = 0.368. \quad (2.5.7)$$

The value of this parameter, as a function of the ionic strength (I) is given in Eqs. (2.4.15–2.4.17).

What does “linearization” of Eq. (2.5.5) mean, and what are the limitations of this simplification? In fact, the basis for such approaches is the expansion of these functions in series, using the following equations:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots, \quad (2.5.8)$$

and:

$$\sinh x = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \cdots. \quad (2.5.9)$$

For the case:

$$\frac{ze\psi}{kT} \ll 1, \quad (2.5.10)$$

one can apply these equations, using only the first terms of the series:

$$e^{-\frac{ze\psi}{kT}} \approx 1 - \frac{ze\psi}{kT}, \text{ and: } \sinh \frac{e\psi}{kT} \approx \frac{e\psi}{kT}. \quad (2.5.11)$$

Let us investigate the scope of this simplification. For $T = 298$ K one can calculate:

$$z\psi \frac{e}{kT} = z\psi \frac{1.60218 \cdot 10^{-19}}{1.3807 \cdot 10^{-23} \cdot 298} = 38.94z\psi. \quad (2.5.12)$$

Using, for example, a potential $\psi = 0.01$ V and a charge number $z = 1$, one gets: $e^{-0.3894} = 0.6774$, and according to Eq. (2.5.8): $1 - 0.3894 = 0.6106$. Correspondingly: $\sinh 0.3894 = 0.3993$. Both calculations, really, indicate small differences, which will become larger if the electrical potential exceeds the value of 10 mV. This is also important for the application of this linearization in the previous Section 2.4.3, as well as in the following considerations.

Beside the calculation of the diffusion layer it is necessary to know the relation between the potentials ψ_0 and ψ_H . In general, the difference between these two parameters will become smaller as the ionic strength of the solution drops. In solutions of extremely low ionic strength, this difference can be neglected.

In fact, besides the screening of fixed charges by mobile ions, other processes must be considered, such as dipole orientations, interactions with water molecules or other processes of adsorption, based on Van-der-Waals interactions. In this way, it is even possible that against electrostatic repulsion, coions were absorbed at the interface, increasing its charge density. In this case the Helmholtz-potential (ψ_H) can become even larger than the surface potential (ψ_0).

We will discuss the relevance of electrical double layers for functions of biological membranes in detail in Section 2.5.5. The most important consequences are local ionic concentrations, including local pH values (Fig. 2.44). This can be calculated according to the Nernst equilibrium [Eq. (3.2.34), Sect. 3.2.2]. The relevance of these local conditions is understandable, considering the function of membrane-bound enzymes.

A number of electro-mechanical interactions occur which are caused by these electrical double layers near charged surfaces, and which are summarized by the term: *electrokinetic phenomena*. On the one hand, an externally applied electric field may induce mechanical effects, like movement or streaming, on the other

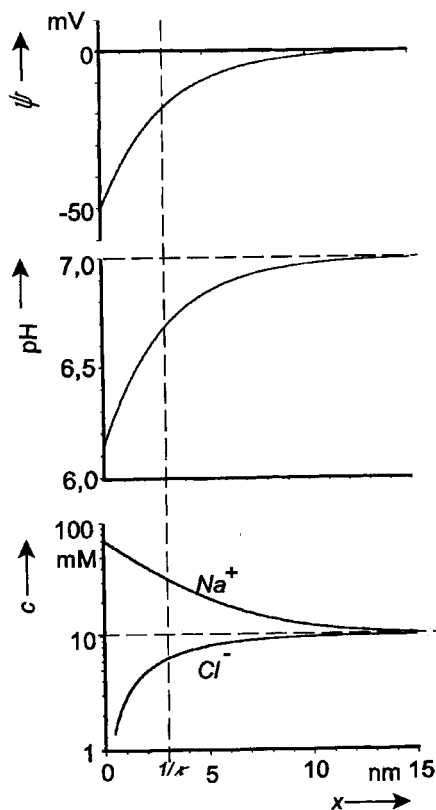


Fig. 2.44. Electrical potential (ψ), pH, as well as Na^+ - and Cl^- -concentrations (c) as a function of the distance (x), near a negatively charged surface in a diffuse double layer. The curves consider the following parameters: $\psi_0 = -50$ mV, $\text{pH}_\infty = 7.0$, $c_\infty = 10$ mM, $T = 298$ K (the index ∞ means: at $x \rightarrow \infty$). In this case the Debye-Hückel length ($1/\kappa$) amounts 3 nm

hand, a mechanical movement may lead to the occurrence of electrical potential differences. In Fig. 2.45 these phenomena are depicted schematically.

Electrophoresis means the movement of charged molecules and particles in an externally applied electric field. The displacement of cells and other particles can be measured by microscopic observations. Their mobility (b) is defined as the relation between their velocity (v) and the electric field strength (E):

$$b = \frac{v}{E}. \quad (2.5.13)$$

The particles move in the opposite direction from their double layers. Because of the increasing frictional force, a stationary velocity, will be reached.

The classical theory of cell electrophoresis is based on the model which was proposed by Smoluchowski, considering the movement of large spherical particles. "Large" in this sense means that the diameter of the particle, and the

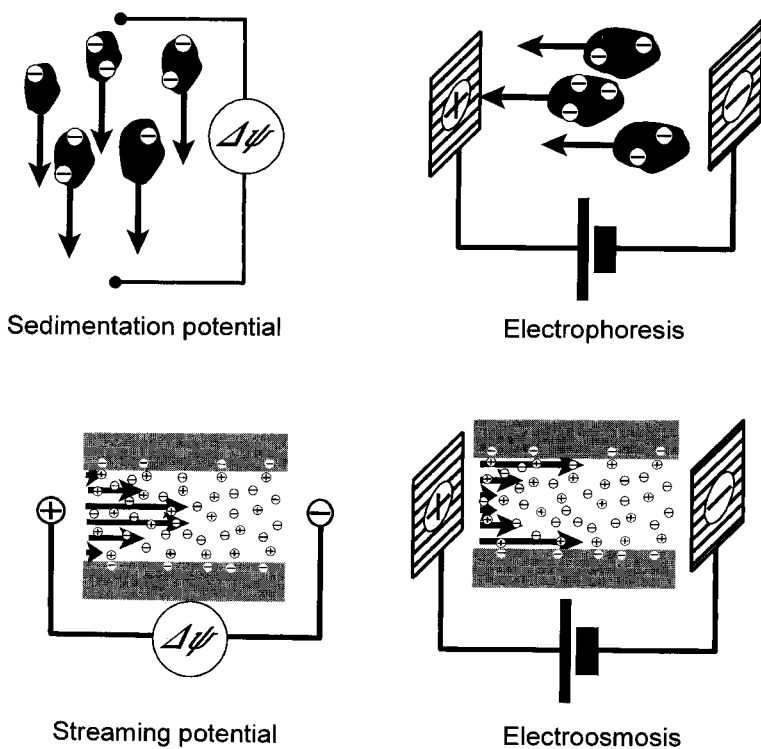


Fig. 2.45. Schematic representation of electrokinetic phenomena of particles (above), and surface-near regions in tubes or capillaries (beneath); *left* the occurrence of electrical potentials as a result of mechanical movement (grey arrows); *right* the induction of movement (above) or streaming (beneath) as a result of an applied electric field (black arrows)

corresponding radius of bending, must be large in relation to the Debye-Hückel parameter ($1/\kappa$). Smoluchovski postulated that a shearing plane exists, caused by the electrokinetically induced streaming. At this plane, the electroneutrality of the system is disturbed. The electrical potential at this distance from the surface is called *electrokinetic potential*, or ζ -potential. It can be calculated by the measured electrophoretic mobility (b) using the following equation:

$$\zeta = \frac{\eta v}{\epsilon_0 \epsilon E} = \frac{\eta b}{\epsilon_0 \epsilon}, \quad (2.5.14)$$

where η is the viscosity of the medium. Surprisingly, the radius (r) of the particle is not included in this equation. Sometimes, this equation contains an additional factor, to take into account stronger deviations from the shape of a sphere.

Of course, the above introduced ζ -potential is not identical with the surface potential (ψ_0). With some approximation, however, it can be considered as equal to the Helmholtz-potential (ψ_H ; see Fig. 2.43). In the same way, as discussed above, the ζ -potential increases, by constant surface charge density (σ_0), if the ionic strength in the solution is decreasing (Fig. 2.46).

It is possible to calculate the surface potential (ψ_0) and the surface charge density (σ_0) from the ζ -potential only with certain approximations. For this, the equation used is derived from the Gouy-Chapman theory of electrical double layers:

$$\sigma_0 = \sqrt{8\epsilon_0 \epsilon RT} \sqrt{I} \sinh \frac{F\zeta}{2RT}. \quad (2.5.15)$$

In Fig. 2.46 the dependence of the ζ -potential is depicted as a function of the ionic strength of a one-one-valent electrolyte, which in this case is equal to its

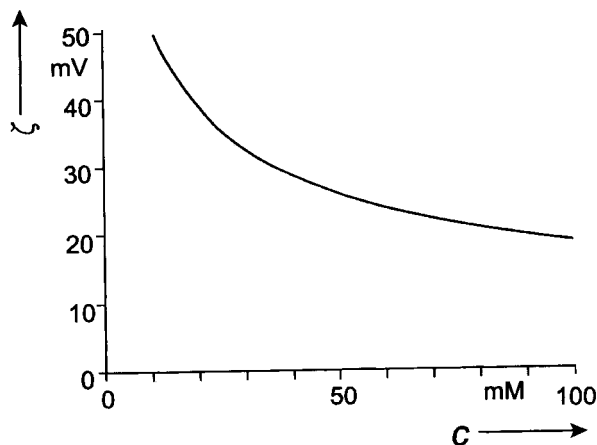


Fig. 2.46. The ζ -potential (depicted as a negative value) as a function of the concentration of a one-one-valent electrolyte for a surface with a constant surface charge density ($\sigma_0 = 0.014 \text{ C m}^{-2}$) according to Eq. (2.5.15)

concentration. In this case a surface charged density of $\sigma_0 = 0.014 \text{ C m}^{-2}$ was chosen which corresponds to that of a human erythrocyte.

Equation (2.5.15) can be simplified by expanding it as series, corresponding to Eq. (2.5.9). If the Debye-Hückel parameter (κ) was used according to Eq. (2.4.15), we have

$$\sigma_0 = \zeta \kappa \epsilon \epsilon_0, \quad (2.5.16)$$

which, when combined with Eq. (2.5.14), results in

$$\sigma_0 = \eta \kappa b. \quad (2.5.17)$$

Introducing this kind of linearization we found that this simplification can be accepted as a reasonable approximation only for potentials $\leq 10 \text{ mV}$. As indicated, however, in Fig. 2.46, this limit is exceeded for most cases of biological membranes. Furthermore, all equations were derived considering the fixed charges as located on a planar boundary. The thickness of the Helmholtz layer, which amounts only to a few tenths of a nanometer explains what the word "planar" means. The roughness in the order of a molecular diameter is always enough to exceed this assumption.

To measure electrophoresis, equipment is used which allows observation of the movement of cells in a homogeneous dc-field microscopically (Fig. 2.47). Usually, for electrophoretic measurements, a field strength in the order of 1 kV m^{-1} is used. The chamber needs a sufficient temperature control because, according to Eq. (2.5.14), the electrophoretic movement depends on the viscosity of the solution. Furthermore, it must be considered that an electro-osmotic flow occurs, caused by the surface charges of the glass, which are mostly negative. Hence, when the chamber is closed this flux must return through the

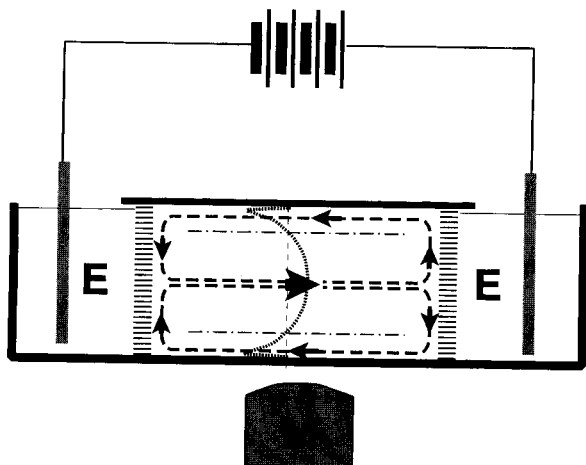


Fig. 2.47. Schematic illustration of equipment to measure cell electrophoresis; M microscope; E chamber for electrodes, which are connected to the measuring chamber by a conducting medium. The arrows indicate the direction of the electro-osmotic flow.

..... - the flow profile in the chamber
 - - - - - the neutral planes

center of the chamber. Between these two flows of opposite direction, neutral planes must exist, where the streaming velocity becomes zero, and where the measurement of the correct electrophoretic mobility of cells is possible. These planes are located at a distance 21.1% of the total thickness of the chamber, on both sides of it (see Fig. 2.47).

Since the development of the method of cell electrophoresis many cells and organisms have been investigated. Under physiological conditions, nearly all biological objects indicate a ζ -potential between -10 and -70 mV. Only at $\text{pH} < 4$, in some cases, do these potentials become positive. This also holds for bacteria, yeast and plant cells, which are surrounded by a cell wall. The cell wall is a thick (in relation to the membrane) porous ion exchange layer with fixed negative charges.

In contrast to cell electrophoresis, which is used routinely to measure surface charges, the reversible effect, i.e. the *sedimentation potential*, some times also called the *Dorn-effect*, seems to be not so important. In this case, a small electrical potential difference occurs in suspensions of quickly sedimenting charged particles.

An electrical potential difference which occurs if a solution is streaming along a charged surface, or inside a capillary with surface charges, is called *streaming potential*. The reason for this is the superposition of the streaming profile and the profile of mobile charges in an electrical double layer. In detail, the theory of streaming potential is quite complex. In the center of a capillary, for example, the streaming velocity is maximum, and the concentration of the coions of the surface charges minimum. Therefore, these coions will flow faster than the counter-ions near the surface. This leads to a certain charge differentiation at the ends of the capillary. The resulting electric field, however, enforces the movement of counter-ions and lowers the movement of these coions. The effect, therefore, is lowered by this reaction. In capillaries with an extremely small diameter, a Donnan-effect must additionally be considered. This means that the amount of surface charge always shifts the total electrolyte composition in this fine capillary toward a Donnan equilibrium (see Sect. 3.2.3). The total amount of counter-ions, therefore, is larger than that of the coions. This can even lead to a streaming potential with opposite polarity. It is possible to measure the streaming potential in biological tissue if a flow of solutions is induced by osmotic gradients. The potential difference measured in such experiments is only in the order of a few millivolts.

The reversible effect of streaming potential is *electro-osmosis*. In this case, based on the same considerations, near a charged surface, or in a capillary, a flow of solution occurs if a dc-field is applied.

The physiological role of electrokinetic phenomena is not very clear. It cannot be ruled out that electrophoresis of charged components in the membrane, or of proteins in intercellular spaces occurs under the influence of potentials generated in vivo (see Sect. 3.5.2). A hypothesis exists that microorganisms can move electrophoretically in their own generated field. Streaming potentials occur in bone during mechanical deformations. In this case, it is induced by the flow in the small canals of the bone, the so-called canaliculi (see Sect. 3.5.2).

Further reading: Theory of electrical double layers: Donath and Voigt 1986; Glaser 1996; Voigt and Donath 1989; cell electrophoresis: Bauer 1994; streaming potential in bones: Guzelsu and Walsh 1993; self-electrophoresis of microorganisms: Lammert et al. 1996

2.5.5

The Electrostatic Structure of the Membrane

After explaining some structural and mechanical properties of biological membranes in Sections 2.5.2 and 2.5.3, and after having discussed ionic properties near fixed charges, we will now introduce the membrane as an electrical structure. This seems to be a crucial point for many biophysical considerations, and it will pervade this whole book. Electrical properties of the biological membrane will be mentioned in the context of many topics. Whereas we introduce here, the electrostatic properties of the membrane, we will continue this discussion later in Sections 3.4.2 and 3.5.2, including electrodynamic processes, such as generation of electrical currents by processes coupled with the cell metabolism. In Section 3.5.3 the explanation of passive membrane properties will be continued, including its behavior in electromagnetic fields. This knowledge will be applied in Section 3.5.4 to introduce techniques of cell manipulations with electric fields. Finally, in Section 4.4 we will use this model to discuss the influence of weak electric and electromagnetic fields on cells and organisms.

In contrast to the external medium and to the cell plasma, the cell membrane has a high electrical resistance and a low dielectric constant. This allows us to consider it as an extremely thin hydrophobic electrically isolating interface between two aqueous phases, behaving like a capacitor with certain capacity (C), and resistance (R). From this point of view, it is possible to consider the passive electric behavior of the cell as an RC -element (see Figs. 3.25 and 3.38).

The specific capacity (C_{sp}) can be calculated from the membrane thickness (Δx) and the dielectric constant (ϵ):

$$C_{sp} = \frac{\epsilon_0 \epsilon}{\Delta x}. \quad (2.5.18)$$

The specific capacity of the cell membrane is relatively constant, because both parameters, the dielectric constant (ϵ), as well as the membrane thickness (Δx) cannot vary significantly. For the normal cell membrane it amounts to approximately 10 mF m^{-2} (for methods of measuring this parameter, see Sects. 3.5.3 and 3.5.4). Equation (2.5.18) allows calculation of the mean dielectric constant of the membrane, assuming: $\Delta x = 8 \cdot 10^{-9} \text{ m}$:

$$\epsilon = \frac{C_{sp} \Delta x}{\epsilon_0} = \frac{10^{-2} \cdot 8 \cdot 10^{-9}}{8.854 \cdot 10^{-12}} = 9.0.$$

This value is surprisingly large, because the dielectric constant of lipids amounts to only $\epsilon = 3.5$. The reason for this difference is the heterogeneity of the membrane, in particular, its protein content.

The specific capacity of the biological membrane is a very important parameter, indicating the relation between the amount of charge (σ , in C m^{-2}) which is required to generate a membrane potential difference ($\Delta\psi$, in V):

$$C_{\text{sp}} = \frac{\sigma}{\Delta\psi}. \quad (2.5.19)$$

In Section 3.4.2 we will use this relation to explain the role of the membrane as an electrochemical accumulator.

In Section 2.5.2 we described the membrane as a dynamic structure containing fixed charges in a 2-dimensional pattern of the membrane area and in a charge profile perpendicular to this. This particular profile along an x -coordinate, cutting the membrane area perpendicularly, is depicted in Fig. 2.48 schematically. This picture does not contain the mobile ions in the double layers, and the ionic distribution which is responsible for generation of the transmembrane potential ($\Delta\psi$). We will discuss the circumstances leading to the occurrence of this transmembrane potential as a Donnan potential, as a diffusion potential, or finally as a result of electrogenic transporters later in Section 3 in detail.

The fixed surface charges of the outer part of the cell membrane are the result of dissociations of carboxyl groups of neuraminic acids (also called sialic acids),

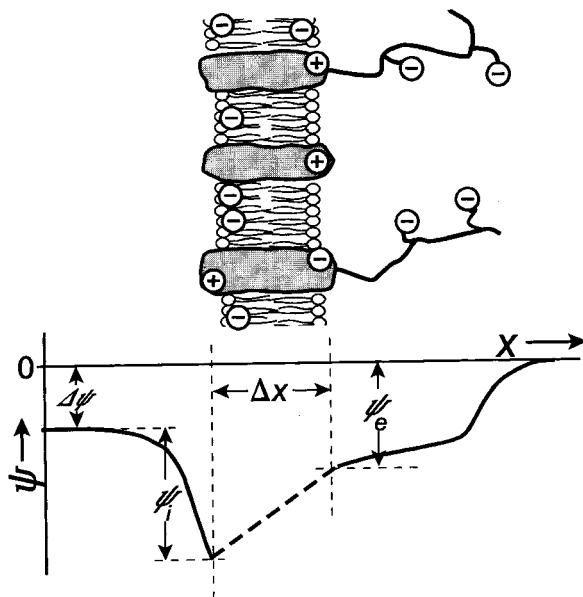


Fig. 2.48. The fixed charges of a cell membrane and the corresponding electrical potential (ψ). The transmembrane potential ($\Delta\psi$) is indicated as the difference between the potentials of the internal and external phases. The parameters ψ_i and ψ_e indicate the internal and external surface potentials; Δx means the thickness of the membrane. Inside the membrane, the function $\psi(x)$ depends on the positions where the membrane is cut. Therefore, the function at this point is marked only by a simple linear connection between the surface potentials

positioned at the ends of the glycoproteins. The isoelectric point of these neuraminic acids is near pH 2.6. For most membranes the negative charges are between -0.01 and -0.02 C m^{-2} . Human erythrocytes carry on their surfaces, which are about $140 \mu\text{m}^2$, approximately 10^7 dissociated neuraminic acid groups.

Additional carriers of electric charges are the polar groups of several phospholipids. The negatively charged phosphatidylserine is of particular importance. In the membrane of human erythrocytes it is located exclusively in the inner leaflet of the membrane. In this case, its surface charge amounts to approximately -0.09 C m^{-2} .

As depicted in Fig. 2.38, proteins are located in the membrane in such a way that their polar groups are orientated towards the water phase. Hence, they also contribute to the surface charge of the membrane by charges of both sign. These charges of membrane proteins are very important for controlling various protein functions, such as ion transport through the membrane.

These considerations indicate that the models of electrical double layers, resulting from charges of a planar surface (Fig. 2.43) are just rough approximations of the real situation on the cell surface. The glycocalyx forms an external layer outside the membrane. The fixed charges of it are to be considered as charges in a space, not on a surface. It must therefore be introduced in the Poisson-Boltzmann equation [see Eq. (2.5.4)] as space density (ρ , in C m^{-3}), instead of the surface charge density (σ , in C m^{-2}):

$$\nabla^2 \psi = -\frac{1}{\epsilon_0 \epsilon} \left(\rho + F \sum_{i=1}^n c_{i0} z_i e^{-\frac{z_i e \psi}{kT}} \right). \quad (2.5.20)$$

Even for the simplest functions, $\rho(x)$, an integration of Eq. (2.5.20) is possible only by iteration. The resulting function $\psi(x)$ (Fig. 2.48), of course differs from that for a classical double layer (Fig. 2.43). The effective range of the electrical potential in this case is predicted firstly by the thickness of the glycocalyx, and only indirectly by the Debye-Hückel length ($1/\kappa$). This thickness, however, is itself determined by electrostatic interactions of its charges. In solutions of high ionic strengths, these charges are more or less shielded, and the glycoproteins are closely attached to the membrane. In solutions with lower ionic strength, however, the fixed charges repulse each other, the glycoprotein filaments are therefore packed more loosely, and the glycocalyx becomes thicker. Hence, the function $\rho(x)$, i.e. the distribution of the charges of the neuraminic acids, itself depends on $\psi(x)$. The surface structure of the cell is therefore controlled by the cell itself, as well as by the conditions of its surroundings.

Not only external, but also internal membrane surfaces carry fixed charges. These are mostly the charges of phosphatidylserine, and additionally, a number of surface-associated proteins, which are not depicted in Fig. 2.48. In contrast to the external surface charges of the membrane, which can be measured by cell electrophoresis, the charges of the cytoplasmatic side cannot be measured easily.

As well as the charge profile in a direction perpendicular to the membrane surface, there is a lateral mosaic of charges, which is partly controlled by the cytoskeleton. This charge distribution has been investigated by electron microscopy, and recently by atomic force microscopy, using electrostatically bound particles of heavy metals. It is also possible to follow the movement of charged proteins, marked by fluorescence labels in strong external fields. The investigation of the dynamics of this lateral electrostatic structure of biological membranes is, however, only at the beginning.

Although we will discuss the role of the membrane electric field, as well as the application of external fields on cell membranes in detail later in Sections 3.4, 3.5, and 4.4.4, we will nevertheless mention here the biological relevance of this electrostatic structure, and its role in several cell physiological processes. In general, on the one hand, the profile of the electrical potential directly creates a profile of specific ion conditions. On the other hand, the electric field strength, as the derivative of the potential, affects some processes directly.

Inside the membrane, and in its vicinity, electric field strengths in the order of 10^7 V m^{-1} occur. In Fig. 2.48 the course of the potential inside the membrane is drawn only with a dotted line as a linear connection between the potentials of both membrane surfaces ψ_i and ψ_e , because the real function $\psi(x)$ in this region strongly depends on the location of the cut in the membrane. This function would be quite different if the x -axis were to cut a protein, or if it were located inside the lipid phase. The local charge densities, as well as the local dielectric constants would differ strongly. Furthermore, it must be remembered that the function $\psi(x)$ will change dramatically if the transmembrane potential $\Delta\psi$ depolarizes during the action potential in a time course of about 1 ms (see Fig. 3.27). Hence, we come to the conclusion that inside and near the membrane, an extremely strong electric field occurs which is not only a 1-dimensional function $\psi(x)$, as depicted in Fig. 2.48, but which must be considered as a 3-dimensional, time dependent field.

It is clear that this field must influence all polar, or polarizable molecules in this region. A number of transport proteins in the membrane are influenced by this field. Nerve cells are the classical example for a feedback circle, generating the action potential: the alteration of the membrane permeability, induced by any kind of excitation, changes the diffusion potential $\Delta\psi$, and therefore the resulting modification of the field strength in the membrane changes the permeability of the Na^+ , and subsequently K^+ -channels, which again influences the membrane potential. We will explain these reactions in detail in Section 3.4.3. Potential sensitive transporters are distributed in all cells.

Proteins and lipids are influenced by the field strength inside the membrane. The field influences the orientation of the polar head groups of the lipids, and modifies in this way the effective area which they occupy in the membrane. We discussed the consequence of this modification in Section 2.5.2. The field may also induce an intrinsic membrane pressure, which may alter the phase transition in the hydrophobic region of the membrane. Such processes may be the basis of mechanisms of electro-mechanical coupling.

Furthermore, the function of the electrical potential near the membrane generates particular micro-conditions near charged groups. Depending on the ionic strength, and therefore on the Debye-Hückel length ($1/\kappa$), a lateral mosaic of local concentrations and local pH values can occur. In Fig. 2.44 the local ionic concentrations and local pH values are illustrated as a function of the distance from a charged surface. It must be stressed that the field changes the concentration of multivalent ions, for example Ca^{++} , to a larger degree than univalent ions like K^+ or Na^+ . In this context, the role of local charges must be considered for the control of membrane-bound enzymes and transporters.

Further reading: Cevc 1990; Glaser 1996; McLaughlin 1989; surface mapping with atomic force microscopy: Heinz and Hoh 1999

Energetics and Dynamics of Biological Systems

The functions of biological systems are based on a highly organized molecular structure. So, in Section 2, we considered firstly the physical principles ruling molecular and supramolecular structures. This was done using approaches of quantum mechanics and statistical thermodynamics.

Now we will come to various functions of cells, tissues, and organs. For this, of course, molecular considerations form an important fundament, but at the same time, we enter the field of so-called continuum physics. Here, we will use parameters such as concentration, volume, viscosity, dielectric constants, conductivity etc., which of course are based on molecular properties of matter, but which have been introduced and defined simply by phenomenology. This means that these parameters are defined for sufficiently large homogeneous phases. This, however, does not meet the conditions of biological structures. In Section 2 we were constantly meeting this problem, considering for example mechanical or electrical properties of biological membranes. Even in the following considerations we must always be aware of this situation. The step from a molecular to phenomenological approach, nevertheless, must be taken not only in the physics of simple materials, but also in biophysics, if one considers larger systems like cells and organisms.

We will come back to this point in general in the context of the electrical structure of organisms, when discussing levels of biological organization in Section 3.5.1 (Fig. 3.30).

3.1

Some Fundamental Concepts of Thermodynamics

One of the first important treatises on the problems of thermodynamics was the famous monograph by Sadi Carnot, entitled *Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance* which was published in 1824. In this book, thermodynamics was introduced as the theory of heat engines. Julius Robert Mayer (1814–1878) was a physician and scientist to whom we owe the first numerically defined correlation between heat and work (1842). He can, therefore, be considered as the discoverer of the first law of thermodynamics. He had already discussed the relation between functioning of the human body and the heat engine as a result of his own physiological observations. Meanwhile, thermodynamics have become the theoretical fund-

ament of all kinds of energy transformation, and consequently, of all kinds of movement.

Thermodynamic considerations of biological processes require an extension of the classical thermodynamics of equilibria towards the direction of thermodynamics of irreversible processes, i.e. towards nonequilibrium thermodynamics. This extension was taken in two steps: firstly, only small deviations away from equilibrium are taken into consideration. In this case, linear relations between forces and rates can be assumed. In contrast to these linear approaches, the thermodynamics of non-linear processes can calculate systems far from equilibrium, where steep gradients of potentials exist. In this case, so-called *dissipative structures* appear, which are stationary states with completely new qualities.

It seems important to stress here that although the far-from-equilibrium condition of an organism represents an absolute precondition of life, nevertheless, subsystems in the organism exist, which can be properly calculated using equilibrium thermodynamics, or thermodynamics of linear approaches. This means that biophysics must span the whole scale of thermodynamic approaches.

3.1.1

Systems, Parameters and State Functions

In Section 2.3.3, the term “system” was introduced with the explanation of the term “structure”. We defined the system as an aggregate of elements with certain interrelations. Furthermore, we explained that systems with interrelations which are not simply relations, but interactions, are called dynamic systems. These are the kinds of systems where thermodynamic approaches can be applied.

The question of what kind of model we should use, what we should consider as a system, and what are its elements, depends exclusively on the particular problem, and the corresponding point of view. An element may be a system itself, extending the problem. An organism, for example, as an element in an ecological system can become a system itself, if we ask a physiological question such as what are the interactions between its organs. The organ can be considered a system of cells, the cell as a system of organelles, and so on.

A dynamic system can be analyzed in different ways. In contrast to system theory, which calculates the kinetic interplay of individual elements, thermodynamics considers a system simply as a continuum which stands in a defined interrelation with its environment. The limit of this continuum does not have to be a wall or a membrane. It can also be a process that changes the quality of the subject of study. Chemical reactions as well as processes of proliferation and evolution are examples of this.

In thermodynamics, systems are classified as follows according to the nature of their boundary against their environment:

- The *isolated system*: this is an idealized system that does not exchange any kind of energy or matter with its environment.

- The *closed system*: this system can exchange all kinds of energy with its environment, but not matter.
- The *open system*: it can exchange both energy and matter with its environment.

The closed system can be influenced by its environment, and can cause changes in its environment. However, it cannot be involved in an exchange of matter.

The state of a system can be described by a number of *state variables*. These are either extensive or intensive parameters. *Intensive* parameters are non additive. They are independent of the size of the system (e.g. temperature, concentration, pressure, density). *Extensive* parameters on the other hand, are additive when two systems are combined (e.g. mass, volume).

Changes in a system are often characterized by differentials of its state variables. A *differential* describes a very small change of a dependent variable (dy), if in a function $y = f(x)$, a small change in the variable (dx) occurs. It can be calculated from the product of the first derivative of the function $f(x)$, multiplied by dx :

$$dy = f'(x)dx. \quad (3.1.1)$$

Most thermodynamic equations are functions with several variables. Hence, the derivatives can be obtained with respect to one variable if the others are kept constant. This procedure is called *partial differentiation*. It has a special notation with the parameters that are to be kept constant put as subscript to the symbols in parentheses. The following example is quoted out of context, to demonstrate this.

$$\left(\frac{\partial G}{\partial n_i}\right)_{p,T,n_j} = \mu_i \quad \text{for: } j \neq i. \quad (3.1.2)$$

The partial derivative of the Gibbs free energy G with respect to the molar number of substance i , when pressure (p), temperature (T), and the molar number of all the other substances (n_j) are kept constant, gives per definition, the chemical potential (μ_i) of the substance i . In general, this is the same procedure as is used when a function with several dependent variables is represented graphically in a 2-dimensional plot against one selected variable, keeping all other variables constant.

Small changes in a state function with several variables can be represented by a so-called *total differential*. For this, all partial differentials of this function must be summarized. These partial differentials are calculated as shown in Eq. (3.1.1), using, however, partial derivatives. The following equation, for example, would apply to the Gibbs free energy $[G(p, T, n_i)]$:

$$dG = \left(\frac{\partial G}{\partial p}\right)_{T,n_j} dp + \left(\frac{\partial G}{\partial T}\right)_{p,n_j} dT + \sum_{i=1}^m \left(\frac{\partial G}{\partial n_i}\right)_{p,T,n_j} dn_i. \quad (3.1.3)$$

The mathematical definition of the total differential is of very great physical importance to thermodynamics. This will be indicated by the following chain of reversible logical deductions:

$$\begin{array}{ccccc}
 \text{dG is a} & & \text{G is a} & & \text{G depends only on the state} \\
 \text{total} & \leftrightarrow & \text{state} & \leftrightarrow & \text{of the system, and not on the} \\
 \text{differential} & & \text{function} & & \text{way in which that state was} \\
 & & & & \text{achieved}
 \end{array}$$

For this reason it is important that this property of a function is able to be mathematically proven.

A differential equation is not always written in the easily followed way shown in Eq. (3.1.3). Often it is presented as a Pfaffian differential equation:

$$dn = L dx + M dy + N dz. \quad (3.1.4)$$

The capital letters here represent any variable. There is a mathematical indicator to determine whether dn is a total differential. This is the so-called *Cauchy condition*, stressing that dn is a total differential when:

$$\frac{\partial L}{\partial y} \stackrel{!}{=} \frac{\partial M}{\partial x}; \quad \frac{\partial M}{\partial z} \stackrel{!}{=} \frac{\partial N}{\partial y}; \quad \frac{\partial L}{\partial z} \stackrel{!}{=} \frac{\partial N}{\partial x}. \quad (3.1.5)$$

If this is applied to Eq. (3.1.3), this means:

$$\frac{\partial(\frac{\partial G}{\partial p})}{\partial T} = \frac{\partial(\frac{\partial G}{\partial T})}{\partial p} = \frac{\partial^2 G}{\partial T \partial p}. \quad (3.1.6)$$

Additionally, it follows from Eq. (3.1.6) that for such functions if they are differentiated several times, the sequence of differentiations is unimportant. We will use this property in a later derivation [Eq. (3.2.6)].

Total differentials do not only result from energetic parameters. This formalism of course can be applied to any state function. Volume changes in mixed phases for example, can be described by the following total differential equation:

$$dV = \bar{V}_1 dn_1 + \bar{V}_2 dn_2 + \bar{V}_3 dn_3 + \cdots + \bar{V}_m dn_m, \quad (3.1.7)$$

where dV is the change in volume that occurs when the molar number of one or more components of the system is changed. Equation (3.1.7) allows us to define the partial mole volume of a given substance i :

$$\bar{V}_i = \left(\frac{\partial V}{\partial n_i} \right)_{n_j} \quad \text{for: } j \neq i. \quad (3.1.8)$$

The partial molar volume has the inverse unit to the concentration, namely: $\text{m}^3 \text{mol}^{-1}$.

3.1.2

Gibbs Fundamental Equation

The scientific basis of thermodynamics is its three principles which are founded on experimentally verifiable, empirical facts. Upon this solid foundation a framework of definitions and relations has been built up which enables far-reaching postulations on all kinds of energy transformations.

The principle of conservation of energy, the so-called *first law of thermodynamics* states that there must exist a physical parameter having the property of a state function, which includes the consequences discussed in Section 3.1.1. Work (W), as a physical parameter does not comply with this condition. General experience shows that a change of a system from state A to state B can be achieved through many ways that differ greatly from one another in the amount of work that is required. Therefore, work cannot be a state function which could be used to characterize the energy state of a system independently of the way in which it was achieved. In specific cases such as the work differential of an ideal gas, this can be proved using the Cauchy condition [Eq. (3.1.5)].

Let a parameter called *internal energy* (U), now be defined as a state function which, as such, has a total differential dU . Furthermore, let the internal energy of a system be increased by dU if a certain amount of heat (dQ) is introduced into the system, and/or if certain work (dW) is done on the system.

$$dU = dQ + dW. \quad (3.1.9)$$

This equation contains the essence of the first principle of thermodynamics. Both the differentials dQ , as well as dW , are reflections of a change of energy. However, according to the second principle of thermodynamics, heat (Q), as a physical parameter, differs from all other forms of energy because it possesses a specific property: any form of energy can be completely transformed into heat, but heat itself can only partly be transformed into work. Entropy (S) makes its appearance in phenomenological thermodynamics as a kind of measure of the quality of heat. Entropy is defined as follows for a quasi-reversible process:

$$dS = \frac{dQ_{\text{rev}}}{T}. \quad (3.1.10)$$

This allows us to find an expression for dQ which can be introduced in Eq. (3.1.9).

Let us now consider the differential of work (dW) in more detail. It can be expressed as a sum of products of sort of work coordinates, extensive parameters indicating a kind of measure of the performed work, and work coefficients, intensive parameters, reflecting the effort required for this. For

example, the work (dW_p) which is done when a gas is compressed by a pressure p resulting in a volume alteration dV will be:

$$dW_p = -p dV, \quad (3.1.11)$$

where p represents the work coefficient, and dV the work coordinate. The sign of this equation depends on the definition of the work differential (see also Sect. 2.1.3). A positive dW means that there is an increase in the work done in favor of the system. In this case work is achieved through the compression, i.e. a negative differential of the volume.

Work can be done in many different ways. An expansion of a material, for example, means elongation (dl) in response to the application of a force (F):

$$dW_l = F dl. \quad (3.1.12)$$

A cell can do work by transporting a certain number of atoms or molecules (dn) against a concentration gradient. At this point, the chemical potential (μ) must be introduced as the work coefficient. We will come back to this parameter in detail later [Eq. (3.1.33)]. In this case the work differential can be written as follows.

$$dW_n = \mu dn. \quad (3.1.13)$$

Finally, let us, from among the many other possible examples, consider charge transport. If a particular amount of charge (dq) is transported against an electric potential (ψ), then the electrical work done will be:

$$dW_q = \psi dq. \quad (3.1.14)$$

Equations (3.1.11) to (3.1.14) can be combined:

$$dW = -p dV + F dl + \mu dn + \psi dq. \quad (3.1.15)$$

Considering that usually in the system a number of m substances are transported, then this equation can be expanded as follows:

$$dW = -p dV + F dl + \sum_{i=1}^m \mu_i dn_i + \psi dq. \quad (3.1.16)$$

This is the more detailed form of the work differential which, together with Eq. (3.1.10), can be introduced in Eq. (3.1.9):

$$dU = T dS - p dV + F dl + \sum_{i=1}^m \mu_i dn_i + \psi dq. \quad (3.1.17)$$

Equation (3.1.17) is a differential form of *Gibbs fundamental equation*. Of course it can be expanded by adding more kinds of work differentials.

Alternatively, this equation will be automatically reduced if certain changes become irrelevant. For example, suppose a defined transformation within a system is not accompanied by a mechanical strain. Then l remains constant, and consequently, $dl = 0$. As a consequence, the corresponding term disappears from the equation.

In Eq. (3.1.17) the Gibbs fundamental equation appears in the form of a Pfaffian differential. Such expressions can be integrated under certain conditions, which apply in this case. This gives:

$$U = TS - pV + Fl + \sum_{i=1}^m \mu_i n_i + \psi q. \quad (3.1.18)$$

It must be noted that the transition from Eq. (3.1.17) to Eq. (3.1.18) does not mean a simple elimination of the differential operators; it is the result of a proper integration which is not described here!

Using the rule given in Eq. (3.1.3), this process of integration can be reversed. It gives:

$$\begin{aligned} dU = T dS + S dT - p dV - V dp + F dl + l dF \\ + \sum_{i=1}^m \mu_i dn_i + \sum_{i=1}^m n_i d\mu_i + \psi dq + q d\psi. \end{aligned} \quad (3.1.19)$$

A comparison of this result with the initial Eq. (3.1.17) shows that the following condition must be satisfied:

$$S dT - V dp + l dF + \sum_{i=1}^m n_i d\mu_i + q d\psi \stackrel{!}{=} 0. \quad (3.1.20)$$

This is the so-called Gibbs-Duham equation. It is useful for some calculations because it allows reduction of the degree of freedom of a system by one variable.

It has proved useful to define not only the internal energy (U), but also three further energy functions. In some books the introduction of these parameters is explained in a physical way, by discussing processes of vapor compression etc., but it seems to be simpler just to accept the definitions of these parameters, and subsequently substantiate their usefulness.

The definitions are:

$$\text{enthalpy,} \quad H = U + pV; \quad (3.1.21)$$

$$\text{Helmholtz free energy,} \quad F = U - TS; \quad (3.1.22)$$

$$\text{Gibbs free energy,} \quad G = H - TS. \quad (3.1.23)$$

The Gibbs fundamental equation [Eq. (3.1.17)] can now easily be written down for these new defined functions. Let us first transform Eq. (3.1.21) into a total differential, according to Eq. (3.1.3). According to the definition [Eq. (3.1.21)], the enthalpy (H) is a function of: U , p , and V . This gives:

$$dH = \left(\frac{\partial H}{\partial U} \right)_{p,V} dU + \left(\frac{\partial H}{\partial p} \right)_{U,V} dp + \left(\frac{\partial H}{\partial V} \right)_{U,p} dV. \quad (3.1.24)$$

From Eq. (3.1.21) follows directly:

$$\left(\frac{\partial H}{\partial U} \right)_{p,V} = 1; \quad \left(\frac{\partial H}{\partial p} \right)_{U,V} = V; \quad \left(\frac{\partial H}{\partial V} \right)_{U,p} = p, \quad (3.1.25)$$

which, when combined with Eq. (3.1.24), results in

$$dH = dU + V dp + p dV. \quad (3.1.26)$$

Combining with this, Eq. (3.1.17) gives the Gibbs fundamental equation for dH :

$$dH = T dS + V dp + F dl + \sum_{i=1}^m \mu_i dn_i + \psi dq. \quad (3.1.27)$$

In the same way it is possible to derive from Eq. (3.1.22) the relations:

$$dF = -S dT - p dV + F dl + \sum_{i=1}^m \mu_i dn_i + \psi dq, \quad (3.1.28)$$

and:

$$dG = -S dT + V dp + F dl + \sum_{i=1}^m \mu_i dn_i + \psi dq. \quad (3.1.29)$$

The choice of whether Eqs. (3.1.17), (3.1.27), (3.1.28), or (3.1.29) should be used to calculate a particular system depends on the external conditions and the question which is being asked. Investigating a system under isobaric conditions ($p = \text{const}$, i.e. $dp = 0$), it is useful to apply the equation for dH [Eq. (3.1.27)], or for dG [Eq. (3.1.29)], because in this case the term volume expansion work ($V dp$) vanishes. This corresponds to the situation in most biological investigations. Therefore, we will use mostly enthalpy (H) and Gibbs free energy (G) in all further biophysical calculations instead of the inner energy (U) and the Helmholtz free energy (F).

If the conditions are isothermal ($dT = 0$), as well as isobar ($dp = 0$), then in Eq. (3.1.29) the terms connected with heat and volume work will vanish. Hence,

dG expresses directly the deviation of the energy content, as a result of work which was done. The gradient of free Gibbs energy therefore indicates the direction of a spontaneous process in the same way as a gradient of potential energy indicates the direction of a rolling sphere on an uneven surface.

The differential dH , on the other hand, is a measure of the thermal characteristics of isobar processes.

All these forms of the Gibbs fundamental function [Eqs. (3.1.27), (3.1.28), (3.1.29)], as well as Eq. (3.1.7) for the partial volume, can be integrated according to Eq. (3.1.17) for dU .

For questions of equilibrium thermodynamics, only the situation at the beginning and at the end of a reaction is of interest. For this, differences between parameters of these states are defined. So, for example, an entropy change of a reaction $\Delta_R S$ is defined as follows:

$$\Delta_R S = S_{\text{product}} - S_{\text{substrate}}. \quad (3.1.30)$$

In a corresponding way, parameters like $\Delta_R U$, $\Delta_R H$, $\Delta_R F$, and $\Delta_R G$ are defined. In contrast to entropy, however, no absolute values exist for energetic parameters. They always need a defined reference value. So, for example, *standard energies of formation* ($\Delta_F U$, $\Delta_F H$, $\Delta_F F$, and $\Delta_F G$) are defined as energetic changes that occur when a substance is formed from its elements under standard conditions ($T = 297 \text{ K}$), or better: *would occur*, because in most cases a direct synthesis of the substance from its elements is impossible. Analogous with Eq. (3.1.30) it results in:

$$\Delta_R G = \Delta_F G_{\text{product}} - \Delta_F G_{\text{substrate}}. \quad (3.1.31)$$

The chemical potential (μ_i) of the substance i is particularly important for the following calculations. It can be easily defined using Eqs. (3.1.17), (3.1.27), (3.1.28), or (3.1.29):

$$\mu_i = \left(\frac{\partial U}{\partial n_i} \right)_{S, V, l, n_j, q} = \left(\frac{\partial H}{\partial n_i} \right)_{S, p, l, n_j, q} = \left(\frac{\partial F}{\partial n_i} \right)_{T, V, l, n_j, q} = \left(\frac{\partial G}{\partial n_i} \right)_{T, p, l, n_j, q}. \quad (3.1.32)$$

The chemical potential, therefore, is a partial expression having the dimensions J mol^{-1} . The chemical potential of the substance i (μ_i) can be calculated from the concentration (c_i), or better from the chemical activity (a_i) of this substance using:

$$\mu_i = \mu_i^0 + RT \ln a_i. \quad (3.1.33)$$

In this equation a_i is just a number, i.e. the number of moles per liter, as the chemical activity is measured. The chemical activity is a kind of effective concentration. Its relation to the concentration is given by:

$$a_i = f_i c_i. \quad (3.1.34)$$

In this equation, f_i is the *coefficient of activity*. In ideal solutions, $f_i = 1$, that is, the activity of the dissolved substance is equal to its concentration c_i . Usually f_i decreases as the concentration in the solution increases (see Fig. 3.1). For dissociating salts, f represents an average activity coefficient for the ions. For example, the ions in a 100 mM solution of NaCl show an average activity coefficient of 0.8. The chemical activity of this solution therefore is equal to an ideal solution with a concentration of only 80 mM. In contrast to the coefficient of activity, which is a dimensionless number, the chemical activity has the same units as the concentration.

In some cases it may be useful to employ the mole fraction as a measure of concentration. The *mole fraction* of a substance i is defined as the number of moles of that substance (n_i), divided by the total number of moles of all substances present:

$$x_i = \frac{n_i}{\sum_{i=1}^m n_i}, \quad (3.1.35)$$

where:

$$\sum_{i=1}^m x_i = 1. \quad (3.1.36)$$

According to Eq. (3.1.34), the mole fraction of a substance can also be expressed as the mole fraction activity (a_{xi}).

The standard potential (μ_i^0) can be easily defined by means of Eq. (3.1.33). It follows from this equation that when $a_i = 1$, $\mu_i = \mu_i^0$. The standard potential

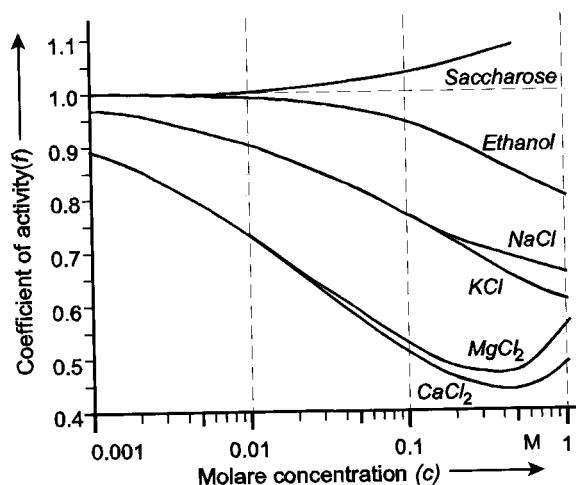


Fig. 3.1. Coefficients of activity (f_i) of various substances as functions of their concentrations (c_i) in aqueous solutions under standard conditions ($T = 297$ K)

therefore is the chemical potential of a substance i in a solution, with an activity of 1 M, if this measure of concentration is applied. In Eq. (3.1.33), using the mol fraction (x_i) of a substance, or its mol fraction activity (a_{xi}) as concentration measure, the chemical standard potential (μ_i^0) is determined by the chemical potential of the pure substance ($a_{ix} = 1$).

A further extension of the Gibbs fundamental equation concerns the term ψdq . The charge of a single charged ion is determined by the Faraday constant (F). The charge on n moles of a z -fold charged ion is obtained as follows:

$$q = znF. \quad (3.1.37)$$

This is a function with one independent variable (n). Therefore it is easily transformed into a differential according to Eq. (3.1.1):

$$dq = \left(\frac{dq}{dn} \right) dn = zF dn. \quad (3.1.38)$$

If more than one ion is in the solution, the charges can be summarized:

$$dq = \sum_{i=1}^m z_i F dn_i. \quad (3.1.39)$$

Introducing this expression into the Gibbs fundamental equation, the equation then gets two terms with the differential dn_i . It is obviously useful to combine these terms. First, consider these two terms of the Gibbs equation in isolation from the other terms of this equation:

$$\psi \sum_{i=1}^m z_i F dn_i + \sum_{i=1}^m \mu_i dn_i = \sum_{i=1}^m (\mu_i + z_i F \psi) dn_i = \sum_{i=1}^m \tilde{\mu}_i dn_i. \quad (3.1.40)$$

Here, the expression inside the brackets taken together is described as the electrochemical potential ($\tilde{\mu}_i$) of the substance i .

$$\tilde{\mu}_i = \mu_i + z_i F \psi. \quad (3.1.41)$$

The electrochemical potential is the basis for most electrochemical calculations, and thus forms an important starting point for further considerations.

3.1.3

Force and Motion

After introduction of the energetic state functions in the previous section we will now consider their application to determine forces leading to any sort of motion.

A sphere is rolling downhill. It is moving spontaneously from a position with a higher potential energy to one with a lower potential. The direction of its

movement follows a force vector ($\mathbf{X}$) and is, consequently, determined by the negative gradient of the energy U :

$$\mathbf{X} = -\text{grad } U. \quad (3.1.42)$$

If consideration of the energy gradient is confined to the direction of the x -coordinate, this equation can be simplified to give:

$$X_x = -\frac{dU}{dx} \mathbf{i}, \quad (3.1.43)$$

where $\mathbf{i}$ is simply a unit vector, i.e. a vector with the amount of 1, and an arrow, directing toward the x -coordinate. For dU , any appropriate energy state function should be substituted as shown in Section 3.1.2. Let us consider for example the force acting on a charge (q) in an electric field in the x -direction [$\mathbf{E} = (-d\psi/dx) \cdot \mathbf{i}$]. Substituting for dU in Eq. (3.1.43), the expression, according Eq. (3.1.17), gives:

$$X_q = -\frac{dU}{dx} \mathbf{i} = -\frac{d\psi}{dx} q \mathbf{i} = E q. \quad (3.1.44)$$

This assumes that there is no other gradient in the x -direction, i.e. that neither p , nor T , nor μ are functions of x . The introduction of the field strength (E) is in accordance with the definition given in Eq. (2.4.5). Equation (4.1.44) is identical to Eq. (2.4.6), which was derived in another way.

Equation (3.1.44) cannot be applied to practical calculations of ion transport because only the transport of a charge is considered. In contrast to the movement of an electron, the transport of ions always means that there is an additional change in concentration. For transport of a noncharged substance (i), the negative gradient of its chemical potential (μ_i) is the driving force:

$$X_i = -\text{grad } \mu_i. \quad (3.1.45)$$

Ions, in contrast, are driven by their gradient of the electrochemical potential ($\tilde{\mu}$) which, according to Eq. (3.1.41), includes the electric potential. Applying the differential operator, the electric potential (ψ) transforms into an electric field strength (E):

$$X_i = -\text{grad } \tilde{\mu}_i = -(\text{grad } \mu_i - z_i F E). \quad (3.1.46)$$

There are many kinds of movement in biological systems that are the concern of biophysics. Their range covers electron transfer, structural changes of molecules, chemical reactions, fluxes of molecules and ions, streaming of liquids in the body, and finally, mechanical movements of limbs and whole bodies. Fluxes occupy a central position in the study of movements in biology, and therefore these will be considered here for the sake of simplicity, as a sort of generalized movement.

The term flux (J_i) stands for the amount of a substance (i) that passes perpendicularly through a unit of surface per unit time. This definition shows that the flux in general is a vector. Often, flux through a membrane is considered, the direction of which is always predicted.

The relation between Flux J_i of the component i , and its velocity v_i is:

$$J_i = c_i v_i. \quad (3.1.47)$$

A system which is traversed by a flux, and where no substance is added or removed to this flow, is called a *conservative* one. In this case the following conditions apply:

$$\text{div } J_i = 0. \quad (3.1.48)$$

The differential operator div ("divergence") can be replaced by the Nabla operator (∇), as explained in Section 2.4.1. In contrast to the operator 'grad', which, applied to a scalar, and results in a vector, the 'div' operator is to be applied to a vector, producing a scalar. The Nabla operator (∇) is applicable to differentiate vectors as well as scalars.

If there is a point in such a system which is traversed by a flux, where $\text{div } J_i > 0$, then there will be a source adding substance to the flux. If $\text{div } J_i < 0$, there will be a removal of some of the substance like a sink. Figure 3.2 illustrates this situation using an example where the flow is simply in the x -direction.

This formalism is applied, describing for example fluxes through a biological tissue. For the transport of ions, this system is conservative, because no

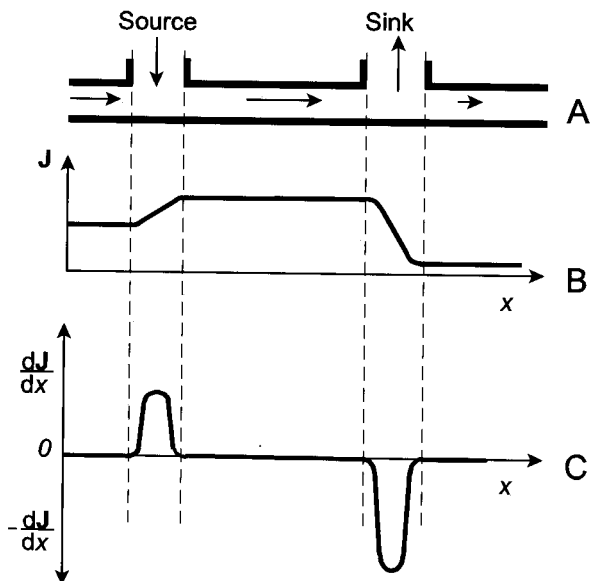


Fig. 3.2. A Graphic representation of a linear flow with a source, and a sink; B the flux (J) as a function of x ; C the change of the flow (dJ/dx) as a function of x

accumulation or depletion of ions occurs. In contrast, for the transport of oxygen the condition $\text{div } J_o < 0$ holds, because the tissue uses oxygen for its own respiration.

If a constant force acts on a body, then the latter accelerates. However, as the velocity of this body increases, the friction is likely to increase too. When both the driving force and the frictional force become the same amount, then the body will move with a constant velocity. This is a special case of a stationary state, the so called *stationary motion*.

It is a fundamental experience in physics that there are regions, characterized by their relations between force and movement, having quite different qualities. If, for example, a comparatively minor force acts on a body, then the body will attain a velocity that is proportional to the force. This is a region where a linear analysis of irreversible thermodynamics is applicable. If the same body is more forcibly moved, then the frictional force will increase in a stronger way, and a non-linear approach is necessary.

This concept, illustrated here by a mechanical example, has a more general application. A linear approach to a general equation of motion can be formulated as follows:

$$J_i = L_i X_i, \quad (3.1.49)$$

where L_i is a coefficient which is a kind of a generalized conductance. In the same way, the following equation can be written:

$$X_i = R_i J_i, \quad (3.1.50)$$

In this case a resistance factor is applied: $R_i = 1/L_i$. Ohms law is a special form of this general expression:

$$U = RI, \quad (3.1.51)$$

where U in this case is the electromotive force, depending on the potential gradient ($\text{grad } \psi$), and I is the electric current.

In a similar way force and velocity can be linearly related. For this it is necessary to introduce a mobility factor (ω), and its corresponding coefficient of friction (f):

$$v = \omega X = \frac{X}{f}. \quad (3.1.52)$$

The flux equation (3.1.47) can now be written as follows:

$$J_i = c_i \omega_i X_i \quad (3.1.53)$$

We introduced the Stokes' equation (2.3.34), when discussing molecular movement in Section 2.3.6. This is also a particular expression of this linear approach. It indicates the frictional force (F) of a sphere with radius (r), moving

in a liquid with viscosity (η) at velocity ($\mathbf{v}$). Under stationary conditions, this friction force is equal to the driving force:

$$\mathbf{F} = 6\pi r\eta\mathbf{v}. \quad (3.1.54)$$

In view of Eq. (3.1.52), the following analogy is obvious:

$$f = \frac{1}{\omega} = 6\pi r\eta. \quad (3.1.55)$$

The definitive mechanical behavior of bodies moving in water will be discussed in detail in Section 3.7. Here it is only being used as an example for the consideration of general principles.

Figure 3.3 shows the frictional coefficient (f) for a sphere with a diameter of 1 cm in water as a function of its velocity ($\mathbf{v}$). The frictional coefficient only remains constant up to a velocity of about 1 mm s^{-1} , corresponding to Eq. (3.1.55). As the velocity increases above this, the frictional factor deviates, at first slightly, and then greatly from the horizontal line. This illustrates the transition of the system from linear to non-linear behavior. This is the transition to far-equilibrium conditions, where non-linear thermodynamic approaches must be applied.

Let us explain the genesis of such a non-linear relation by a simple example. Let a phenomenological coefficient be a sum of a constant term (L_i), and a variable term L'_i , which is proportional to the flux (J_i). In this case, the linear approach of Eq. (4.1.49) transforms into:

$$J_i = (L_i + L'_i J_i) X_i. \quad (3.1.56)$$

Solving this equation for J_i , it gives:

$$J_i = \frac{L_i X_i}{1 - L'_i X_i}. \quad (3.1.57)$$

Hence we have a non-linear function $J_i(X_i)$.

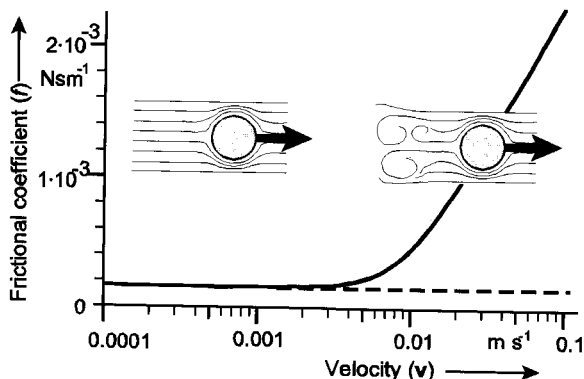


Fig. 3.3. The coefficient of friction ($f = F/v$) of a sphere with a diameter of 0.01 m in water as a function of its velocity (v)

The qualitative consequences of these non-linear approaches will be considered further in the next section. Let us at present remain in the field of linear thermodynamics.

If different kinds of motions and forces occur simultaneously in a system, they will influence each other. Let us consider again the flux (J_i) as a generalized kind of motion. The nonequilibrium thermodynamics in its scope of linear approaches allows us to write down a set of phenomenological equations forming a flux matrix. It includes the idea whereby all forces and fluxes in the system are coupled with each other. The simple Eq. (3.1.49), therefore, will be expanded to the following set of equations:

$$\begin{aligned} J_1 &= L_{11}X_1 + L_{12}X_2 + L_{13}X_3 + \cdots + L_{1n}X_n \\ J_2 &= L_{21}X_1 + L_{22}X_2 + L_{23}X_3 + \cdots + L_{2n}X_n \\ J_3 &= L_{31}X_1 + L_{32}X_2 + L_{33}X_3 + \cdots + L_{3n}X_n \\ &\quad \dots \\ J_n &= L_{n1}X_1 + L_{n2}X_2 + L_{n3}X_3 + \cdots + L_{nn}X_n. \end{aligned} \quad (3.1.58)$$

In these equations the vector notation of fluxes and forces is still retained, regardless of whether, in rare cases (which we will discuss in the following) they may really be scalars.

The parameters L_{mn} are phenomenological coefficients, also called *coupling coefficients*, *cross coefficients*, or *Onsager coefficients*. (This equation was first used by L. Onsager in 1931). This general set of equations in reality reduces, because a flux J_m is coupled with a force X_m only when $L_{mn} \neq 0$.

Equation (3.1.58) shows that n forces with their corresponding fluxes require a set of equations with n^2 coupling coefficients. Onsager, however, was able to show that this matrix is symmetric. It means that near the equilibrium the following relation holds:

$$L_{mn} = L_{nm} \quad \text{for: } n \neq m. \quad (3.1.59)$$

This is *Onsager's law on the reciprocal relation*. It leads to a significant reduction of the coefficients in the matrix from n^2 , down to the number of $n(n+1)/2$.

Directly linked pairs of forces and fluxes, as for example J_1 and X_1 , J_2 and X_2 , ..., J_n and X_n , are called *conjugated*. The coefficients (L_{nn}), linking these pairs, are always positive. If two fluxes really are coupled, the following condition must hold:

$$L_{mm} \cdot L_{nn} \geq L_{mn}^2. \quad (3.1.60)$$

From this, a *degree of coupling* (q_{mn}) can be defined:

$$q_{mn} = \frac{L_{mn}}{\sqrt{L_{mm}L_{nn}}}. \quad (3.1.61)$$

This degree of coupling can vary as follows: $1 \geq q_{mn} \geq 0$. When $q_{mn} = 0$, the fluxes are completely independent of each other, when $q_{mn} = 1$, there is maximal coupling.

As mentioned before in Eq. (3.1.58) all fluxes and forces are printed in bold letters as vector parameters. At the same time we mentioned that we will consider here fluxes in a very general sense, symbolizing all kinds of motion. This means that true fluxes of matter, which really are vectors going in a particular direction, as well as, for example, chemical reactions can be considered. The flux, as mentioned in this equation, therefore, can also mean a production of a substance, or a disappearance of it, by a chemical reaction. In this case, however, the flux does not remain a vector, but becomes a scalar. How can we include these scalar fluxes into a matrix together with vectors, without violating mathematical rules?

Let us consider a simple set of flux equations, including transport of a substance (J_i) and a chemical reaction, the rate of which we will denote by the scalar flux J_R . Formal mathematics appears to allow only the following possibility:

$$\begin{aligned} J_R &= L_{RR}X_R + L_{Ri}X_i \\ J_i &= L_{iR}X_R + L_{ii}X_i. \end{aligned} \quad (3.1.62)$$

In the first equation, the product of two vectors ($L_{Ri}X_i$) gives a scalar, as well as the product of scalars ($L_{RR}X_R$), and so this equation is a sum of scalars resulting in a scalar flux (J_R). In contrast, in the second equation all the terms of the sum are vectors. In both equations, therefore, the mathematical requirement for homogeneity has been satisfied.

But: what is the meaning of a vectorial coupling coefficient? Introducing this parameter, we declared it as a sort of conductivity. What does a conductivity vector mean? In fact, in so-called *anisotropic systems* vectorial coefficients can appear. In an isotropic system, for example, in an aqueous solution, the mobility of an ion in all directions is equal. The parameter L , therefore is a scalar. Considering, however, the same ion in a pore of a membrane, its movement is possible only in a predetermined direction, and its conductivity consequently becomes a vector.

These circumstances are considered in the so-called *Curie-Prigogine principle*. It states that the direct coupling of scalar and vectorial fluxes is possible only in anisotropic systems. This principle is important in biophysical considerations of active ion transport. In this case a hydrolysis of ATP is coupled to the transport of an ion against an electrochemical gradient.

3.1.4

Entropy and Stability

The second principle of thermodynamics states that an isolated system moves spontaneously towards a maximum in its entropy. When this state is achieved, then it is in thermodynamic equilibrium. In the same way, the decrease of the

free energy down to a minimum can be considered as the way towards the equilibrium in the sense of the second principle.

Any movement as a result of energy transformation leads to an increase in the entropy of the system or its environment. The term *entropy production* ($\sigma = dS/dt$) has been introduced to characterize this process. The entropy production is always positive, but can approach zero asymptotically. The condition: $\sigma = 0$ would mean an idealized reversible process. Thermodynamically, a process is defined as being reversible if it can be repeated an arbitrary number of times without requiring the supply of additional energy.

To prevent misunderstanding, the different meanings of the term 'reversible' in physics, chemistry, and biology must be pointed out. In physics, the term 'reversible' is used according to the thermodynamic definition, i.e. connected with the condition: $\sigma = 0$. When a chemist speaks about a 'reversible reaction', or a 'reversible electrode', he or she only means processes that in principle could run in both directions, independently of the required energy. Finally, biologists say that a change of a biological system is 'reversible' when it is able to reverse an induced change so that no irreparable damage is caused (for example: reversible inhibition of metabolic processes).

Considering the total entropy balance of a system, it must be pointed out that, in contrast to a closed system, where as a result of energy transformations only an entropy increase is possible, in an open system an entropy decrease is even possible. This can be done if substances are incorporated which have a low entropy content, in exchange for entropy-rich substances which are being extruded. To characterize this process, an entropy flux (J_S) is formulated which penetrates the whole system. Hence, the total entropy balance of the system can be written as follows:

$$\frac{dS}{dt} = -\nabla J_S + \sigma. \quad (3.1.63)$$

The overall change of entropy in an open system (dS/dt), therefore, results as a sum of the always positive entropy production (σ), and the divergence of the entropy flux ($\nabla J_S \equiv \text{div } J_S$), penetrating the system. In reference to the definition in Section 3.1.3 (see Fig. 3.2), the system in fact is not conservative in relation to this entropy flux. Depending on the relation of the two terms in the sum [Eq. (3.1.63)], the total energy change (dS/dt) can become positive as well as negative. The control of the term ∇J_S can be considered as the work of a Maxwell demon, as described in Section 2.3.2 (Fig. 2.9).

For further considerations it may be useful to introduce a thermodynamically based classification of the various kinds of stationary states. We define a *stationary state* as a state in which the structure and parameters are time independent. The reasons leading to this quality can be quite different. The water level of a lake, for example, can be time independent, i.e. constant, either because there is no inflow into the lake, and no outflow, or because inflow and outflow are equal to each other. These two kinds of stationary states can be distinguished by their entropy production. In the first case no energy is required

to maintain this state, therefore there is no entropy production, the system is in *thermodynamic equilibrium* ($\sigma \doteq 0$). In contrast, the lake with exactly the same in- and outflow is in a *steady state*. This is a stationary state with entropy production ($\sigma > 0$). The thermodynamic definition of the steady state is the only possible one. A steady state cannot be defined by its kinetic properties.

Let us illustrate this statement by an example: using radio-nuclides it is possible to demonstrate that human erythrocytes exchange chloride as well as potassium ions with their environment. With this method, it is possible to measure directly the corresponding exchange rates. This kinetic method of analysis may give the impression that both ions, Cl^- , as well as K^+ , are in a steady state because in both cases the unidirectional fluxes, outside $\rightarrow$ in and inside $\rightarrow$ out, are equal. This, however, is an incorrect conclusion. The chloride ions in fact are distributed passively between the external medium and the cytoplasm, according to their thermodynamic equilibrium. The observed exchange of radioactive chloride results from their self diffusion by thermal motion. This process is not accomplished by entropy production because no energy is converted. It is like a stochastic exchange of water molecules between the two vessels in Fig. 3.4A. In contrast to this, potassium is pumped actively into the cell by use of metabolic energy against its electrochemical gradient, and diffuses passively back, driven by this gradient (Fig. 3.4B). Both are true fluxes in the thermodynamic sense, producing entropy. Potassium, therefore, in contrast to chloride, really is in a stationary state. This example indicates that the above described kinetic experiment of compartment analyses is unsuitable for distinguishing between an equilibrium state and a steady state. We will come back to this problem later in detail (Sects. 3.4.1 and 5.2.1).

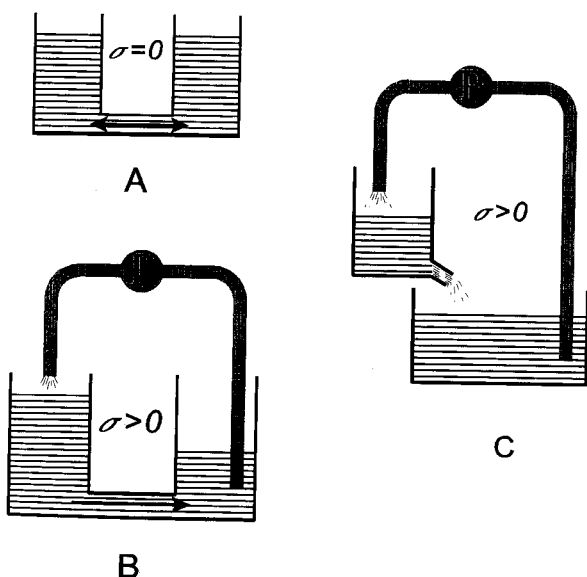


Fig. 3.4. Stationary states in hydraulic models: A thermodynamic equilibrium ($\sigma = 0$); B a steady state system ($\sigma > 0$), which becomes an equilibrium (A), if the pump (P) is stopped; C a steady state system ($\sigma > 0$), where the stop of the pump (P) would lead to a complete outflow of the liquid from the upper vessel

As we will see later (Sect. 3.3.3), the steady state of potassium can be transformed into an equilibrium state if the pump is inhibited. In this case, a Donnan equilibrium will be established which means an equilibration of the electrochemical potentials. The steady state of sodium and potassium in the cell therefore resembles the case B in Fig. 3.5, changing into the equilibrium state (Fig. 3.5A), when the active transport is inhibited. In contrast to this, various substances do not show the possibility of an equilibrium distribution. If the influx is stopped in these cases, the substance disappears completely (Fig. 3.4C).

An important property of all stationary states is their kind of stability. Let us illustrate this again by a mechanical example, using the behavior of a sphere on a surface (Fig. 3.5). The requirement for a stationary state in this case simply means an equilibrium and means the kind of position of the sphere on any small, but horizontal part of the surface. In the case of an *indifferent* state, the surface has a large horizontal region. In this case the energy of the sphere will not be changed by a change of its position. In case of a *stable* state, every change of the position leads to an increase of the energy of the sphere, and generates a force driving the sphere back to its original state. In contrast, an *unstable state* is characterized by a situation where even small changes of the position release forces that cause the system to be deflected even more. Additionally, sometimes so-called metastable states are considered. As *metastable*, a stable state can be considered which is delimited from another one by a small barrier which can be easily overcome.

Figure 3.6 indicates all possible kinds of stationary states. First of all the presence, or the absence of entropy production indicates whether the given stationary state is a thermodynamic equilibrium ($\sigma = 0$), or whether it is a steady state ($\sigma > 0$). Furthermore, in the case of thermodynamic equilibrium, global and local equilibria must be distinguished between. In case of a *global equilibrium*, the function of free energy indicates only one minimum. This means that no alteration, how ever strong it may be, can bring the system into another equilibrium state. As an example, the equilibrium distribution of particular kinds of ions between the cell and its environment can be mentioned. In the case of *local equilibrium*, the energetic function indicates two or more minima which are separated by more or less large energy barriers. As an example of this, reaction isotherms can be considered, as illustrated in Figs. 2.12 and 2.32. The stability of such local equilibria is as high as the energy barrier

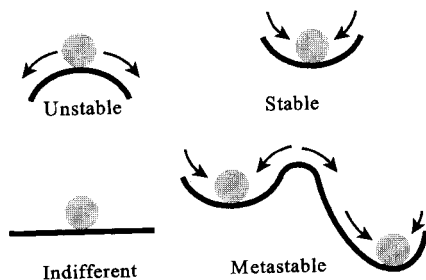


Fig. 3.5. Various kinds of stability conditions

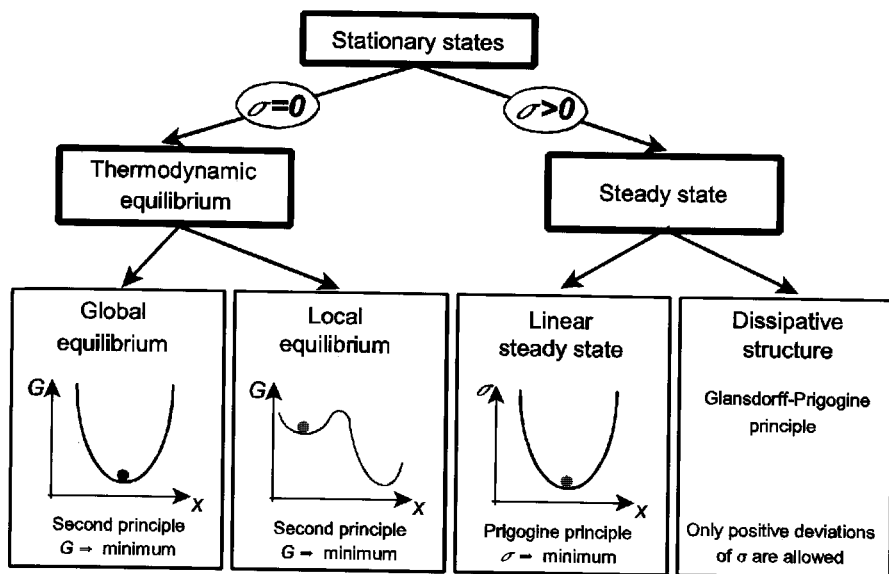


Fig. 3.6. Possible kinds of stationary states and conditions of their stability

between these minima is. If a barrier between two minima is so small that thermal fluctuations can lead to quick transitions, the state is called metastable. This, for example, is the typical situation for enzyme-substrate complexes. For the schemes in Fig. 3.6 it must be considered that in reality G is not simply a function of a single reaction coordinate (x), but rather a hyperplane in a n -dimensional space.

To consider the two kinds of stationary states ($\sigma > 0$) from Fig. 3.6 in detail, we must learn something more about entropy production. Entropy production is not only an indicator for distinguishing between the equilibrium state and steady state, but is also important for some stability considerations of the latter.

In concordance with its definition, entropy production has the measure: $\text{J K}^{-1} \text{s}^{-1}$. In some cases, additionally, this parameter is related to the mass in kg or to the molar mass.

Multiplying entropy production by temperature, one gets *Rayleighs dissipation function* (Φ), which can be calculated from fluxes and forces as follows:

$$\Phi = \sigma T = \sum_{i=1}^m J_i X_i. \quad (3.1.64)$$

This equation holds for the region of linear, as well as of non-linear flux-force relations. Particularly for the linear region, I. Prigogine was able to show that systems tend to develop towards a reduced entropy production. This is the *Prigogine principle of minimal entropy production*. Systems which are not far from thermodynamic equilibrium, and which are kept in imbalance by

continuously acting forces consequently may move towards a steady state, the stability of which is included in this criterion.

The dissipation function reflects the specific heat generation of a system, and can be measured calorimetrically. Therefore, the heat production of systems has been discussed in context with this Prigogine principle of minimal entropy production. Simple biological processes as well as highly complex events such as ontogenesis, wound healing, tumor development and others have been investigated in this way.

Figure 3.7 shows some observations made in mice. While the total heat production ($\dot{Q}$) of the animals increases in accordance with their age and mass (m), the specific heat production, that is the heat production relative to the body weight, as the dissipation function (Φ), reaches a maximum and then decreases. There are some short-term deviations in the course of this development which are reflected in variations around the average. Similar measurements have been made during the development of amphibia, and give rise to qualitatively similar curves. If an organism is injured during the phase of decreasing entropy production, then the regeneration processes lead to a temporary increase in the slope of the curve. In fact the entropy production reflects the metabolic activity of the organism and is therefore proportional to its specific oxygen consumption.

Despite these results, whereby after passing through a phase of rapid development, the system moves towards a stationary state of minimal entropy production, it is still controversial as to whether the Prigogine principle can be applied to large systems which include a great number of different subsystems, particularly those which are far from equilibrium.

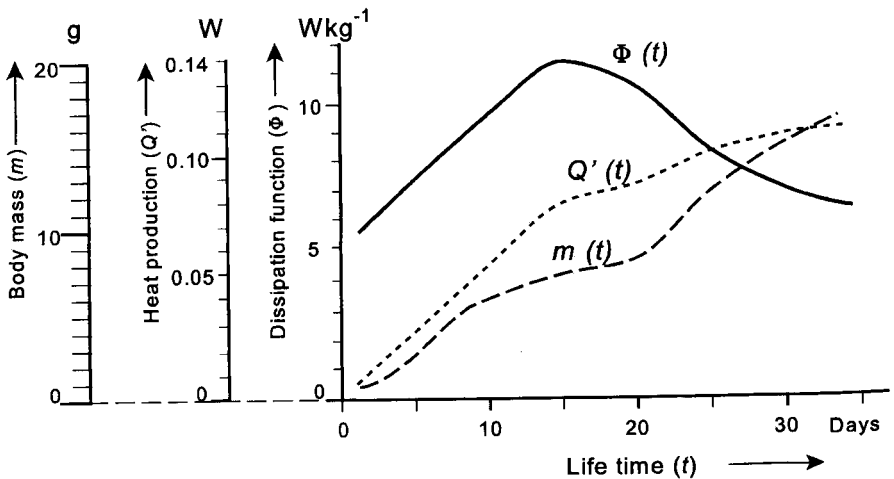


Fig. 3.7. Dissipation function (Φ), heat production ($\dot{Q}$), and mass (m) of mice as a function of their life span from the time of birth, up to the 35th day of life. (Data from Lamprecht and Zotin 1978)

If a system deviates from the region of linear approaches, then the Prigogine principle is no longer valid. In contrast to steady states in the scope of linear thermodynamic approaches which are always stable and do not show any kinds of metastability, systems in the region of non-linear approaches show more complicated behavior.

Considering non-linear systems we must at first generalize the term 'stationary'. We already mentioned in context with the Stockes law [Eq. (3.1.54)] the so-called stationary movement as a movement with constant velocity where the frictional force is equal to the driving force. This sort of movement is also a kind of stationary, i.e. time independent state. The term 'stationary state' can also be applied to states that are not at rest, but show repetitive periodic movements in a stationary manner. For example, cardiac function can be called 'stationary', if the frequency and amplitude of the heart beat does not change during the period of observation.

In the case of a linear steady state, fluctuations of the system parameters produce only positive deviations of entropy production, bringing the system back to the stationary state, which is therefore stable in any case. In contrast to this, in the region of non-linear approaches, far from equilibrium, fluctuations of a stationary state can also lead to negative deviations of the entropy production, leading to a destabilization of the system. The system may jump from one into another stationary state. The steady states in the region of non-linear approaches therefore are mostly metastable (see Fig. 3.5). Their stability condition requires the occurrence of only positive deviation of the entropy production. This is the so-called *Glansdorff-Prigogine principle*.

The transition from linear to non-linear approaches of thermodynamics is not only associated with an increase in the coefficient of friction, or with the occurrence of several stationary states, but also with the spontaneous development of so-called *dissipative structures*. The general concept of the diversity of structures has already been discussed in Section 2.3.3. We postulated that there are basically two different types of structures: equilibrium structures ($\sigma = 0$), and dissipative structures ($\sigma > 0$). In contrast to equilibrium structures which are always structures in space, dissipative structures can also be structures in time, or in a space-time continuum.

In order to illustrate this, Fig. 3.3 shows the function $f(v)$ of a sphere, and the pattern of flow close to it. The laminar flow, which occurs at a low velocity, can be compared with the turbulent flow, which appears when the non-linear region of the function is attained. When the flux-force relation changes from the laminar to the non laminar region, then the unstructured laminar flow becomes unstable and vortices appear which, in the terminology of thermodynamics, are dissipative structures (for details of streaming behavior, see Sect. 3.7.1).

There is extensive literature on the theory of dissipative structures and on their occurrence in nature. Most of these dissipative structures are periodic structures in space, such as cloud patterns, flow phenomena in liquids with an applied temperature gradient, so-called Benard cells, plasma waves in electron beam tubes etc. In addition, there are many time structures, including for

example all kinds of sound production, from the electronic organ to vibrating strings, and to the wind instruments.

In biological systems, in spite of many speculations, dissipative structures in space have not been unequivocally demonstrated. On the other hand, time patterns showing this property have been found to occur quite often. Examples of this include oscillations in metabolism, periodical changes in the membrane potential of certain cells, for example in the cells of the cardiac pacemaker in the sino-auricular node, and finally more complex, oscillatory movements in ecosystems. Sometimes, such oscillations in local concentrations of certain substances become 'frozen', so that structures arise which do not require entropy producing processes to sustain them, but which originally had been built as dissipative structures. One can conceive, for example, the genesis of the first viable bio-macromolecule occurring in this way. Similar processes could form the basis of morphogenesis (for further detail, see Sects. 5.3.2 and 5.3.3).

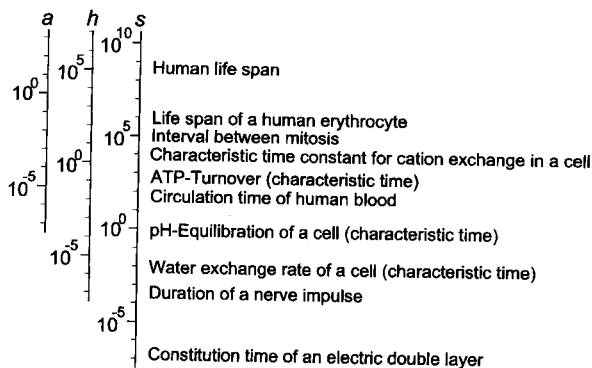
Let us come now to a further aspect of stationary states. If a system is considered as stationary, this is always in relation to a defined period of time. An organism could be said to be in a stationary state for a number of hours, or for days. That is, its volume, the content of certain substances, its shape, temperature etc. are constant within defined tolerance for this period of time. Nevertheless, the organism in fact is ageing or growing. If a longer time period is considered, then the changes in these parameters will exceed the defined tolerance limits.

A biological system consists of sub-systems, each of which is associated with a different time constant. For example, the rate of aging in man is slow, compared with the rate of mitosis of a hemopoietic cell. Thus, the conditions in bone marrow can be regarded as stationary during the course of several cycles of mitosis in spite of the ageing processes affecting the organism as a whole. In vivo, the life span of a human erythrocyte is about 100 days. The ionic composition of juvenile erythrocytes differs somewhat from that of the mature cells. If one is interested in the ionic regulation of these cells, then, because of the temporal characteristics of such processes, experiments lasting a few hours will be sufficient to provide the required information. Within such short time periods the ionic concentration can be regarded as stationary. Figure 3.8 shows some characteristic times on a logarithmic scale that will serve to extend this list of examples.

These considerations hint at the existence of a *time hierarchy* of stationary states, related to their time constants which range over several orders of magnitude. The kinetic consequences of this circumstance will be considered in Section 3.2.3 and especially 5.2.2. The following concept, however, is important for our further thermodynamic considerations.

The living organism as a whole, when considered within a limited period of time, is in a stationary state with entropy production, i.e. in a steady state. It is made up of a great number of subsystems which are ordered in a defined time hierarchy. The steady state of the system as a whole does not imply that all of the sub-systems are also in a steady state. A large proportion of them, particularly those with a short time constant, are in thermodynamic equilibrium. If the

Fig. 3.8. Characteristic time constants (s – seconds, h – hours, a – years) of various processes in a logarithmic scale, to illustrate the time hierarchy in biological systems



system as a whole changes its parameters slowly, then these sub-systems are capable of following such changes quickly, so that they almost completely adapt within their characteristic time, and thus are always in a stationary state. This is sometimes called a *quasi-stationary*, or *quasi-equilibrium* state.

The following example will illustrate this: the water content of a tissue depends on the ionic composition of its cells. Sodium and potassium ions are being actively transported against passive fluxes, giving rise to a steady state. In this way the active transport, and the corresponding passive fluxes regulate the osmotic properties of the cells. The characteristic time of the water flux is much shorter than that of the cations (see Fig. 3.8). As a result, the water in the interior of the cells is always in osmotic equilibrium with the surrounding medium. In Section 3.2.3 we will discuss this example in detail in context with the Donnan-osmotic quasi-equilibrium.

3.1.5

Thermodynamic Basis of Biochemical Reactions

In the previous formulation of the Gibbs equation [Eqs. (3.1.17), (3.1.27)–(3.1.29)], the chemical reaction was not explicitly enclosed. In fact, a chemical reaction can be analyzed by referring to the chemical potential of its components. The reaction:



can be considered as a replacement of the substances A and B, by the substances C and D.

Considering this process for isobaric ($dp = 0$) and isothermic ($dT = 0$) conditions, where only concentration changes take place ($dn_i \neq 0$, $dq = 0$, $dl = 0$), the equation for Gibbs free energy [Eq. (3.1.29)] reduces to:

$$dG = \sum_{i=1}^m \mu_i dn_i. \quad (3.1.66)$$

The change in the Gibbs free energy ($\Delta_R G$) according to the definition (3.1.31) then gives:

$$\Delta_R G = v_c \mu_c + v_d \mu_d - v_a \mu_a - v_b \mu_b. \quad (3.1.67)$$

Introducing in this equation the expression for the chemical potential [Eq. (3.1.33)], it follows:

$$\begin{aligned} \Delta_R G = v_c \mu_c^0 + v_d \mu_d^0 - v_a \mu_a^0 - v_b \mu_b^0 \\ + RT(v_c \ln a_c + v_d \ln a_d - v_a \ln a_a - v_b \ln a_b). \end{aligned} \quad (3.1.68)$$

Now, we can introduce a molar free standard reaction energy:

$$\Delta_R G^0 = v_c \mu_c^0 + v_d \mu_d^0 - v_a \mu_a^0 - v_b \mu_b^0. \quad (3.1.69)$$

Introducing this in Eq. (3.1.68), and combining the logarithmic terms, we obtain *Van't Hoffs equation*:

$$\Delta_R G = \Delta_R G^0 + RT \ln \frac{a_c^{v_c} a_d^{v_d}}{a_a^{v_a} a_b^{v_b}}. \quad (3.1.70)$$

In the case of thermodynamic equilibrium: $\Delta_R G = 0$. If the symbol a_i^0 stands for activities of the system in equilibrium, we can introduce the equilibrium constant (K_p) for isobaric processes as follows:

$$K_p \equiv \frac{a_c^{0v_c} a_d^{0v_d}}{a_a^{0v_a} a_b^{0v_b}} = e^{-\frac{\Delta_R G^0}{RT}}. \quad (3.1.71)$$

Introducing this expression into Eq. (3.1.70) it gives:

$$\Delta_R G^0 = -RT \ln K_p. \quad (3.1.72)$$

The molar free standard Gibbs energy of a chemical reaction ($\Delta_R G^0$) can be calculated from the standard energies of formation ($\Delta_F G^0$) obtained from tables, using Eq. (3.1.31). Here, we use the property of Gibbs energy as a state function, as discussed in the introduction (Sect. 3.1.1), whereas its value is independent of the way in which the state was achieved.

If a reaction is not in equilibrium, then the direction it will take can be determined by calculating $\Delta_R G$ [Eq. (3.1.70)] from the given activities of the components. Spontaneously, the reaction can run in the direction which is indicated in Eq. (3.1.65) only in the case where: $\Delta_R G < 0$, giving a reduction of the free energy during this process.

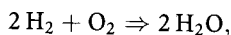
So, we have considered chemical reactions simply as a changes in concentrations of their compounds, not differentiating whether the substance was introduced to the system by a transport process, or by a chemical reaction.

This, in fact, is correct and in most cases useful. In order to analyze systems in which chemical reactions take place in addition to real transport, this approach can lead to misunderstandings. In these cases it is useful to introduce a special term into the Gibbs equation, called the *degree of advancement* of a chemical reaction ($d\xi$):

$$d\xi = \frac{1}{\nu_i} dn_i. \quad (3.1.73)$$

A positive value of $d\xi$ represents a particular step in the reaction from left to right, whereas $dn_i > 0$ means an increase, and $dn_i < 0$ a decrease of the molar amount of the substance. In order to keep to these sign definitions, the stoichiometric coefficients (ν_i) of the initial substrates must become negative, and those of the end products, positive.

Taking for example the reaction:



the following applies:

$$d\xi = -\frac{1}{2} dn_{\text{H}_2} = -dn_{\text{O}_2} = \frac{1}{2} dn_{\text{H}_2\text{O}}. \quad (3.1.74)$$

Now, a work coefficient according to our discussion in Section 3.1.2, the *chemical affinity* (A) is introduced:

$$A = - \sum_{i=1}^m \nu_i \mu_i, \quad (3.1.75)$$

where $i = 1 \dots m$ mean the components of the given reaction. Using Eq. (3.1.71), the following relation is obtained:

$$\sum_{i=1}^m \mu_i dn_i = -A d\xi. \quad (3.1.76)$$

Both expressions can be present in the Gibbs equation simultaneously if transport processes and chemical reactions are considered together. This may be important when calculating processes of active ion transport through membranes.

Affinity, from its definition, proves to be an energy difference and, as such, the driving force of a chemical reaction. Consequently, it can be directly inserted into the flux matrix [Eq. (3.1.58)] instead of a generalized force ($\mathbf{X}$).

In contrast to $\Delta_R G$, which determines the direction of a chemical reaction, the molar reaction enthalpy $\Delta_R H$ indicates the heat of a reaction. This means:

$$\begin{aligned} \Delta_R H > 0 & \quad \text{– endothermic reaction,} \\ \Delta_R H < 0 & \quad \text{– exothermic reaction.} \end{aligned}$$

According to the definition (2.1.23), for isothermal systems:

$$\Delta_R G = \Delta_R H - T \Delta_R S. \quad (3.1.77)$$

We already pointed out that if a reaction proceeds spontaneously then $\Delta_R G$ must apply. According to Eq. (3.1.77), in the case of an endothermic reaction this will only occur when:

$$\Delta_R H < T \Delta_R S. \quad (3.1.78)$$

The direction of such a reaction is therefore determined by the rise in entropy of the system. These sorts of processes, therefore, are called *entropy driven reactions*. The classical example of such a reaction is the melting of ice. In Section 2.4.2 we discussed various biomolecular reactions of this type in the context of water structure.

Equation (3.1.77) at the same time enables the determination of the temperature at which the two phases of a system would be in equilibrium. This would be the melting point (T_M) of ice, or the temperature at which collective phase transitions take place in macromolecular systems.

If, at equilibrium $\Delta_R G = 0$, then it follows from Eq. (3.1.77):

$$T_M = \frac{\Delta_R H}{\Delta_R S}. \quad (3.1.79)$$

In this way, the measurement of the thermal characteristics of a phase change enables the concomitant change in entropy to be determined.

Further reading: There is a very extensive literature on the thermodynamics of irreversible processes. For basic problems and approaches in calculating fluxes, see: Katchalsky and Curran (1965); Prigogine (1967); Schnakenberg (1981). For the problems of thermodynamics and evolution see: Glansdorf and Prigogine (1971); Haken (1983); Lamprecht and Zotin 1985. For entropy driven processes: Lauffer (1975)

3.2

The Aqueous and Ionic Equilibrium of the Living Cell

From the point of view of thermodynamics, a living cell can be considered as a system in non-equilibrium. This non-equilibrium state is maintained by permanent processes of energy transformation. In spite of the state of the whole system, however, some of its subsystems nevertheless may be in thermodynamic equilibrium. Such equilibria, or quasi-equilibria have already been discussed in Section 3.1.4, in connection with the time hierarchy of biological systems. In this section we will direct our attention to these equilibrium states, especially equilibrium distributions of charged and uncharged components.

3.2.1

Osmotic Pressure

The concept of osmotic pressure is very important for understanding a number of cell physiological processes. Unfortunately, osmotic pressure is often confused with hydrostatic pressure, especially with the turgor pressure of plants. In this section it will be shown that osmotic pressure is just a property of a solution or a suspension and that although it can be the origin of the hydrostatic pressure in the cell, it is not at all identical to it.

The effect of osmotic pressure can be demonstrated best by a *Pfeffer cell* (see Fig. 3.20). This consists of a glass bell with a vertical tube or a manometer at the top. The mouth of the bell is closed by a semipermeable membrane. The bell is filled with a solution and submerged in a vessel with pure water. The membrane allows the water, but not the molecules of the solute, to pass through. In his experiments in 1877, the botanist W. Pfeffer used a sheet of porous pottery to close the bell. The pores of this "membrane" were covered by a precipitated layer of cupric(II)-hexacyanoferrate(II). In this experiment the water penetrates the membrane, passes into the glass bell, and increases the internal hydrostatic pressure. Eventually an equilibrium will be reached when the internal pressure balances the osmotic force which drives the water into the Pfeffer cell. If the membrane is truly semipermeable, and only if pure solvent is outside the glass bell, then, and only then, does the hydrostatic pressure difference between the internal and the external phase in this equilibrium equal the osmotic pressure of the solution inside the Pfeffer cell.

In order to analyze this situation let us assume two phases (*I* and *II*) separated from each other by a membrane (see Fig. 3.9). Let the membrane be semipermeable and thus allow the solvent (*w*), but not the solute (*s*), to pass through. Because of solvent flux through the membrane, the volumes of both phases and their concentrations will change. This will also alter the chemical potentials (μ_i) of the components (*i*).

The concentration dependence of the chemical potential [see Eq. (3.1.33)], using the mole fraction (x_i) as a measure of concentration is:

$$\mu_i = \mu_{ix}^0 + RT \ln x_i. \quad (3.2.1)$$

To achieve thermodynamic equilibrium under isothermal conditions, the chemical potential of the exchangeable components of both phases *I* and *II*

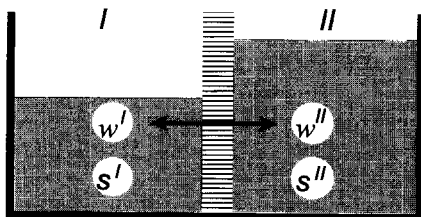


Fig. 3.9. Derivation of osmotic pressure: phases *I* and *II* are separated from each other by a semipermeable membrane. Only the solvent (*w*), but not the solute (*s*), can penetrate this membrane. The hydrostatic pressure of both sides of the membrane is different

must be equal. In the example considered here, this will only apply to the solvent (w). This requires

$$\mu_w^I \stackrel{!}{=} \mu_w^{II}, \quad (3.2.2)$$

which means that

$$\mu_{wx}^{0I} + RT \ln x_w^I = \mu_{wx}^{0II} + RT \ln x_w^{II}, \quad (3.2.3)$$

or

$$RT \ln \frac{x_w^I}{x_w^{II}} = \mu_{wx}^{0II} - \mu_{wx}^{0I}. \quad (3.2.4)$$

Now, let us first direct our attention to the difference in the standard chemical potentials on the right-hand side of Eq. (3.2.4). Because of the pressure difference induced by the flow of the solvent, $\mu_{wx}^{0I} \neq \mu_{wx}^{0II}$. The pressure dependence of the standard chemical potentials therefore must be studied in more detail.

The following definition of the standard chemical potential can be derived from Eq. (3.1.32):

$$\mu_i^0 = \left(\frac{\partial G^0}{\partial n_i} \right)_{T,p,n_j}. \quad (3.2.5)$$

The pressure dependence of μ_i^0 can therefore be expressed as

$$\frac{\partial \mu_i^0}{\partial p} = \frac{\partial \left(\frac{\partial G^0}{\partial n_i} \right)}{\partial p} = \frac{\partial \left(\frac{\partial G^0}{\partial p} \right)}{\partial n_i}. \quad (3.2.6)$$

The way in which the sequence of the derivation steps in a total differential can be changed was discussed in Section 3.1.1 [see also Eq. (3.1.6)].

From the equation of Gibbs free energy [Eq. (3.1.29)] we obtain:

$$dG^0 = V dp \quad \text{for: } dT = 0, \quad dl = 0, \quad dq = 0, \quad dn = 0, \quad (3.2.7)$$

and

$$\frac{\partial G_i^0}{\partial p} = V_i. \quad (3.2.8)$$

If this is substituted into Eq. (3.2.6), then

$$\frac{\partial \mu_i^0}{\partial p} = \frac{\partial V_i}{\partial n_i} = \bar{V}_i. \quad (3.2.9)$$

We are already acquainted with the variable $\bar{V}$ which is the partial molar volume of the substance i [Eq. (3.1.8)]. Now, the differential $d\mu_i^0$ can be calculated. This is a small change in μ_i^0 that occurs when there is a small change in the pressure dp . Following the general rule (3.1.1), we obtain:

$$d\mu_i^0 = \left(\frac{\partial \mu_i^0}{\partial p} \right) dp = \bar{V}_i dp. \quad (3.2.10)$$

To obtain an expression for the difference in the standard potentials due to alterations in pressure, the equation must be integrated between the corresponding limits:

$$\int_{\mu_i^{0I}}^{\mu_i^{0II}} d\mu_i^0 = \int_{p^I}^{p^{II}} \bar{V}_i dp, \quad (3.2.11)$$

$$\mu_i^{0II} - \mu_i^{0I} = \bar{V}_i (p^{II} - p^I). \quad (3.2.12)$$

(This manner of integration only applies when $\bar{V} = \text{const}$, that is, when the partial molar volume is independent of the pressure. This is only the case for ideal solutions.)

Now, it is possible to substitute Eq. (3.2.12), related to the solvent (w) into Eq. (3.2.4):

$$p^{II} - p^I = \frac{RT}{\bar{V}_w} \ln \frac{x_w^I}{x_w^{II}}. \quad (3.2.13)$$

This equation indicates the hydrostatic pressure difference ($p^{II} - p^I$) induced in our system at thermodynamic equilibrium for different values of the mole fractions in the two phases $x_w^I \neq x_w^{II}$. If phase I contains only pure solvent ($x_w^I = 1$), this pressure difference is called osmotic pressure (π), therefore:

$$\pi = \frac{RT}{\bar{V}_w} \ln \frac{1}{x_w^{II}} = -\frac{RT}{\bar{V}_w} \ln x_w^{II}. \quad (3.2.14)$$

This equation and its derivation allow us to describe osmotic pressure as a parameter reflecting a property of a solution. Under isothermal conditions the osmotic pressure of a solution is equal to the hydrostatic pressure, which is required to alter the chemical potential of the pure solvent in such a way that it will be equal to the chemical potential of the solvent in this solution.

Equation (3.2.14) is a precise expression for calculating the osmotic pressure of a solution, provided the mole fraction activity of the solvent x_w^{II} is used to express the concentration of the solvent (w) in the solution. Some simplifications can be made for dilute solutions. First, the mole fraction *concentration* can

be used instead of the mole fraction *activity*. According to Eq. (3.1.35), this means:

$$x_w = \frac{n_w}{n_s + n_w}. \quad (3.2.15)$$

Thus, if the sum of the mole fractions of all components of a solution equals 1 [Eq. (3.1.36)], then

$$x_w = 1 - x_s. \quad (3.2.16)$$

The number of molecules of solvent in diluted solutions is much larger than the number of molecules of the solute ($n_w \gg n_s$). A 0.1 molar solution of the substance (s) in water (w), for example, contains $n_s = 0.1$ mol of solute, but $n_w = 55.6$ mol of water (n_w means moles water per 1000 g solution, therefore, $1000/18 = 55.6$). In this case, combined with Eqs. (3.2.16) and (3.2.15) becomes simply

$$x_w = 1 - \frac{n_s}{n_s + n_w} \approx 1 - \frac{n_s}{n_w}. \quad (3.2.17)$$

In addition, the following holds for dilute solutions:

$$\bar{V}_w = \frac{\partial V_w}{\partial n_w} \approx \frac{V_w}{n_w}. \quad (3.2.18)$$

Thus, $n_w = V_w/\bar{V}_w$. One can also introduce the molar concentration of the solute (s), using $n_s/V = c_s$, and the total volume of the solution ($V \approx V_w$). Substituting this in Eq. (3.2.17), we obtain:

$$x_w = 1 - \frac{n_s \bar{V}_w}{V} = 1 - c_s \bar{V}_w. \quad (3.2.19)$$

Let us substitute this expression in Eq. (3.2.14):

$$\pi = -\frac{RT}{\bar{V}_w} \ln(1 - c_s^H \bar{V}_w). \quad (3.2.20)$$

Now, we use the following rule to expand logarithmic expressions in series. For any number $|q| < 1$, it holds:

$$\ln(1 - q) = -q - \frac{q^2}{2} - \frac{q^3}{3} - \frac{q^4}{4} - \dots \quad (3.2.21)$$

For q we use the expression $c_s^H \bar{V}_w$, which is much smaller than 1. In this case we are justified in retaining only the first term of this series. This gives:

$$\pi = -\frac{RT}{\bar{V}_w} (-c_s^H \bar{V}_w), \quad (3.2.22)$$

and

$$\pi = RTc_s^H. \quad (3.2.23)$$

This is the *Van't Hoff equation* for osmotic pressure.

It is interesting to note that, in 1877, Van't Hoff derived this relation not in the way shown here, but by considering the analogy with the state equation for an ideal gas. He assumed that 1 mol of an ideal gas, when confined to a volume of 1 liter, would exert a pressure of 2.27 MPa on the walls of the vessel. He reasoned that the molecules of a solute, when dissolved at a concentration of 1 mol/l should behave in the same way as the particles of the gas. This pressure he called "osmotic".

The equation of state for an ideal gas is:

$$p = \frac{n}{V}RT. \quad (3.2.24)$$

This can be transformed into:

$$\pi = \frac{n}{V}RT = cRT. \quad (3.2.25)$$

Both the thermodynamic derivation as well as the analogy used by Van't Hoff show that Eq. (3.2.23) is only applicable for ideal solutions, or with some approximation for solutions which are very diluted. This restriction can be overcome using a correction factor. This is the so-called *osmotic coefficient* (g):

$$\pi = gcRT. \quad (3.2.26)$$

This factor must not be confused with the activity coefficient (f), which was introduced in Section 3.1.2 [Eq. (3.1.34)]. This will be easily understood if f is directly introduced into the thermodynamic derivation of the Van't Hoff equation [say, in Eq. (3.2.3)]. The relation between f and g is somewhat complicated and will not be discussed here.

The osmotic coefficients for NaCl, KCl and sucrose are given in Fig. 3.10 as functions of concentration. With the increase in concentration, the osmotic coefficient (g) deviates more and more from the value 1. This deviation, however, is smaller as for the activity coefficient (f) (see Fig. 3.1).

In addition to the non-ideal behavior of a solution, it must be noted that the osmotic pressure exerted by dissociable substances results from the sum of the osmotic pressures of the ions formed by this dissociation. Thus, if a monovalent salt (such as NaCl) is assumed to be completely dissociated into Na^+ and Cl^- then the osmotically active concentration is twice as great as the salt concentration. Therefore one must distinguish between the *osmolarity* and the *molarity* of a solution. The osmolarity of a 0.1 M solution of NaCl (at 25 °C) is therefore: $2 \times 0.1 \times g = 2 \times 0.1 \times 0.932 = 0.1864$ osmolar. The osmolarity can be pH-dependent for polyelectrolyte solutions because it will change according to the degree of dissociation.

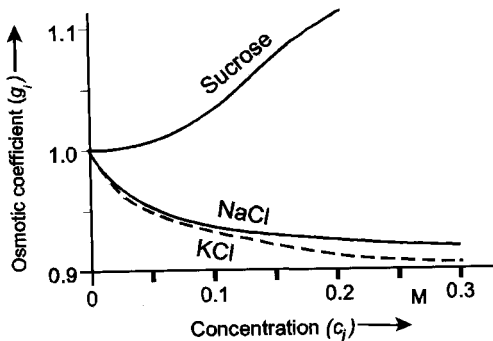


Fig. 3.10. The osmotic coefficient (g) as function of the concentration (c_s) in aqueous solutions

The osmotic pressure of a solution is usually measured in an indirect way. Because the vapor pressure and the freezing point of a solution are basically determined by the same set of laws, instruments which can measure these properties are also convenient for the determination of osmotic pressure. True solutions are most simply and most accurately measured cryoscopically. The use of a vapor pressure osmometer is better for solutions of larger molecules or for polydisperse suspensions.

Osmotic pressure is not only a property of true solutions, but also of colloidal solutions and to some extent also of suspensions. In this context it is referred to as *colloid-osmotic pressure*. In gels, osmotic pressure acts to oppose the tendency of the molecular framework to contract (Parsegian, in Leopold 1986). In this context it should be mentioned that even the water in membrane pores is in osmotic equilibrium with the bulk water. Therefore the diameter of a pore becomes controlled osmotically (Zimmerberg and Parsegian 1986). In view of the fixed charges of all biological surfaces (see Sect. 2.5.4), however, the osmotic behavior of these microphases must be considered as Donnan-osmotic systems, which we will consider later (Sect. 3.2.3).

In cell physiological considerations the hydrostatic pressure, which arises from differences in osmotic pressure, is of special interest. Only in the case of the thermodynamic equilibrium of water in this system, and only if the membrane is really semipermeable in relation to all components of the solution, does the difference in the osmotic pressure equal the difference in the generated hydrostatic pressure.

For solutions with several components and with various permeability properties, one can use the following relation between osmotic ($\Delta\pi$) and hydrostatic (Δp) differences:

$$\Delta p = \sum_{i=1}^n \sigma_i \Delta \pi_i. \quad (3.2.27)$$

This equation takes into account that in the system n substances, each with an osmotic difference $\Delta \pi_i = \pi_{i(\text{internal})} - \pi_{i(\text{external})}$, determine the osmotic conditions. Their effectiveness, with regard to the development of hydrostatic pressure,

depends on the value of a factor (σ_i) which is known as *Staverman's reflection coefficient*. In contrast to the 'classical' approach, this model takes into account that the membrane is not semipermeable, but permselective. This means that all components of the solution can more or less penetrate the membrane. We will analyze this situation in detail later using the approaches of nonequilibrium thermodynamics (Sect. 3.3.1). We will see that from consideration of the corresponding flux matrix [Eq. (3.3.23)] the following relation can be derived:

$$\sigma_s = \frac{v_w - v_s}{v_w}. \quad (3.2.28)$$

Using the indices of the Van't Hoff's equation, v_w and v_s represent the rate of movement of the solvent (water) and the solute in the membrane. In the case of $v_s \rightarrow 0$, the reflection coefficient becomes $\sigma_s \rightarrow 1$. This is the 'classical' situation of semipermeability, and therefore $\Delta p \rightarrow \Delta \pi$. However, when $v_s \rightarrow v_w$, then $\sigma_i \rightarrow 0$ and hence $\Delta p \rightarrow 0$. This occurs in the case of a membrane which allows the osmotically active substance as well as the solvent to pass through. In this case, no hydrostatic pressure can develop even at initial differences in osmotic pressure. In such a system a thermodynamic equilibrium would finally lead to a homogeneous distribution of the substance i (see Fig. 3.20).

In general, the reflection coefficient for disaccharides, such as sucrose, and for larger molecules equals nearly 1. Smaller molecules, especially those which can directly penetrate the lipid layer of a biological membrane, show lower values (see Table 3.1).

In plant cells, osmotic differences generate the so-called *turgor pressure*, which can be as high as several hundred kPa. This intracellular pressure plays a large role in the mechanical stabilization of plants. It forces the cell membrane of

Table 3.1. Typical values of reflection coefficients of non-electrolytes for human erythrocytes (Giebisch et al. 1979) and for *Nitella flexilis* (Zimmermann and Steudle 1978). Values in parentheses: Levitt and Mlekoday 1983

	Erythrocytes	<i>Nitella</i>
Urea	0.79 (0.95)	0.91
Thiourea	0.91	
Ethylene glycol	0.86 (1.0)	0.94
Glycerol	0.88	0.80
Acetamide	0.80	0.91
Propionamide	0.84	
Malonamide	1.00	
Sucrose		0.97
Glucose		0.96
Methanol		0.31
Ethanol		0.34
Isopropanol		0.35
<i>n</i> -Propanol		0.17

the plant cell against the mechanically stabilizing cell wall, which itself is freely permeable for ions and small nonelectrolytes. The turgor pressure can be directly measured by means of special pressure probes that can be introduced into the cell (Zimmermann 1989).

In contrast to plant cells, animal cells (and also plant protoplasts) have no ability to resist an internal hydrostatic pressure. Within certain limits, osmotic changes in the environment can be compensated by alterations in cellular volume, thus maintaining a constant membrane area. Cell swelling, for example, can cause the forming of a sphere or smoothing of the cell surface. As described in Section 2.5.3 (Fig. 2.41), a mechanical expansion of the cell membrane, without introduction of additional molecules, however, is nearly impossible. An osmotic difference of only 1 mosmol can generate a maximum internal pressure of 2.27 kPa. Measurements made on erythrocyte membranes have shown that their tension can, at the most, only withstand an internal pressure of 0.1 kPa. For this reason, these cells have complicated mechanisms for osmoregulation, including a number of coupled receptors and transport systems.

Formally, the swelling of a cell with a constant content of osmotically active substances can be expressed by the following relation:

$$\pi(V - V_i) = \pi'(V' - V_i). \quad (3.2.29)$$

When there is a change in the osmotic pressure of the external medium from π' to π then the cell volume changes from V' to V . The parameter V_i represents an apparent, osmotically inert i.e. constant volume. Solving Eq. (3.2.29) for V , we obtain:

$$V = \frac{1}{\pi} \pi'(V' - V_i) + V_i. \quad (3.2.30)$$

This is a phenomenological expression of the cellular volume (V) as a function of external osmotic pressure (π). If V is plotted against $1/\pi$, a straight line is obtained with a slope equal to $\pi'(V' - V_i)$, intersecting the ordinate at the point V_i . This osmotic dependence of the volume has been confirmed, with reasonable accuracy, for a number of cells. In Fig. 3.11 it can be seen that for human erythrocytes the osmotically inert volume V_i amounts to 48% of the total volume of the cells under isotonic conditions. This value is too high, because these cells contain 71% water. In fact, this apparent volume is the result of the interplay between different processes. Water loss causes an increase in internal concentrations of all components of the cell. This leads automatically to changes in the charge density of the proteins, and thus to changes in ionic conditions and pH, having a retroactive effect on the osmotic pressure. Consequently, it is not possible to explain the volume regulation of a cell without considering the complete ionic metabolism. In addition, it must be taken into account that the osmotic coefficient of the proteins itself is to a great extent concentration dependent (see Fig. 3.12). We will come back to this problem in connection with the Donnan equilibrium (Sect. 3.2.3, Fig. 3.17).

Fig. 3.11. The relative volume (V_{rel}) of human erythrocytes in a NaCl-phosphate solution in relation to the isotonic condition ($\pi_{isoton} = 0.295$ osmol, $V_{isoton} = 1$), plotted as a function of $1/\pi$ (after Cook 1967)

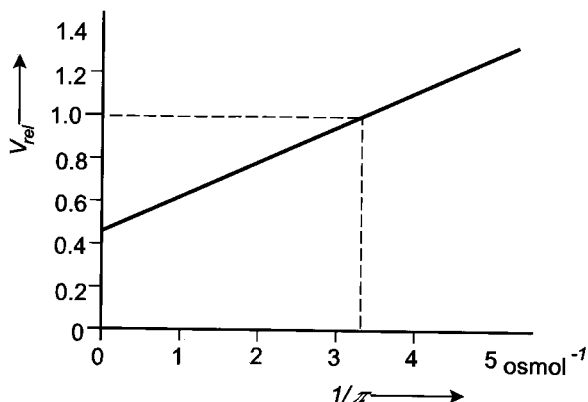
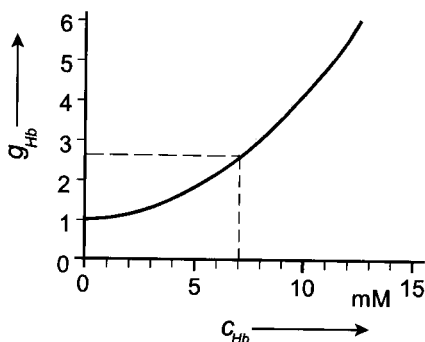


Fig. 3.12. The osmotic coefficient of hemoglobin (g_{Hb}). The dashed lines indicate the region of the in vivo concentration. The curve reflects measurements from sheep and cow hemoglobin of Adair as fitted by Freedman and Hoffman (1979)



The swelling behavior, as described by Eq. (3.2.29), is also intrinsically demonstrable for mitochondria and chloroplasts. Certainly, the reflection coefficients of their membranes are very different from that of the cell membrane. Additionally, even with minor fluctuations of the transport processes of these extremely small organelles with a relatively large surface area, the internal osmolarity may alter quickly.

Further reading: The philosophy of osmosis: Ogston 1966. Swelling of erythrocytes: Freedman and Hoffman 1979; Glaser and Donath 1984; Solomon et al. 1986. Osmoregulation of animal cells: Hoffmann and Simonsen 1989. Osmotic pressure in plants: Zimmermann 1978; Zimmermann and Steudle 1978

3.2.2

Electrochemical Equilibrium – The Nernst Equation

We will now consider a system consisting of two phases, each of which contains a solution with a salt AB , but with different concentrations in the phases I and II

(Fig. 3.13). Let the salt be completely dissociated into its ions A and B . The membrane is semipermeable in that it allows the ion A , but not the ion B , to pass through.

Before analyzing conditions of thermodynamic equilibrium of this system, we will consider the situation qualitatively. Let us suppose that there is an osmotic equilibration between both phases, compensated by electroneutral components. Thus, alteration of concentrations, induced by volume changes, can be neglected. In this system the electrostatic equilibrium will be disturbed because only the ion A , driven by its concentration gradient, but not its counterpart, the ion B can penetrate the membrane. This leads to an increase in the electrical potential difference across the membrane. Eventually, a strong electric field hinders a further diffusion of ion A . Ion A , consequently, will be subject to two opposing forces: on the one hand, the driving force, induced by the concentration gradient, i.e. the gradient of its chemical potential and, on the other hand, an opposing electrostatic force which only arises as a result of its own diffusion. The following equilibrium will be established: a few ions cross the membrane, inducing an electric field which stops further diffusion.

The basic condition for calculating this equilibrium is the equality of the electrochemical potentials of ion A between phases I and II :

$$\tilde{\mu}_A^I = \tilde{\mu}_A^{II}.$$

Substituting the expressions for the electrochemical potentials according to Eq. (3.1.41), one obtains

$$\mu_A^{oI} + RT \ln a_A^I + z_A F \psi^I = \mu_A^{oII} + RT \ln a_A^{II} + z_A F \psi^{II}. \quad (3.2.31)$$

(We will assume isothermal conditions, i.e. $T^I = T^{II} = T$.)

In contrast to the derivation of the equation for the osmotic pressure, in the present case, the standard potentials of these components of phases I and II are equal, because there is no pressure difference ($\mu_A^{oI} = \mu_A^{oII}$).

Taking this into account, and re-arranging Eq. (3.2.31), it gives

$$z_A F (\psi^I - \psi^{II}) = RT (\ln a_A^{II} - \ln a_A^I), \quad (3.2.32)$$

and therefore

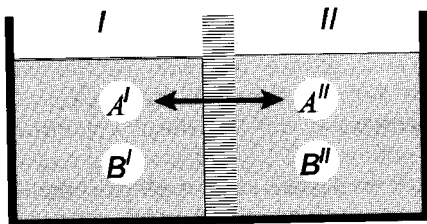


Fig. 3.13. Diagram illustrating the derivation of the Nernst equation. Phases I and II are separated from each other by a membrane which is permeable only for the ion A , but not B , of the salt AB

$$\Delta\psi \equiv (\psi^I - \psi^{II}) = \frac{RT}{z_A F} \ln \frac{a_A^{II}}{a_A^I}. \quad (3.2.33)$$

This is the *Nernst-Equation*. It gives the electrical potential difference ($\Delta\psi$) across the membrane as a function of the chemical activities of the permeating ion in both phases at thermodynamic equilibrium.

Equation (3.2.33) can be re-arranged to show the difference of ion activities (a^I and a^{II}) which builds up in two phases, with a given potential difference ($\Delta\psi$) in between:

$$a_A^I = a_A^{II} e^{-\frac{z_A F \Delta\psi}{RT}}. \quad (3.2.34)$$

Such a relation has already been derived and employed using the Boltzmann equation (Sect. 2.3.15), and applying it to calculate local ion concentrations near charged particles [Eq. (2.4.10)], or in electric double layers [Eq. (2.5.6)]. In these cases, however, the concentrations (c_i) were used instead of the chemical activities (a_i). This is allowed only for ideal, i.e. diluted, solutions, or when the activity coefficients (f_i) are equal in both phases.

The Nernst equation, therefore, permits the calculation on the one hand, of the distribution of ions as a function of the electrical potential [Eq. (3.2.34)] and, on the other hand, the electrical potential, which is induced by an unequal distribution of ions [Eq. (3.2.33)]. For both cases, however, thermodynamic equilibrium is required!

The separation of the two phases of the system by a semipermeable membrane, as discussed here, reflects just one special example. In the case of ion distribution in electric double layers, or ionic clouds, as described in Sections 2.5.4 and 2.5.5, the electric potential gradient is predicted by the fixed charges, and the distributions of the ions are not limited by a membrane. Here, both the anions and the cations in the case of equilibrium, are distributed according to this equation.

All equations derived in the present section are applicable only for thermodynamic equilibria. This means that the Nernst equation cannot be used to calculate the membrane potential of a living cell. Actually, the membrane potential of a living cell is either a diffusion potential (liquid junction potential), or it is generated by electrogenic ion pumps (see Sects. 3.4.2 and 3.4.3). On the other hand, despite the nonequilibrium distribution of the ions in the cell in general, it is quite possible that some types of ions may be distributed passively and then they are actually in equilibrium.

An example of this is the chloride concentration of most animal cells. Because its membrane permeability is rather fast, and because no chloride pumps in the membrane exist, its distribution is mostly passive, and predicted by the existing transmembrane potential. The Nernst equation, therefore, allows the calculation of the internal chloride concentration, if the external chloride concentration and the transmembrane potential are known. On the other hand, knowing, for example, the distribution of chloride inside and outside the cells, one can calculate the transmembrane potential of the cell. In this context, however, it is

important to stress that in this case the chloride distribution is just an *indicator* of the membrane potential, but not the *reason* for it!

This consideration allows the establishment of a method for measuring the transmembrane potential of cells without microelectrodes. The chloride distribution can be easily determined using the radioisotope ^{36}Cl . Sometimes one can use other small charged organic molecules, which penetrate the membrane quickly, and are labeled by ^3H or ^{14}C . Knowing the distribution of these ions, the membrane potential ($\Delta\psi$) can be calculated according to Eq. (3.2.33).

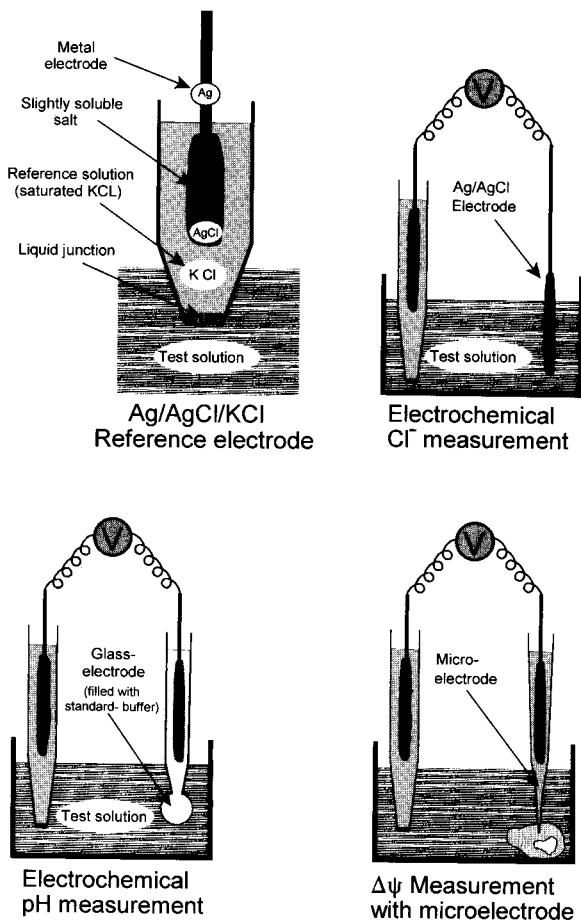
Furthermore, the Nernst equation allows the calculation of electrode potentials. If a metal is dipped into an electrolyte solution, then cations are detached from the metal lattice and an electrochemical equilibrium is obtained. In general, this is the same situation as discussed above. The resulting electrical potential difference between the metal and the solution is called *electrode potential*. If the cation from the metal of the electrode forms a poorly soluble salt with an anion of the solution, and if the concentration of this anion is much greater than that of the cation, then the electrode potential, linked through the soluble product of the salt, is directly determined by the chemical activity of this anion. In this way, for example, a silver electrode covered by AgCl can be used to measure the Cl^- activity of a solution (see Fig. 3.14). The voltage of this electrode with respect to a reference electrode, according to the Nernst equation [Eq. (3.2.33)], is proportional to the logarithm of the Cl^- activity in the solution. It is important to emphasize that these kinds of electrochemical methods allow the measurement of ion activities (a_i), in contrast to most chemical methods, indicating its concentrations (c_i).

The electrochemical measurement of ion activity and of many other chemical substances has become an important and universal technique. Using special semipermeable membranes, or water impermeable layers containing special ionophores, electrodes can be made which are highly selective in measuring the activity of special chemical components. In this case, these membranes or microphases separate a reference solution of known composition from the test solution. The potential difference between the reference solution and the test solution is measured as the voltage between two reference electrodes dipped in both phases.

A typical example of this kind of measurement is the usual pH electrode (see Fig. 3.14). In this case, a thin membrane made of special glass allows the protons to equilibrate between both phases. If, as a reference phase, the pH electrode is filled by a buffer holding the pH constant, then the voltage between the pH electrode and the reference electrode indicates the pH of the test solution. Usually, combinations between pH and reference electrodes are used, consisting of one single glass body.

For electrochemical methods of analysis, a pair of electrodes is always necessary. The electromotive force will be measured between a selective measuring electrode and a so-called reference electrode. A *reference electrode* is an electrode that exhibits a constant electrode potential which is independent of the composition of the solution into which it is immersed. If a pair of identical reference electrodes is used, then the electrode potentials will mutually oppose

Fig. 3.14. The construction of an Ag/AgCl/KCl-reference electrode, as well as its application for electrochemical measurements of Cl^- activity and pH, and as a reference electrode in electrophysiology



each other and therefore the direct potential difference, i.e. the actual electromotive force between the two phases can be measured. If two different reference electrodes are used, a constant $\Delta\psi$ is superimposed, which can be subtracted from the measured value.

To make a reference electrode, a metal electrode is covered with a slightly soluble salt of this metal. This is immersed into a reference solution of constant composition, which is connected to the test solution by means of a pathway, where little, if any, convection can take place (liquid junction). This liquid junction is formed in different ways. It can be a tube, filled by an agar gel, however, in industrially produced electrodes, it is mostly just an opening in the glass wall, covered by a small filter made from sintered glass. Sometimes a ground glass stopper is used with a liquid film in between.

The most common types of reference electrodes are Ag/AgCl/KCl electrodes (see Fig. 3.14). In this case, a saturated solution containing 4.6 M KCl as a

reference solution is used. KCl is chosen to avoid diffusion potentials at the tip of the electrodes (for diffusion potential, see Sect. 3.3.3). In any case, a concentration gradient exists between the reference solution and the test solution. This could become a source of an electrical potential difference if the mobilities of both ions are different. The mobility of the K^+ and Cl^- ions in aqueous solutions, however, is almost equal. This means that even for strong concentration gradients between the reference solution and the test solution, no diffusion potential can occur. This would be not the case if, for example, NaCl was used as the reference solution.

Unfortunately, this useful property of K^+ and Cl^- ions is valid only in pure solutions. If a reference electrode is immersed, for example, in suspensions of charged particles, a so-called *suspension effect* occurs. Under the influence of the electric double layers of these particles, the mobility of the ions may change. Therefore small diffusion potentials can occur. This effect, for example, is important in pH measurement in blood. Moreover, it can be the source of errors in measurements with microelectrodes in cells.

Microelectrodes, as applied in electrophysiology, are usually Ag/AgCl/KCl electrodes. They are constructed from glass tubes which are drawn out at one end to a very fine capillary (diameter $< 1 \mu m$). In this case, no further diffusion limitation in the tip is necessary. Because of the electric charge of the glass and the extremely thin glass wall at the tip, so-called *tip potentials* of microelectrodes occur which in worst cases can be as large as several millivolts. Microelectrodes can also be sealed with ion-selective materials so that intracellular ionic activity can be measured directly.

Further reading: Electrodes in general: Varma and Selman 1991; microelectrodes: Amman 1986

3.2.3

The Donnan Equilibrium

The Donnan state represents an equilibrium between two phases, containing not only small anions (A) and cations (C), both of which can penetrate the membrane, but also charged molecules or particles (M) for which the membrane is impermeable (see Fig. 3.15). This type of equilibrium was investigated by F. G. Donnan in 1911.

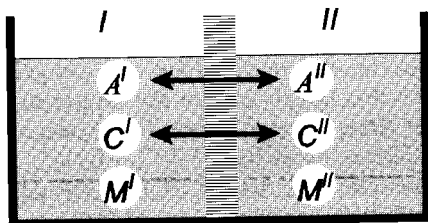


Fig. 3.15. The derivation of the Donnan equilibrium. Phases I and II are separated from each other by a membrane that is permeable for the anions (A) and the cations (C), but not for the charged molecules M

The Donnan equilibrium, like the Nernst equilibrium which was explained in the previous section, is important for calculating the distribution of ions in the living cell. It requires that these ions are distributed according to the thermodynamic equilibrium. In each case it must be decided which sorts of charged components are exchangeable (like A and C), and which of them are nonexchangeable (like M).

If the cell membrane is opened for ions by an ionophore, by a toxin, or by any other influence, then only organic molecules, like proteins, are in the class of nonexchangeable charged components (M). A living cell will also shift slowly toward a Donnan equilibrium, if, as a result of low temperature or a chemical blocker, the ATP-driven ion pumps are inhibited. An example of this is the distribution of ions in erythrocytes in blood stored over a long period of time.

A Donnan equilibrium can be established also as a sort of quasi-equilibrium state (see Sect. 3.1.4). Figure 3.8 demonstrates the time hierarchy of characteristic rate constants of various biological processes. For illustration, let us consider the following situation. In human erythrocytes, for example, the distribution of Cl^- ions between the internal and external media is passive, i.e. not influenced by an active pump. The same holds for the equilibration of the pH. These processes of equilibration are very fast. The whole erythrocyte itself, however, in vivo is not in equilibrium, because sodium and potassium are pumped by Na-K-ATPase. Nevertheless, the distribution of the quickly exchangeable Cl^- ions and the pH can be calculated according to the Donnan equilibrium, if Na^+ and K^+ are considered as quasi nonpermeable charges (M). This state is called a "quasi" equilibrium state, because actually for a longer time interval, the concentration of potassium and sodium may shift. This type of time hierarchy was discussed already in Section 3.1.4 (Fig. 3.8). The equilibrium of Cl^- and the pH, therefore, follows the shifting Na^+ and K^+ concentration and can be considered as constant only for a limited time of observation.

Let us now consider such a system, consisting of permeable ions (index i), and nonpermeable charged components (index m). Let us denote the concentration, or the activity of the components inside the cell, with c^I or a^I , and in the external solution with c^{II} , or a^{II} respectively. The parameter z_m denotes the number and the sign of charges of the nonpermeable molecule.

It must be decided whether an isobar ($\Delta p = 0$) or whether an isochor ($\Delta V = 0$) system will be considered. The first case is suitable for investigating an animal cell, which is able to compensate for changes in the volume by swelling or shrinking. The second situation, to some extent, can be applied to plant cells, with the possibility of increasing the internal pressure considerably. Further, let us assume Staverman coefficients of the nonpermeable components are $\sigma_i = 1$.

In the case of animal cells, which is easier to handle, the Donnan equilibrium is determined by the following conditions:

- All permeable ions (i) are distributed in equilibrium according to Eq. (3.2.34):

$$a_i^I = a_i^{II} e^{-\frac{z_i F \Delta \psi}{RT}}. \quad (3.2.35)$$

- In both phases the sum of charges is zero (electroneutrality condition):

$$\sum z_i c_i + \sum z_m c_m = 0. \quad (3.2.36)$$

- Water between both phases is distributed according to its equilibrium.

$$\Delta\pi = 0 \quad (\text{for: } \Delta p = 0). \quad (3.2.37)$$

These conditions allow us now to create equations for particular cases and to solve them for the corresponding parameters. The conditions (3.2.35) allow the calculation of the relation of the activities of the permeating ions inside and outside the cell. If the activities of univalent cations are denoted by a_{iC} (for $z_{iC} = +1$), and correspondingly for anions a_{iA} (for $z_{iA} = -1$), it follows:

$$\frac{a_A^I}{a_A^{II}} = \frac{a_C^{II}}{a_C^I} = e^{\frac{F\Delta\psi}{RT}} \equiv r. \quad (3.2.38)$$

The parameter r is known as the *Donnan ratio*. According to Eq. (3.2.38), it is related to the Donnan potential in the following way:

$$\Delta\psi = \frac{RT}{F} \ln r. \quad (3.2.39)$$

The Donnan potential ($\Delta\psi$) is substantially determined by the amount of nonpermeable charged components of the cell, as reflected in the condition (3.2.36). Let us now consider the situation with just a single kind of cation (C, with $z_C = +1$), and a single kind of anion (A, with $z_A = -1$), and a singly charged component (M, with z_M) inside the cell. In this case, the equation of electroneutrality of both phases, according Eq. (3.2.36) can be written easily:

$$\begin{cases} c_C^I - c_A^I + z_M^I c_M^I = 0 \\ c_C^{II} - c_A^{II} = 0 \end{cases}. \quad (3.2.40)$$

This system of equations can be solved by reorganizing it and dividing one by the other:

$$\frac{c_A^I}{c_A^{II}} = \frac{c_C^I + z_M^I c_M^I}{c_C^{II}}. \quad (3.2.41)$$

If the activity coefficients of the ions in both phases are equal, one can substitute the Donnan ratio [Eq. (3.2.38)] in this equation:

$$r = \frac{1}{r} + \frac{z_M^I c_M^I}{c_C^{II}}. \quad (3.2.42)$$

Re-arranging this, a simple quadratic equation for r is obtained:

$$r^2 - \frac{z_M^I c_M^I}{c_C^{II}} r - 1 = 0, \quad (3.2.43)$$

which can be solved in the usual way:

$$r = \frac{z_M^I c_M^I}{2c_C^{II}} \pm \sqrt{\left(\frac{z_M^I c_M^I}{2c_C^{II}}\right)^2 + 1}. \quad (3.2.44)$$

The dependence of the Donnan ratio (r) on the expression $z_M^I c_M^I / 2c_C^{II}$ is shown in Fig. 3.16. Considering the corresponding definition (3.2.38), it is clear that the negative values of the Donnan ratio (r) have no real meaning. Therefore, only the positive roots of Eq. (3.2.44) are of interest. For the case $z_M^I c_M^I = 0$ it becomes $r = 1$ and Eq. (3.2.39) gives $\Delta\psi = 0$. If the impermeable components are negatively charged ($z_M^I < 0$), then $r < 1$. This gives a negative membrane potential ($\psi < 0$). On the other hand, the value $z_M^I > 0$ means $r > 1$ and therefore $\Delta\psi > 0$.

A reduction of the ionic strength in the external solution, i.e. a reduction of c_C^{II} , leads to an increase in the absolute value of the ratio $z_M^I c_M^I / 2c_C^{II}$. If $z_M^I < 0$, this will lead to a shift of $\Delta\psi$ in a negative direction, and vice versa, if $z_M^I > 0$, a reduction of the ionic strength causes an increase in the Donnan potential. The absolute value of the Donnan potential, therefore, increases when the charge concentration ($z_M^I c_M^I$) is increased, as well as when the ionic strength in the external solution (c_C^{II}) is reduced.

Equation (3.2.44) allows us to understand the behavior of a Donnan system in general, but in fact, it reflects an extremely simplified system. It has not been

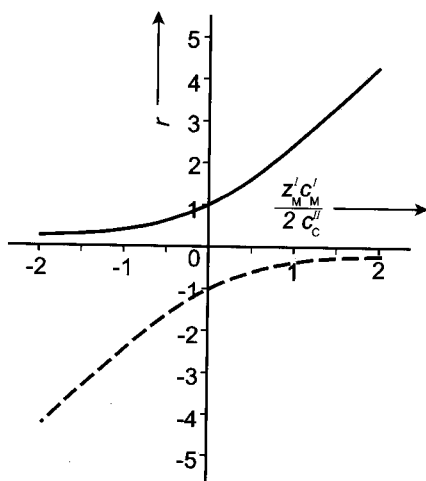


Fig. 3.16. Plot of Eq. (3.2.44). Solid curve positive, dashed curve negative values of the root expression

taken into account that according to the condition of iso-osmolarity [Eq. (3.2.37)], alterations of the cell volume could occur. This would lead to alterations of all intracellular concentrations.

In addition, as determined by the Donnan potential, a pH equilibrium between the two phases in this system could be obtained. This effect has already been discussed in connection with the local pH values at charged surfaces (Sect. 2.5.4). Retroactively, this pH alteration can influence the Donnan potential, because the charges of organic molecules are usually pH-dependent (see Sect. 2.4.6, Figs. 2.34 and 2.35). In the simplest way, this dependence can be expressed as:

$$z_M = -z_{M0}(\text{pH} - \text{pH}_{\text{iso}}). \quad (3.2.45)$$

Near the isoelectric point ($\text{pH} = \text{pH}_{\text{iso}}$) the total molecule is uncharged ($z_M = 0$). Below this point ($\text{pH} < \text{pH}_{\text{iso}}$), z_M becomes positive, and above it ($\text{pH} > \text{pH}_{\text{iso}}$), z_M becomes negative.

Considering all of these relations, one obtains a system of non-linear equations that can be solved by iteration. Some basic properties of such a system will be demonstrated for the case of Donnan equilibrium of human erythrocytes. As already mentioned, this state can occur under conditions of blood storage or as a result of other treatment.

The most important non-penetrating charge component in erythrocytes is hemoglobin. In vivo, it has a concentration of 7 mM. Its isoelectric point $\text{pH}_{\text{iso}} = 6.8$ (at 25 °C) and $z_{M0} = 10.5$ eq/mol. The Donnan potential of erythrocytes in solutions of various NaCl concentrations containing sucrose to correct differences in the osmotic pressure, is shown in Fig. 3.17. The intracellular pH of the erythrocytes depends on the pH value of the external solution which is indicated on the abscissa, and on the Donnan potential itself. At the isoelectric point of hemoglobin, the Donnan potential: $\Delta\psi = 0$. The highest potentials, both negative and positive, are obtained at the greatest distances from the isoelectric point, and in solutions with the lowest NaCl concentrations.

The volume changes in these cells are shown in Fig. 3.18. The volume, as the percentage of the in vivo volume of erythrocytes, is indicated as a function of

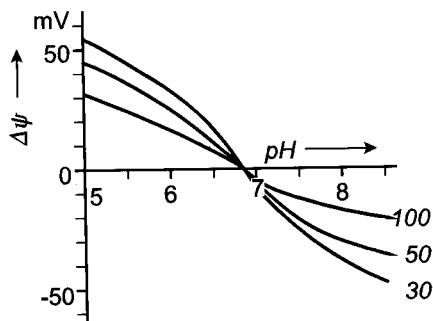


Fig. 3.17. The Donnan potential ($\Delta\psi$) of human erythrocytes in isotonic NaCl-sucrose solutions as a function of the external pH. The curves represent situations for NaCl solutions of 30, 50, and 100 mM ($\pi = 300$ mosmol, $T = 25$ °C). (After Glaser et al. 1980)

the NaCl concentration of the external solution at constant pH = 7.4. It is important to note that independently of the external NaCl concentration, the osmotic pressure is always adjusted to the isotonic osmotic pressure, $\pi = 300$ mosmol. In spite of this, the cells shrink in solutions of low ionic strength! An isotonic condition of an incubation medium, therefore, is no guarantee for the maintenance of a normal cell volume! A normal volume is achieved only in a solution containing about 20 mosmol sucrose and 140 mM NaCl. In this case, the sucrose just compensates the osmotic pressure of the hemoglobin, which has a comparatively high osmotic coefficient (see Fig. 3.12). Furthermore, it is interesting that the volume curve rises steeply at high ionic strengths. Mathematically, no results can be obtained from the equations assuming conditions of pure 152 mM NaCl solutions. In this case, the volume would become infinitely large. The experiments indicate that erythrocytes in solutions of pure electrolytes undergo hemolysis, if the membrane becomes permeable for these ions. This is known as *Donnan osmotic hemolysis*. As indicated in Fig. 3.18, this can occur even in isotonic solutions.

To determine experimentally whether a cell is in a state of Donnan equilibrium, the relation between the internal and external ionic activities must be ascertained. For a Donnan state to be present, the Donnan ratios from Eq. (3.2.38) must correspond. For most cells in vivo, the ratio obtained using the sum of the sodium and potassium ions $[(a_K^I + a_{Na}^I)/(a_K^{II} + a_{Na}^{II})]$ may be close to that for a Donnan equilibrium, but this would not be the case for either of the individual ions. Active transport changes their relative concentrations in opposing directions.

Further reading: Overbeek 1956; Glaser and Donath 1984

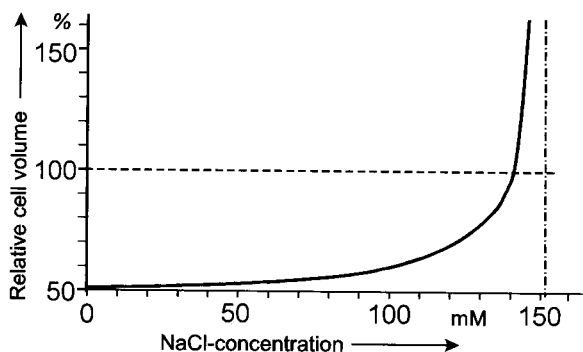


Fig. 3.18. Donnan-osmotic alterations of the relative volume (V) of human erythrocytes as a function of external NaCl concentration in isotonic NaCl-sucrose solutions (pH = 7.4, $T = 25^\circ\text{C}$). The osmotic pressure of the solutions with different NaCl-concentrations is always balanced by sucrose according to $\pi = 300$ mosmol. At $c = 152$ mM, the solution contains only NaCl, with no sucrose. The volume is given as the percentage of the in vivo volume of erythrocytes. (After Glaser et al. 1980, redrawn)

3.3

The Thermodynamic Analysis of Fluxes

Before we discuss the particular properties of the transport of molecules and ions through biological membranes, we must consider some general thermodynamic equations which underlie all diffusion processes. Later we will see that these equations are also applicable for biological membranes, although simple diffusion processes can be considered as rather exceptional cases in biological membranes.

A flux, as defined in Section 3.1.3, is the amount of a substance which passes in a perpendicular direction through a definite area of a surface in a unit of time. Its basic measure, therefore, is: $\text{kg s}^{-1} \text{ m}^{-2}$, or better: $\text{mol s}^{-1} \text{ m}^{-2}$. In the following discussion, the molar flux of a substance will be denoted by the symbol J .

When considering fluxes through cell membranes, difficulties may arise in some cases if the exact surface area of a cell is not known. In these cases modified units are used, such as: 'mol s^{-1} per cell', 'mol s^{-1} per liter of cells', 'mol s^{-1} per liter of cell-water' etc. As unit of time, instead of seconds, minutes, hours, or even days are frequently used. Such units are quite convenient for some physiological investigations. However, it is not possible to substitute such units directly into the thermodynamic equations.

3.3.1

The Flux of Uncharged Substances

Let us first consider the diffusion of an uncharged substance (i), neglecting any coupling with other fluxes. Using Eqs. (3.1.45) and (3.1.49), gives

$$J_i = L_i X_i = -L_i \text{grad } \mu_i = -L_i \text{grad } (\mu_i^0 + RT \ln a_i). \quad (3.3.1)$$

In order to calculate the gradient of the chemical potential, the dependence of the parameters on their position in space must be considered. Assuming constant temperature ($\text{grad } T = 0$), and constant pressure ($\text{grad } p = 0$), the gradient of the standard potential ($\text{grad } \mu_i^0$) also becomes zero. Equation (3.3.1) therefore can be written as:

$$J_i = -L_i RT \text{grad } \ln a_i = -\frac{L_i RT}{a_i} \text{grad } a_i. \quad (3.3.2)$$

(To explain this rearrangement: the differential operator 'grad' can be handled like a deviation d/dx . Then the rules of sequential differentiation are applied.)

As has already been discussed in Section 3.1.3, a flux can also be expressed by the parameters concentration (c_i) and velocity (v_i) [Eq. (3.1.47)]. From this we came to an equation which includes mobility (ω_i) [Eq. (3.1.53)]. If we consider, furthermore, that X_i is not a simple mechanical force, but results from the gradient of a chemical potential determined by molar concentrations, the Avogadro number (N) must be included.

$$J_i = \frac{c_i w_i}{N} X_i. \quad (3.3.3)$$

By comparing Eq. (3.3.1) and (3.3.3), we obtain:

$$L_i = \frac{c_i w_i}{N}. \quad (3.3.4)$$

Suppose: $c_i \approx a_i$, then the combination of Eq. (3.3.4) with Eq. (3.3.2) gives:

$$J_i = -\frac{w_i}{N} RT \text{grad } c_i = -w_i kT \text{grad } c_i. \quad (3.3.5)$$

Introducing the diffusion coefficient, one gets *Fick's First Law of Diffusion*:

$$J_i = -D_i \text{grad } c_i. \quad (3.3.6)$$

If there is only a 1-dimensional concentration gradient in the x -direction, this equation simplifies to:

$$J_{ix} = -D_i \frac{dc_i}{dx}. \quad (3.3.7)$$

The mobility (ω_i) of an almost spherical molecule can be interpreted mechanically by referring to Stokes' law [Eq. (3.1.55)]. By definition it has the units: $\text{m s}^{-1} \text{N}^{-1}$. In Section 2.3.6 we introduced the diffusion coefficient (D_i) in the context of the translational movement of molecules. This led to relations between the diffusion coefficient, mobility (ω_i), and molar mass (M_i) [see: Eqs. (2.3.38) to (2.3.40)].

We introduced the diffusion coefficient as: $D_i = \omega_i kT$. Its unit, therefore, is: $\text{m}^2 \text{s}^{-1}$. Using the measure: $\text{mol m}^{-3} \text{m}^{-1} = \text{mol m}^{-4}$ for the concentration gradient in Eq. (3.3.6), or (3.3.7) one gets the units: $\text{mol s}^{-1} \text{m}^{-2}$ for flux, according to its definition.

In addition to Fick's first law of diffusion, which gives the flux as a function of the concentration gradient, *Fick's second law* allows us to determine the establishment of a concentration gradient of a given substance (c_i) as a function of time (t). It is a partial differential equation of the second order which, for diffusion in one direction, is:

$$\left(\frac{\partial c_i}{\partial t} \right)_x = D \left(\frac{\partial^2 c_i}{\partial x^2} \right)_t. \quad (3.3.8)$$

This equation is used to calculate a concentration gradient which occurs when diffusion takes place in a homogeneous phase.

In contrast to this case where a continuous gradient of concentration occurs, in membrane systems various discontinuities in the concentration are to be expected. This means that the function $c_i(x)$ could become rather complicated. Schematically, this is illustrated in Fig. 3.19. The simplest case is shown by the

solid line where phase I contains a solution with concentration c^I and correspondingly, c^{II} is the concentration in phase II. The concentration in the membranes falls linearly with a slope of $\Delta c/\Delta x$.

The broken line in Fig. 3.19 shows an irregular concentration pattern. In this case effects are considered which can occur at the membrane surface as well as in its interior. The deviation A marks an adsorption of ions in an electrical double layer. The effective thickness of this layer is very small, being less than 10 nm. Inside the membrane (Fig. 3.19B), deviations of the linear concentration profile can occur by differences of the mobility of the substance in the x -direction or, in the case of ion transport, even by dielectric inhomogeneities.

In special cases *diffusion layers*, also called *unstirred* or *boundary layers*, may occur near the surface of membranes (Fig. 3.19C). These are near membrane regions without streaming or convection. In these regions, substances transported through the membrane, or involved in a chemical reaction, can move only by diffusion. This can produce an increase or a decrease in local concentrations. It depends simply on the relationship between the transport or reaction rate through the membrane, introducing substance into this region, and the rate of taking it away by diffusion. In this case a stationary concentration gradient is built up.

In systems with artificial ion exchange membranes, large diffusional layers can be indicated by special interference methods or by microelectrodes. In contrast to some speculations in earlier papers, *in vivo* such layers are mostly much lower than $1\text{ }\mu\text{m}$. These layers may substantially affect biochemical reactions or transport processes of biological membranes. They can become important especially in cases where the cell surface is covered by microvilli or special caverns, or where reactions take place in the intercellular space. Even the occurrence of diffusional layers of protons near membranes is discussed.

Let us now consider the simplest case, represented by the solid line in Fig. 3.19 in more detail. Let the concentration gradient (dc_i/dx) inside the membrane be constant and equal $\Delta c_i/\Delta x$, whereas: $\Delta c_i = c_i^{II} - c_i^I$. In this case Eq. (3.3.7) becomes

$$J_i = -D_i \frac{\Delta c_i}{\Delta x} \equiv -P_i \Delta c_i. \quad (3.3.9)$$

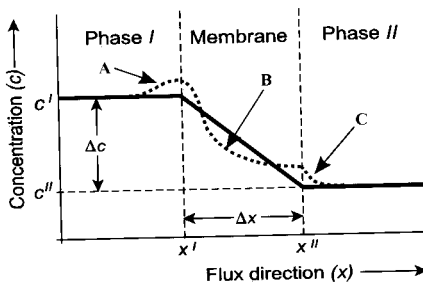


Fig. 3.19. Possible functions $c(x)$ in a membrane system. — – ideal case, ---- – disturbed case, caused by the following reasons: A adsorption of substance at the membrane surface; B differences in the mobility of the substance inside the membrane; C diffusion layer caused by imperfect stirring at the surface

The parameter $P_i = D_i/\Delta x$ is the permeability coefficient measured in m s^{-1} . The same parameter will be used in Section 3.3.3 to calculate fluxes of ions. It is important to stress that this is the same parameter with an identical definition.

Let us now consider a system with flux interactions. In this case the flux coupling must be taken into consideration as discussed in Section 3.1.3. To demonstrate these approaches, we will consider only the simplest case of a binary system, represented for example by a flux of an uncharged substance (J_s) and its solvent, the water (J_w). The driving forces of both fluxes are the negative gradients of their chemical potentials. To simplify the derivation we will use simply their differences $\Delta\mu_s$ and $\Delta\mu_w$ as driving forces.

In a first step we must write down the equation for the dissipation function according to the rule of Eq. (3.1.64):

$$\Phi = J_w \Delta\mu_w + J_s \Delta\mu_s. \quad (3.3.10)$$

In this case Φ is an integral parameter for the whole membrane thickness Δx .

In the next step we will modify this equation in such a way that instead of the parameters $\Delta\mu_s$ and $\Delta\mu_w$, forces are introduced which are directly measurable.

Let us first consider the difference of chemical potential of the water ($\Delta\mu_w$):

$$\Delta\mu_w = \Delta\mu_{wx}^0 + RT \ln \frac{x_w^I}{x_w^{II}}. \quad (3.3.11)$$

In Section 3.2.1 we discussed the chemical potential of water in detail and considered in particular its dependence on the mole fraction (x_w) as well as on pressure (p). Now we will make use of these derivations. Using Eqs. (3.2.12) and (3.2.13), as well as the definition of the osmotic pressure, we come to:

$$\Delta\mu_w^0 = \bar{V}_w \Delta p \quad \text{and} \quad RT \ln \frac{x_w^I}{x_w^{II}} = -\bar{V}_w \Delta\pi, \quad (3.3.12)$$

where $\bar{V}$ is the partial volume of water, Δp the difference of hydrostatic pressure and $\Delta\pi$ the difference of osmotic pressure. Equation (3.3.11) therefore can be rewritten as:

$$\Delta\mu_w = \bar{V}_w (\Delta p - \Delta\pi). \quad (3.3.13)$$

Let us now consider the other chemical potential difference in Eq. (3.3.10), namely $\Delta\mu_s$. This parameter depends on pressure difference in the same way as the difference of the chemical potential of water. According to Eqs. (3.3.11) and (3.3.12) one can write:

$$\Delta\mu_s = \bar{V}_s \Delta p + RT \ln \frac{c_s^I}{c_s^{II}}. \quad (3.3.14)$$

The second term of this equation can be expanded as a series using the common rule for parameters $x > 0$:

$$\ln x = 2 \left[\left(\frac{x-1}{x+1} \right) + \frac{1}{3} \left(\frac{x-1}{x+1} \right)^3 + \frac{1}{5} \left(\frac{x-1}{x+1} \right)^5 + \dots \right]. \quad (3.3.15)$$

Now we substitute for x the value c^I/c^II . With good approximation we are justified in retaining only the first term of this series:

$$\ln \frac{c^I}{c^{II}} = 2 \left(\frac{\frac{c^I}{c^{II}} - 1}{\frac{c^I}{c^{II}} + 1} \right) = \frac{c^I - c^{II}}{\frac{c^I + c^{II}}{2}} = \frac{\Delta c}{\bar{c}}. \quad (3.3.16)$$

This shows that the logarithm of the ratio of the two concentrations, or even activities, can be replaced by the difference in concentration (Δc) divided by the arithmetic mean ($\bar{c}$) of the concentrations of the two phases. This connection is not only true approximately, as seen by this expansion of the series, but it can be proved mathematically that it is exactly equal.

Relation (3.3.16), applied to Eq. (3.3.14), together with the Van't Hoff equation for the osmotic pressure: $\Delta\pi = RT\Delta c$ [Eq. (3.2.23)], gives:

$$\Delta\mu_s = \bar{V}_s \Delta p + \frac{1}{\bar{c}_s} \Delta\pi. \quad (3.3.17)$$

Now we have really found reasonable expressions for the differences of the chemical potentials, and using them we can rearrange the equation for the dissipation function [Eq. (3.3.10)]. Introducing Eqs. (3.3.14) and (3.3.17) into Eq. (3.3.10) we get:

$$\Phi = J_w \bar{V}_w (\Delta p - \Delta\pi) + J_s \left(\bar{V}_s \Delta p + \frac{1}{\bar{c}_s} \Delta\pi \right), \quad (3.3.18)$$

and after rearrangement:

$$\Phi = \Delta p (J_w \bar{V}_w + J_s \bar{V}_s) + \Delta\pi \left(\frac{J_s}{\bar{c}_s} - J_w \bar{V}_w \right). \quad (3.3.19)$$

The expressions in parentheses can be regarded as new flux variables. Flux was defined as the amount of a substance in moles that traverses a surface in a unit of time. Multiplying this parameter by the partial molar volume ($\bar{V}$), one gets a volume flow. Let us define a total volume flux (J_V) as the sum of individual volume fluxes of all components:

$$J_V = J_w \bar{V}_w + J_s \bar{V}_s. \quad (3.3.20)$$

To illustrate the meaning of the second term of Eq. (3.3.19), let us remember the relation $J_i = c_i \mathbf{v}_i$ [Eq. (3.1.47)]. The term: $J_s/\bar{c}_s$ in Eq. (3.3.19) therefore is an expression for the velocity of the substance ($\mathbf{v}_s$). The volume flow of the solvent ($J_w \bar{V}_w$) can also be considered as the velocity of it ($\mathbf{v}_w$). From this it follows:

$$\frac{J_s}{\bar{c}_s} - J_w \bar{V}_w = \mathbf{v}_s - \mathbf{v}_w \equiv J_D. \quad (3.3.21)$$

The parameter J_D , called *exchange flux* is therefore the difference between the velocity of solute relative to the solvent.

As a result of these transformations we finally get a dissipation function which contains measurable parameters:

$$\Phi = J_V \Delta p + J_D \Delta \pi. \quad (3.3.22)$$

This makes it possible to write down a flux matrix following the pattern of Eq. (3.1.58):

$$\begin{aligned} J_V &= L_V \Delta p + L_{VD} \Delta \pi \\ J_D &= L_{DV} \Delta p + L_D \Delta \pi. \end{aligned} \quad (3.3.23)$$

This equation shows that in case of the transport of a solution with a single uncharged substance, the membrane permeability is determined by four coefficients: L_V , L_{VD} , L_{DV} and L_D . The driving forces are the osmotic ($\Delta \pi$), and the hydrostatic (Δp) pressures.

Some of the coefficients used in these phenomenological equations can easily be illustrated: the parameter L_V shows for example how fast the solution passes through the membrane in response to a hydrostatic pressure difference (Δp). Introducing the conditions $\Delta \pi = 0$ and $\Delta p > 0$ in the upper part of Eq. (3.3.23) it becomes: $L_V = J_V / \Delta p$. This means that L_V is a mechanical *filtration coefficient* or a kind of hydraulic conductivity.

Under the same conditions the substance can also be forced through a filter. In this case the second line in Eq. (3.3.23) gives $L_{DV} = J_D / \Delta p$. Because of this, L_{DV} is called *ultrafiltration coefficient*.

The flux matrix (3.3.23) makes it possible to describe the time dependence of an osmotic system. Figure 3.20 shows how the pressure in a Pfeffer's cell,

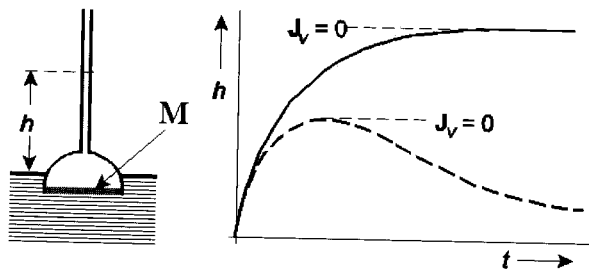


Fig. 3.20. The height of the water column in a Pfeffer osmometer depending on time in two experiments with different conditions: Pfeffer's cell is closed by a semipermeable membrane (M) (—) and solvent as well as solute can penetrate the membrane (- - -). The conditions with stationary pressure in this manometer ($dh/dt = 0$, i.e. $dp/dt = 0$) simultaneously means stationary volume ($dV/dt = J_V = 0$)

measured by the height of the water column in the vertical tube, changes with time. Only in the case of a semipermeable membrane (solid line), will a thermodynamic equilibrium with a constant pressure difference be achieved. If, in addition to the solvent, the membrane allows some of the solute to pass through then there will be a decline in the pressure after the initial rise. The osmotic pressure inside the osmometer will continuously fall. In this case a state without any volume flux ($dV/dt = J_V = 0$) is achieved only for a short time. For this situation it follows from the first line of the flux matrix [Eq. (3.3.23)]:

$$L_V \Delta p + L_{VD} \Delta \pi = 0, \quad (3.3.24)$$

and furthermore:

$$(\Delta p)_{J_V=0} = -\frac{L_{VD}}{L_V} \Delta \pi. \quad (3.3.25)$$

In Section 3.2.1 we introduced Staverman's reflection coefficient (σ). Relating Eq. (3.2.27) with Eq. (3.3.25), the following connection between the reflection coefficient and the coupling coefficients results:

$$\sigma = -\frac{L_{VD}}{L_V}. \quad (3.3.26)$$

For a semipermeable membrane with: $\sigma = 1$, it therefore holds that:

$$L_{VD} = -L_V.$$

A better illustration of this parameter allows the following consideration: a solution will be forced through a membrane by a hydrostatic pressure ($\Delta p > 0$) without any osmotic difference ($\Delta \pi = 0$). Using these conditions and dividing the second equation of the flux matrix (3.3.23) by the first, we obtain:

$$\frac{J_D}{J_V} = \frac{L_{DV}}{L_V} = -\sigma. \quad (3.3.27)$$

According to Eq. (3.3.21), J_D can be replaced by the difference ($\mathbf{v}_s - \mathbf{v}_w$). For very dilute solutions it holds: $J_w \bar{V}_w \gg J_s \bar{V}_s$. In this case Eq. (3.2.27) can be written as:

$$\sigma = \frac{\mathbf{v}_w - \mathbf{v}_s}{\mathbf{v}_w}. \quad (3.3.28)$$

This equation has already been discussed in Section 3.2.1.

The deviations shown here clearly demonstrate that the consideration of more complex systems, for example solutions with more components, would lead to a huge number of parameters and relations.

This theory of coupled fluxes has been widely used to explain the stationary volume and stationary pressure observed in experiments with cells under non-equilibrium conditions. Such experiments have been carried out mostly on red blood cells and on plant cells. Human red blood cells, placed initially in an isotonic solution, shrink immediately when the osmotic pressure of the external solution is increased by adding a permeable substance i ($\sigma_i < 1$). Then, however, if the substance passes through the membrane, the cell usually reverts to its initial volume, as illustrated in Fig. 3.20 by the dashed line. Such experiments with fast volume changes are undertaken to determine the reflection coefficient using stop-flow techniques.

Further reading: Katchalsky and Curran 1965; Kotyk et al. 1988; Stein 1990; Zimmermann 1989. Recent papers on unstirred diffusion layers: Barry 1998; Evtodienko et al. 1998; Pohl et al. 1998

3.3.2

Fluxes of Electrolytes

The diffusion of ions is governed by the same fundamental laws as fluxes of uncharged substances [see Eq. (3.3.1)].

$$J_i = L_i X_i = \frac{c_i w_i}{N} X_i = -\frac{c_i w_i}{N} \text{grad } \tilde{\mu}_i. \quad (3.3.29)$$

In the case of electrolyte fluxes, the driving force is induced by the negative gradient of the electrochemical potential ($\tilde{\mu}_i$). The coupling coefficient (L_i) can again be considered as a product of concentration (c_i), and mobility (ω_i/N).

Let us first consider the expression 'grad $\tilde{\mu}_i$ '. For a concentration gradient only in the x -direction, instead of the differential operator 'grad', the differential quotient can be applied:

$$\frac{d\tilde{\mu}_i}{dx} = \frac{d}{dx} (\mu_i^0 + RT \ln a_i + z_i F \psi). \quad (3.3.30)$$

In order to simplify the equation, let the solution be close to ideal, i.e. let the activity coefficient be: $f_i \approx 1$, and therefore $a_i \approx c_i$. Furthermore, let the system be under isobaric (grad $p = 0$), and isothermic (grad $T = 0$) conditions. In this case Eq. (3.3.30) becomes

$$\frac{d\tilde{\mu}_i}{dx} = \frac{RT}{c_i} \frac{dc_i}{dx} + z_i F \frac{d\psi}{dx}, \quad (3.3.31)$$

which, when combined with Eq. (3.3.29), results in

$$J_i = -\frac{c_i w_i}{N} \left(\frac{RT}{c_i} \frac{dc_i}{dx} + z_i F \frac{d\psi}{dx} \right), \quad (3.3.32)$$

or, modified using the diffusion coefficient according to Eq. (3.3.6):

$$J_i = -D \left(\frac{dc_i}{dx} + \frac{z_i F c_i}{RT} \frac{d\psi}{dx} \right). \quad (3.3.33)$$

This is the *Nernst-Planck equation*. It contains the differential quotients of the concentration $[c_i(x)]$, and of the electrical potential $[\psi(x)]$. These differential quotients can be integrated only if the corresponding functions are known. This problem has already been discussed with regard to the concentration $[c_i(x)]$ in Section 3.3.1 (see Fig. 3.19). In contrast to Fick's equation, here the function $\psi(x)$ must be known in addition.

The simplest approach is the consideration of linear gradients. This corresponds, for example, to a membrane with large, water filled, and non charged pores where the ions can move freely as in bulk water. Integrating Eq. (3.3.33) for these conditions, the differential quotients are replaced by simple ratios of differences, and instead of the concentration c_i , the mean concentration of both phases $[\bar{c}_i = (c_i^I + c_i^{II})/2]$ appears.

$$J_{ix} = -D \left(\frac{\Delta c_i}{\Delta x} + \frac{z_i F \bar{c}_i}{RT} \frac{\Delta \psi}{\Delta x} \right). \quad (3.3.34)$$

Or, using the permeability coefficient: $P_i = D/\Delta x$ [see Eq. (3.3.9)]:

$$J_{ix} = -P_i \left(\Delta c_i + \frac{z_i F \bar{c}_i}{RT} \Delta \psi \right). \quad (3.3.35)$$

In 1943 D. E. Goldman integrated the Nernst-Planck equation, supposing only the so-called *constant field conditions*, i.e., assuming: $E = -\text{grad } \psi = \text{const}$. The concentration profile results from the bulk concentrations in both phases and from its passive distribution in the electric field. Integrating the Nernst-Planck equation (3.3.33) with these conditions, one gets the following expression:

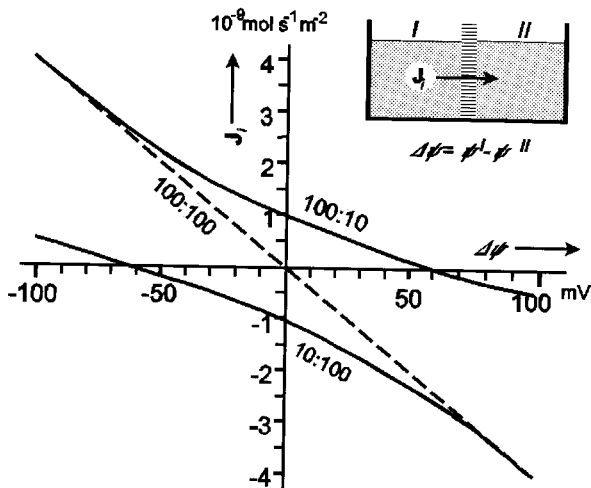
$$J_i = -P_i \beta \frac{c_i^I - c_i^{II} e^{\beta}}{1 - e^{\beta}} \quad \text{with: } \beta = \frac{z_i F}{RT} \Delta \psi. \quad (3.3.36)$$

The function $J = f(\Delta \psi)$ is illustrated in Fig. 3.21. It considers the flux of a monovalent cation ($z_i = +1$) penetrating the membrane with a permeability $P_i = 10^{-7} \text{ m s}^{-1}$. Let the flux (J_i) be positive if it is directed phase I $\Rightarrow$ phase II. A negative J_i , therefore, means that the flux is in the opposite direction. The potential gradient is negative per definition, if ψ decreases from phase I to phase II.

The dotted line in Fig. 3.21 represents a cation flux which is driven only by the electrical potential. ($c_i^I = c_i^{II} = 100 \text{ mM}$). In this case, the flux vanishes ($J_i = 0$) if there is no potential difference ($\Delta \psi = 0$).

If there is an additional driving force resulting from a concentration gradient, the curve is displaced towards either of the solid lines. In this case an ion flux exists even if $\Delta \psi = 0$. If $c_i^I = 100 \text{ mM}$, and $c_i^{II} = 10 \text{ mM}$ then, when $\Delta \psi = 0$, there

Fig. 3.21. The flux (J_i) of a univalent cation ($z_i = +1$, $P_i = 10^{-7} \text{ m s}^{-1}$) driven by a concentration, as well as an electrical potential gradient according to the Goldman equation (3.3.36). The concentrations in the phases *I* and *II* are listed on the curves. Description in the text



is a flux from *I* to *II* (i.e. $J_i > 0$). When the concentrations are reversed, J_i becomes negative. The solid lines cut the abscissa at +60 mV and -60 mV respectively. These points mark where the driving force due to the electric field exactly balances the driving force due to the concentration gradient. This corresponds to the equilibrium situation with $J_i = 0$, described by the Nernst equation [Eq. (3.2.34)].

In fact, the Goldman equation is a good approach even in cases where the linearity of the function $\psi(x)$ is not exactly realized. At the end of the following Section we will consider the role of surface charges.

Let us consider at this point an equation that makes it possible to calculate the ratios of ionic fluxes independently of the functions $c(x)$ and $\psi(x)$. In this context a term must be defined which we need again in further considerations: the flux which is measurable directly by changes of chemical concentrations is called the *net flux* (J). Using radioactive isotopes it is possible to indicate that this net flux in fact results from the difference between two opposite *unidirectional fluxes* J_{12} and J_{21} :

$$J = J_{12} - J_{21}. \quad (3.3.37)$$

In contrast to the unidirectional fluxes, the directions of which are indicated by subscripts (for example: J_{12} , J_{21}), the net flux will be defined as positive, as in the example of Fig. 3.22, if it takes place in the direction corresponding to Eq. (3.3.37).

It must be stressed that these are examples of simple diffusion processes based on thermodynamic fluctuations. Later in Section 3.4.1 (see Fig. 3.23) we will consider systems of co-transport through biological membranes, which show the same kinetics but which must be treated using completely different thermodynamic approaches. These two kinds of transport processes therefore must not be confused.

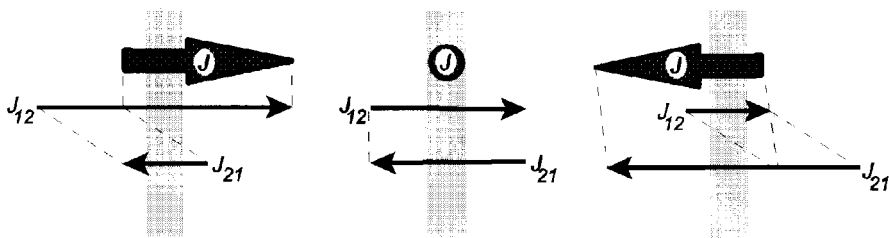


Fig. 3.22. The net flux (J), as the sum of the unidirectional fluxes J_{12} and J_{21}

Considering Eq. (3.3.37), one can write Fick's equation [Eq. (3.3.9)] in the following way:

$$J = J_{12} - J_{21} = -P\Delta c = P(c^I - c^{II}) = Pc^I - Pc^{II}. \quad (3.3.38)$$

From this we can conclude with certain justification:

$$J_{12} = Pc^I; \quad J_{21} = Pc^{II}. \quad (3.3.39)$$

(We will use these formulations again in Section 5.2.1 for compartment analysis.)

To calculate unidirectional fluxes of ions one could use the same approaches as in Eqs. (3.3.39), but just introducing a kind of electrochemical concentration ($\tilde{c}$). This parameter, as we will indicate later, has no further importance. The definition results from the following equation:

$$\tilde{\mu} = \mu^0 + RT \ln c + zF\psi \stackrel{!}{=} \mu^0 + RT \ln \tilde{c} + zF\psi^0. \quad (3.3.40)$$

Therefore: $\tilde{c} = c$, if $\psi^0 = \psi$. This means that a kind of zero-potential will be established, whatever it is. From this definition follows:

$$\ln \tilde{c} = \ln c + \frac{zF}{RT}(\psi - \psi^0), \quad (3.3.41)$$

and:

$$\tilde{c} = ce^{\frac{zF}{RT}(\psi - \psi^0)}. \quad (3.3.42)$$

This equation is of little use since the reference potential ψ^0 is unknown. However, if this relation is substituted in the Eqs. (3.3.39), and if the ratio of the unidirectional fluxes are calculated, all unknown parameters cancel:

$$\frac{J_{12}}{J_{21}} = \frac{c^I}{c^{II}} e^{\frac{zF}{RT}(\psi^I - \psi^{II})}. \quad (3.3.43)$$

(To avoid the accumulation of subscripts we ignored in these derivations the index for the specific substance i .)

This formula is known as *Ussing's equation* which relates unidirectional fluxes to concentrations and electrical potential differences. All parameters of this equation can be measured and the validity of the equation can therefore be checked experimentally. As preconditions for this equation only concentration gradients and electrical potential gradients are considered. If the relation calculated by these gradients, then additional driving forces are involved, such as in processes of active transport.

Further reading: Friedman 1986; Goldman 1943; Kotyk 1988; Stein 1986, 1990; Syganow and vonKlitzing 1999

3.3.3

The Diffusion Potential

The diffusion of an electrolyte, in general, can be considered as a separate diffusion of the dissociated ions along their own electrochemical gradient. Cations and anions, however, may have different mobilities (w_i). The slower diffusion types of ions will lag behind the faster ones. A local potential difference called *diffusion potential* builds up, retarding the faster ions and speeding up the slower ones. Diffusion potentials can occur in homogenous solutions as well as between two phases separated by a membrane which is permeable for the ions.

An equation for the diffusion potential can be derived, postulating the electroneutrality of the sum of all ion fluxes. In this case the cation flux (J_C) induces a flow of charge ($J_C z_C$). In the case of electroneutrality, it must be compensated by the charge transport of anions ($J_A z_A$):

$$J_C z_C + J_A z_A = 0. \quad (3.3.44)$$

Now, we can use the flux equations derived in the previous Section 3.3.2. If linear functions $c(x)$ and $\psi(x)$ are proposed, the simplest flux equations [Eq. (3.3.35)] can be used. Inserting them into Eq. (3.3.44), one gets:

$$z_C P_C \Delta c_C + \frac{z_C^2 F \bar{c}_C P_C}{RT} \Delta \psi + z_A P_A \Delta c_A + \frac{z_A^2 F \bar{c}_A P_A}{RT} \Delta \psi = 0. \quad (3.3.45)$$

Let us take into account that the concentration of the ions (c_C , c_A) depends on the concentration of the salt (c), whereas $c_C = v_C c$, and $c_A = v_A c$. Introducing this, one can rearrange Eq. (3.3.45) and resolve it for $\Delta \psi$ in the following way:

$$\Delta \psi = -\frac{RT}{F} \left(\frac{z_C P_C v_C + z_A P_A v_A}{z_C^2 P_C v_C + z_A^2 P_A v_A} \right) \frac{\Delta c}{c}, \quad (3.3.46)$$

or:

$$\Delta\psi = \frac{RT}{F} \left(\frac{z_C P_C v_C + z_A P_A v_A}{z_C^2 P_C v_C + z_A^2 P_A v_A} \right) \ln \frac{c^I}{c^II}. \quad (3.3.47)$$

The relation: $\Delta c/\bar{c} = \ln(c^I/c^II)$ has already been introduced in Section 3.3.1 in relation to Eq. (3.3.16). It is easy to understand that Eq. (3.3.47) will be transformed into the Nernst Equation [Eq. (3.2.33)], if the membrane becomes semipermeable, i.e. if $P_C = 0$, or if $P_A = 0$.

A better approach for the conditions of membranes will be the flux equation, derived by Goldman for conditions of constant electric field (see Sect. 3.3.2). Introducing this Goldman equation [Eq. (3.3.36)] into the equation for electro-neutrality of fluxes [Eq. (3.3.44)], one gets the following expression:

$$P_C \frac{F\Delta\psi}{RT} \left(\frac{c_C^I - c_C^{II} e^{\frac{F\Delta\psi}{RT}}}{1 - e^{\frac{F\Delta\psi}{RT}}} \right) + P_A \frac{F\Delta\psi}{RT} \left(\frac{c_A^I - c_A^{II} e^{-\frac{F\Delta\psi}{RT}}}{1 - e^{-\frac{F\Delta\psi}{RT}}} \right) = 0. \quad (3.3.48)$$

This equation can also be rearranged and solved for $\Delta\psi$. For this we first transform the denominators of the fractions, using the expression:

$$1 - e^{-x} = -e^{-x}(1 - e^x).$$

This leads to:

$$\frac{F\Delta\psi}{RT(1 - e^{\frac{F\Delta\psi}{RT}})} \left[P_C \left(c_C^I - c_C^{II} e^{\frac{F\Delta\psi}{RT}} \right) - P_A e^{\frac{F\Delta\psi}{RT}} \left(c_A^I - c_A^{II} e^{-\frac{F\Delta\psi}{RT}} \right) \right] = 0. \quad (3.3.49)$$

When $\Delta\psi \Rightarrow 0$ the expression in front of the square brackets will not approach zero. Therefore the sum inside the parentheses must be equal to zero:

$$P_C c_C^I - P_C c_C^{II} e^{\frac{F\Delta\psi}{RT}} - P_A c_A^I e^{\frac{F\Delta\psi}{RT}} + P_A c_A^{II} = 0. \quad (3.3.50)$$

Which gives, after some rearrangements:

$$e^{\frac{F\Delta\psi}{RT}} = \frac{P_A c_A^{II} + P_C c_C^I}{P_A c_A^I + P_C c_C^{II}}, \quad (3.3.51)$$

and:

$$\Delta\psi = \frac{RT}{F} \ln \frac{P_A c_A^{II} + P_C c_C^I}{P_A c_A^I + P_C c_C^{II}}. \quad (3.3.52)$$

This is the *Goldman-Hodgkin-Katz equation* which is commonly used in electrophysiology to calculate diffusion potentials in living cells (mostly it is just

named the *Goldman equation*). This expression also becomes a Nernst equation [Eq. (3.2.33)], introducing the conditions of a semipermeable membrane ($P_A = 0$, or $P_C = 0$).

It is possible to extend this equation to systems containing more than one salt, i.e. various monovalent ions. To take into account the non-ideal character of solutions, chemical activities (a_i) instead of concentrations (c_i) can be used. In this case the *Goldman-Hodgkin-Katz equation* can be written as follows:

$$\Delta\psi = \frac{RT}{F} \ln \frac{\sum_{\text{Anions}} P_A a_A^i + \sum_{\text{Cations}} P_C a_C^e}{\sum_{\text{Anions}} P_A a_A^e + \sum_{\text{Cations}} P_C a_C^i}. \quad (3.3.53)$$

When considering cells, the superscript i in this formula means “internal”, the superscript e “external” concentrations. Correspondingly: $\Delta\psi = \psi^i - \psi^e$.

In a large number of papers experiments are described indicating the applicability of the Goldman equation for various cells and membranes. However, some limitations become obvious when one tries to apply this equation in relation to real cellular conditions. Mostly these are already included in the conditions of the applied flux equation.

The most important limitation comes from the assumption considering a free diffusion of the ions in a homogeneous and constant electric field. Even in the case of large pores in the membrane this assumption is valid only with some approximations. It must be taken into account that the thickness of a biological membrane is just as large as the Debye-Hückel radius of an ion. Additionally, the coefficients of permeability are defined for large systems and can be used for considerations of molecular systems only approximately.

In order to calculate fluxes and transmembrane potentials, there are approaches that take into account the local concentrations of ions in the electrical double layer at the membrane boundary. If the surface potential ψ_0 is known, it is possible to introduce surface concentrations into the Goldman equation instead of the concentration of ions in the bulk phase, using Eqs. (2.4.10) or (3.2.34). In this case, however, the value $\Delta\psi$ of Eq. (3.3.53) no longer means the potential difference between the two bulk phases, as measured in electrophysiology by microelectrodes, but just between the inner and outer surface (see Fig. 2.48). Thus, the diffusion potential of a membrane can be controlled by local surface charges.

Further reading: Goldman 1943; Katchalsky and Curran 1965; Syganow and vonKlitzing 1999

3.4

The Nonequilibrium Distribution of Ions in Cells and Organelles

In recent decades, our knowledge on transport mechanisms in biological membranes has developed rapidly. New insights into structures and molecular mechanisms of transport proteins and ionophores show the occurrence of

complicated systems of control and regulation. Furthermore, the large number of different transport processes in a single cell appear to be closely meshed together. By pH and volume regulation, and by control of internal Ca^{++} concentrations etc. transport processes may furthermore control complex cellular functions, such as protein expression, cell proliferation etc. In this respect, the transmembrane potential also becomes a subject of particular interest, regarding its generation as well as its role in cellular processes.

3.4.1

Ion Transport in Biological Membranes

Figure 3.23 illustrates a classification of various types of ion transport mechanisms in the membranes of cells and organelles. It shows that the simple diffusion of ions, as described in Section 3.3 occurs only in special cases. Processes of simple electrodiffusion are postulated for example for Na^+ and K^+ channels in membranes of nerve and muscle cells. Certainly, to some extent they occur also as *leak flux* in other cells, and of course in the extracellular space of tissues. Furthermore, membrane pores produced by antibiotics, toxins, virus penetrations etc. produce diffusion paths which sometimes indicate a high degree of selectivity. We will come back to the functional role of selective changes of ion permeability of biological membranes later in Section 3.4.2.

Ion transport in biological membranes is mainly achieved through proteins or protein complexes which transport simultaneously two or more ions in a constant stoichiometrical relationship. Such stoichiometrically coupled fluxes are called *co-transport*s. There are two kinds of co-transport systems: in the case of the *symport*, a strongly coupled flux of anions and cations occurs in the same direction. An example of this could be a complex which transfers simultaneously one Cl^- and one K^+ ion through the membrane in the same direction. An *antiport*, in contrast to this, is a system transporting simultaneously two ions of identical charges in opposite directions, for example one K^+ , against one H^+ . If, in a co-transport system an equal number of charges is transported, then the process becomes electroneutral, and therefore does not depend directly on electric field conditions. If not, an electrical current will be the result of the transport. We will call this sort of process *rheogenic*, i.e. "current producing". Rheogenic co-transport processes can be recognized by their electrical conductivity, a property which they have in common with simple diffusion processes. They can be controlled by electric fields.

An *active transport* is a sort of pump, transporting ions or even uncharged molecules against their own chemical or electrochemical gradients. It is "uphill transport", which uses metabolic energy (ΔG , in Fig. 3.23). In most cases this involves so-called transport ATPases, using the energy of the hydrolytic reaction: $\text{ATP} \Rightarrow \text{ADP}$. This mechanism can also run in the opposite direction. In chloroplasts and mitochondria, for example, ATP is synthesized by a "downhill" proton flux (see Sect. 2.2.3, Fig. 2.7).

	Passive transport				Active transport	
	Diffusion		Cotransport		Transport ATPase	
	PORE	CHANNEL	SYMPORT	ANTIORT		
Electro-neutral $\mu_1 z_1 = \mu_2 z_2$			$\text{Na}^+, \text{K}^+ - 2\text{Cl}^-$ $\text{K}^+ - \text{Cl}^-$ $\text{Na}^+ - \text{Cl}^-$	$\text{Cl}^- - \text{HCO}_3^-$ $\text{Cl}^- - \text{Cl}^-$ $\text{Na}^+ - \text{H}^+$ $\text{K}^+ - \text{H}^+$		$\text{Na}^+ - \text{H}^+ - \text{ATPase}$
Rheogen $\mu_1 z_1 \neq \mu_2 z_2$	$\text{Na}^+, \text{K}^+, \text{Cl}^-$	$\text{Na}^+, \text{K}^+, \text{Cl}^-$		$3\text{Na}^+ - \text{Ca}^{++}$ $\text{H}^+ - \text{Ca}^{++}$	$\text{H}^+ - \text{ATPase}$ $\text{Ca}^{++} - \text{ATPase}$	$3\text{Na}^+ - 2\text{K}^+ - \text{ATPase}$

Fig. 3.23. Classification of various systems of ion transporters in biological membranes

Active transport can also be rheogenic. In this case the transport directly induces electric currents and fields (see Sect. 3.5.2, Fig. 3.32). Frequently such types of transport are also called *electrogenic* which means: “generating an electrical membrane potential”. Looking at these two terms accurately, they are not identical! Even an electro-neutral pump can be “electrogenic” if it produces a concentration gradient of ions which subsequently generates a diffusion potential. On the other hand, a rheogenic pump may influence the transmembrane potential only if there is a sufficiently high membrane resistance.

The number of different transport paths in a single membrane of a cell can be rather high. Let us illustrate this for the case of cells of a renal proximal tubule (Fig. 3.24). It is obvious that the fluxes are coupled with each other by the concentrations of their common ions. Co-transport systems not only couple together fluxes of particular ions, but additionally in some cases they are responsible for permeation of small uncharged molecules. This, for example, is the case in the uptake of monosaccharides in the epithelium of the small intestine. In this case a rheogenic Na^+ glucose symport occurs. In a similar way, fluxes of amino acids are coupled with transport of Na^+ , or H^+ ions. In many cases the electrochemical gradient produced by active transporters is the source of energy driving the uphill transport of metabolites.

Modern biotechnological approaches allow new possibilities for studying the molecular mechanisms of these processes. It is possible to isolate the corresponding proteins, to clone them, and to modify specific groups. A large number of kinetic models have been proposed to explain these transport mechanisms. These models treat the processes of transport like enzymatic

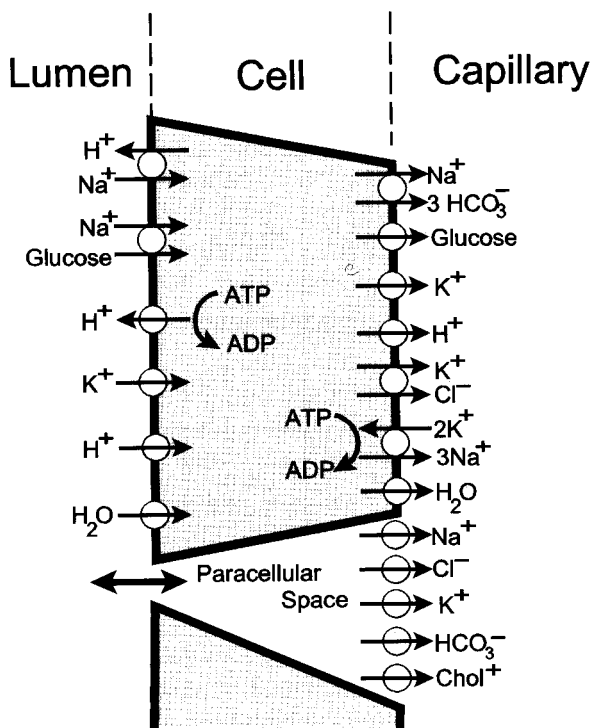


Fig. 3.24. Various types of ion transporters in cells of a renal proximal tubule and in the paracellular space. (After Verkman and Alpern 1987)

reactions. They propose a sequence of binding, of molecular conformation, and of releasing processes. Mechanistic approaches, like a revolving door, or a ferry boat, in fact are not realistic. This would require the rotation or the transmembrane shift of a large molecule, or at least of a part of it in the membrane during the transport process. More likely, periodic oscillations of small side chains of the transporting molecule complexes would occur. Depending on the kind of transport, 1 to 10^6 ions could be translocated per second. The side chains which are responsible for the transport therefore must oscillate in a frequency range up to 1 MHz. We will come back to this number later, discussing effects of external AC fields (Sect. 4.4.5).

The existence of transporters with strongly fixed stoichiometry forces us to rethink the requirement of flux electroneutrality which we postulated in Section 3.3.3 [Eq. (3.3.44)]. Considering rheogenic symports, it is not the electroneutrality of a single flux which is required, but rather the electroneutrality of all fluxes in a membrane of a single cell together. The calculation of the balance of charges and ions in a cell therefore is possible only considering all fluxes. Additionally, changes of fixed charges inside the cell induced by internal pH alterations should be taken into account.

The existence of co-transporters in cells rather than simple diffusion is a sign of optimization. If ionic transport was based on simple processes of electro-

diffusion, then an occasional alteration of the membrane potential would cause an immediate change of all electrolyte fluxes, and subsequently a shift of the internal concentration of all ions in the cell. In contrast, the system of electroneutral co-transporters is independent of the transmembrane potential and will protect the cell against such disturbances.

Simple balance calculations can be applied to describe active transport thermodynamically. Such approaches, of course, do not help to clarify the molecular mechanisms, but they give some insights into the processes of energy transformation. Let us illustrate this, asking the question: how much power is required for an ion pump to maintain a steady state of ions in the cell?

A system like that of Fig. 3.4B is determined by the active transport (J_A) as well as by the passive flux (J_i). In Section 3.1.4 we introduced the dissipation function $\Phi = \sigma T$ [Eq. (3.1.64)], which must be larger than 0. According to Eq. (3.1.64) for our system it amounts to:

$$\Phi = J_A X_A + J_i X_i \quad \text{for:} \quad \Phi > 0. \quad (3.4.1)$$

Let us look at a real example. The glucose uptake (J_G) of a cell in the intestine is driven by the passive influx of sodium (J_{Na}). Using Eq. (3.4.1) we obtain:

$$J_G X_G + J_{Na} X_{Na} > 0. \quad (3.4.2)$$

If v equivalents of sodium ions are transported for each mole of glucose then:

$$J_G = v J_{Na}.$$

Introducing this into Eq. (3.4.2) and assuming that both fluxes are not equal to zero, it follows that:

$$v X_G + X_{Na} > 0, \quad (3.4.3)$$

and thus:

$$X_{Na} > -v X_G. \quad (3.4.4)$$

Let us now replace the forces (X) by the differences of the corresponding chemical or electrochemical potential (see Sect. 3.3.1), we obtain:

$$-v \Delta \mu_G < \Delta \tilde{\mu}_{Na}. \quad (3.4.5)$$

Using Eqs. (3.1.33) and (3.1.41), and the conditions: $\Delta T = 0$ and $\Delta p = 0$, we get:

$$v RT \ln \frac{a_G^i}{a_G^e} < - \left(RT \ln \frac{a_{Na}^i}{a_{Na}^e} + F \Delta \psi \right), \quad (3.4.6)$$

where $\Delta \psi = \psi_i - \psi_e$, and after rearrangement:

$$\left(\frac{a_G^i}{a_G^e}\right)^v < \frac{a_{Na}^e}{a_{Na}^i} e^{-\frac{F\Delta\psi}{RT}}. \quad (3.4.7)$$

This equation allows us to calculate the maximal rate of enrichment of glucose in the cell which can be achieved for a given electrochemical gradient of sodium ions. Assuming that the membrane potential of the cell is: $\Delta\psi = -50$ mV, and the relation of sodium ions: $a_{Na}^e/a_{Na}^i = 10$ ($T = 300$ K), it follows:

$$\left(\frac{a_G^i}{a_G^e}\right)^v < 69. \quad (3.4.8)$$

If the fluxes are coupled 1:1 ($v = 1$), this process can give a maximum enrichment of glucose by a factor of 69, when the pump is performing optimally.

Similar calculations can also be applied to types of active transport which are driven by chemical reactions, as for example transport ATPases. In this case in the equation of the dissipation function [Eq. (3.4.1)], the reaction rate (as a sort of scalar flux) and the chemical affinity of the energy supplying reaction [Eq. (3.1.75)] must be included.

This calculation shows that the intensity of the pump which is necessary to build up a certain concentration gradient depends both on the coupling stoichiometry of the fluxes and on the passive back flow. This means that it is not only the power of the pump which is responsible for the achieved level of steady state, but also the conductivity or the permeability of the substance being considered, allowing it to flow backwards. This is illustrated in the scheme of Fig. 3.4B: the power of the pump must be higher if a greater difference in the levels of the vessels is required, or if the outflow is to be faster.

Let us briefly consider a problem which has been discussed for a long time, but which is not yet resolvable because of lack of experimental approaches. Can the condition: $\text{grad } T = 0$ really be valid for biological membranes? It is interesting to remember that very small temperature differences between the two sides of a cell membrane may cause a considerable temperature gradient. Let us suppose that a fast exothermic enzymatic reaction on one side of the membrane continuously produces heat, and in this way increases the local temperature by only 0.01°C in relation to the opposite membrane surface. In this case a temperature gradient of $0.01/10^{-8} = 10^6 \text{ K m}^{-1}$ is built up. This could be sufficient to explain active transport as a kind of thermodiffusion. A pressure difference of 130 kPa could be induced in this way (Spanner 1954). These are, of course, just interesting speculations because despite some model experiments, we cannot yet check this experimentally (see Harth and Vaupel 1971; Tasaka 1975).

Further reading: Bernhardt 1994; Fleischer et al. 1989; Hille 1992; Hoffmann and Simonsen 1989; Kotyk et al. 1988; Lauger 1991; Stein 1986, 1990; Yeagle 1991

3.4.2

The Network of Cellular Transporters. The Cell as an Accumulator of Electrochemical Energy

If a cell were only a poly-electrolyte system without metabolically driven ion pumps it would remain in a state of Donnan equilibrium. This means that there would be a Donnan distribution of the mobile ions, and as a result, a Donnan osmotic pressure (see Sect. 3.2.3). In the living cell, however, active transport systems driven by metabolic energy (Fig. 3.23), modify this ionic composition, as shown schematically in the model of Fig. 3.4B. The living cell therefore reaches a steady state, i.e. a stationary state of nonequilibrium (see Fig. 3.6).

The function of this particular ionic state in the process of life can be considered from several aspects. One of these is that the ion pumps raise the electrochemical energy of the cell to a higher level. Thus the cell can be regarded as a kind of electrochemical energy store which can be easily tapped. Furthermore, the shift of a system from the equilibrium into a nonequilibrium state is the precondition for its homeostatic regulation. This, by the way, is also the reason for the increased temperature in homeothermic animals. The setting up of a concentration gradient of ions across the membrane makes the cells able to control and regulate an intracellular environment which is the precondition of various cellular processes.

What are the immediate effects of ionic pumps on the cell?

- They control and regulate the internal ionic milieu. In this way, steep gradients of the electrochemical potentials of particular ions are built up, essentially without changing the total internal ionic concentration. The internal potassium concentration of animal cells, for example, is usually much higher than the external one. Simultaneously, however, the sodium concentration is lower to the same degree. The sum of both of these ions in the cytoplasm, taken together, is nearly the same as in the external medium.
- In the case of rheogenic pumps, they directly induce transmembrane potentials. In this case the pumps are called electrogenic.
- They can produce a direct osmotic effect, changing the concentration of osmotically active substances.
- They can establish particular internal ionic conditions, controlling, for example, the extremely low intracellular calcium concentration.

Some direct consequences of the active transport processes can be construed from the effects of stopping the pump by the use of specific inhibitors. In this case effects can be observed like Donnan-osmotic swelling, internal pH shifts, an increase in the internal calcium concentration, a change of transmembrane potential etc. Mostly, using such inhibitors, the overall internal ionic conditions are altered.

In Fig. 3.24 for example, the multitude of transporters is illustrated for kidney tubule cells. There are 13 different transport systems shown which determine the cellular milieu and another five other fluxes between the luminal and serosal

surfaces of the epithelium across the paracellular gap. This picture in fact is incomplete as, for example, Ca^{++} fluxes are not shown, and the diagram does not include the intracellular organelles with their own transporters.

Using this example, we will illustrate the interconnections of these transport properties qualitatively, following for example one particular path: transport ATPases pump protons out of the cell, others decrease the internal sodium content, and in the same way enrich the cytoplasm with potassium. Extruding positive charges, both processes induce an inside negative transmembrane potential. Simultaneously, an electrochemical sodium gradient is generated which drives a "passive" sodium flux outside-in. This influx, however, is made by a glucose-sodium co-transporter. It therefore leads automatically to a glucose accumulation in the cell. The glucose finally diffuses in its own concentration gradient on the opposite side of the cell layer from the cytoplasm into the capillary.

All these manifold transporters occurring in a single cell respond to different stimulants. Some of them become active only if a particular pH exists, others if the internal calcium concentration is raised. There are voltage sensitive transporters responding to particular transmembrane potentials, or others which respond to mechanical stress. The electroneutral $\text{Na}^+ \text{H}^+$ antiporter which is present in most animal cells, merits particular attention. Under physiological conditions, at neutral pH_i it is inactive. However, if the internal pH increases, it becomes activated. This property qualifies it to be a volume regulating system. This mechanism was demonstrated in the case of lymphocytes. It has also been shown that this $\text{Na}^+ \text{H}^+$ antiporter can be activated by a multitude of substances including hormones, growth factors, lectins etc. These substances alter the above mentioned pH threshold. This seems to be an important control mechanism for the regulation of complex biological phenomena.

After this overview of some more or less cell-based physiological considerations we will now return to biophysical aspects and will look into the origin of the transmembrane potential of living cells. We started in Section 2.4.1 with the general definition of the electrical potential. According to this, the electrical potential $[\psi(x, y, z)]$ is a scalar state parameter in 3-dimensional space, similar to temperature (T), or pressure (p). The function $\psi(x)$ is mostly used to characterize the function of the potential along a line, perpendicular through the membrane (Figs. 2.21, 2.48). As *transmembrane potential* ($\Delta\psi$) the potential difference is defined between two points, one on the inside, the other on the outside of the membrane, each at a suitable distance from it (Fig. 2.48). The sign of this difference results from its definition:

$$\Delta\psi = \psi_i - \psi_e. \quad (3.4.9)$$

Names such as Donnan potential, diffusion potential, Nernst potential, are just expressions describing the mechanisms which can give rise to the electrical transmembrane potential and do not refer in any way to different kinds of potentials which might exist simultaneously. There is only one electrical potential $\psi(x, y, z, t)$ at a given point in space (x, y, z), and at a given time (t). In Fig. 2.46, the function $\psi(x)$ illustrates this function in a very simplified way. It

includes the transmembrane potential and the two surface potentials at both boundaries.

We have already learned that processes of active transport can be rheogenic (Fig. 3.23). If the transported charges can be rapidly neutralized by other fluxes, as for example by the Cl^- exchange in the membrane of human erythrocytes, then a rheogenic pump has no direct electrical consequences for the cell. If, however, no such short-circuit flux exists, the transported net charges build up a transmembrane potential, and the rheogenic pump becomes electrogenic.

Classical electrophysiology assumed that the transmembrane potential was essentially a K^+ diffusion potential. This is likely to be true for some cells, however, it has been established that for many cells electrogenic pumps are exclusively responsible for the membrane potential. Inhibition of the pumps in this case immediately leads to changes of $\Delta\psi$. Additionally, in such cells transmembrane potentials appear to be independent of external potassium concentrations (see: Bashford and Pasternak 1986).

In any case, the $\text{Na}^+ \text{K}^+$ ATPase, occurring in nearly all cell membranes, generates an electrochemical gradient of sodium and potassium. For most animal cells a relation near 1:10 occurs for $a_{\text{K}}^i > a_{\text{K}}^e$, and $a_{\text{Na}}^i < a_{\text{Na}}^e$. Chloride ions are distributed mostly passively, according to the Nernst equation. This nonequilibrium distribution of the cations can lead to a diffusion potential which can be calculated by the Goldman equation [Eq. (3.3.53)] as follows:

$$\Delta\psi = \frac{RT}{F} \ln \frac{P_{\text{Cl}} a_{\text{Cl}}^i + P_{\text{K}} a_{\text{K}}^e + P_{\text{Na}} a_{\text{Na}}^e}{P_{\text{Cl}} a_{\text{Cl}}^e + P_{\text{K}} a_{\text{K}}^i + P_{\text{Na}} a_{\text{Na}}^i}. \quad (3.4.10)$$

Even if the internal ion activities a_{K}^i and a_{Na}^i remain constant, the diffusion potential ($\Delta\psi$) can vary widely because of changing permeabilities (P_i). The limits of such variations can be easily obtained from Eq. (3.4.10):

For $P_{\text{K}} \gg P_{\text{Na}}, P_{\text{Cl}}$ Eq. (3.4.10) reduces to:

$$\Delta\psi_{\text{K}} = \frac{RT}{F} \ln \frac{a_{\text{K}}^e}{a_{\text{K}}^i}, \quad (3.4.11)$$

and for $P_{\text{Na}} \gg P_{\text{K}}, P_{\text{Cl}}$ it follows that

$$\Delta\psi_{\text{Na}} = \frac{RT}{F} \ln \frac{a_{\text{Na}}^e}{a_{\text{Na}}^i}. \quad (3.4.12)$$

The Goldman equation [Eq. (3.4.10)], therefore, reduces to a Nernst equation [Eq. (3.2.33)] which was derived for such kinds of semi-permeable membranes. If the typical relations of activities for sodium and potassium, as mentioned before, are inserted into Eqs. (3.4.11) and (3.4.12), then it is easy to understand that $\Delta\psi_{\text{K}} < 0$, and $\Delta\psi_{\text{Na}} > 0$.

This situation is illustrated in Fig. 3.25. The electrochemical gradients of potassium and sodium which are generated using metabolic energy can be